

(those found in the body fluids), but it can also respond to individual foreign proteins. The **cellular immune system** destroys host cells infected by viruses and also destroys some parasites and foreign tissues.

At the heart of the humoral immune response are soluble proteins called **antibodies** or **immunoglobulins**, often abbreviated **Ig**. Immunoglobulins bind bacteria, viruses, or large molecules identified as foreign and target them for destruction. Making up 20% of blood protein, the immunoglobulins are produced by **B lymphocytes**, or **B cells**, so named because they complete their development in the bone marrow.

The agents at the heart of the cellular immune response are a class of **T lymphocytes**, or **T cells** (so called because the latter stages of their development occur in the thymus), known as **cytotoxic T cells** (**T_C cells**, also called killer T cells). Recognition of infected cells or parasites involves proteins called **T-cell receptors** on the surface of T_C cells. Receptors are proteins, usually found on the outer surface of cells and extending through the plasma membrane, that recognize and bind extracellular ligands, thus triggering changes inside the cell.

In addition to cytotoxic T cells, there are **helper T cells** (**T_H cells**), whose function it is to produce soluble signaling proteins called cytokines, which include the interleukins. T_H cells interact with macrophages. The T_H cells participate only indirectly in the destruction of infected cells and pathogens, stimulating the selective proliferation of those T_C and B cells that can bind to a particular antigen. This process, called **clonal selection**, increases the number of immune system cells that can respond to a particular pathogen. A host organism needs time, often days, to mount an immune response against a new antigen, but **memory cells** permit a rapid response to pathogens previously encountered. A vaccine to protect against a particular viral infection often consists of weakened or killed virus or isolated proteins from a viral or bacterial protein coat. When injected into a person, the vaccine generally does not cause an infection and illness, but it effectively “teaches” the immune system what the viral particles look like, stimulating the production of memory cells. On subsequent infection, these cells can bind to the virus and trigger a rapid immune response.

The importance of T_H cells is dramatically illustrated by the epidemic produced by HIV (human immunodeficiency virus), the virus that causes AIDS (acquired immune deficiency syndrome). T_H cells are the primary targets of HIV infection; elimination of these cells progressively incapacitates the entire immune system.

P1 Each recognition protein of the immune system, either a T-cell receptor or an antibody produced by a B cell, specifically binds some particular chemical structure, distinguishing it from virtually all others. Humans are capable of producing more than 10^8 different antibodies with distinct binding specificities. Given this extraordinary diversity, any chemical structure on

the surface of a virus or an invading cell will most likely be recognized and bound by one or more antibodies. Antibody diversity is derived from random reassembly of a set of immunoglobulin gene segments through genetic recombination mechanisms that are discussed in Chapter 25 (see Fig. 25-42).

A specialized lexicon is used to describe the unique interactions between antibodies or T-cell receptors and the molecules they bind. Any molecule or pathogen capable of eliciting an immune response is called an **antigen**. An antigen may be a virus, a bacterial cell wall, or an individual protein or other macromolecule. A complex antigen may be bound by several different antibodies. An individual antibody or T-cell receptor binds only a particular molecular structure within the antigen, called its **antigenic determinant** or **epitope**.

It would be unproductive for the immune system to respond to small molecules that are common intermediates and products of cellular metabolism. Molecules of $M_r < 5,000$ are generally not antigenic. However, when small molecules are covalently attached to large proteins in the laboratory, they can be used to elicit an immune response. These small molecules are called **haptens**. The antibodies produced in response to protein-linked haptens will then bind to the same small molecules in their free form. Such antibodies are sometimes used in the development of analytical tests described later in this chapter or as a specific ligand in affinity chromatography (see Fig. 3-17c). We now turn to a more detailed description of antibodies and their binding properties.

Antibodies Have Two Identical Antigen-Binding Sites

Immunoglobulin G (IgG) is the major class of antibody molecule and one of the most abundant proteins in the blood serum. IgG has four polypeptide chains: two large ones, called heavy chains, and two smaller ones, called light chains, linked by noncovalent and disulfide bonds into a complex of M_r 150,000. The heavy chains of an IgG molecule interact at one end, then branch to interact separately with the light chains, forming a Y-shaped molecule (Fig. 5-20). At the “hinges” separating the base of an IgG molecule from its branches, the immunoglobulin can be cleaved with proteases. Cleavage with the protease papain liberates the basal fragment, called **Fc** because it usually crystallizes readily, and the two branches, called **Fab**, the antigen-binding fragments. Each branch has a single antigen-binding site.

The fundamental structure of immunoglobulins was first established by Gerald Edelman and Rodney Porter in the 1960s. Each chain is made up of identifiable domains. Some are constant in sequence and structure from one IgG to the next; others are variable. The constant domains have a characteristic structure known as the **immunoglobulin fold**, a well-conserved structural motif in the $\alpha\beta$ class of proteins (Chapter 4). There are three of these constant domains in each heavy chain and one in

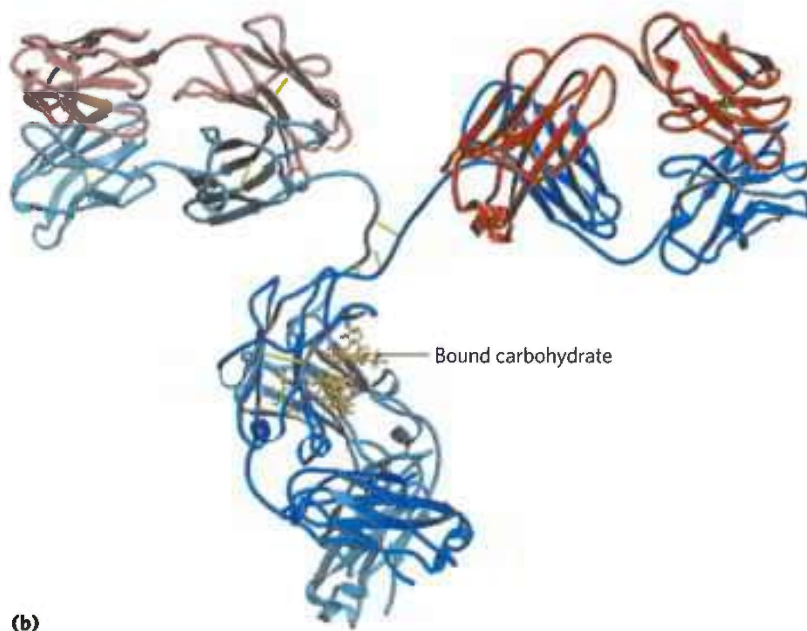
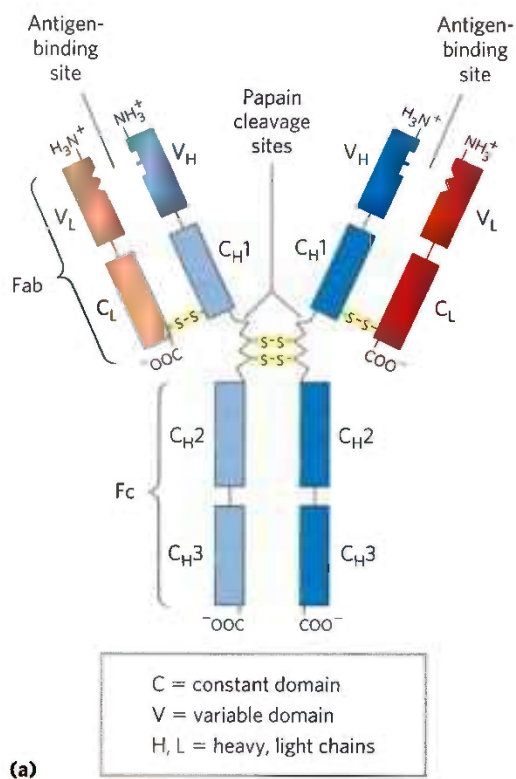


FIGURE 5-20 Immunoglobulin G. (a) Pairs of heavy and light chains combine to form a Y-shaped molecule. Two antigen-binding sites are formed by the combination of variable domains from one light (V_L) chain and one heavy (V_H) chain. Cleavage with papain separates the Fab and Fc portions of the protein in the hinge region. The Fc portion also contains bound

carbohydrate (shown in (b)). (b) A ribbon model of an IgG molecule. Although the molecule has two identical heavy chains (two shades of blue) and two identical light chains (two shades of red), it crystallized in the asymmetric conformation shown here. Conformational flexibility is important to the function of immunoglobulins. [(b) Data from PDB ID 1IGT, L. J. Harris et al., *Biochemistry* 36:1581, 1997.]

each light chain. The heavy and light chains also have one variable domain each, in which most of the variability in amino acid sequence is found. The variable domains associate to create the antigen-binding site (Fig. 5-20), allowing formation of an antigen-antibody complex (Fig. 5-21).

In many vertebrates, IgG is but one of five classes of immunoglobulins. Each class has a characteristic type of heavy chain, denoted α , δ , ϵ , γ , and μ for IgA, IgD, IgE, IgG, and IgM, respectively. Two types of light chain, κ and λ , occur in all classes of immunoglobulins. The overall structures of IgD and IgE are similar to that of IgG. IgM occurs either in a monomeric, membrane-bound form or in a secreted form that is a cross-linked pentamer of this basic structure (Fig. 5-22). IgA, found principally in secretions such as saliva, tears, and milk, can be a monomer, a dimer, or a trimer. IgM is the first antibody to be made by B lymphocytes and the major antibody in the early stages of a primary immune response. Some B cells soon begin to produce IgD (with the same antigen-binding site as the IgM produced by the same cell), but the particular function of IgD is less clear.

The IgG described above is the major antibody in secondary immune responses, which are initiated by a class of B cells called memory B cells. As part of the organism's ongoing immunity to antigens already encountered

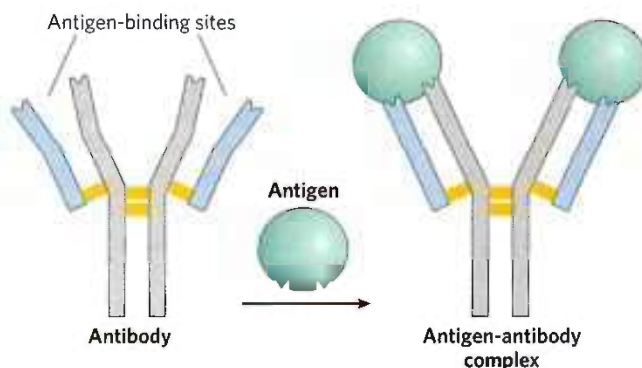


FIGURE 5-21 Binding of IgG to an antigen. To generate an optimal fit for the antigen, the binding sites of IgG often undergo slight conformational changes. Such induced fit is common to many protein-ligand interactions.

and dealt with, IgG is the most abundant immunoglobulin in the blood. When IgG binds to an invading bacterium or virus, it activates certain leukocytes such as macrophages to engulf and destroy the invader, and it also activates some other parts of the immune response. Receptors on the macrophage surface recognize and bind the Fc region of IgG. When these Fc receptors bind an

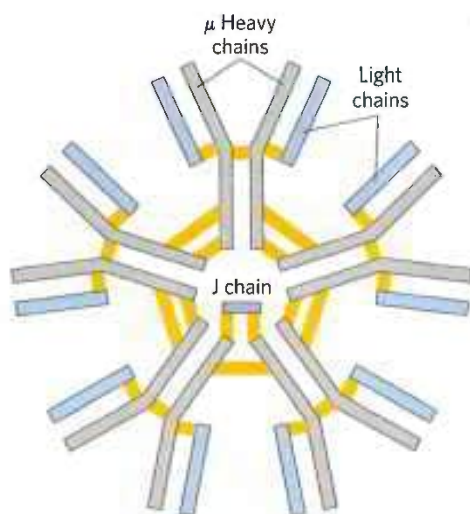


FIGURE 5-22 IgM pentamer of immunoglobulin units. The pentamer is cross-linked with disulfide bonds (yellow). The J chain is a polypeptide of M_r 20,000 found in both IgA and IgM.

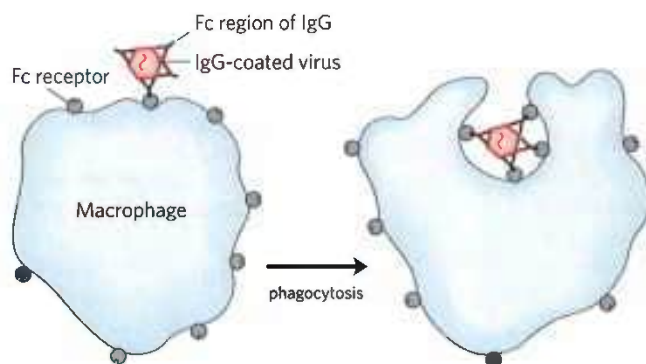


FIGURE 5-23 Phagocytosis of an antibody-bound virus by a macrophage. The Fc regions of antibodies bound to the virus now bind to Fc receptors on the surface of a macrophage, triggering the macrophage to engulf and destroy the virus.

antibody-pathogen complex, the macrophage engulfs the complex by phagocytosis (Fig. 5-23).

 IgE plays an important role in the allergic response, interacting with basophils (phagocytic leukocytes) in the blood and with histamine-secreting cells called mast cells, which are widely distributed in tissues. This immunoglobulin binds, through its Fc region, to special Fc receptors on the basophils or mast cells. In this form, IgE serves as a receptor for antigen. If antigen is bound, the cells are induced to secrete histamine and other biologically active amines that cause the dilation and increased permeability of blood vessels. These effects on the blood vessels are thought to facilitate the movement of immune system cells and proteins to sites of inflammation. They also produce the symptoms normally associated with allergies. Pollen and other allergens are recognized as foreign, triggering an immune response normally reserved for pathogens. ■

Antibodies Bind Tightly and Specifically to Antigen

P2 The binding specificity of an antibody is determined by the amino acid residues in the variable domains of its heavy and light chains. Some of those residues, particularly those lining the antigen-binding site, are hypervariable—especially likely to differ. Specificity is conferred by chemical complementarity between the antigen and its specific binding site. For example, a binding site with a negatively charged group may bind an antigen with a positive charge in the complementary position. In many instances, complementarity is achieved interactively as the structures of antigen and binding site influence each other as they come closer together. Conformational changes in the antibody and/or the antigen then allow the complementary groups to interact fully. This is an example of induced fit. The complex of a peptide derived from HIV (a model antigen) and an Fab molecule, shown in Figure 5-24, illustrates some of these properties. The changes in structure observed on antigen binding are particularly striking in this example.

A typical antibody-antigen interaction is quite strong, characterized by K_d values as low as 10^{-10} M (recall that a lower K_d corresponds to a stronger binding interaction; see Table 5-1). The K_d reflects the energy derived from the hydrophobic effect and the various ionic, hydrogen-bonding, and van der Waals interactions that stabilize the binding. The binding energy required to produce a K_d of 10^{-10} M is about 65 kJ/mol.

The Antibody-Antigen Interaction Is the Basis for a Variety of Important Analytical Procedures

The extraordinary binding affinity and specificity of antibodies make them valuable analytical reagents. Two types of antibody preparations are in use: polyclonal and monoclonal. **Polyclonal antibodies** are those produced by many different B lymphocytes responding to one antigen, such as a protein injected into an animal. Cells in the population of B lymphocytes produce antibodies that bind specific, different epitopes within the antigen. Thus, polyclonal preparations contain a mixture of antibodies that recognize different parts of the protein. **Monoclonal antibodies**, in contrast, are synthesized by a population of identical B cells (a **clone**) grown in cell culture. These antibodies are homogeneous, all recognizing the same epitope.

The specificity of antibodies has practical uses. In a versatile analytical technique, an antibody is attached to a reagent that makes it easy to detect (Fig. 5-25). When the antibody binds the target protein, the label reveals the presence of the protein in a solution or its location in a gel, or even in a living cell (Fig. 5-25a). In one application of this technique, an **immunoblot** or **Western blot** assay (Fig. 5-26b), proteins that have been separated by gel electrophoresis are transferred electrophoretically to a nitrocellulose membrane.

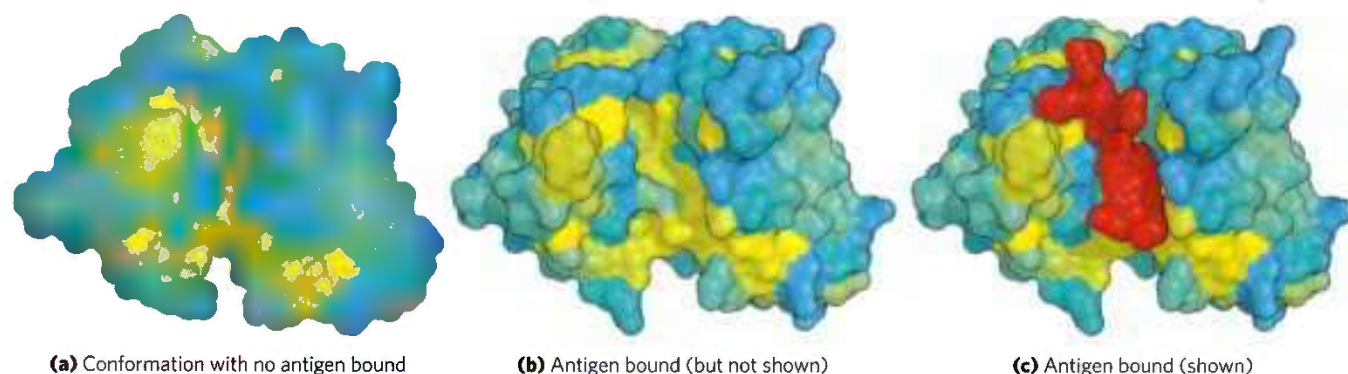


FIGURE 5-24 Induced fit in the binding of an antigen to IgG.

The Fab fragment of an IgG molecule is shown here with the surface contour colored to represent hydrophobicity. Hydrophobic surfaces are yellow and hydrophilic surfaces are blue, with shades of blue to green to yellow in between. (a) View of the Fab fragment in the absence of antigen (a small peptide derived from HIV), looking down on the antigen binding site. (b) The

same view, but with the Fab fragment in the "bound" conformation with the antigen omitted to provide an unobstructed view of the altered binding site. Note how the hydrophobic binding cavity has enlarged and several groups have shifted position. (c) The same view as (b) but with the antigen (red) in the binding site. [Data from (a) PDB ID 1GGC, R. L. Stanfield et al., *Structure* 1:83, 1993; (b, c) PDB ID 1GGI, J. M. Rini et al., *Proc. Natl. Acad. Sci. USA* 90:6325, 1993.]

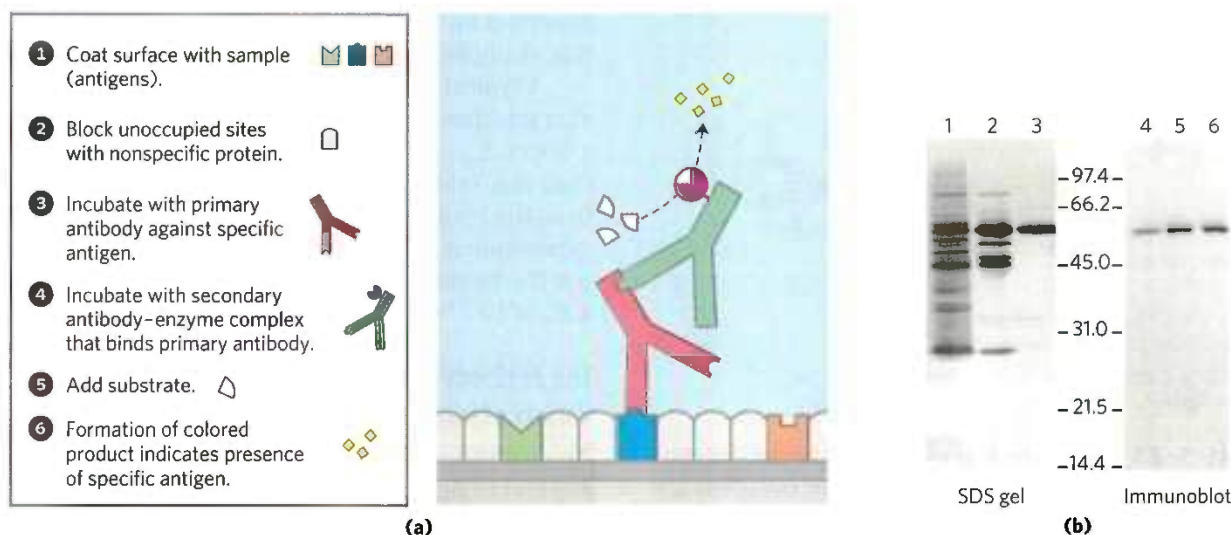


FIGURE 5-25 Antibodies as analytical reagents. (a) A schematic representation of the use of the specific reaction of an antibody with its antigen to identify and quantify a specific protein in a complex sample. (b) One common application is the immunoblot. Lanes 1 to 3 are from an SDS gel; samples from successive stages in the purification of a protein kinase were separated

and stained with Coomassie blue. Lanes 4 to 6 show the same samples, but these were electrophoretically transferred to a nitrocellulose membrane after separation on an SDS gel. The membrane was then "probed" with antibody against the protein kinase. The numbers between the SDS gel and the immunoblot indicate M_r in thousands. [(b) State of Wisconsin Lab of Hygiene, Madison, WI.]

After washing, the membrane is treated successively with primary antibody, secondary antibody linked to enzyme, and substrate. A colored precipitate forms only along the band containing the protein of interest. Immunoblotting allows the detection of a minor component in a sample and provides an approximation of its molecular weight.

We will encounter other aspects of antibodies in later chapters. They are extremely important in medicine and can tell us much about the structure of proteins and the action of genes.

SUMMARY 5.2 Complementary Interactions between Proteins and Ligands: The Immune System and Immunoglobulins

- The immune response is mediated by interactions among an array of specialized leukocytes and their associated proteins. T lymphocytes produce T-cell receptors. B lymphocytes produce immunoglobulins.
- Humans have five classes of immunoglobulins, each with different biological functions. The most abundant class is IgG, a Y-shaped protein with two heavy chains

and two light chains. The domains near the upper ends of the Y are hypervariable within the broad population of IgGs and form two antigen-binding sites.

■ A given immunoglobulin generally binds to only a part, called the epitope, of a large antigen. Binding often involves a conformational change in the IgG, an induced fit to the antigen.

■ The exquisite binding specificity of immunoglobulins is exploited in analytical techniques such as immunoblotting.

5.3 Protein Interactions Modulated by Chemical Energy: Actin, Myosin, and Molecular Motors

Organisms move. Cells move. Organelles and macromolecules within cells move. Most of these movements arise from the activity of a fascinating class of protein-based molecular motors. Fueled by chemical energy, usually derived from ATP, large aggregates of motor proteins undergo cyclic conformational changes that accumulate into a unified, directional force—the tiny force that pulls apart chromosomes in a dividing cell, and the immense force that levers a pouncing, quarter-ton jungle cat into the air.

The interactions among motor proteins, as you might predict, feature complementary arrangements of ionic, hydrogen-bonding, and hydrophobic groups at protein binding sites. In motor proteins, however, the resulting interactions achieve exceptionally high levels of spatial and temporal organization.

Motor proteins underlie the migration of organelles along microtubules, the motion of eukaryotic and bacterial flagella, and the movement of some proteins along DNA. Here, we focus on the well-studied example of the contractile proteins of vertebrate skeletal muscle as a paradigm for how proteins translate chemical energy into motion.

The Major Proteins of Muscle Are Myosin and Actin

The contractile force of muscle is generated by the interaction of two proteins, myosin and actin. These proteins are arranged in filaments that undergo transient interactions and slide past each other to bring about contraction. Together, actin and myosin make up more than 80% of the protein mass of muscle.

Myosin (M_r 520,000) has six subunits: two heavy chains (each of M_r 220,000) and four light chains (each of M_r 20,000). The heavy chains account for much of the overall structure. At their carboxyl termini, they are arranged as extended α helices, wrapped around each other in a fibrous, left-handed coiled coil similar to that of α -keratin (Fig. 5-26). At its amino terminus, each heavy

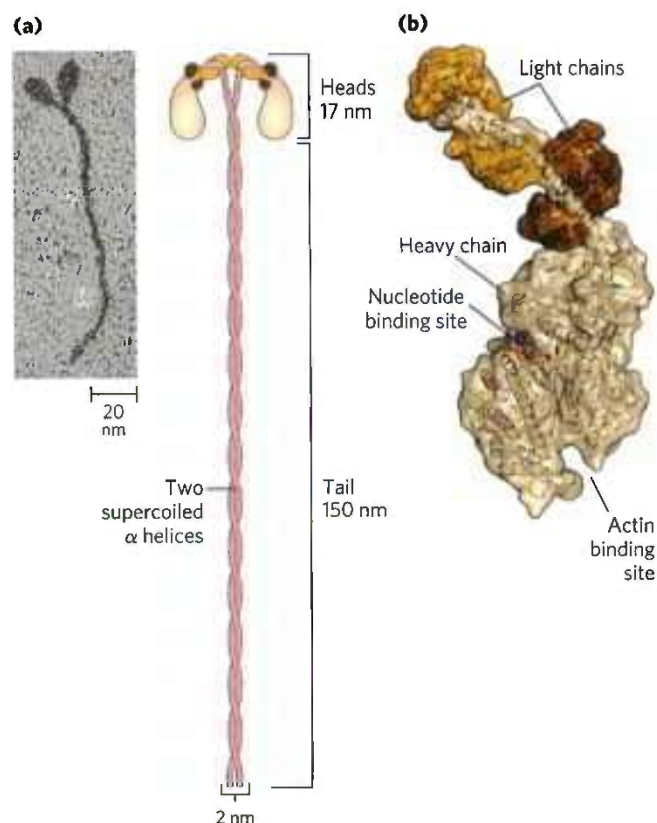


FIGURE 5-26 Myosin. (a) Myosin has two heavy chains: the carboxyl termini forming an extended coiled coil tail and the amino termini having clublike globular head domains. Two myosin light chains are associated with each myosin head. (b) A three-dimensional representation of the myosin head, showing the binding sites for nucleotide (ATP) and actin. [(a) Research from Takeshi Katayama, et al. "Stimulatory effects of arachidonic acid on myosin ATPase activity and contraction of smooth muscle via myosin motor domain," *Am. J. Physiol. Heart Circ. Physiol.* Vol 298, Issue 2, pp. H505–H514, February 2010, Fig. 6b. (b) Data from PDB ID 2MYS, I. Rayment et al., *Science* 261:50, 1993.]

chain has a large globular domain containing a site where ATP is hydrolyzed. The light chains are associated with the globular domains.

In muscle cells, molecules of myosin aggregate to form structures called **thick filaments** (Fig. 5-27a). These rodlike structures are the core of the contractile unit. Within a thick filament, several hundred myosin molecules are arranged with their fibrous "tails" associated to form a long, bipolar structure. The globular domains project from either end of this structure, in regular stacked arrays.

The second major muscle protein, **actin**, is abundant in almost all eukaryotic cells. In muscle, molecules of monomeric actin, called G-actin (globular actin; M_r 42,000), associate to form a long polymer called F-actin (filamentous actin). The **thin filament** consists of F-actin (Fig. 5-27b), along with the proteins troponin and tropomyosin (discussed below). The filamentous parts of thin filaments assemble as successive monomeric actin

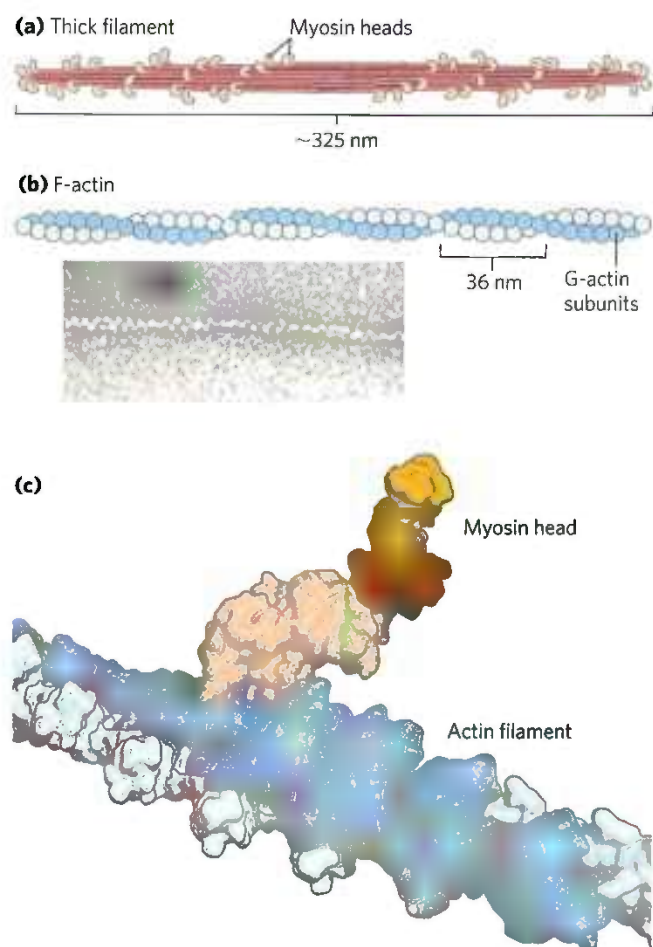


FIGURE 5-27 The major components of muscle. (a) Myosin aggregates to form a bipolar structure called a thick filament. (b) F-actin is a filamentous assemblage of G-actin monomers that polymerize two by two, giving the appearance of two filaments spiraling about one another in a right-handed fashion. (c) Space-filling model of an actin filament with one myosin head bound to an actin monomer within the filament. [(b) Research from Dr. Roger W. Craig PhD, University of Massachusetts Medical School. (c) Data from PDB ID 2MYS, I. Rayment et al., *Science* 261:50, 1993; PDB ID 6BNQ, P. S. Gurel et al., *eLife* 6, 2017, doi 10.7554/eLife.31125]

molecules add to one end. Upon addition, each monomer binds ATP, then hydrolyzes it to ADP, so every actin molecule in the filament is complexed to ADP. This ATP hydrolysis by actin functions only in the assembly of the filaments; it does not contribute directly to the energy expended in muscle contraction. Each actin monomer in the thin filament can bind tightly and specifically to one myosin head group (Fig. 5-27c).

Additional Proteins Organize the Thin and Thick Filaments into Ordered Structures

Skeletal muscle consists of parallel bundles of **muscle fibers**, each fiber a single, very large, multinucleated cell, 20 to 100 μm in diameter, formed from

many cells fused together; a single fiber often spans the length of the muscle. Each fiber contains about 1,000 **myofibrils**, 2 μm in diameter, each consisting of a vast number of regularly arrayed thick and thin filaments complexed to other proteins (Fig. 5-28). A system of flat membranous vesicles called the **sarcoplasmic reticulum** surrounds each myofibril. Examined under the electron microscope, muscle fibers reveal alternating regions of high electron density and low electron density, called the **A bands** and **I bands** (Fig. 5-28b, c). The A and I bands arise from the arrangement of thick and thin filaments, which are aligned and partially overlapping. The I band is the region of the bundle that in cross section would contain only thin filaments. The darker A band stretches the length of the thick filament and includes the region where parallel thick and thin filaments overlap. Bisecting the I band is a thin structure called the **Z disk**, perpendicular to the thin filaments and serving as an anchor to which the thin filaments are attached. The A band, too, is bisected by a thin line, the **M line** or M disk, a region of high electron density in the middle of the thick filaments. The entire contractile unit, consisting of bundles of thick filaments interleaved at either end with bundles of thin filaments, is called the **sarcomere**. The arrangement of interleaved bundles allows the thick and thin filaments to slide past each other (by a mechanism discussed below), causing a progressive shortening of each sarcomere (Fig. 5-28d).

Myosin Thick Filaments Slide along Actin Thin Filaments

P1 The interaction between actin and myosin, like that between all proteins and ligands, involves weak bonds. When ATP is not bound to myosin, a face on the myosin head group binds tightly to actin (Fig. 5-29). When ATP binds to myosin and is hydrolyzed to ADP and phosphate, a coordinated and cyclic series of conformational changes occurs in which myosin releases the F-actin subunit and binds another subunit farther along the thin filament.

The cycle has four major steps (Fig. 5-29). In step ①, ATP binds to myosin and a cleft in the myosin molecule opens, disrupting the actin-myosin interaction so that the bound actin is released. ATP is then hydrolyzed in step ②, causing a conformational change in the protein to a “high-energy” state that moves the myosin head and changes its orientation in relation to the actin thin filament. Myosin then binds weakly to an F-actin subunit closer to the Z disk than the one just released. As the phosphate product of ATP hydrolysis is released from myosin in step ③, another conformational change occurs in which the myosin cleft closes, strengthening the myosin-actin

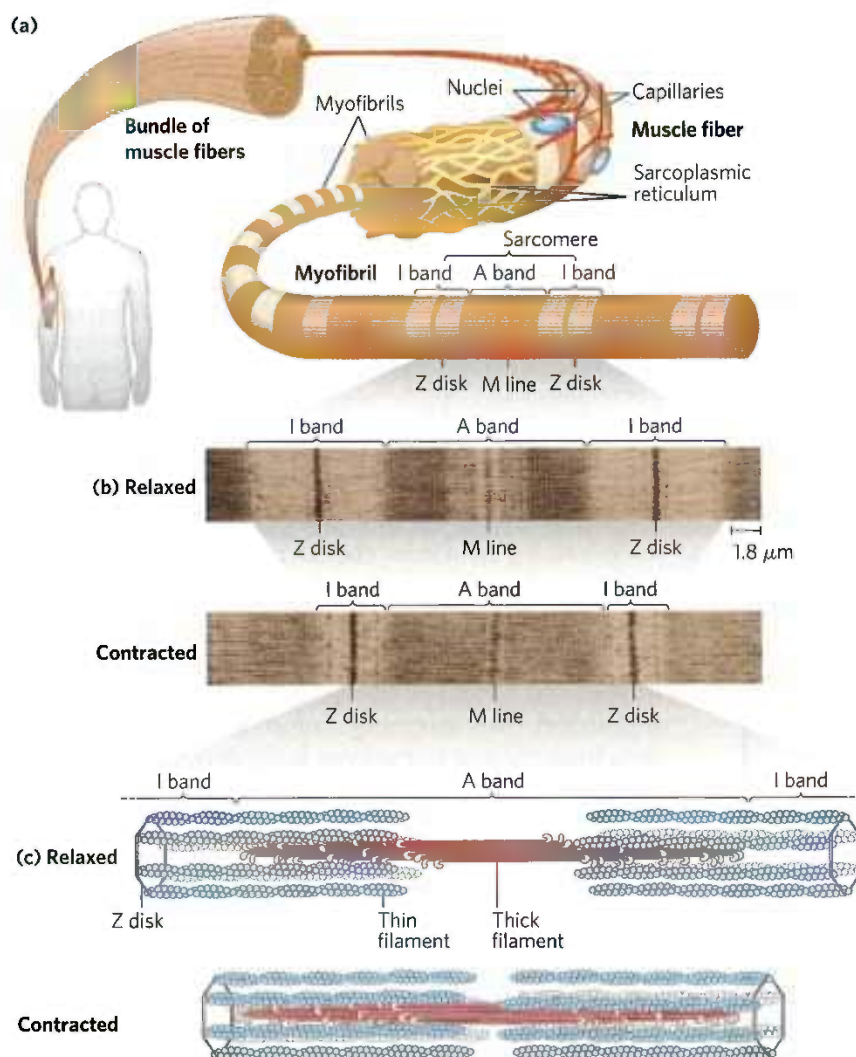


FIGURE 5-28 Skeletal muscle. (a) Muscle fibers consist of single, elongated, multinucleated cells that arise from the fusion of many precursor cells. The fibers are made up of many myofibrils (only six are shown here for simplicity) surrounded by the membranous sarcoplasmic reticulum. The organization of thick and thin filaments in a myofibril gives it a striated appearance. When muscle contracts, the I bands narrow and the Z disks move closer together, as seen in electron micrographs of (b) relaxed muscle and (c) contracted muscle. (d) Thick filaments are bipolar structures created by the association of many myosin molecules. Muscle contraction occurs by the sliding of the thick and thin filaments past each other so that the Z disks in neighboring I bands draw closer together. The thick and thin filaments are interleaved such that each thick filament is surrounded by six thin filaments. [(b) Medical Images/James E. Dennis.]

binding. This is followed quickly by step 4, a “power stroke” during which the conformation of the myosin head returns to the original resting state, its orientation relative to the bound actin changing so as to pull the tail of the myosin toward the Z disk. ADP is then released to complete the cycle. Each cycle generates about 3 to 4 pN (piconewtons) of force and moves the thick filament 5 to 10 nm relative to the thin filament.

Because there are many myosin heads in a thick filament, at any given moment some (probably 1% to 3%) are bound to thin filaments. This prevents thick filaments from slipping backward when an individual myosin head releases the actin subunit to which it was bound. The thick filament thus actively slides forward past the adjacent thin filaments. This process, coordinated among the many sarcomeres in a muscle fiber, brings about muscle contraction.

The interaction between actin and myosin must be regulated so that contraction occurs only in response to appropriate signals from the nervous system. The regulation is mediated by a complex of two proteins, **tropomyosin** and **troponin** (Fig. 5-30). Tropomyosin binds to the thin filament, blocking the attachment sites for the myosin head groups. Troponin is a Ca^{2+} -binding protein. A nerve impulse causes release of Ca^{2+} ions from the sarcoplasmic reticulum. The released Ca^{2+} binds to troponin (another protein-ligand interaction) and causes a conformational change in the tropomyosin-troponin complexes, exposing the myosin-binding sites on the thin filaments. Contraction follows.

Working skeletal muscle requires two types of molecular functions that are common in proteins: binding and catalysis. The actin-myosin interaction, a protein-ligand interaction like that of immunoglobulins with antigens, is reversible and leaves the participants

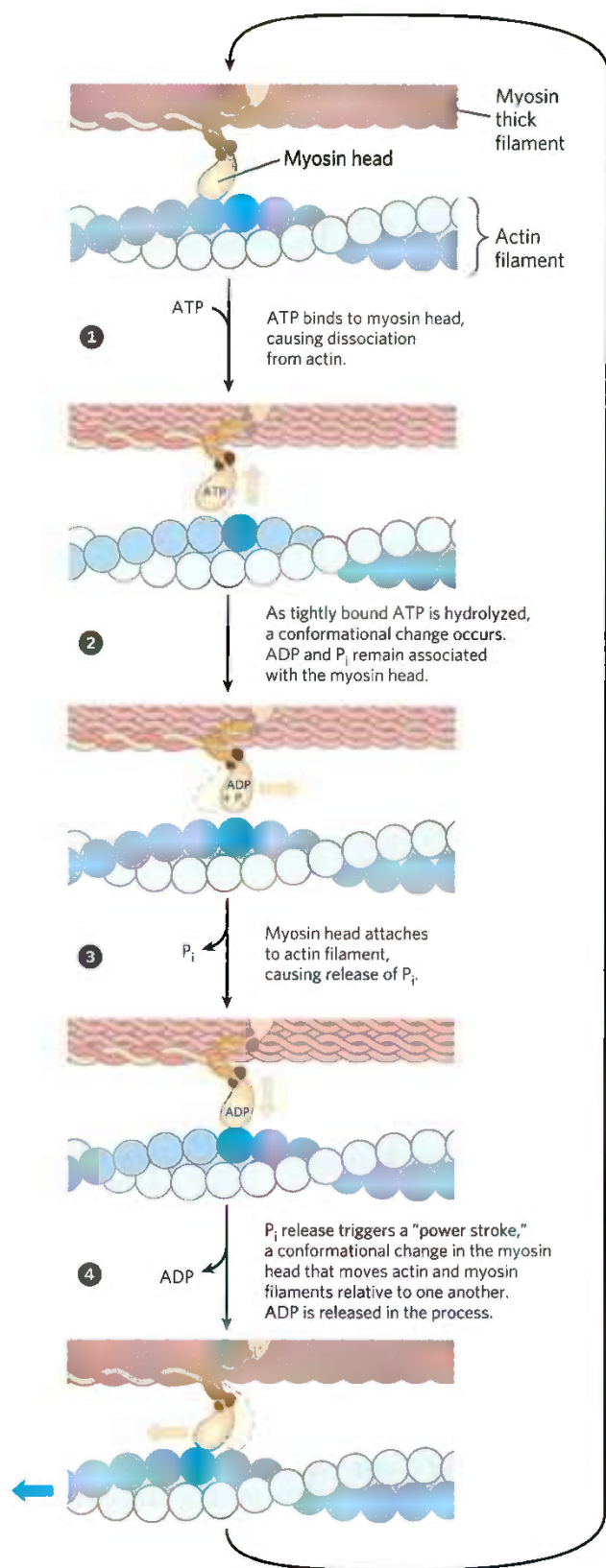


FIGURE 5-29 Molecular mechanism of muscle contraction. Conformational changes in the myosin head that are coupled to stages in the ATP hydrolytic cycle cause myosin to successively dissociate from one actin subunit, then associate with another farther along the actin filament. In this way, the myosin heads slide along the thin filaments, drawing the thick filament array into the thin filament array (see Fig. 5-28).

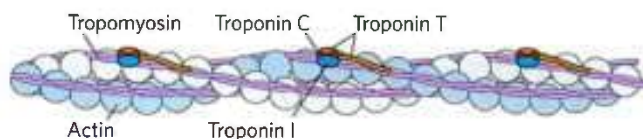


FIGURE 5-30 Regulation of muscle contraction by tropomyosin and troponin. Tropomyosin and troponin are bound to F-actin in the thin filaments. In the relaxed muscle, these two proteins are arranged around the actin filaments so as to block the binding sites for myosin. Tropomyosin is a two-stranded coiled coil of α helices, the same structural motif as in α -keratin (see Fig. 4-11). It forms head-to-tail polymers twisting around the two actin chains. Troponin is attached to the actin-tropomyosin complex at regular intervals of 38.5 nm. Troponin consists of three different subunits: I, C, and T. Troponin I prevents binding of the myosin head to actin; troponin C has a binding site for Ca^{2+} ; and troponin T links the entire troponin complex to tropomyosin. When the muscle receives a neural signal to initiate contraction, Ca^{2+} is released from the sarcoplasmic reticulum (see Fig. 5-28a) and binds to troponin C. This causes a conformational change in troponin C, which alters the positions of troponin I and tropomyosin so as to relieve the inhibition by troponin I and allow muscle contraction.

unchanged. This interaction illustrates why reversibility is important; a permanent actin-myosin interaction defines the state of rigor mortis, a state we all want to avoid as long as possible. When ATP binds myosin, however, it is hydrolyzed to ADP and P_i (inorganic phosphate). Myosin is not only an actin-binding protein, but also an ATPase—an enzyme. The function of enzymes in catalyzing chemical transformations is the topic of the next chapter.

SUMMARY 5.3 Protein Interactions Modulated by Chemical Energy: Actin, Myosin, and Molecular Motors

- Protein-ligand interactions achieve a special degree of spatial and temporal organization in motor proteins. Muscle contraction results from choreographed interactions between myosin and actin, coupled to the hydrolysis of ATP by myosin.
- Myosin consists of two heavy chains and four light chains, forming a fibrous coiled coil (tail) domain and a globular (head) domain. Myosin molecules are organized into thick filaments, which slide past thin filaments composed largely of actin. ATP hydrolysis in myosin is coupled to a series of conformational changes in the myosin head, leading to dissociation of myosin from one F-actin subunit and its eventual reassociation with another, farther along the thin filament. The myosin thus slides along the actin filaments.
- Muscle contraction is stimulated by the release of Ca^{2+} from the sarcoplasmic reticulum. The Ca^{2+} binds to the protein troponin, leading to a conformational change in a troponin-tropomyosin complex that triggers the cycle of actin-myosin interactions.

KEY TERMS

Terms in bold are defined in the glossary.

ligand 147	immune response 164
binding site 147	lymphocytes 164
induced fit 147	antibody 165
hemoglobin 148	immunoglobulin (Ig) 165
heme 148	B lymphocyte (B cell) 165
porphyrin 148	T lymphocyte (T cell) 165
globins 149	antigen 165
globin fold 149	epitope 165
equilibrium expression 150	hapten 165
association constant, K_a 150	immunoglobulin fold 165
dissociation constant, K_d 150	polyclonal antibodies 167
allosteric protein 157	monoclonal antibodies 167
modulator 157	immunoblotting 167
homotropic 157	Western blotting 167
heterotropic 157	myosin 169
Hill equation 157	actin 169
Hill coefficient 157	myofibril 170
Bohr effect 160	sarcomere 170

PROBLEMS

1. Relationship between Affinity and Dissociation Constant Protein A has a binding site for ligand X with a K_d of 3.0×10^{-7} M. Protein B has a binding site for ligand X with a K_d of 4.0×10^{-8} M. Calculate the K_a for each protein. Which protein has a higher affinity for ligand X? Explain your reasoning.

2. Modeling Apparent Negative Cooperativity Which of these situations would produce a Hill plot with $n_H < 1.0$? Explain your reasoning in each case.

- The protein has multiple subunits, each with a single ligand-binding site. Ligand binding to one site decreases the binding affinity of other sites for the ligand.
- The protein is a single polypeptide with two ligand-binding sites, each having a different affinity for the ligand.
- The protein is a single polypeptide with a single ligand-binding site. As purified, the protein preparation is heterogeneous, containing some protein molecules that are partially denatured and thus have a lower binding affinity for the ligand.
- The protein has multiple subunits, each with a single ligand-binding site. Ligands bind independently to each site, do not affect the binding affinity of other sites, and bind with identical affinities.

3. Reversible Ligand Binding I The protein calcineurin binds to the protein calmodulin with an association rate of $8.9 \times 10^3 \text{ M}^{-1} \text{ s}^{-1}$ and an overall dissociation constant, K_d , of 10 nM. Calculate the dissociation rate, k_d , including appropriate units.

4. Reversible Ligand Binding II The *E. coli* nickel-binding protein binds to its ligand, Ni^{2+} , with a K_d of 100 nM. Calculate the Ni^{2+} concentration when the fraction of binding sites occupied by the ligand (Y) is (a) 0.25, (b) 0.6, (c) 0.95.

5. Reversible Ligand Binding III You are a technician in a biochemistry lab running receptor binding experiments. The target membrane-bound receptor has been partially purified from mouse, rat, and human cell lines. Using various concentrations of the same radioactive ligand for each receptor in a saturation binding assay, you generate the binding data shown in the table. The dependent variable, Y , is the fraction of binding sites occupied by the ligand.

Ligand concentration (nM)	Y		
	Mouse receptor	Rat receptor	Human receptor
0.2	0.048	0.29	0.17
0.5	0.11	0.50	0.33
1.0	0.20	0.67	0.50
4.0	0.50	0.89	0.80
10	0.71	0.95	0.91
20	0.83	0.97	0.95
50	0.93	0.99	0.98

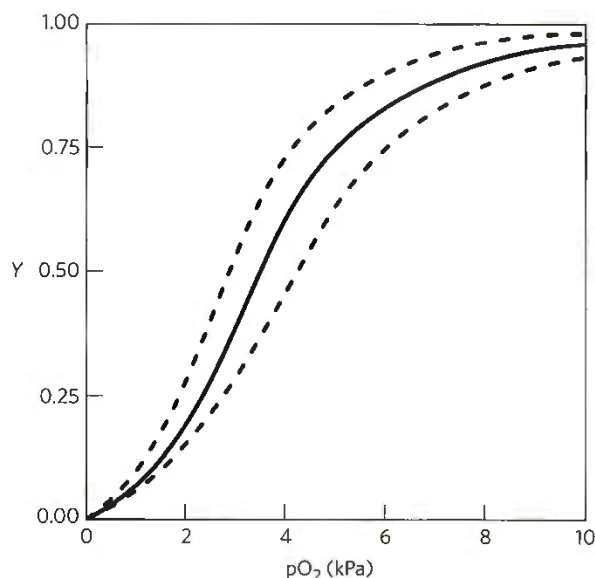
- Determine the mouse receptor K_d in this binding experiment.
- Which receptor binds most tightly to this ligand?



6. Reversible Ligand Binding IV Exposure to carbon monoxide can lead to unconsciousness and ultimately death. Suffocation occurs when hemoglobin is half-saturated with CO—that is, when only two of every four oxygen-binding sites are occupied with CO. Explain why death may occur at this point, even though half of the oxygen-binding sites are still available to transport O_2 . (Hint: See Box 5-1.)

7. Cooperativity in Hemoglobin Under appropriate conditions, hemoglobin dissociates into its four subunits. The isolated α subunit binds oxygen, but the O_2 -saturation curve is hyperbolic rather than sigmoid. In addition, the binding of oxygen to the isolated α subunit is not affected by the presence of H^+ , CO_2 , or BPG. What do these observations indicate about the source of the cooperativity in hemoglobin?

8. Oxygen Binding to Hemoglobin The solid curve in the plot shown is an O_2 -binding curve for human hemoglobin. For each condition, indicate whether the stated physiological change would shift the curve to the left (dashed curve), produce no change (black curve), or shift the curve to the right (dashed curve).



- (a) an increase in the concentration of CO_2
- (b) an increase in the proton concentration (decrease in pH)
- (c) an increase in the concentration of 2,3-bisphosphoglycerate (BPG)

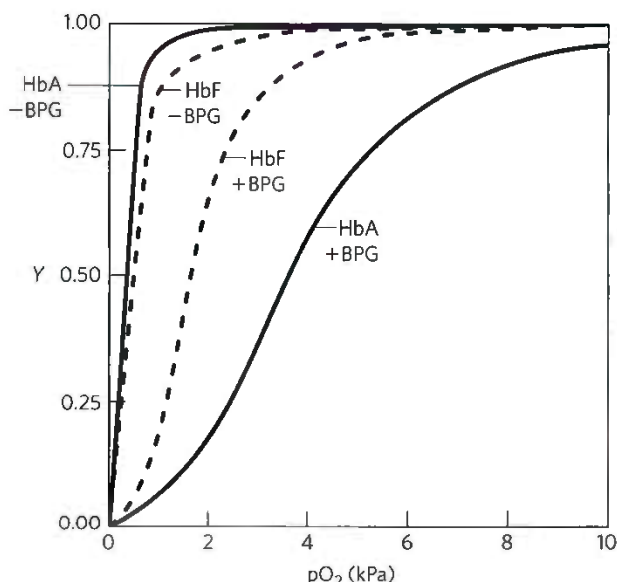
9. Comparison of Fetal and Maternal Hemoglobins

Studies of oxygen transport in pregnant mammals show that the O_2 -saturation curves of fetal and maternal blood are markedly different when measured under the same conditions. Fetal erythrocytes contain a structural variant of hemoglobin, HbF, consisting of two α and two γ subunits ($\alpha_2\gamma_2$), whereas maternal erythrocytes contain HbA ($\alpha_2\beta_2$).

(a) Which hemoglobin has a higher affinity for oxygen under physiologic conditions?

(b) What is the physiological significance of the different O_2 affinities?

When all the BPG is carefully removed from samples of HbA and HbF, the measured O_2 -saturation curves (and consequently the O_2 affinities) are displaced to the left. However, HbA now has a greater affinity for oxygen than does HbF. When BPG is reintroduced, the O_2 -saturation curves return to normal, as shown in the graph.



(c) What is the effect of BPG on the O_2 affinity of hemoglobin? How can this information be used to explain the different O_2 affinities of fetal and maternal hemoglobin?

10. Hemoglobin Variants There are almost 500 naturally occurring variants of hemoglobin. Most are the result of a single amino acid substitution in a globin polypeptide chain. Some variants produce clinical illness, though not all variants have deleterious effects. A brief sample of hemoglobin variants is shown here.

HbS (sickle cell Hb): substitutes a Val for a Glu on the surface

Hb Cowtown: eliminates an ion pair involved in T-state stabilization

Hb Memphis: substitutes one uncharged polar residue for another of similar size on the surface

Hb Bibba: substitutes a Pro for a Leu involved in an α helix

Hb Milwaukee: substitutes a Glu for a Val

Hb Providence: substitutes an Asn for a Lys that normally projects into the central cavity of the tetramer

Hb Philly: substitutes a Phe for a Tyr, disrupting hydrogen bonding at the $\alpha_1\beta_1$ interface

Select the hemoglobin variants that are described by each statement.

- (a) The Hb variant *least* likely to cause pathological symptoms
- (b) The variant(s) most likely to show pI values different from that of HbA on an isoelectric focusing gel
- (c) The variant(s) most likely to show a decrease in BPG binding and an increase in the overall affinity of the hemoglobin for oxygen

11. Oxygen Binding and Hemoglobin Structure A team of biochemists uses genetic engineering to modify the interface region between hemoglobin subunits. The resulting hemoglobin variants exist in solution primarily as $\alpha\beta$ dimers (few, if any, $\alpha_2\beta_2$ tetramers form). Are these variants likely to bind oxygen more weakly or more tightly? Explain your answer.

12. Reversible (and Tight) Binding to an Antibody An antibody with high affinity for its antigen has a K_d in the low nanomolar range. Assume an antibody binds an antigen with a K_d of 5×10^{-8} M. Calculate the antigen concentration when Y, the fraction of binding sites occupied by the ligand, is (a) 0.4, (b) 0.5, (c) 0.8, (d) 0.9.

13. Using Antibodies to Probe Structure-Function Relationships in Proteins A monoclonal antibody binds to G-actin but not to F-actin. What does this tell you about the epitope recognized by the antibody?



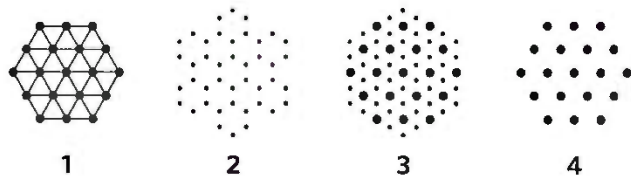
14. The Immune System and Vaccines Some pathogens have developed mechanisms to evade the immune system, making it difficult or impossible to develop effective vaccines against them.

(a) African sleeping sickness is caused by a protozoan called *Trypanosoma brucei*, carried by the tsetse fly. The trypanosome surface is dominated by one coat protein, the variable surface glycoprotein (VSG). The trypanosome genome encodes over 1,000 different versions of VSG. All of the cells in an initial infection feature the same VSG coat on their surfaces, and this is readily recognized as foreign by the immune system. However, an individual trypanosome in the broader population will switch and randomly begin expressing a different variant of the VSG coat. All the descendants of that cell will have the new and different protein on their surface. As the population with the second VSG coat increases, an individual cell will then switch to a third VSG protein coat, and so on.

(b) The human immunodeficiency virus (HIV) has an error-prone system for replicating its genome, effectively introducing mutations at an unusually high rate. Many of the mutations affect the viral protein coat. Describe how each pathogen can survive the immune response of its host.

15. How We Become a "Stiff" When a vertebrate dies, its muscles stiffen as they are deprived of ATP, a state called rigor mortis. Using your knowledge of the catalytic cycle of myosin in muscle contraction, explain the molecular basis of the rigor state.

16. Sarcomeres from Another Point of View The symmetry of thick and thin filaments in a sarcomere is such that six thin filaments ordinarily surround each thick filament in a hexagonal array. Match each cross section (transverse cut) image of a sarcomere with the correct viewpoint.



- (a) at the M line
- (b) through the I band
- (c) through the dense region of the A band
- (d) through the less dense region of the A band, adjacent to the M line (see Fig. 5-28b, c)

BIOCHEMISTRY ONLINE

17. IgG and Lysozyme Crystal Structure To fully appreciate how proteins function in a cell, it is helpful to have a three-dimensional view of how proteins interact with other cellular components. Fortunately, this is possible using online protein databases and three-dimensional molecular viewing utilities such as JSmol, a free and user-friendly molecular viewer that is compatible with most browsers and operating systems.

In this exercise, examine the interactions between the enzyme lysozyme and the Fab portion of the antilysozyme antibody. Use the PDB identifier 1FDL to explore the structure of the IgG1 Fab fragment-lysozyme complex (antibody-antigen complex). To answer the questions, use the information on the Structure Summary page at the Protein Data Bank (www.rcsb.org), and view the structure using JSmol or a similar viewer.

- (a) Which chains in the three-dimensional model correspond to the antibody fragment, and which correspond to the antigen, lysozyme?
- (b) What type of secondary structure predominates in this Fab fragment?
- (c) How many amino acid residues are in the heavy and light chains of the Fab fragment? In lysozyme? Estimate the percentage of the lysozyme that interacts with the antigen-binding site of the antibody fragment.
- (d) Identify the specific amino acid residues in lysozyme and in the variable regions of the Fab heavy and light chains that are situated at the antigen-antibody interface. Are the residues contiguous in the primary sequence of the polypeptide chains?

18. Exploring Antibodies in the Protein Data Bank The PDB-101 Molecule of the Month article on "Antibodies" (<http://pdb101.rcsb.org/motm/21>) summarizes what you have read in this chapter regarding antibody structure and function. To paraphrase the article, a variety of antibodies, on the order of one hundred million different types, are always circulating in our bloodstream, searching for foreign invaders to attack. Once an invader is discovered, the antibody binds the invader with its

flexible arms, containing the Fab region. Thin, flexible chains connect these flexible arms to the antibody base, called the Fc region. This base determines which class the antibody belongs to, as some antibodies have four or ten binding sites due to their structural formation.

(a) How many specific antigen-binding sites are there on the first immunoglobulin image (derived from PDB ID 1IGT) in the article?

(b) When a virus enters your lungs, how long does it take for you to produce one or more antibodies that bind to it?

(c) Approximately how many types of different antibodies are present in your blood?

(d) Explore the structure of the immunoglobulin molecule (PDB ID 1IGT) by clicking the link in the article or by using a search engine to find the structure summary for PDB ID 1IGT. Use one of the 3D viewers on the PDB site to view a ribbon structure for this immunoglobulin. Identify the two light chains and two heavy chains (use the viewer controls to distinguish them by color).

DATA ANALYSIS PROBLEM

19. Protein Function During the 1980s, the structures of actin and myosin were known only at the resolution shown in Figure 5-26a. Although researchers knew that the globular head portion of myosin bound to actin and hydrolyzed ATP, there was a substantial debate about where in the myosin molecule the contractile force was generated. At the time, two competing models were proposed for the mechanism of force generation in myosin.

In the "hinge" model, the head bound to actin, but the pulling force was generated by contraction of the "hinge region" in the myosin tail. The hinge region is in the heavy meromyosin portion of the myosin molecule; this is roughly the point labeled "Two supercoiled α helices" in Figure 5-26. In the "S1" model (S1 being a name used to describe the head), the pulling force was generated in the S1 "head" itself and the tail was just for structural support.

Many experiments were performed but provided no conclusive evidence. Then, in 1987, James Spudich and his colleagues at Stanford University published a study that, although not conclusive, went a long way toward resolving this controversy.

Recombinant DNA techniques were not sufficiently developed to address this issue in vivo, so Spudich and colleagues used an interesting in vitro motility assay. The alga *Nitella* has extremely long cells, often several centimeters long and about 1 mm in diameter. These cells have actin fibers that run along their long axes, and the cells can be cut open along their length to expose the actin fibers. Spudich and his group had observed that plastic beads coated with myosin would "walk" along these fibers in the presence of ATP, just as myosin would do in contracting muscle.

For these experiments, the researchers used a more well-defined method for attaching the myosin to the beads. The "beads" were clumps of killed bacterial (*Staphylococcus aureus*) cells. These cells have a protein on their surface that binds to the Fc region of antibody molecules (Fig. 5-20a). The antibodies, in turn, bind to several (unknown) places along the tail of the myosin molecule. When bead-antibody-myosin

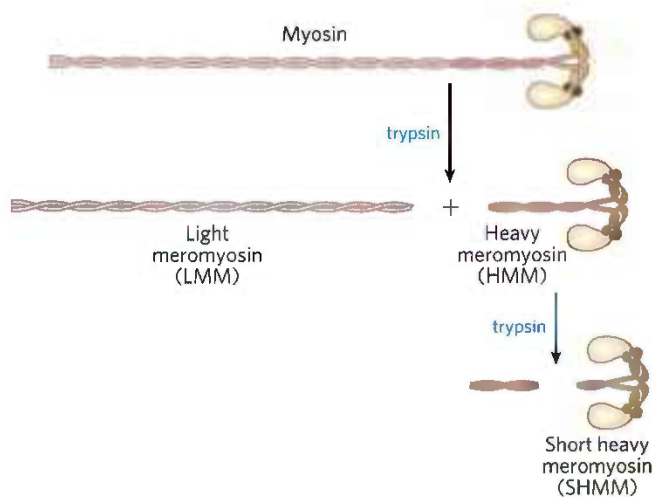
complexes were prepared with intact myosin molecules, they would move along *Nitella* actin fibers in the presence of ATP.

(a) Sketch a diagram showing what a bead-antibody-myosin complex might look like at the molecular level.

(b) Why was ATP required for the beads to move along the actin fibers?

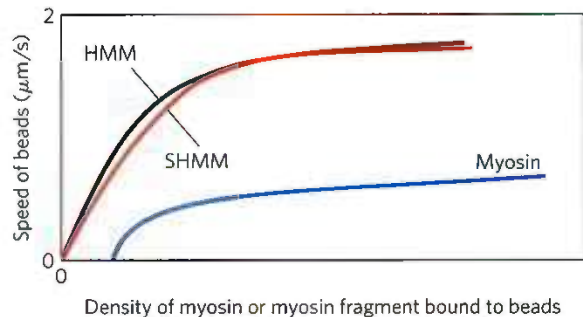
(c) Spudich and coworkers used antibodies that bound to the myosin tail. Why would this experiment have failed if they had used an antibody that bound to the part of S1 that normally bound to actin? Why would this experiment have failed if they had used an antibody that bound to actin?

To help focus on the part of myosin responsible for force production, Spudich and colleagues used trypsin to produce two partial myosin molecules: heavy meromyosin (HMM) and light meromyosin (LMM), by cleavage of a single specific peptide bond in the myosin tail. Additional incubation with trypsin produced an additional cleavage, eliminating more of the tail and the hinge region to generate short heavy meromyosin (SHMM).



(d) Why might trypsin attack this peptide bond first, rather than other peptide bonds in myosin?

Spudich and colleagues prepared bead-antibody-myosin complexes with varying amounts of myosin, HMM, or SHMM and measured their speed of movement along *Nitella* actin fibers in the presence of ATP. The graph below sketches their results.



(e) Which model ("S1" or "hinge") is consistent with these results? Explain your reasoning.

(f) Provide a plausible explanation for the increased speed of the beads with increasing myosin density.

(g) Provide a plausible explanation for the plateauing of the speed of the beads at high myosin density.

Reference

Hynes, T.R., S.M. Block, B.T. White, and J.A. Spudich.

1987. Movement of myosin fragments in vitro: domains involved in force production. *Cell* 48:953-963.

ENZYMES

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There are two fundamental conditions for life. First, the organism must be able to self-replicate (a topic to be considered in Part III); second, it must be able to catalyze chemical reactions efficiently and selectively. The central importance of catalysis may seem surprising, but it is easy to demonstrate. As described in Chapter 1, living systems make use of energy from the environment. Many of us, for example, consume substantial amounts of sucrose—common table sugar—as a kind of fuel, usually in the form of sweetened foods and drinks. The conversion of sucrose to CO_2 and H_2O in the presence of oxygen is a highly exergonic process, releasing free energy that we can use to think, move, taste, and see. However, a bag of sugar can remain on the shelf for years without any obvious conversion to CO_2 and H_2O . Although this chemical process is thermodynamically favorable, it is very slow. Yet when sucrose is consumed by a human (or almost any other organism), it releases its chemical energy in seconds. The difference is catalysis. Without catalysis, chemical reactions such as sucrose oxidation could not occur on a useful time scale, and thus could not sustain life. Essentially all reactions that occur in cells are, and must be, catalyzed by enzymes, the most remarkable and highly specialized of the proteins.

Our discussion of enzymes is organized around five principles:

P1 Enzymes are powerful biological catalysts.

Rate accelerations by enzymes are often far greater than those by synthetic or inorganic catalysts. Like all catalysts, enzymes increase reaction rates, lowering reaction activation barriers. Enzymes do not affect the equilibria of reactions.

P2 Enzymes exhibit a very high degree of specificity. Each enzyme catalyzes only one chemical reaction, or sometimes a few closely related reactions. Reaction activation barriers are thus lowered selectively.

P3 Enzymatic reactions occur in specialized pockets called active sites. These pockets are similar to ligand binding sites, except that a reaction occurs there—the conversion of a substrate, a molecule that is acted on by an enzyme, to a product.

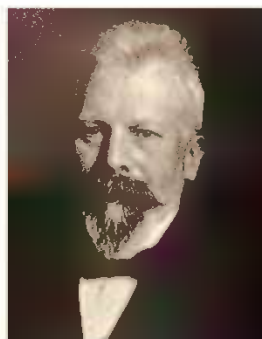
P4 Two concepts explain the catalytic power of enzymes. First, enzymes bind most tightly to the transition state of the catalyzed reaction, using binding energy to lower the activation barrier. Second, enzyme active sites are organized by evolution to facilitate multiple mechanisms of chemical catalysis simultaneously.

P5 Many enzymes are regulated. Regulatory mechanisms include reversible covalent modification, binding of allosteric modulators, proteolytic activation, noncovalent binding to regulatory proteins, and elaborate regulatory cascades. Enzymes are often subject to multiple methods of regulation, which allows for exquisite control of every chemical process that occurs in a cell.

The study of enzymes has immense practical importance. Some diseases, especially inheritable genetic disorders, are the result of a deficiency or even a total absence of one or more enzymes. Other disease conditions may be caused by excessive activity of an enzyme. Measurements of the activities of enzymes in blood plasma, erythrocytes, or tissue samples are important in diagnosing certain illnesses. Many drugs act through interactions with enzymes. Enzymes are also important practical tools in chemical engineering, food technology, and agriculture. Virtually every process studied in a biochemical laboratory involves one or often many enzymes. We now turn to a broader description of these remarkable catalysts.

6.1 An Introduction to Enzymes

Much of the history of biochemistry is the history of enzyme research. Biological catalysis was first recognized and described in the late 1700s, in studies on the digestion of meat by secretions of the stomach. The science of biochemistry can be traced to an experiment by Eduard Buchner in 1897, which demonstrated that cell-free yeast extracts could ferment sugar to alcohol. Buchner thus proved that fermentation was promoted by molecules that continued to function when removed from cells. This work marked the end of vitalistic notions advanced by Louis Pasteur that biological catalysis was a process inseparable from living systems. Frederick W. Kühne later gave the name **enzymes** (from the Greek *enzymos*, “leavened”) to the molecules detected by Buchner.



Eduard Buchner, 1860–1917
[Science Museum/Science & Society
Picture Library]

Not until the late 1920s did it become clear that enzymes were proteins. In 1926, James Sumner provided the first breakthrough, isolating and crystallizing the enzyme urease. Further work by Northrop, Kunitz, and others led to general acceptance of the enzyme-protein association by the early 1930s. Since the latter part of the twentieth century, thousands of enzymes have been purified, their structures elucidated, and their mechanisms explained.

The power of enzyme catalysts is often astonishing. The enzyme orotidine phosphate decarboxylase, an enzyme involved in the biosynthesis of pyrimidine nucleotides, provides a special example, with a rate enhancement of 10^{17} . **P1** The uncatalyzed reaction has a half-life of 78 million years. On the enzyme, the reaction occurs on a time scale of milliseconds.

Most Enzymes Are Proteins

With the exception of a few classes of catalytic RNA molecules (Chapter 26), enzymes are proteins. Their catalytic activity depends on the integrity of their native protein conformation. If an enzyme is denatured or dissociated into its subunits, catalytic activity is usually lost. The catalytic activity of each enzyme is intimately linked to its primary, secondary, tertiary, and quaternary protein structure.

Enzymes, like other proteins, have molecular weights ranging from about 12,000 to more than 1 million. Some enzymes require no chemical groups for activity other than their amino acid residues. Others require an additional chemical component called a **cofactor**—either one or more inorganic ions, such as Fe^{2+} , Mg^{2+} , Mn^{2+} , or Zn^{2+} (Table 6-1), or a complex organic or metalloorganic molecule called a **coenzyme**. Coenzymes act as transient carriers of specific functional groups (Table 6-2). Most are derived from vitamins, organic nutrients required in small amounts in the diet. We consider coenzymes in more

TABLE 6-1 Some Inorganic Ions That Serve as Cofactors for Enzymes

Ions	Enzymes
Cu^{2+}	Cytochrome oxidase
Fe^{2+} or Fe^{3+}	Cytochrome oxidase, catalase, peroxidase
K^{+}	Pyruvate kinase
Mg^{2+}	Hexokinase, glucose 6-phosphatase, pyruvate kinase
Mn^{2+}	Arginase, ribonucleotide reductase
Mo	Dinitrogenase
Ni^{2+}	Urease
Zn^{2+}	Carbonic anhydrase, alcohol dehydrogenase, carboxypeptidases A and B

TABLE 6-2 Some Coenzymes That Serve as Transient Carriers of Specific Atoms or Functional Groups

Coenzyme	Examples of chemical groups transferred	Dietary precursor in mammals
Biocytin	CO_2	Biotin (vitamin B_7)
Coenzyme A	Acyl groups	Pantothenic acid (vitamin B_5) and other compounds
5'-Deoxyadenosylcobalamin (coenzyme B_{12})	H atoms and alkyl groups	Vitamin B_{12}
Flavin adenine dinucleotide	Electrons	Riboflavin (vitamin B_2)
Lipoate	Electrons and acyl groups	Not required in diet
Nicotinamide adenine dinucleotide	Hydride ion (H^-)	Nicotinic acid (niacin, vitamin B_3)
Pyridoxal phosphate	Amino groups	Pyridoxine (vitamin B_6)
Tetrahydrofolate	One-carbon groups	Folate (vitamin B_9)
Thiamine pyrophosphate	Aldehydes	Thiamine (vitamin B_1)

Note: The structures and modes of action of these coenzymes are described in Part II.

TABLE 6-3 International Classification of Enzymes

Class number	Class name	Type of reaction catalyzed
1	Oxidoreductases	Transfer of electrons (hydride ions or H atoms)
2	Transferases	Group transfer
3	Hydrolases	Hydrolysis (transfer of functional groups to water)
4	Lyases	Cleavage of C—C, C—O, C—N, or other bonds by elimination, leaving double bonds or rings, or addition of groups to double bonds
5	Isomerases	Transfer of groups within molecules to yield isomeric forms
6	Ligases	Formation of C—C, C—S, C—O, and C—N bonds by condensation reactions coupled to cleavage of ATP or similar cofactor
7	Translocases	Movement of molecules or ions across membranes or their separation within membranes

detail as we encounter them in the metabolic pathways discussed in Part II. Some enzymes require *both* a coenzyme and one or more metal ions for activity. A coenzyme or metal ion that is very tightly or even covalently bound to the enzyme protein is called a **prosthetic group**. A complete, catalytically active enzyme together with its bound coenzyme and/or metal ions is called a **holoenzyme**. The protein part of such an enzyme is called the **apoenzyme** or **apoprotein**. Finally, some enzyme proteins are modified covalently by phosphorylation, glycosylation, and other processes. Many of these alterations are involved in the regulation of enzyme activity.

Enzymes Are Classified by the Reactions They Catalyze

Many enzymes have been named by adding the suffix “-ase” to the name of their substrate or to a word or phrase describing their activity. Thus, urease catalyzes hydrolysis of urea, and DNA polymerase catalyzes the polymerization of nucleotides to form DNA. Other enzymes were named by their discoverers for a broad function, before the specific reaction catalyzed was known. For example, an enzyme known to act in the digestion of foods was named pepsin, from the Greek *pepsis*, “digestion,” and lysozyme was named for its ability to lyse (break down) bacterial cell walls. Still others were named for their source: trypsin, named in part from the Greek *tryein*, “to wear down,” was obtained by rubbing pancreatic tissue with glycerin. Sometimes the same enzyme has two or more names, or two different enzymes have the same name. To limit ambiguity, biochemists worldwide have adopted a system for naming and classifying enzymes. This system divides enzymes into seven classes, each with subclasses, based on the type of reaction catalyzed (Table 6-3). Each enzyme is assigned a four-part classification number and a systematic name, which identifies the reaction it catalyzes. As an example, the formal systematic name of the enzyme catalyzing the reaction



is ATP:D-hexose 6-phosphotransferase, which indicates that it catalyzes the transfer of a phosphoryl group from

ATP to glucose. Its Enzyme Commission number (E.C. number) is 2.7.1.1. The first number (2) denotes the class name (transferase); the second number (7), the subclass (phosphotransferase); the third number (1), a phosphotransferase with a hydroxyl group as acceptor; and the fourth number (1), D-glucose as the phosphoryl group acceptor. For many enzymes, a trivial name is more frequently used—in this case, hexokinase. A complete list and description of the thousands of known enzymes is maintained by the Nomenclature Committee of the International Union of Biochemistry and Molecular Biology (www.qmul.ac.uk/sbcs/iubmb).

SUMMARY 6.1 An Introduction to Enzymes

- Life depends on powerful and specific catalysts: enzymes. Almost every biochemical reaction is catalyzed by an enzyme.
- With the exception of a few catalytic RNAs, all known enzymes are proteins. Many require nonprotein coenzymes or cofactors for their catalytic function.
- Enzymes are classified according to the type of reaction they catalyze. All enzymes have formal E.C. numbers and names, and most have trivial names.

6.2 How Enzymes Work

The enzymatic catalysis of reactions is essential to living systems. Under biologically relevant conditions, uncatalyzed reactions tend to be slow—most biological molecules are quite stable in the neutral-pH, mild-temperature, aqueous environment inside cells. Reactions required to digest food, send nerve signals, or contract a muscle simply do not occur at a useful rate without catalysis.

P3 The distinguishing feature of an enzyme-catalyzed reaction is that it takes place within the confines of a pocket on the enzyme called the **active site** (Fig. 6-1). The active site provides a specific environment, customized by evolution, in which a given reaction can occur more rapidly. The molecule that is bound in the active site and acted upon

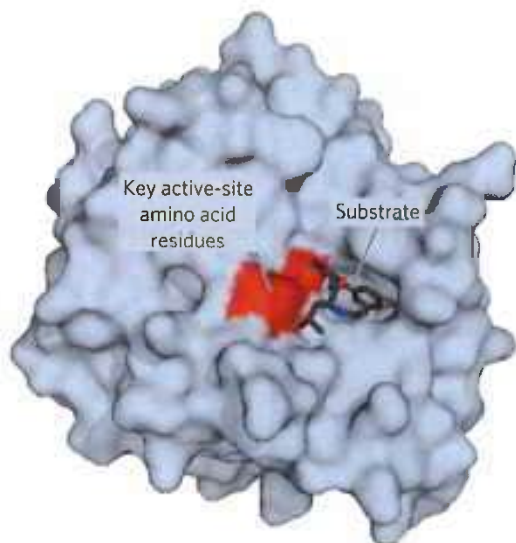
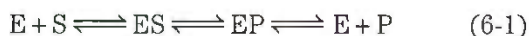


FIGURE 6-1 Binding of a substrate to an enzyme at the active site. The enzyme chymotrypsin with bound substrate. Some key active-site amino acid residues appear as a red splotch on the enzyme surface. [Data from PDB ID 7GCH, K. Brady et al., *Biochemistry* 29:7600, 1990.]

by the enzyme is called the **substrate**. The surface of the active site is lined with amino acid residues with substituent groups that bind the substrate and catalyze its chemical transformation. Often, the active site encloses a substrate, sequestering it completely from solution. The enzyme-substrate complex is central to the action of enzymes. It is also the starting point for mathematical treatments that define the kinetic behavior of enzyme-catalyzed reactions and for theoretical descriptions of enzyme mechanisms.

Enzymes Affect Reaction Rates, Not Equilibria

A simple enzymatic reaction might be written



where E, S, and P represent the enzyme, substrate, and product; ES and EP are transient complexes of the enzyme with the substrate and with the product.

To understand catalysis, we must first appreciate the important distinction between reaction equilibria and reaction rates. **P1** The function of a catalyst is to increase the *rate* of a reaction. Catalysts do not affect reaction *equilibria*. (Recall that a reaction is at equilibrium when there is no net change in the concentrations of reactants or products.) Any reaction, such as $S \rightleftharpoons P$, can be described by a reaction coordinate diagram (Fig. 6-2), a picture of the energy changes during the reaction. As discussed in Chapter 1, energy in biological systems is described in terms of free energy, G . In the coordinate diagram, the free energy of the system is plotted against the progress of the reaction (the reaction coordinate). The starting point for either the forward reaction or the reverse reaction is called the **ground state**, the contribution to the free energy of the system by an average molecule (S or P) under a given set of conditions.

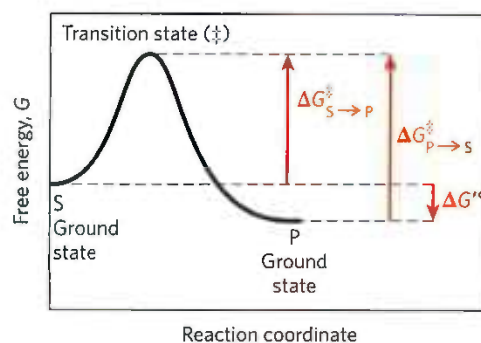


FIGURE 6-2 Reaction coordinate diagram. The free energy of the system is plotted against the progress of the reaction $S \rightarrow P$. Such a diagram describes the energy changes during the reaction. The horizontal axis (reaction coordinate) reflects the progressive chemical changes (e.g., bond breakage or formation) as S is converted to P. The activation energies, $\Delta G^\ddagger$, for the $S \rightarrow P$ and $P \rightarrow S$ reactions are indicated. ΔG° is the overall standard free-energy change in the direction $S \rightarrow P$.

KEY CONVENTION To describe the free-energy changes for reactions, chemists define a standard set of conditions (temperature of 298 K; partial pressure of each gas, 1 atm, or 101.3 kPa; concentration of each solute, 1 M), and express the free-energy change for a reacting system under these conditions as ΔG° , the **standard free-energy change**. Because biochemical systems commonly involve H^+ concentrations far below 1 M, biochemists define a **biochemical standard free-energy change**, $\Delta G'^\circ$, the standard free-energy change at pH 7.0; we employ this definition throughout the book. A more complete definition of $\Delta G'^\circ$ is given in Chapter 13. ■

The equilibrium between S and P reflects the difference in the free energies of their ground states. In the example shown in Figure 6-2, the free energy of the ground state of P is lower than that of S, so ΔG° for the reaction is negative (the reaction is exergonic) and at equilibrium there is more P than S (the equilibrium favors P).

A favorable equilibrium does not mean that the $S \rightarrow P$ conversion will occur at a rapid or even detectable rate. The *rate* of a reaction is instead dependent on an entirely different parameter, the energy barrier between S and P. That barrier consists of the energy required for alignment of reacting groups, formation of transient unstable charges, bond rearrangements, and other transformations required for the reaction to proceed in either direction. This is illustrated by the energy “hill” in Figures 6-2 and 6-3. To undergo reaction, the molecules must overcome this barrier and therefore must be raised to a higher energy level. At the top of the energy hill is a point at which decay to the S or P state is equally probable (it is downhill either way). This is called the **transition state**, often symbolized by a double dagger ($\ddagger$). The transition state is not a chemical species with any significant stability and should not be confused with a reaction intermediate (such as ES or EP). It is simply a fleeting molecular moment in which events such as bond breakage, bond formation, and charge development have proceeded to

the precise point at which decay to substrate and decay to product are equally likely.

The difference between the energy level of the ground state and the energy level of the transition state is the **activation energy**, $\Delta G^\ddagger$. The rate of a reaction reflects this activation energy: a higher activation energy corresponds to a slower reaction. Lowering the activation energy increases the rate of reaction. Reaction rates can be increased by raising the temperature and/or pressure, thereby increasing the number of molecules with sufficient energy to overcome the energy barrier. Alternatively, the activation energy can be lowered, and reaction rate increased, by adding a catalyst (**Fig. 6-3**).

Catalysts do not affect reaction equilibria. The bidirectional arrows in Equation 6-1 make this important point: any enzyme that catalyzes the reaction $S \rightarrow P$ also catalyzes the reaction $P \rightarrow S$. The role of enzymes is to *accelerate* the interconversion of S and P. The enzyme is not used up in the process, and the equilibrium point is unaffected. However, the reaction reaches equilibrium much faster when the appropriate enzyme is present, because the rate of the reaction is increased.

This relationship between reaction rate and activation energy is illustrated by the process introduced at the beginning of the chapter: the conversion of sucrose and oxygen to carbon dioxide and water:



This conversion, which takes place through a series of separate reactions, has a very large and negative ΔG° , and at equilibrium the amount of sucrose present is negligible. Yet sucrose is a stable compound, because the activation energy barrier that must be overcome before sucrose reacts with oxygen is quite high. Sucrose can be stored in a container with oxygen almost indefinitely without reacting. In cells, however, sucrose is readily broken down to CO_2 and H_2O in a series of reactions catalyzed by enzymes. These enzymes not only accelerate

the reactions, they organize and control them so that much of the energy released is recovered in other chemical forms and made available to the cell for other tasks. The reaction pathway by which sucrose (and other sugars) is broken down is the primary energy-yielding pathway for cells, and the enzymes of this pathway allow the reaction sequence to proceed on a biologically useful (millisecond) time scale.

Any reaction may have several steps, involving the formation and decay of transient chemical species called **reaction intermediates**.* A reaction intermediate is any species on the reaction pathway that has a finite chemical lifetime (longer than a molecular vibration, $\sim 10^{-13}$ second). When the $S \rightleftharpoons P$ reaction is catalyzed by an enzyme, the ES and EP complexes can be considered intermediates, even though S and P are stable chemical species (Eqn 6-1); the ES and EP complexes occupy valleys in the reaction coordinate diagram (**Fig. 6-3**). Additional, less-stable chemical intermediates often exist in the course of an enzyme-catalyzed reaction. The interconversion of two sequential reaction intermediates thus constitutes a reaction step. When several steps occur in a reaction, the overall rate is determined by the step (or steps) with the highest activation energy; this is called the **rate-limiting step**. In a simple case, the rate-limiting step is the highest-energy point in the diagram for interconversion of S and P. In practice, the rate-limiting step can vary with reaction conditions, and for many enzymes several steps may have similar activation energies, which means they are all partially rate-limiting.

Activation energies are energy barriers to chemical reactions. These barriers are crucial to life itself. Without such energy barriers, complex macromolecules would revert spontaneously to much simpler molecular forms, and the complex and highly ordered structures and metabolic processes of cells could not exist. **P2** Over the course of evolution, enzymes have developed to lower activation energies *selectively*, and thus increase rates, for reactions that are needed for cell survival.

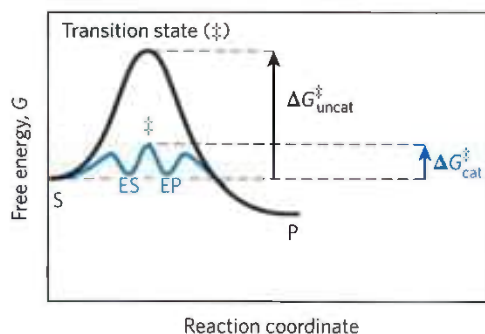


FIGURE 6-3 Reaction coordinate diagram comparing enzyme-catalyzed and uncatalyzed reactions. In the reaction $S \rightarrow P$, the ES and EP intermediates occupy minima in the energy progress curve of the enzyme-catalyzed reaction. The terms $\Delta G^\ddagger_{\text{uncat}}$ and $\Delta G^\ddagger_{\text{cat}}$ correspond to the activation energy for the uncatalyzed reaction and the overall activation energy for the catalyzed reaction, respectively. The activation energy is lower when the enzyme catalyzes the reaction.

Reaction Rates and Equilibria Have Precise Thermodynamic Definitions

P1 Reaction *equilibria* are inextricably linked to the standard free-energy change for the reaction, ΔG° , and reaction *rates* are linked to the activation energy, $\Delta G^\ddagger$. A basic introduction to these thermodynamic relationships is the next step in understanding how enzymes work.

* In this chapter, *step* and *intermediate* refer to chemical reactions and chemical species in the reaction pathway of a single enzyme-catalyzed reaction. In the context of metabolic pathways involving many enzymes (discussed in Part II), these terms are used somewhat differently: an entire enzymatic reaction is often referred to as a “step” in a pathway, and the product of one enzymatic reaction (which is the substrate for the next enzyme in the pathway) is referred to as a pathway “intermediate.”

An equilibrium such as $S \rightleftharpoons P$ is described by an **equilibrium constant**, K_{eq} , or simply K (p. 23). Under the standard conditions used to compare biochemical processes, an equilibrium constant is denoted K'_{eq} (or K'):

$$K'_{eq} = \frac{[P]}{[S]} \quad (6-2)$$

From thermodynamics, the relationship between K'_{eq} and $\Delta G'^{\circ}$ can be described by the expression

$$\Delta G'^{\circ} = -RT \ln K'_{eq} \quad (6-3)$$

where R is the gas constant, $8.315 \text{ J/mol} \cdot \text{K}$, and T is the absolute temperature, 298 K (25°C). Equation 6-3 is developed and discussed in more detail in Chapter 13. The important point here is that the equilibrium constant is directly related to the overall standard free-energy change for the reaction (Table 6-4). A large negative value for $\Delta G'^{\circ}$ reflects a favorable reaction equilibrium (one in which there is much more product than substrate at equilibrium)—but as already noted, this does not mean the reaction will proceed at a rapid rate.

The rate of any reaction is determined by the concentration of the reactant (or reactants) and by a **rate constant**, usually denoted by k . For the unimolecular reaction $S \rightarrow P$, the rate (or velocity) of the reaction, V —representing the amount of S that reacts per unit of time—is expressed by a **rate equation**:

$$V = k[S] \quad (6-4)$$

In this reaction, the rate depends only on the concentration of S . This is called a first-order reaction. The factor k is a proportionality constant that reflects the probability of reaction under a given set of conditions (pH, temperature, and so forth). Here, k is a first-order rate constant and has units of reciprocal time, such as s^{-1} . If a

first-order reaction has a rate constant k of 0.03 s^{-1} , this may be interpreted (qualitatively) to mean that 3% of the available S will be converted to P in 1 second. A reaction with a rate constant of $2,000 \text{ s}^{-1}$ will be over in a small fraction of a second. If a reaction rate depends on the concentration of two different compounds, or if the reaction is between two molecules of the same compound, the reaction is second order and k is a second-order rate constant, with units of $\text{M}^{-1}\text{s}^{-1}$. The rate equation then becomes

$$V = k[S_1][S_2] \quad (6-5)$$

From transition-state theory we can derive an expression that relates the magnitude of a rate constant to the activation energy:

$$k = \frac{kT}{h} e^{-\Delta G^{\ddagger}/RT} \quad (6-6)$$

where k is the Boltzmann constant and h is Planck's constant. The important point is that the relationship between the rate constant k and the activation energy $\Delta G^{\ddagger}$ is inverse and exponential. In simplified terms, this is the basis for the statement that **P1** a lower activation energy means a faster reaction rate—and lowering activation energies is what enzymes do.

Now we turn from *what* enzymes do to *how* they do it.

A Few Principles Explain the Catalytic Power and Specificity of Enzymes

Enzymes are extraordinary catalysts. The rate enhancements they bring about are in the range of 5 to 17 orders of magnitude (Table 6-5). Enzymes are also very specific, readily discriminating between substrates with quite similar structures. How can these enormous and highly selective rate enhancements be explained? What is the source of the energy for the dramatic lowering of the activation energies for specific reactions?

The answer to these questions has two distinct but interwoven parts. The first lies in the *noncovalent* interactions between enzyme and substrate. What

TABLE 6-4 Relationship between K'_{eq} and $\Delta G'^{\circ}$

K'_{eq}	$\Delta G'^{\circ}$ (kJ/mol)
10^{-6}	34.2
10^{-5}	28.5
10^{-4}	22.8
10^{-3}	17.1
10^{-2}	11.4
10^{-1}	5.7
1	0.0
10^1	-5.7
10^2	-11.4
10^3	-17.1

Note: The relationship is calculated from $\Delta G'^{\circ} = -RT \ln K'_{eq}$ (Eqn 6-3).

TABLE 6-5 Some Rate Enhancements Produced by Enzymes

Cyclophilin	10^5
Carbonic anhydrase	10^7
Triose phosphate isomerase	10^9
Carboxypeptidase A	10^{11}
Phosphoglucomutase	10^{12}
Succinyl-CoA transferase	10^{13}
Urease	10^{14}
Orotidine monophosphate decarboxylase	10^{17}

really sets enzymes apart from most other catalysts is the formation of a specific ES complex. The interaction between substrate and enzyme in this complex is mediated by the same forces that stabilize protein structure, including hydrogen bonds, ionic interactions, and the hydrophobic effect (Chapter 4). Formation of each weak interaction in the ES complex is accompanied by release of a small amount of free energy that stabilizes the interaction. The energy derived from noncovalent enzyme-substrate interaction is called **binding energy, ΔG_B** . Its significance extends beyond a simple stabilization of the enzyme-substrate interaction. As we will see, **P4** *binding energy is a major source of free energy used by enzymes to lower the activation energies of reactions.*

The second part of the explanation lies in covalent interactions between enzyme and substrate, plus a few additional chemical catalytic mechanisms. Chemical reactions of many types take place between substrates and enzymes' functional groups (specific amino acid side chains, metal ions, and coenzymes). Catalytic functional groups on an enzyme may form a transient covalent bond with a substrate and activate it for reaction, or a group may be transiently transferred from the substrate to the enzyme. In many cases, these reactions occur only in the enzyme active site. Covalent interactions between enzymes and substrates lower the activation energy (and thereby accelerate the reaction) by providing an alternative, lower-energy reaction path. Metal ions facilitate additional mechanisms of catalysis that do not involve covalent interactions.

We now consider noncovalent contributions to catalysis and the additional chemical mechanisms in turn.

Noncovalent Interactions between Enzyme and Substrate Are Optimized in the Transition State

How does an enzyme use noncovalent binding energy to lower the activation energy for a reaction? Formation of the ES complex is not the explanation in itself, although some of the earliest considerations of enzyme mechanisms began with this idea. Studies on enzyme specificity carried out by Emil Fischer led him to propose, in 1894, that enzymes were structurally complementary to their substrates, so that they fit together like a lock and key (**Fig. 6-4**). This elegant idea, that a specific (exclusive) interaction between two biological molecules is mediated by molecular surfaces with complementary shapes, has greatly influenced the development of biochemistry. However, the "lock and key" hypothesis can be misleading when applied to enzymatic catalysis. An enzyme completely complementary to its substrate would be a very poor enzyme, as we can demonstrate.

Consider an imaginary reaction, the breaking of a magnetized metal stick. The uncatalyzed reaction

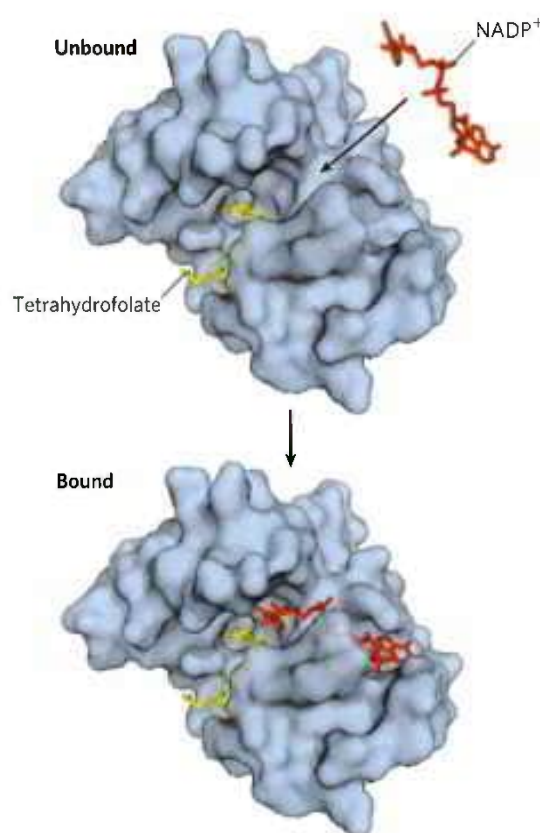


FIGURE 6-4 Complementary shapes of a substrate and its binding site on an enzyme. The enzyme dihydrofolate reductase with its substrate NADP⁺, unbound and bound; another bound substrate, tetrahydrofolate, is also visible. In this model, the NADP⁺ binds to a pocket that is complementary to it in shape and ionic properties, an illustration of Emil Fischer's "lock and key" hypothesis of enzyme action. In reality, the complementarity between protein and ligand (in this case, substrate) is rarely perfect, as we saw in Chapter 5.

[Data from PDB ID 1RA2, M. R. Sawaya and J. Kraut, *Biochemistry* 36:586, 1997.]

is shown in **Figure 6-5a**. Let's examine two imaginary enzymes—two "stickases"—that could catalyze this reaction, both of which employ magnetic forces as a paradigm for the binding energy used by real enzymes. We first design an enzyme perfectly complementary to the substrate (**Fig. 6-5b**). The active site of this stickase is a pocket lined with magnets. To react (break), the stick must reach the bent transition state of the reaction. However, the snug fit of the stick in this active site means that magnets hinder the required bending. Such an enzyme *impedes* the reaction, stabilizing the substrate instead. In a reaction coordinate diagram (**Fig. 6-5b**), this kind of ES complex would correspond to an energy trough from which the substrate would have difficulty escaping. Such an enzyme would be useless.

The modern notion of enzymatic catalysis, first proposed by Michael Polanyi (1921) and J. B. S. Haldane (1930), was elaborated by Linus Pauling in 1946 and by William P. Jencks in the 1970s: in order to catalyze reactions, an enzyme must be complementary to the

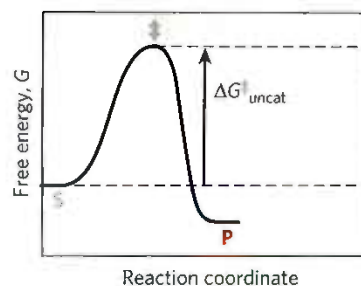
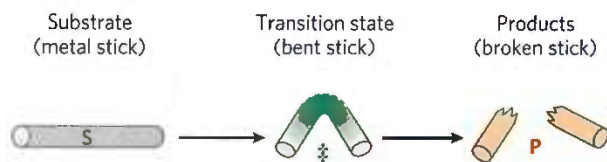
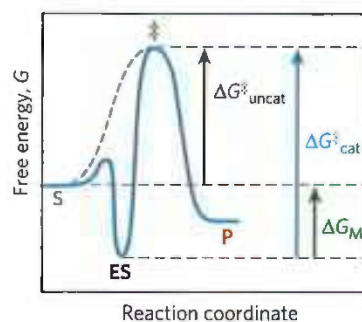
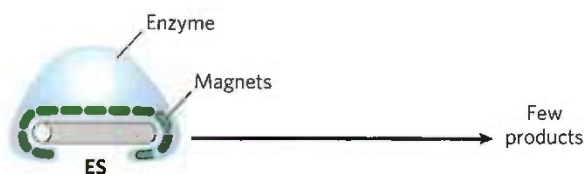
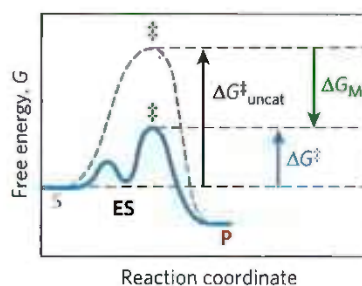
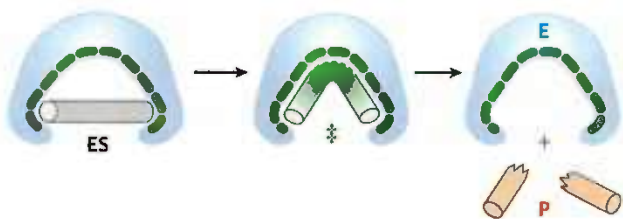
(a) No enzyme**(b) Enzyme complementary to substrate****(c) Enzyme complementary to transition state**

FIGURE 6-5 An imaginary enzyme (stickase) designed to catalyze breakage of a metal stick. (a) Before the stick is broken, it must first be bent (the transition state). In both stickase examples, magnetic interactions take the place of weak bonding interactions between enzyme and substrate. (b) A stickase with a magnet-lined pocket complementary in structure to the stick (the substrate) stabilizes the substrate. Bending is impeded by the magnetic attraction between stick and stickase. (c) An enzyme with a pocket complementary to the reaction transition state helps to destabilize the stick, contributing to catalysis of the reaction. The binding energy of the magnetic

interactions compensates for the increase in free energy required to bend the stick. Reaction coordinate diagrams (right) show the energy consequences of complementarity to substrate versus complementarity to transition state (EP complexes are omitted). ΔG_M , the difference between the transition-state energies of the uncatalyzed and catalyzed reactions, is contributed by the magnetic interactions between the stick and stickase. When the enzyme is complementary to the substrate (b), the ES complex is more stable and has less free energy in the ground state than substrate alone. The result is an *increase* in the activation energy.

reaction transition state. This means that optimal interactions between substrate and enzyme occur only in the transition state. Figure 6-5c demonstrates how such an enzyme can work. The metal stick binds to the stickase, but only a few of the possible magnetic interactions are used in forming the ES complex. The bound substrate must still undergo the increase in free energy needed to reach the transition state. Now, however, the increase in free energy required to draw the stick into a bent and partially broken conformation is offset, or “paid for,” by the magnetic interactions that form between our imaginary enzyme and substrate (analogous to the binding energy in a real enzyme) in the transition state. Many of these interactions involve parts of the stick that are distant from the point of

breakage; thus, interactions between the stickase and nonreacting parts of the stick provide some of the energy needed to catalyze stick breakage. This “energy payment” translates into a lower net activation energy and a faster reaction rate.

Real enzymes work on an analogous principle. **P4** Some weak interactions are formed in the ES complex, but the full complement of such interactions between substrate and enzyme is formed only when the substrate reaches the transition state. The free energy (binding energy) released by the formation of these interactions partially offsets the energy required to reach the top of the energy hill. The summation of the unfavorable (positive) activation energy $\Delta G^{\ddagger}$ and the favorable (negative) binding energy ΔG_B results

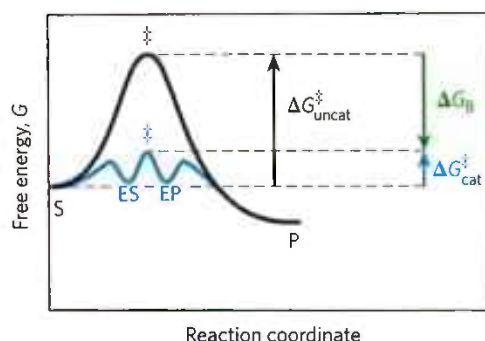


FIGURE 6-6 Role of binding energy in catalysis. To lower the activation energy for a reaction, the system must acquire an amount of energy equivalent to the amount by which $\Delta G^\ddagger$ is lowered. Much of this energy comes from binding energy, ΔG_B , contributed by formation of weak noncovalent interactions between substrate and enzyme in the transition state. The role of ΔG_B is analogous to that of ΔG_M in Figure 6-5.

in a lower *net* activation energy (Fig. 6-6). Even on the enzyme, the transition state is not a stable species but a brief point in time that the substrate spends atop an energy hill. However, the enzyme-catalyzed reaction is much faster than the uncatalyzed process because the hill is much smaller. **P4** The important principle is that *weak binding interactions between the enzyme and the substrate provide a substantial driving force for enzymatic catalysis*. The groups on the substrate that are involved in these weak interactions can be at some distance from the bonds that are broken or changed. The weak interactions formed only in the transition state are those that make the primary contribution to catalysis.

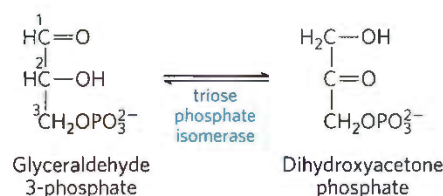
The requirement for multiple weak interactions to drive catalysis is one reason why enzymes (and some coenzymes) are so large. An enzyme must provide functional groups for ionic, hydrogen-bond, and other interactions, and also must precisely position these groups so that binding energy is optimized in the transition state. **P3** Adequate binding is accomplished most readily by positioning a substrate in a cavity (the active site) where it is effectively removed from water. The size of proteins reflects the need for superstructure to keep interacting groups properly positioned and to keep the cavity from collapsing.

Can we demonstrate quantitatively that binding energy accounts for the huge rate accelerations brought about by enzymes? Yes. As a point of reference, Equation 6-6 allows us to calculate that $\Delta G^\ddagger$ must be lowered by about 5.7 kJ/mol to accelerate a first-order reaction by a factor of 10, under conditions commonly found in cells. The energy available from formation of a single weak interaction is generally estimated to be 4 to 30 kJ/mol. The overall energy available from many such interactions is therefore sufficient to lower activation energies by the 60 to 100 kJ/mol required to explain the large rate enhancements observed for many enzymes.

P2 The same binding energy that provides energy for catalysis also gives an enzyme its **specificity**, the

ability to discriminate between a substrate and a competing molecule. Conceptually, specificity is easy to distinguish from catalysis, but this distinction is much more difficult to make experimentally, because catalysis and specificity arise from the same phenomenon. If an enzyme active site has functional groups arranged optimally to form a variety of weak interactions with a particular substrate in the transition state, the enzyme will not be able to interact to the same degree with any other molecule. For example, if the substrate has a hydroxyl group that forms a hydrogen bond with a specific Glu residue on the enzyme, any molecule lacking a hydroxyl group at that particular position will be a poorer substrate for the enzyme. **P2** In general, *specificity* is derived from the formation of many weak interactions between the enzyme and its specific substrate molecule.

The importance of binding energy to catalysis can be readily demonstrated. For example, the glycolytic enzyme triose phosphate isomerase catalyzes the interconversion of glyceraldehyde 3-phosphate and dihydroxyacetone phosphate:



This reaction rearranges the carbonyl and hydroxyl groups on carbons 1 and 2. However, more than 80% of the enzymatic rate acceleration has been traced to enzyme-substrate interactions involving the phosphate group on carbon 3 of the substrate. This was determined by comparing the enzyme-catalyzed reactions with glyceraldehyde 3-phosphate and with glyceraldehyde (no phosphate group at position 3) as substrate.

The general principles outlined above can be illustrated by a variety of recognized catalytic mechanisms. These mechanisms are not mutually exclusive, and a given enzyme might incorporate several catalytic strategies in its overall mechanism of action.

Consider what needs to occur for a reaction to take place. Prominent physical and thermodynamic factors contributing to $\Delta G^\ddagger$, the barrier to reaction, might include (1) the entropy of molecules in solution, which reduces the possibility that they will react together; (2) the solvation shell of hydrogen-bonded water that surrounds and helps to stabilize most biomolecules in aqueous solution; (3) the distortion of substrates that must occur in many reactions; and (4) the need for proper alignment of catalytic functional groups on the enzyme. Binding energy can be used to overcome all these barriers.

First, a large restriction in the relative motions of two substrates that are to react, or **entropy reduction**,

is one obvious benefit of binding them to an enzyme. Binding energy constrains the substrates in the proper orientation to react—a substantial contribution to catalysis, because productive collisions between molecules in solution can be exceedingly rare. Substrates can be precisely aligned on the enzyme, with many weak interactions between each substrate and strategically located groups on the enzyme clamping the substrate molecules into the proper positions. Studies have shown that simply constraining the motion of two reactants can produce rate enhancements of many orders of magnitude (Fig. 6-7).

Second, in water, many organic molecules are surrounded by a solvation shell of structured water that can hinder reactions (see Fig. 2-8). Formation of weak bonds between substrate and enzyme results in **desolvation** of the substrate. Enzyme-substrate interactions replace most or all of the hydrogen bonds between the substrate and water that would otherwise impede reaction.

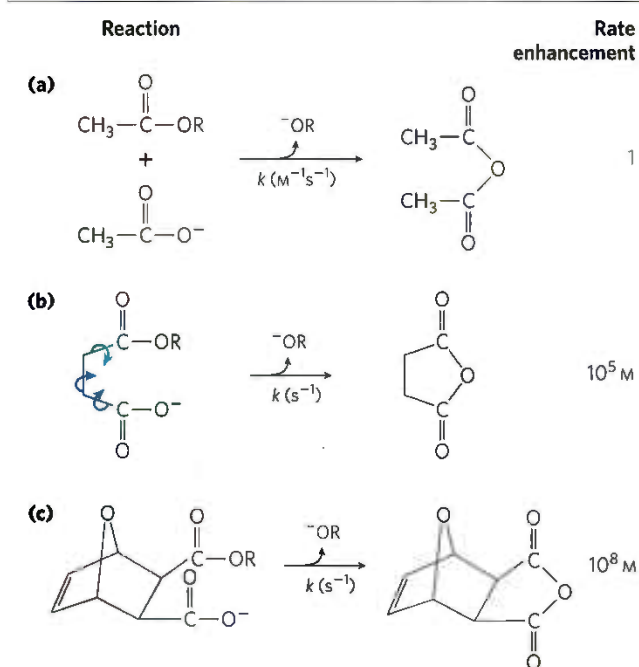


FIGURE 6-7 Rate enhancement by entropy reduction. Shown here are reactions of an ester with a carboxylate group to form an anhydride. The R group is the same in each case. (a) For this bimolecular reaction, the rate constant k is second order, with units of $\text{M}^{-1}\text{s}^{-1}$. (b) When the two reacting groups are in a single molecule, and thus have less freedom of motion, the reaction is much faster. For this unimolecular reaction, k has units of s^{-1} . Dividing the rate constant for (b) by the rate constant for (a) gives a rate enhancement of about 10^5 M . (The enhancement has units of molarity because we are comparing a unimolecular reaction and a bimolecular reaction.) Put another way, if the reactant in (b) were present at a concentration of 1 M , the reacting groups would behave as though they were present at a concentration of 10^5 M . Note that the reactant in (b) has freedom of rotation about three bonds (shown with curved arrows), but this still represents a substantial reduction of entropy over (a). If the bonds that rotate in (b) are constrained as in (c), the entropy is reduced further and the reaction exhibits a rate enhancement of 10^8 M relative to (a).

Third, binding energy involving weak interactions that are formed only in the reaction transition state helps to compensate thermodynamically for the unfavorable free-energy change associated with any distortion, primarily electron redistribution, that the substrate must undergo to react.

Finally, the enzyme itself usually undergoes a change in conformation when the substrate binds, induced by multiple weak interactions with the substrate. This is referred to as **induced fit**, a mechanism postulated by Daniel Koshland in 1958. The motions can affect a small part of the enzyme near the active site or can involve changes in the positioning of entire domains. Typically, a network of coupled motions occurs throughout the enzyme that ultimately brings about the required changes in the active site. Induced fit serves to bring specific functional groups on the enzyme into the proper position to catalyze the reaction. The conformational change also permits formation of additional weak bonding interactions in the transition state. In either case, the new enzyme conformation has enhanced catalytic properties. As we have seen, induced fit is a common feature of the reversible binding of ligands to proteins (Chapter 5). Induced fit is also important in the interaction of almost every enzyme with its substrate.

Covalent Interactions and Metal Ions Contribute to Catalysis

In most enzymes, noncovalent binding energy is just one of several contributors to the overall catalytic mechanism. **P4** Once a substrate is bound to an enzyme, properly positioned catalytic functional groups aid in the cleavage and formation of bonds by a variety of mechanisms, including general acid-base catalysis, covalent catalysis, and metal ion catalysis. The first two of these mechanisms are distinct from those based on binding energy, because they generally involve transient *covalent* interaction with a substrate or group transfer to or from a substrate.

General Acid-Base Catalysis Transfer of a proton from one molecule to another is the single most common reaction in biochemistry. One or, often, many proton transfers occur in the course of most reactions that take place in cells. Many biochemical reactions occur through the formation of unstable charged intermediates that tend to break down rapidly to their constituent reactant species, impeding the forward reaction (Fig. 6-8). Charged intermediates can often be stabilized by the transfer of protons to form a species that breaks down more readily to products. These protons are transferred between an enzyme and a substrate or intermediate.

The effects of catalysis by acids and bases are often studied using nonenzymatic model reactions, in which the proton donors or acceptors are either the constituents of water alone or other weak acids and bases.

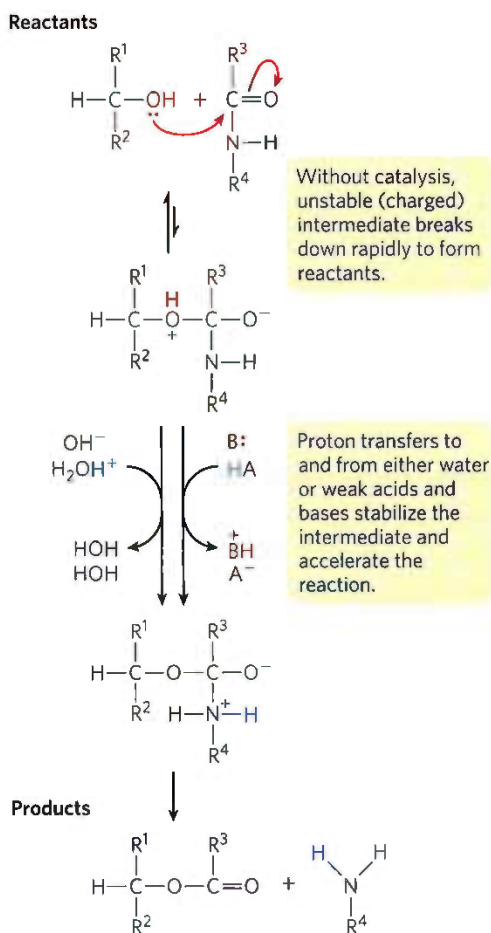


FIGURE 6-8 How a catalyst circumvents unfavorable charge development during cleavage of an amide. The hydrolysis of an amide bond, shown here, is the same reaction as that catalyzed by chymotrypsin and other proteases. Charge development is unfavorable and can be circumvented by donation of a proton by H_3O^+ (specific acid catalysis) or HA (general acid catalysis), where HA represents any acid. Similarly, charge can be neutralized by proton abstraction by OH^- (specific base catalysis) or $\text{B}:$ (general base catalysis), where $\text{B}:$ represents any base.

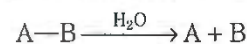
Catalysis of the type that uses only the H^+ (H_3O^+) or OH^- ions present in water is referred to as **specific acid-base catalysis**. However, a further increase in rate can often be afforded by the addition of weak acids and bases to the reaction. Many weak organic acids can supplement water as proton donors, or weak organic bases can supplement water as proton acceptors. The term **general acid-base catalysis** refers to proton transfers mediated by weak acids and bases other than water.

General acid-base catalysis becomes crucial in the active site of an enzyme, where water may not be available as a proton donor or acceptor. Several amino acid side chains can and do take on the role of proton donors and acceptors (Fig. 6-9). These groups can be precisely positioned in an enzyme active site to allow proton transfers, providing rate enhancements of the order of 10^2 to 10^5 . This type of catalysis occurs on the vast majority of enzymes.

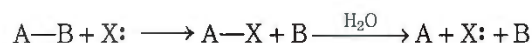
Amino acid residues	General acid form (proton donor)	General base form (proton acceptor)
Glu, Asp	$\text{R}-\text{COOH}$	$\text{R}-\text{COO}^-$
Lys, Arg	$\text{R}-\text{NH}_3^+$	$\text{R}-\text{NH}_2$
Cys	$\text{R}-\text{SH}$	$\text{R}-\text{S}^-$
His	$\text{R}-\text{C}_5\text{H}_4\text{N}^+\text{H}$	$\text{R}-\text{C}_5\text{H}_4\text{N}$
Ser	$\text{R}-\text{OH}$	$\text{R}-\text{O}^-$
Tyr	$\text{R}-\text{C}_6\text{H}_4\text{OH}$	$\text{R}-\text{C}_6\text{H}_4\text{O}^-$

FIGURE 6-9 Amino acids in general acid-base catalysis. Many organic reactions that are used to model biochemical processes are promoted by proton donors (general acids) or proton acceptors (general bases). The active sites of some enzymes contain amino acid functional groups, such as those shown here, that can participate in the catalytic process as proton donors or proton acceptors.

Covalent Catalysis In **covalent catalysis**, a transient covalent bond is formed between the enzyme and the substrate. Consider the hydrolysis of a bond between groups A and B:



In the presence of a covalent catalyst (an enzyme with a nucleophilic group $\text{X}:$) the reaction becomes



Formation and breakdown of a covalent intermediate creates a new pathway for the reaction, but catalysis results *only* when the new pathway has a lower activation energy than the uncatalyzed pathway. Both of the new steps must be faster than the uncatalyzed reaction. Several amino acid side chains, including all those in Figure 6-9, and the functional groups of some enzyme cofactors can serve as nucleophiles in the formation of covalent bonds with substrates. These covalent complexes always undergo further reaction to regenerate the free enzyme. The covalent bond formed between the enzyme and the substrate can activate a substrate for further reaction in a manner that is usually specific to the particular group or coenzyme.

Metal Ion Catalysis Metals, whether tightly bound to the enzyme or taken up from solution along with the substrate, can participate in catalysis in several ways. Ionic interactions between an enzyme-bound metal and a substrate can help orient the substrate for reaction or stabilize charged reaction transition states. This use of weak bonding interactions between metal and substrate is similar to some of the uses of enzyme-substrate

binding energy described earlier, and it can contribute to enzyme–transition state complementarity. Metals can also mediate oxidation-reduction reactions by reversible changes in the metal ion's oxidation state. Nearly a third of all known enzymes require one or more metal ions for catalytic activity.

Most enzymes combine several catalytic strategies to bring about a rate enhancement. A good example is the use of covalent catalysis, general acid-base catalysis, and transition-state stabilization in the reaction catalyzed by chymotrypsin, detailed in Section 6.4.

SUMMARY 6.2 How Enzymes Work

- Enzymes are highly effective catalysts, commonly enhancing reaction rates by a factor of 10^5 to 10^{17} . Enzyme-catalyzed reactions are characterized by the formation of a complex between substrate and enzyme (an ES complex). Substrate binding occurs in a pocket on the enzyme called the active site.

- The function of enzymes and other catalysts is to lower the activation energy, $\Delta G^\ddagger$, for a reaction and thereby enhance the reaction rate. The equilibrium of a reaction is unaffected by the enzyme.

- Reaction equilibria are described by equilibrium constants, K_{eq} , which are related to the biochemical standard free-energy change $\Delta G'^\circ$. Reaction rates are described by rate constants, k , which are related to the activation energy $\Delta G^\ddagger$.

- The extraordinary rate accelerations provided by enzymes are due to noncovalent binding energy supplemented by covalent interactions or metal ion catalysis.

- Noncovalent binding energy, ΔG_B , is maximized in the transition state of the catalyzed reaction. Enzyme–transition state complementarity is a fundamental principle of enzymatic catalysis. Noncovalent interactions may facilitate the path to the transition state, offsetting the energy required for activation, $\Delta G^\ddagger$, by lowering substrate entropy, causing substrate desolvation, or causing a conformational change in the enzyme (induced fit). Binding energy also accounts for the exquisite specificity of enzymes for their substrates.

- General acid-base catalysis and covalent catalysis mechanisms contribute to the catalytic power of enzymes. Covalent interactions between the substrate and the enzyme, group transfers to and from the enzyme, and interactions with metal ions can provide a new, lower-energy reaction path.

6.3 Enzyme Kinetics as an Approach to Understanding Mechanism

Biochemists commonly use several approaches to study the mechanism of action of purified enzymes. The three-dimensional structure of the protein provides

important information, which is enhanced by traditional protein chemistry and modern methods of site-directed mutagenesis (changing the amino acid sequence of a protein by genetic engineering; see Fig. 9-10). These technologies permit enzymologists to examine the role of individual amino acids in enzyme structure and action. However, the oldest approach to understanding enzyme mechanisms, and one that remains very important, is to determine the *rate* of a reaction and how it changes in response to changes in experimental parameters, a discipline known as **enzyme kinetics**. We provide here a basic introduction to the kinetics of enzyme-catalyzed reactions.

Substrate Concentration Affects the Rate of Enzyme-Catalyzed Reactions

At any given instant in an enzyme-catalyzed reaction, the enzyme exists in two forms: the free or uncombined form E and the substrate-combined form ES. When the enzyme is first mixed with a large excess of substrate, there is an initial transient period, the **pre-steady state**, during which the concentration of ES builds up. For most enzymatic reactions, this period is very brief. The pre-steady state is frequently too short to be observed easily, lasting only the time (often microseconds) required to convert one molecule of substrate to product (one enzymatic turnover). The reaction quickly achieves a **steady state** in which [ES] (and the concentrations of any other intermediates) remains approximately constant for much of the remainder of the reaction (Fig. 6-10). As most of the reaction reflects the steady state, the traditional analysis of reaction rates is referred to as **steady-state kinetics**.

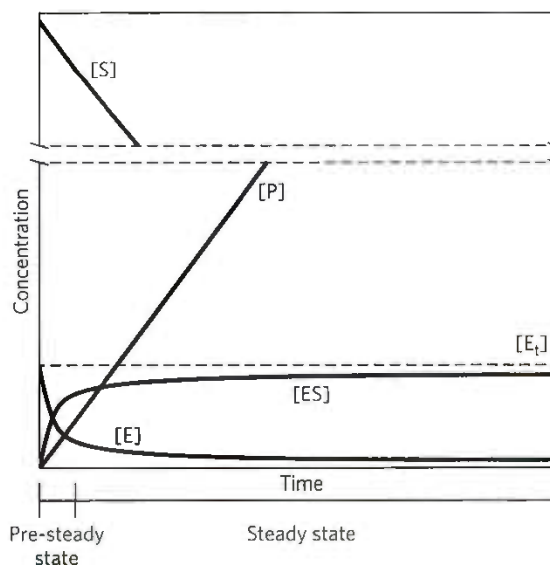


FIGURE 6-10 The course of an enzyme-catalyzed reaction. In a typical reaction, product will increase as substrate declines. The concentration of free enzyme, E, declines rapidly, as the concentration of the ES complex increases and reaches a steady state. The steady-state concentration of ES remains nearly constant for much of the remainder of the reaction.

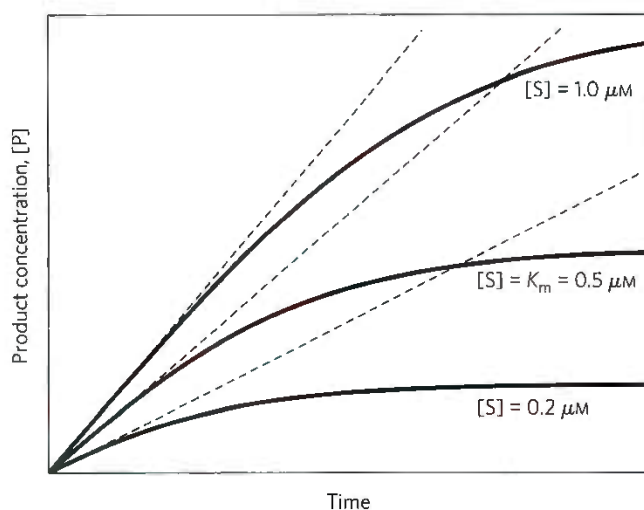


FIGURE 6-11 Initial velocities of enzyme-catalyzed reactions.

A theoretical enzyme that catalyzes the reaction $S \rightleftharpoons P$ is present at a concentration of S sufficient to catalyze the reaction at a maximum velocity, defined as $V_{\max}$, of $1 \mu\text{M}/\text{min}$. The rate observed at a given concentration of S depends on the Michaelis constant, K_m , explained in more detail in the next section. Here, the K_m is $0.5 \mu\text{M}$. Progress curves are shown for substrate concentrations below, at, and above the K_m . The rate of an enzyme-catalyzed reaction declines as substrate is converted to product. A tangent to each curve taken at time = 0 (dashed line) defines the initial velocity, V_0 , of each reaction.

A key factor affecting the rate of a reaction catalyzed by an enzyme is the concentration of substrate, $[S]$. However, studying the effects of substrate concentration is complicated by the fact that $[S]$ changes during the course of an *in vitro* reaction as substrate is converted to product. One simplifying approach is to measure the **initial rate** (or **initial velocity**), designated V_0 (Fig. 6-11). In a typical reaction, the enzyme may be present in nanomolar quantities, whereas $[S]$ may be five or six orders of magnitude higher. If only the beginning of the reaction is monitored, over a period in which only a small percentage of the available substrate (<2%–3%) is converted to product, $[S]$ can be regarded as constant, to a reasonable approximation. V_0 can then be explored as a function of $[S]$, which is adjusted by the investigator. Note that even the initial rate reflects a steady state.

The effect on V_0 of varying $[S]$ when the enzyme concentration is held constant is shown in Figure 6-12. At relatively low concentrations of substrate, V_0 increases almost linearly with an increase in $[S]$. At higher substrate concentrations, V_0 increases by smaller and smaller amounts in response to increases in $[S]$. Finally, a point is reached beyond which increases in V_0 are vanishingly small as $[S]$ increases. This plateau-like V_0 region is close to the **maximum velocity**, $V_{\max}$.

The ES complex is the key to understanding this kinetic behavior, just as it was a starting point for our discussion of catalysis. The kinetic pattern in Figure 6-12 led Victor Henri, following a proposal by Adolphe Wurtz a few decades earlier, to propose in 1903 that the combination of an enzyme with its substrate molecule to form an ES complex is a necessary step in enzymatic catalysis.

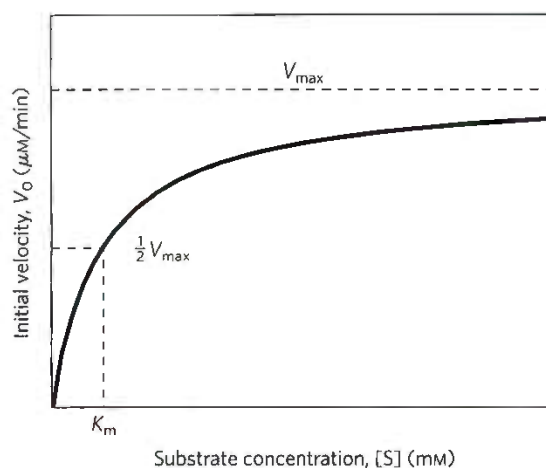


FIGURE 6-12 Effect of substrate concentration on the initial velocity of an enzyme-catalyzed reaction.

The maximum velocity, $V_{\max}$, is extrapolated from the plot, because V_0 approaches but never quite reaches $V_{\max}$. The substrate concentration at which V_0 is half-maximal is K_m , the Michaelis constant. The concentration of enzyme in an experiment such as this is generally so low that $[S] \gg [E]$ even when $[S]$ is described as low or relatively low. The units shown are typical for enzyme-catalyzed reactions and are given only to help illustrate the meaning of V_0 and $[S]$. (Note that the curve describes part of a rectangular hyperbola, with one asymptote at $V_{\max}$. If the curve were continued below $[S] = 0$, it would approach a vertical asymptote at $[S] = -K_m$.)

This idea was expanded into a general theory of enzyme action, particularly by Leonor Michaelis and Maud Menten in 1913. They postulated that the enzyme first combines reversibly with its substrate to form an enzyme-substrate complex in a relatively fast reversible step:



The ES complex then breaks down in a slower second step to yield the free enzyme and the reaction product P:



Because the slower second reaction (Eqn 6-8) must limit the rate of the overall reaction, the overall rate must be proportional to the concentration of the species that reacts in the second step—that is, ES.



Leonor Michaelis, 1875–1949
[Rockefeller Archive Center.]



Maud Menten, 1879–1960
[Archives & Special Collections, University of Pittsburgh Library System.]

At low $[S]$, most of the enzyme is in the uncombined form E . Here, the rate is proportional to $[S]$ because the equilibrium of Equation 6-7 is pushed toward formation of more ES as $[S]$ increases. The maximum initial rate of the catalyzed reaction ($V_{\max}$) is observed when virtually all the enzyme is present as the ES complex and $[E]$ is vanishingly small. Under these conditions, the enzyme is "saturated" with its substrate, so that further increases in $[S]$ have no effect on rate. This condition exists when $[S]$ is sufficiently high that essentially all the free enzyme has been converted to the ES form. After the ES complex breaks down to yield the product P , the enzyme is free to catalyze the reaction of another molecule of substrate (and will do so rapidly under saturating conditions). The saturation effect is a distinguishing characteristic of enzymatic catalysts and is responsible for the plateau observed in Figure 6-12, and the pattern seen in the figure is sometimes referred to as saturation kinetics.

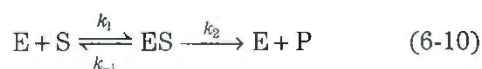
The Relationship between Substrate Concentration and Reaction Rate Can Be Expressed with the Michaelis-Menten Equation

The curve expressing the relationship between $[S]$ and V_0 (Fig. 6-12) has the same general shape for most enzymes (it approaches a rectangular hyperbola), which can be expressed algebraically by the Michaelis-Menten equation. Michaelis and Menten derived this equation starting from their basic hypothesis that the rate-limiting step in enzymatic reactions is the breakdown of the ES complex to product and free enzyme. The equation is

$$V_0 = \frac{V_{\max} [S]}{K_m + [S]} \quad (6-9)$$

This is the **Michaelis-Menten equation**, the **rate equation** for a one-substrate enzyme-catalyzed reaction. It is a statement of the quantitative relationship between the initial velocity V_0 , the maximum velocity $V_{\max}$, and the initial substrate concentration $[S]$, all related through a constant, K_m , called the Michaelis constant. All these terms — $[S]$, V_0 , $V_{\max}$, and K_m — are readily measured experimentally.

Here we develop the basic logic and the algebraic steps in a modern derivation of the Michaelis-Menten equation, which includes the steady-state assumption, a concept introduced by G. E. Briggs and J. B. S. Haldane in 1925. The derivation starts with the two basic steps of the formation and breakdown of ES (Eqns 6-7 and 6-8). Early in the reaction, the concentration of the product, $[P]$, is negligible, and we make the simplifying assumption that the reverse reaction, $P \rightarrow S$ (described by k_{-2}), can be ignored. This assumption is not critical but it simplifies our task. The overall reaction then reduces to



V_0 is determined by the breakdown of ES to form product, which is determined by $[ES]$:

$$V_0 = k_2 [ES] \quad (6-11)$$

Because $[ES]$ in Equation 6-11 is not easily measured experimentally, we must begin by finding an alternative expression for this term. First, we introduce the term $[E_t]$, representing the total enzyme concentration (the sum of free and substrate-bound enzyme). Free or unbound enzyme $[E]$ can then be represented by $[E_t] - [ES]$. Also, because $[S]$ is ordinarily far greater than $[E_t]$, the amount of substrate bound by the enzyme at any given time is negligible compared with the total $[S]$. With these conditions in mind, the following steps lead us to an expression for V_0 in terms of easily measurable parameters.

Step 1 The rates of formation and breakdown of ES are determined by the steps governed by the rate constants k_1 (formation) and $k_{-1} + k_2$ (breakdown to reactants and products, respectively), according to the expressions

$$\text{Rate of ES formation} = k_1 ([E_t] - [ES])[S] \quad (6-12)$$

$$\text{Rate of ES breakdown} = k_{-1}[ES] + k_2[ES] \quad (6-13)$$

Step 2 We now make an important assumption: that the initial rate of reaction reflects a steady state in which $[ES]$ is constant — that is, the rate of formation of ES is equal to the rate of its breakdown. This is called the **steady-state assumption**. The expressions in Equations 6-12 and 6-13 can be equated for the steady state, giving

$$k_1 ([E_t] - [ES])[S] = k_{-1}[ES] + k_2[ES] \quad (6-14)$$

Step 3 In a series of algebraic steps, we now solve Equation 6-14 for $[ES]$. First, the left side is multiplied out and the right side simplified to give

$$k_1 [E_t][S] - k_1 [ES][S] = (k_{-1} + k_2)[ES] \quad (6-15)$$

Adding the term $k_1 [ES][S]$ to both sides of the equation and simplifying gives

$$k_1 [E_t][S] = (k_1 [S] + k_{-1} + k_2)[ES] \quad (6-16)$$

We then solve this equation for $[ES]$:

$$[ES] = \frac{k_1 [E_t][S]}{k_1 [S] + k_{-1} + k_2} \quad (6-17)$$

This can now be simplified further, combining the rate constants into one expression:

$$[ES] = \frac{[E_t][S]}{[S] + (k_{-1} + k_2)/k_1} \quad (6-18)$$

The term $(k_{-1} + k_2)/k_1$ is defined as the **Michaelis constant**, K_m . Substituting this into Equation 6-18 simplifies the expression to

$$[ES] = \frac{[E_t][S]}{K_m + [S]} \quad (6-19)$$

Step 4 We can now express V_0 in terms of $[ES]$. Substituting the right side of Equation 6-19 for $[ES]$ in Equation 6-11 gives

$$V_0 = \frac{k_2[E_t][S]}{K_m + [S]} \quad (6-20)$$

This equation can be simplified further. Because the maximum velocity occurs when the enzyme is saturated (that is, when $[ES] = [E_t]$), $V_{\max}$ can be defined as $k_2[E_t]$. Substituting this in Equation 6-20 gives Equation 6-9:

$$V_0 = \frac{V_{\max}[S]}{K_m + [S]}$$

Note that K_m has units of molar concentration. Does the equation fit experimental observations? Yes; we can confirm this by considering the limiting situations where $[S]$ is very low or very high, as shown in **Figure 6-13**.

An important numerical relationship emerges from the Michaelis-Menten equation in the special case when V_0 is exactly one-half $V_{\max}$ (Fig. 6-13). Then

$$\frac{V_{\max}}{2} = \frac{V_{\max}[S]}{K_m + [S]} \quad (6-21)$$

On dividing by $V_{\max}$, we obtain

$$\frac{1}{2} = \frac{[S]}{K_m + [S]} \quad (6-22)$$

Solving for K_m , we get $K_m + [S] = 2[S]$, or

$$K_m = [S], \quad \text{when} \quad V_0 = \frac{1}{2}V_{\max} \quad (6-23)$$

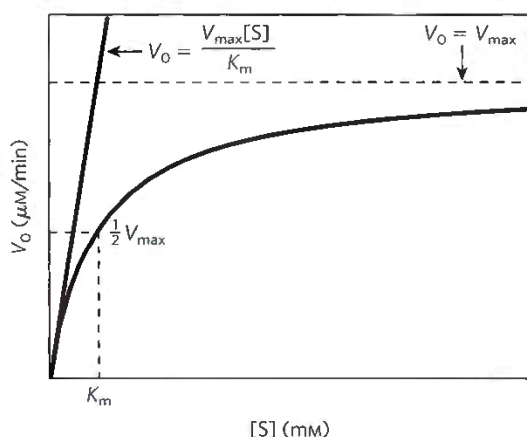


FIGURE 6-13 Dependence of initial velocity on substrate concentration. This graph shows the kinetic parameters that define the limits of the curve at low $[S]$ and high $[S]$. At low $[S]$, $K_m \gg [S]$, and the $[S]$ term in the denominator of the Michaelis-Menten equation (Eqn 6-9) becomes insignificant. The equation simplifies to $V_0 = V_{\max}[S]/K_m$, and V_0 exhibits a linear dependence on $[S]$, as observed here. At high $[S]$, where $[S] \gg K_m$, the K_m term in the denominator of the Michaelis-Menten equation becomes insignificant and the equation simplifies to $V_0 = V_{\max}$; this is consistent with the plateau observed at high $[S]$. The Michaelis-Menten equation is therefore consistent with the observed dependence of V_0 on $[S]$, and the shape of the curve is defined by the terms $V_{\max}/K_m$ at low $[S]$ and $V_{\max}$ at high $[S]$.

This is a very useful, practical definition of K_m : K_m is equivalent to the substrate concentration at which V_0 is one-half $V_{\max}$.

Michaelis-Menten Kinetics Can Be Analyzed Quantitatively

There are several ways to determine $V_{\max}$ and K_m from a plot relating V_0 to $[S]$. A traditional approach is to algebraically transform the Michaelis-Menten equation into equations that convert the hyperbolic curve in the V_0 versus $[S]$ plot into a linear form from which values of $V_{\max}$ and K_m may be obtained by extrapolation. The most common of these approaches takes the reciprocal of both sides of the Michaelis-Menten equation (Eqn 6-9):

$$\begin{aligned} V_0 &= \frac{V_{\max}[S]}{K_m + [S]} \\ \frac{1}{V_0} &= \frac{K_m + [S]}{V_{\max}[S]} \end{aligned} \quad (6-24)$$

Separating the components of the numerator on the right side of the equation gives

$$\frac{1}{V_0} = \frac{K_m}{V_{\max}[S]} + \frac{[S]}{V_{\max}[S]} \quad (6-25)$$

which simplifies to

$$\frac{1}{V_0} = \frac{K_m}{V_{\max}[S]} + \frac{1}{V_{\max}} \quad (6-26)$$

This form of the Michaelis-Menten equation is called the **Lineweaver-Burk equation**. For enzymes obeying the Michaelis-Menten relationship, a plot of $1/V_0$ versus $1/[S]$ (the “double reciprocal” of the V_0 versus $[S]$ plot we have been using up to this point) yields a straight line (**Fig. 6-14**). This line has a slope of $K_m/V_{\max}$, an intercept of $1/V_{\max}$ on the $1/V_0$ axis, and an intercept of $-1/K_m$ on the $1/[S]$ axis. The Lineweaver-Burk plot is a useful way to

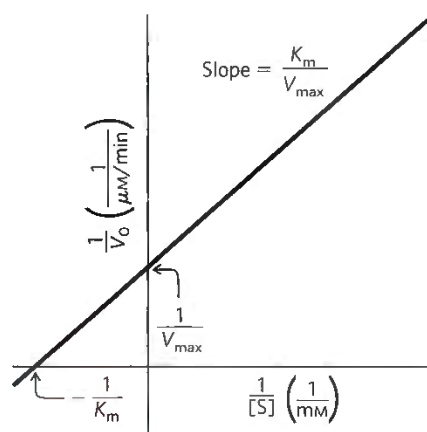


FIGURE 6-14 A double-reciprocal, or Lineweaver-Burk, plot. Plotting $1/V_0$ versus $1/[S]$ puts the data into a linear form. Intercepts on the $1/V_0$ and $1/[S]$ axes are $1/V_{\max}$ and $-1/K_m$, respectively.

display data and can provide some mechanistic information, as we will see. However, the double-reciprocal transformation tends to give undue weight to data obtained at low substrate concentration, and can distort errors in the extrapolated values of $V_{\max}$ and K_m .

More commonly, the parameters $V_{\max}$ and K_m are derived directly from the V_0 versus $[S]$ plot via nonlinear regression, using one of a multitude of curve-fitting programs available online. These are generally easy to use and offer accuracy superior to Lineweaver-Burk or related approaches.

Kinetic Parameters Are Used to Compare Enzyme Activities

It is important to distinguish between the Michaelis-Menten equation and the specific kinetic mechanism on which the equation was originally based. The Michaelis-Menten equation does not depend on the relatively simple two-step reaction mechanism proposed by Michaelis and Menten (Eqn 6-10). All enzymes that exhibit a hyperbolic dependence of V_0 on $[S]$ are said to follow **Michaelis-Menten kinetics**. Many enzymes that follow Michaelis-Menten kinetics have quite different reaction mechanisms, and enzymes that catalyze reactions with six or eight identifiable steps often exhibit the same steady-state kinetic behavior. The practical rule that $K_m = [S]$ when $V_0 = \frac{1}{2}V_{\max}$ (Eqn 6-23) holds for all enzymes that follow Michaelis-Menten kinetics. The most important exceptions to Michaelis-Menten kinetics are the regulatory enzymes, discussed in Section 6.5.

The parameters K_m and $V_{\max}$ can be obtained experimentally for any given enzyme, and their measurement is often a key first step in enzyme characterization. However, the mechanistic insight they offer is limited. Obtaining information about the number, rates, or chemical nature of discrete steps in the reaction usually requires additional complementary approaches. Steady-state kinetics nevertheless is the standard language through which biochemists compare and characterize the catalytic efficiencies of enzymes.

Interpreting K_m and $V_{\max}$ Both the magnitude and the meaning of K_m and $V_{\max}$ can vary greatly from enzyme to enzyme.

K_m can vary even for different substrates of the same enzyme (Table 6-6). The term is sometimes used (often inappropriately) as an indicator of the affinity of an enzyme for its substrate. The actual meaning of K_m depends on specific aspects of the reaction mechanism, such as the number and relative rates of the individual steps. For reactions with two steps,

$$K_m = \frac{k_2 + k_{-1}}{k_1} \quad (6-27)$$

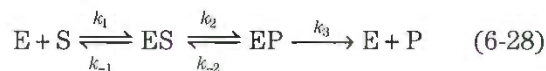
When k_2 is rate-limiting, $k_2 \ll k_{-1}$, and K_m reduces to k_{-1}/k_1 , which is defined as the **dissociation constant**, K_d , of the ES complex. Where these conditions hold,

TABLE 6-6 K_m for Some Enzymes and Substrates

Enzyme	Substrate	K_m (mM)
Hexokinase (brain)	ATP	0.4
	D-Glucose	0.05
	D-Fructose	1.5
Carbonic anhydrase	HCO ₃ ⁻	26
Chymotrypsin	Glycyltyrosinylglycine	108
	N-Benzoyltyrosinamide	2.5
β -Galactosidase	D-Lactose	4.0
Threonine dehydratase	L-Threonine	5.0

K_m does represent a measure of the affinity of the enzyme for its substrate in the ES complex. However, this scenario does not apply for most enzymes. Sometimes $k_2 \gg k_{-1}$, and then $K_m = k_2/k_1$. In other cases, k_2 and k_{-1} are comparable, and K_m remains a more complex function of all three rate constants (Eqn 6-27). The Michaelis-Menten equation and the characteristic saturation behavior of the enzyme still apply, but K_m cannot be considered a simple measure of substrate affinity. Even more common are cases in which the reaction goes through several steps after formation of ES; K_m can then become a very complex function of many rate constants.

The quantity $V_{\max}$ depends upon the rate-limiting step of the enzyme-catalyzed reaction. If an enzyme reacts by the two-step Michaelis-Menten mechanism, $V_{\max} = k_2[E_t]$, where k_2 is rate-limiting. However, the number of reaction steps and the identity of the rate-limiting step(s) can vary from enzyme to enzyme. For example, consider the common situation where product release, $EP \rightarrow E + P$, is rate-limiting. Early in the reaction (when $[P]$ is low), the overall reaction can be adequately described by the scheme



In this case, most of the enzyme is in the EP form at saturation, and $V_{\max} = k_3[E_t]$.

It is useful to define a more general rate constant, k_{cat} , to describe the limiting rate of any enzyme-catalyzed reaction at saturation. If the reaction has several steps, one of which is clearly rate-limiting, k_{cat} is equivalent to the rate constant for that limiting step. For the simple reaction of Equation 6-10, $k_{\text{cat}} = k_2$. For the reaction of Equation 6-28, when product release is clearly rate-limiting, $k_{\text{cat}} = k_3$. When several steps are partially rate-limiting, k_{cat} can become a complex function of several of the rate constants that define each individual reaction step. In the Michaelis-Menten equation, $k_{\text{cat}} = V_{\max}/[E_t]$. Rearranged, $V_{\max} = k_{\text{cat}}[E_t]$, and Equation 6-9 becomes

$$V_0 = \frac{k_{\text{cat}}[E_t][S]}{K_m + [S]} \quad (6-29)$$

TABLE 6-7 Turnover Number, k_{cat} , of Some Enzymes

Enzyme	Substrate	k_{cat} (s^{-1})
Catalase	H_2O_2	40,000,000
Carbonic anhydrase	HCO_3^-	400,000
Acetylcholinesterase	Acetylcholine	14,000
β -Lactamase	Benzylpenicillin	2,000
Fumarase	Fumarate	800
RecA protein (an ATPase)	ATP	0.5

The constant k_{cat} is a first-order rate constant and hence has units of reciprocal time. It is also called the **turnover number**. It is equivalent to the number of substrate molecules converted to product in a given unit of time on a single enzyme molecule when the enzyme is saturated with substrate. The turnover numbers of several enzymes are given in **Table 6-7**.

Comparing Catalytic Mechanisms and Efficiencies The kinetic parameters k_{cat} and K_m are useful for the study and comparison of different enzymes, whether their reaction mechanisms are simple or complex. Each enzyme has values of k_{cat} and K_m that reflect the cellular environment, the concentration of substrate normally encountered in vivo by the enzyme, and the chemistry of the reaction being catalyzed.

The parameters k_{cat} and K_m also allow us to evaluate the kinetic efficiency of enzymes, but either parameter alone is insufficient for this task. Two enzymes catalyzing different reactions may have the same k_{cat} (turnover number), yet the rates of the uncatalyzed reactions may be different and thus the rate enhancements brought about by the enzymes may differ greatly. Experimentally, the K_m for an enzyme tends to be similar to the cellular concentration of its substrate. An enzyme that acts on a substrate present at a very low concentration in the cell usually has a lower K_m than an enzyme that acts on a substrate that is more abundant.

The best way to compare the catalytic efficiencies of different enzymes or the turnover of different substrates

by the same enzyme is to compare the ratio k_{cat}/K_m for the two reactions. This parameter, sometimes called the **specificity constant**, is the rate constant for the conversion of $\text{E} + \text{S}$ to $\text{E} + \text{P}$. When $[\text{S}] \ll K_m$, Equation 6-29 reduces to the form

$$V_0 = \frac{k_{\text{cat}}}{K_m} [\text{E}_t] [\text{S}] \quad (6-30)$$

V_0 in this case depends on the concentration of two reactants, $[\text{E}_t]$ and $[\text{S}]$, so this is a second-order rate equation, and the constant k_{cat}/K_m is a second-order rate constant with units of $\text{M}^{-1}\text{s}^{-1}$. There is an upper limit to k_{cat}/K_m , imposed by the rate at which E and S can diffuse together in an aqueous solution. This diffusion-controlled limit is 10^8 to $10^9 \text{ M}^{-1}\text{s}^{-1}$, and many enzymes have a k_{cat}/K_m near this range (**Table 6-8**). Such enzymes are said to have achieved catalytic perfection. Note that different values of k_{cat} and K_m can produce the maximum ratio.

WORKED EXAMPLE 6-1 Determination of K_m

An enzyme is discovered that catalyzes the chemical reaction



A team of motivated researchers sets out to study the enzyme, which they call happyase. They find that the k_{cat} for happyase is 600 s^{-1} and they carry out several additional experiments.

When $[\text{E}_t] = 20 \text{ nM}$ and $[\text{SAD}] = 40 \text{ }\mu\text{M}$, the reaction velocity, V_0 , is $9.6 \text{ }\mu\text{M s}^{-1}$. Calculate K_m for the substrate SAD.

SOLUTION: We know k_{cat} , $[\text{E}_t]$, $[\text{S}]$, and V_0 . We want to solve for K_m . We will first derive an expression for K_m , beginning with the Michaelis-Menten equation (Eqn 6-9). We will calculate V_{max} by equating it to $k_{\text{cat}}[\text{E}_t]$. We can then substitute the known values to calculate K_m .

$$V_0 = \frac{V_{\text{max}} [\text{S}]}{(K_m + [\text{S}])}$$

First, multiply both sides by $K_m + [\text{S}]$:

$$V_0(K_m + [\text{S}]) = V_{\text{max}} [\text{S}]$$

$$V_0 K_m + V_0 [\text{S}] = V_{\text{max}} [\text{S}]$$

TABLE 6-8 Enzymes for Which k_{cat}/K_m Is Close to the Diffusion-Controlled Limit (10^8 to $10^9 \text{ M}^{-1} \text{ s}^{-1}$)

Enzyme	Substrate	$k_{\text{cat}} (\text{s}^{-1})$	$K_m (\text{M})$	$k_{\text{cat}}/K_m (\text{M}^{-1} \text{ s}^{-1})$
Acetylcholinesterase	Acetylcholine	1.4×10^4	9×10^{-5}	1.6×10^8
Carbonic anhydrase	CO_2	1×10^6	1.2×10^{-2}	8.3×10^7
	HCO_3^-	4×10^5	2.6×10^{-2}	1.5×10^7
Catalase	H_2O_2	4×10^7	1.1×10^0	4×10^7
Crotonase	Crotonyl-CoA	5.7×10^3	2×10^{-5}	2.8×10^8
Fumarase	Fumarate	8×10^2	5×10^{-6}	1.6×10^8
	Malate	9×10^2	2.5×10^{-5}	3.6×10^7
β -Lactamase	Benzylpenicillin	2.0×10^3	2×10^{-5}	1×10^8

Information from A. Fersht, *Structure and Mechanism in Protein Science*, p. 166, W. H. Freeman and Company, 1999.

Subtract $V_0[S]$ from both sides to give

$$V_0 K_m = V_{\max} [S] - V_0 [S]$$

Factor out $[S]$ and divide both sides by V_0 .

$$K_m = \frac{(V_{\max} - V_0)[S]}{V_0}$$

Now, equate $V_{\max}$ to $k_{\text{cat}}[E_t]$, and substitute the given values to obtain K_m .

$$V_{\max} = k_{\text{cat}}[E_t] = (600 \text{ s}^{-1})(0.02 \text{ } \mu\text{M}) = 12 \text{ } \mu\text{M s}^{-1}$$

$$K_m = (12 \text{ } \mu\text{M s}^{-1} - 9.6 \text{ } \mu\text{M s}^{-1})40 \text{ } \mu\text{M} / 9.6 \text{ } \mu\text{M s}^{-1}$$

$$= 96 \text{ } \mu\text{M}^2 \text{ s}^{-1} / 9.6 \text{ } \mu\text{M s}^{-1} = 10 \text{ } \mu\text{M}$$

Once you have worked with this equation, you will recognize shortcuts to solve problems like this. For example, rearranging Equation 6-9 by simply dividing both sides by $V_{\max}$ gives

$$\frac{V_0}{V_{\max}} = \frac{[S]}{K_m + [S]}$$

Thus, the ratio $V_0/V_{\max} = 9.6 \text{ } \mu\text{M s}^{-1} / 12 \text{ } \mu\text{M s}^{-1} = [S]/(K_m + [S])$. This sometimes simplifies the process of solving for K_m , in this case, giving $0.25[S]$, or $10 \text{ } \mu\text{M}$.

WORKED EXAMPLE 6-2 Determination of $[S]$

In a separate happyase experiment using $[E_t] = 10 \text{ mM}$, the reaction velocity, V_0 , is measured as $3 \text{ } \mu\text{M s}^{-1}$. What is the $[S]$ used in this experiment?

SOLUTION: Using the same logic as in Worked Example 6-1 — equating $V_{\max}$ to $k_{\text{cat}}[E_t]$ — we see that $V_{\max}$ for this enzyme concentration is $6 \text{ } \mu\text{M s}^{-1}$. Note that V_0 is exactly half of $V_{\max}$. Recall that K_m is by definition equal to the $[S]$ at which $V_0 = \frac{1}{2}V_{\max}$. Thus, in this example, the $[S]$ must be the same as K_m , or $10 \text{ } \mu\text{M}$. If V_0 were anything other than $\frac{1}{2}V_{\max}$, it would be simplest to use the expression $V_0/V_{\max} = [S]/(K_m + [S])$ to solve for $[S]$.

Many Enzymes Catalyze Reactions with Two or More Substrates

We have seen how $[S]$ affects the rate of a simple enzymatic reaction with only one substrate molecule ($S \rightarrow P$). In most enzymatic reactions, however, two (and sometimes more) different substrate molecules bind to the enzyme and participate in the reaction. Nearly two-thirds of all enzymatic reactions have two substrates and two products. These are generally reactions in which a group is transferred from one substrate to the other, or one substrate is oxidized while the other is reduced. For example, in the reaction catalyzed by hexokinase, ATP and glucose are the substrate molecules, and ADP and glucose 6-phosphate are the products:

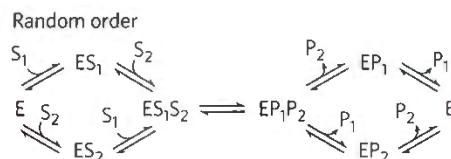


A phosphoryl group is transferred from ATP to glucose. The rates of such bisubstrate reactions can also be analyzed by

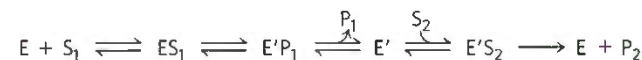
the Michaelis-Menten approach. Hexokinase has a characteristic K_m for each of its substrates (Table 6-6).

Enzymatic reactions with two substrates proceed by one of several different types of pathways. In some cases, both substrates are bound to the enzyme concurrently at some point in the course of the reaction, forming a noncovalent ternary complex (Fig. 6-15a); the substrates bind in a random sequence or in a specific order. Ordered binding occurs when binding of the first substrate creates

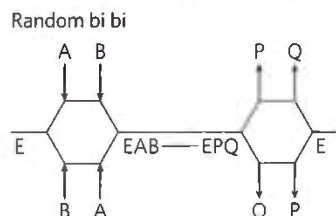
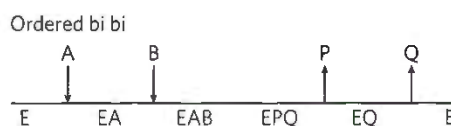
(a) Enzyme reaction involving a ternary complex



(b) Enzyme reaction in which no ternary complex is formed



(c) Cleland nomenclature



(d) Ping-Pong in Cleland nomenclature

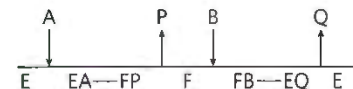


FIGURE 6-15 Common mechanisms for enzyme-catalyzed bisubstrate reactions. (a) The enzyme and both substrates come together to form a ternary complex. In ordered binding, substrate 1 must bind before substrate 2 can bind productively. In random binding, the substrates can bind in either order. Product dissociation can also be ordered or random. (b) An enzyme-substrate complex forms, a product leaves the complex, the altered enzyme forms a second complex with another substrate molecule, and the second product leaves, regenerating the enzyme. Substrate 1 may transfer a functional group to the enzyme (to form the covalently modified E'), which is subsequently transferred to substrate 2. This is called a Ping-Pong or double-displacement mechanism. (c) Ternary complex formation depicted using Cleland nomenclature. In the ordered bi bi and random bi bi reactions shown here, the release of product follows the same pattern as the binding of substrate — both ordered or both random. (d) The Ping-Pong or double-displacement reaction described with Cleland nomenclature.

a condition, often a conformation change, required for the second substrate to bind. In other cases, the first substrate is converted to product and dissociates before the second substrate binds, so no ternary complex is formed. An example of this is the Ping-Pong, or double-displacement, mechanism (Fig. 6-15b).

A shorthand notation developed by W. W. Cleland can be helpful in describing reactions with multiple substrates and products. In this system, referred to as **Cleland nomenclature**, substrates are denoted A, B, C, and D, in the order in which they bind to the enzyme, and products are denoted P, Q, S, and T, in the order in which they dissociate. Enzymatic reactions with one, two, three, or four substrates are referred to as uni, bi, ter, and quad, respectively. The enzyme is, as usual, denoted E, but if it is modified in the course of the reaction, successive forms are denoted F, G, and so on. The progress of the reaction is indicated with a horizontal line, with successive chemical species indicated below it. If there is an alternative in the reaction path, the horizontal line is bifurcated. Steps involving binding and dissociating substrates and products are indicated with vertical lines.

Common reactions with two substrates and two products (bi bi) are described with the shorthand forms illustrated in Figure 6-15c for an ordered bi bi reaction and a random bi bi reaction. In the latter example, the release of product is also random, as indicated by the two sets of bifurcations. Rarely, the binding of substrates is ordered and the release of products is random, or vice versa, eliminating the bifurcation at one end or the other of the progress line. In a Ping-Pong reaction, lacking a ternary complex, the pathway has a transient second form of the enzyme, F (Fig. 6-15d). This is the form in which a group has been transferred from the first substrate, A, to create a transient covalent attachment to the enzyme. As noted above, such reactions are often called double-displacement reactions, as a group is transferred first from substrate A to the enzyme and then from the enzyme to substrate B. Substrates A and B do not encounter each other on the enzyme.

Michaelis-Menten steady-state kinetics can provide only limited information about the number of steps and intermediates in an enzymatic reaction, but the approach can be used to distinguish between pathways that have a ternary intermediate and pathways—including Ping-Pong pathways—that do not (Fig. 6-16). As we will see when we consider enzyme inhibition, steady-state kinetics can also distinguish between ordered and random binding of substrates and products in reactions with ternary intermediates.

Enzyme Activity Depends on pH

In general, steady-state kinetics provides information required to characterize an enzyme and assess its catalytic efficiency. Additional information can be gained by

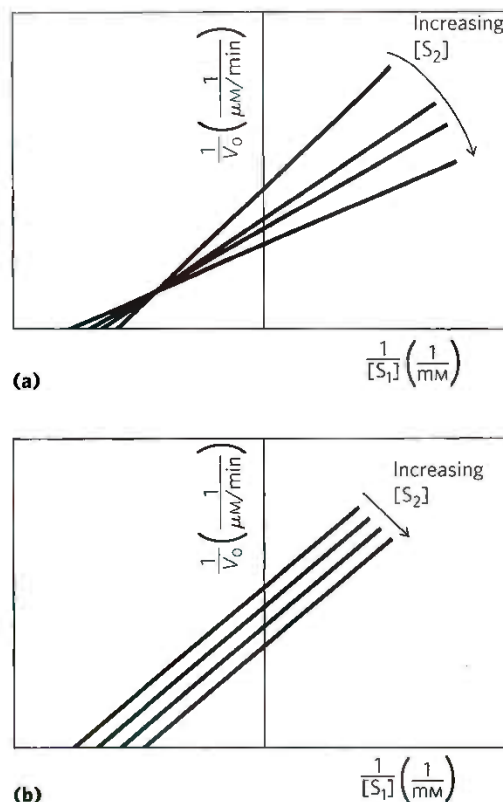


FIGURE 6-16 Steady-state kinetic analysis of bisubstrate reactions. In these double-reciprocal plots, the concentration of substrate 1 is varied while the concentration of substrate 2 is held constant. This is repeated for several values of $[S_2]$, generating several separate lines. (a) Intersecting lines indicate that a ternary complex is formed in the reaction; (b) parallel lines indicate a Ping-Pong (double-displacement) pathway.

examination of how the key experimental parameters k_{cat} and k_{cat}/K_m change when reaction conditions change, particularly pH. Enzymes have an optimum pH (or pH range) at which their activity is maximal (Fig. 6-17); at higher or lower pH, activity decreases. This is not surprising. Amino acid side chains in the active site may act as weak acids and bases only if they maintain a certain state of ionization. Elsewhere in the protein, removing a proton from a His residue, for example, might eliminate an ionic interaction that is essential for stabilizing the active conformation of the enzyme. A less common cause of pH sensitivity is titration of a group on the substrate.

The pH range over which an enzyme undergoes changes in activity can provide a clue to the type of amino acid residue involved (see Table 3-1). A change in activity near pH 7.0, for example, often reflects titration of a His residue. The effects of pH must be interpreted with some caution, however. In the closely packed environment of a protein, the $\text{p}K_a$ of amino acid side chains can be significantly altered. For example, a nearby positive charge can lower the $\text{p}K_a$ of a Lys residue, and a nearby negative charge can increase it. Such effects sometimes result in a $\text{p}K_a$ that is shifted by several pH units from its

value in the free amino acid. In the enzyme acetoacetate decarboxylase, for example, one Lys residue has a pK_a of 6.6 (compared with 10.5 in free lysine) due to electrostatic effects of nearby positive charges.

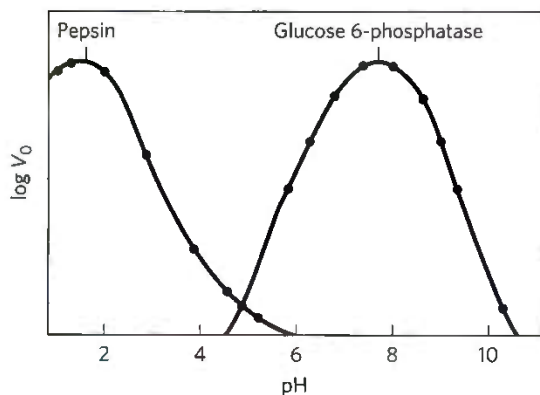
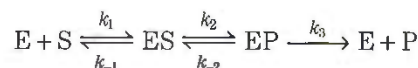


FIGURE 6-17 The pH-activity profiles of two enzymes. These curves are constructed from measurements of initial velocities when the reaction is carried out in buffers of different pH. Because pH is a logarithmic scale reflecting 10-fold changes in $[H^+]$, the changes in V_0 are also plotted on a logarithmic scale. The pH optimum for the activity of an enzyme is generally close to the pH of the environment in which the enzyme is normally found. Pepsin, a peptidase found in the stomach, has a pH optimum of about 1.6. The pH of gastric juice is between 1 and 2. Glucose 6-phosphatase of hepatocytes (liver cells), with a pH optimum of about 7.8, is responsible for releasing glucose into the blood. The normal pH of the cytosol of hepatocytes is about 7.2.

Pre-Steady State Kinetics Can Provide Evidence for Specific Reaction Steps

The mechanistic insight provided by steady-state kinetics can be augmented, sometimes dramatically, by an examination of the pre-steady state. Consider an enzyme with a reaction mechanism that conforms to the scheme in Equation 6-28, featuring three steps:



Overall catalytic efficiency for this reaction can be assessed with steady-state kinetics, but the rates of the individual steps cannot be determined in this way, and the slow (rate-limiting) step can rarely be identified. For the rate constants of individual steps to be measured, the reaction must be studied during its pre-steady state. The first turnover of an enzyme-catalyzed reaction often occurs in seconds or milliseconds, so researchers use special equipment that allows mixing and sampling on this time scale (**Fig. 6-18a**). Reactions are stopped and protein-bound products are quantified, after the timed addition and rapid mixing of an acid that denatures the protein and releases all bound molecules. A detailed description of pre-steady state kinetics is beyond the scope of this text, but we can illustrate the power of this approach by a simple example of an enzyme that uses the pathway

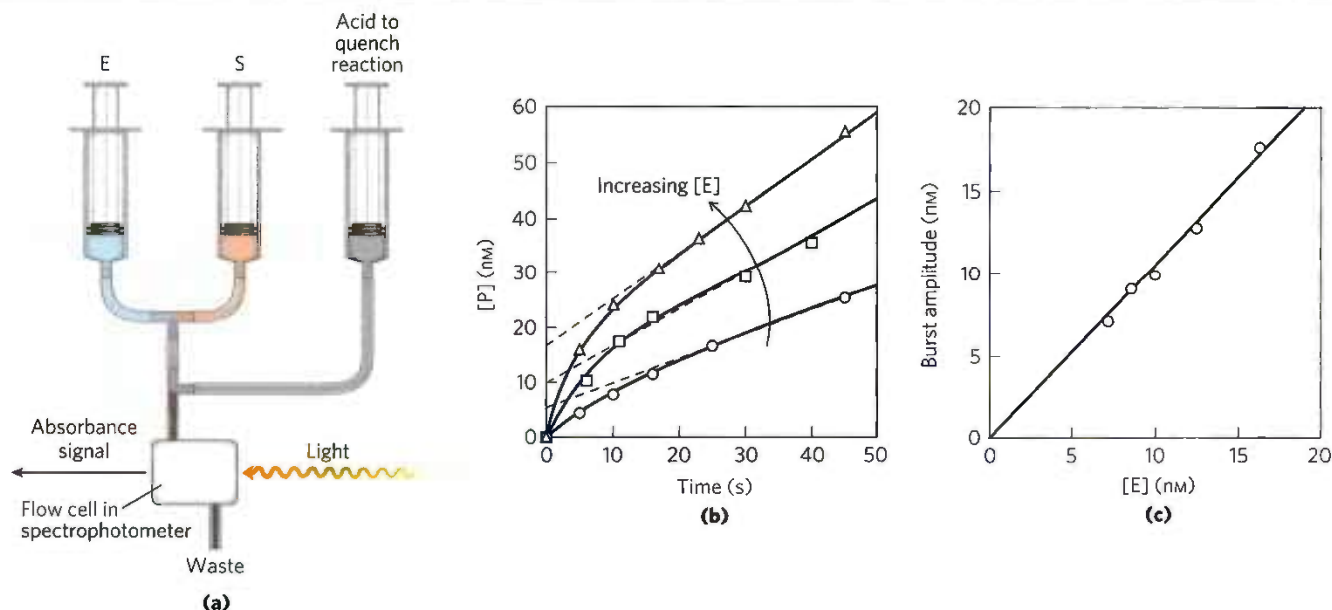


FIGURE 6-18 Pre-steady state kinetics. The transient phase that constitutes the pre-steady state often exists for mere seconds or milliseconds, requiring specialized equipment to monitor it. (a) A simple schematic for a rapid-mixing device, called a stopped-flow device. Enzyme (E) and substrate (S) are mixed with the aid of mechanically operated syringes. The reaction is quenched at a programmed time by adding a denaturing acid through another syringe, and the amount of product formed is measured, in this case with a spectrophotometer. (b) Experimental data for an enzyme reaction show the pre-steady state occurring in the first 5 to 10 seconds. This is a relatively slow reaction and is used as an example because the steady state can be conveniently monitored. The slope of the lines after 15 seconds reflects the steady state. Extrapolating this slope back to zero time (dashed

lines) gives the amplitude of the burst phase. The progress of the reaction during the pre-steady state primarily reflects the chemical steps in the reaction (details of which are not shown). The presence of a burst implies that a step following the chemical step that produces P is rate-limiting—in this case, the product-release step. Notice that the extrapolated intercept at time = 0 increases as [E] increases. (c) A plot of burst amplitude (the intercepts from (b)) versus [E] shows that one molecule of P is formed in each active site during the burst (pre-steady state) phase. This provides evidence that product release is the rate-limiting step, because it is the only step following product formation in this simple enzymatic reaction. The enzyme used in this experiment was RNase P, one of the catalytic RNAs described in Chapter 26.

[(b, c) Data from J. Hsieh et al., *RNA* 15:224, 2009.]

shown in Equation 6-28. This example also involves an enzyme that catalyzes a relatively slow reaction, so the pre-steady state is more conveniently observed.

For many enzymes, dissociation of product is rate-limiting. In this example (Fig. 6-18b, c), the rate of dissociation of the product (k_3) is slower than the rate of its formation (k_2). Product dissociation therefore dictates the rates observed in the steady state. How do we know that k_3 is rate-limiting? A slow k_3 gives rise to a burst of product formation in the pre-steady state, because the preceding steps are relatively fast. The burst reflects the rapid conversion of one molecule of substrate to one molecule of product at each enzyme active site. The observed rate of product formation soon slows to the steady-state rate as the bound product is slowly released. Each enzymatic turnover after the first one must proceed through the slow product-release step. However, the rapid generation of product in that first turnover provides much information. The amplitude of the burst—when one molecule of product is generated per molecule of enzyme present (Fig. 6-18c), measured by extrapolating the steady-state progress line back to zero time—is the highest amplitude possible. This provides one piece of evidence that product release is, indeed, rate-limiting. The rate constant for the chemical reaction step, k_2 , can be derived from the observed rate of the burst phase.

Of course, enzymes do not always conform to the simple reaction scheme of Equation 6-28. Formally, the observation of a burst indicates that a rate-limiting step (typically, product release, or an enzyme conformational change, or another chemical step) occurs after formation of the product being monitored. Additional experiments and analysis can often define the rates of each step in a multistep enzymatic reaction. Some examples of the application of pre-steady state kinetics are included in the descriptions of specific enzymes in Section 6.4.

Enzymes Are Subject to Reversible or Irreversible Inhibition

Enzyme inhibitors are molecules that interfere with catalysis, slowing or halting enzymatic reactions. Enzymes catalyze virtually all cellular processes, so it should not be surprising that enzyme inhibitors are among the most important pharmaceutical agents known. For example, aspirin (acetylsalicylate) inhibits the enzyme that catalyzes the first step in the synthesis of prostaglandins, compounds involved in many processes, including some that produce pain. The study of enzyme inhibitors also has provided valuable information about enzyme mechanisms and has helped define some metabolic pathways. There are two broad classes of enzyme inhibitors: reversible and irreversible.

Reversible Inhibition One common type of **reversible inhibition** is called competitive (Fig. 6-19a). A **competitive inhibitor** competes with the substrate for the active site of an enzyme. While the inhibitor (I) occupies the active site,

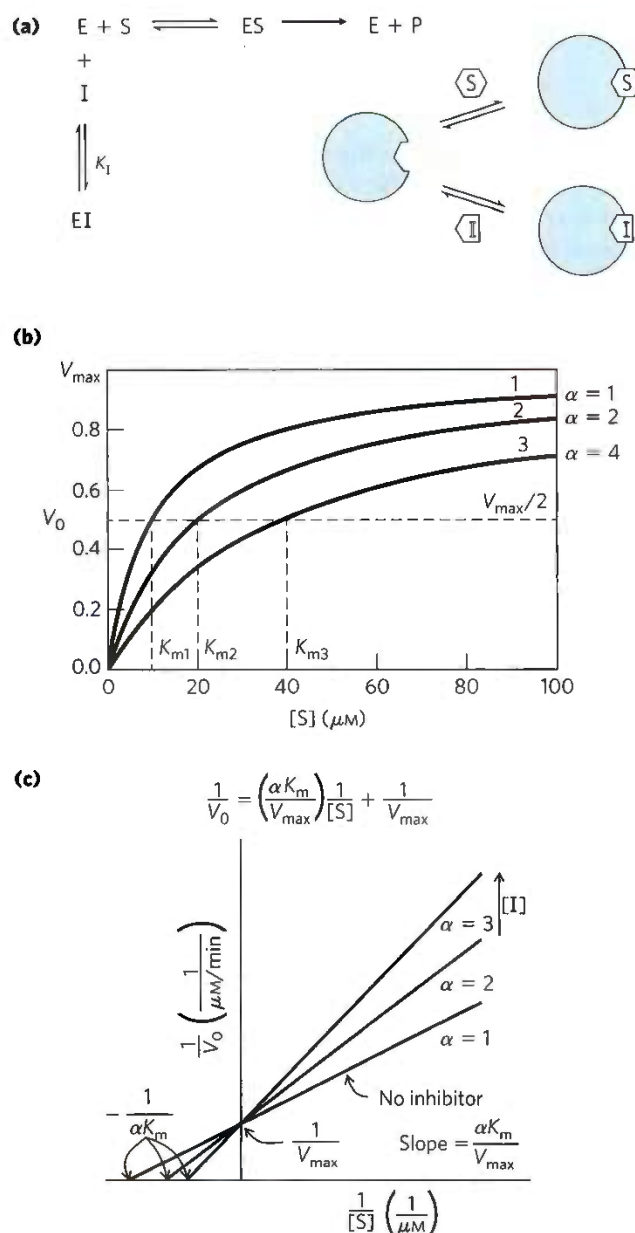


FIGURE 6-19 Competitive inhibition. (a) Competitive inhibitors bind to the enzyme's active site; K_I is the equilibrium dissociation constant for inhibitor binding to E. (b) Competitive inhibitors affect the observed K_m , but not the $V_{\max}$. This is readily evident in the plot of V_0 versus $[S]$. (c) In a Lineweaver-Burk plot, the lines generated + and - inhibitor intersect on the y axis, which reflects $1/V_{\max}$. As the observed K_m increases in the presence of an inhibitor, the intercept on the x axis ($-1/K_m$) moves to the right.

the substrate is excluded, and vice versa. Many competitive inhibitors are structurally similar to the substrate and combine with the enzyme to form an unreactive EI complex. Even fleeting combinations of this type will reduce the efficiency of an enzyme. Competitive inhibition can be analyzed quantitatively by steady-state kinetics. In the presence of a competitive inhibitor, the Michaelis-Menten equation (Eqn 6-9) becomes

$$V_0 = \frac{V_{\max}[S]}{\alpha K_m + [S]} \quad (6-31)$$

where

$$\alpha = 1 + \frac{[I]}{K_I} \quad \text{and} \quad K_I = \frac{[E][I]}{[EI]}$$

Equation 6-31 describes the important features of competitive inhibition. The experimentally determined variable αK_m , the K_m observed in the presence of the inhibitor, is often called the "apparent" K_m .

Bound inhibitor does not inactivate the enzyme. When the inhibitor dissociates, substrate can bind and react. Because the inhibitor binds reversibly to the enzyme, the competition can be biased to favor the substrate simply by adding more substrate. When $[S]$ far exceeds $[I]$, the probability that an inhibitor molecule will bind to the enzyme is minimized and the reaction exhibits its normal V_{max} . However, in the presence of inhibitor, the $[S]$ at which $V_0 = \frac{1}{2} V_{max}$, the apparent K_m , increases in the presence of inhibitor by the factor α (Fig. 6-19b). This effect on apparent K_m , combined with the absence of an effect on V_{max} , is diagnostic of competitive inhibition and is readily revealed in a double-reciprocal plot (Fig. 6-19c). The equilibrium constant for inhibitor binding, K_I , can be obtained from the same plot.

 A medical therapy based on competition at the active site is used to treat patients who have ingested methanol, a solvent found in gas-line antifreeze. The liver enzyme alcohol dehydrogenase converts methanol to formaldehyde, which is damaging to many tissues. Blindness is a common result of methanol ingestion, because the eyes are particularly sensitive to formaldehyde. Ethanol competes effectively with methanol as an alternative substrate for alcohol dehydrogenase. The effect of ethanol is much like that of a competitive inhibitor, with the distinction that ethanol is also a substrate for alcohol dehydrogenase and its concentration will decrease over time as the enzyme converts it to acetaldehyde. The therapy for methanol poisoning is slow intravenous infusion of ethanol, at a rate that maintains a controlled concentration in the blood for several hours. This slows the formation of formaldehyde, lessening the danger while the kidneys filter out the methanol to be excreted harmlessly in the urine. ■

Two other types of reversible inhibition, uncompetitive and mixed, can be defined in terms of one-substrate enzymes, but in practice are observed only with enzymes having two or more substrates. An **uncompetitive inhibitor** (Fig. 6-20a) binds at a site distinct from the substrate active site and, unlike a competitive inhibitor, binds only to the ES complex. In the presence of an uncompetitive inhibitor, the Michaelis-Menten equation is altered to

$$V_0 = \frac{V_{max}[S]}{K_m + \alpha'[S]} \quad (6-32)$$

where

$$\alpha' = 1 + \frac{[I]}{K'_I} \quad \text{and} \quad K'_I = \frac{[ES][I]}{[ESI]}$$

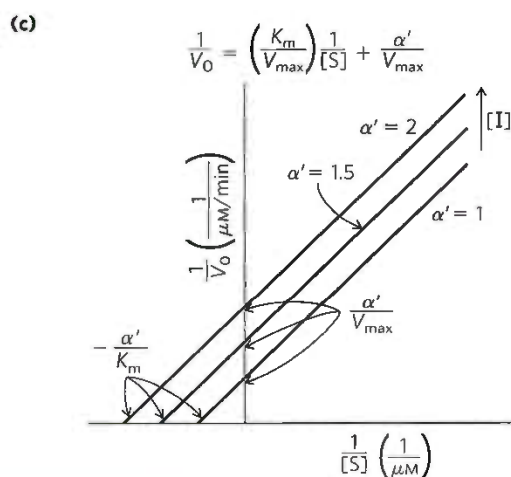
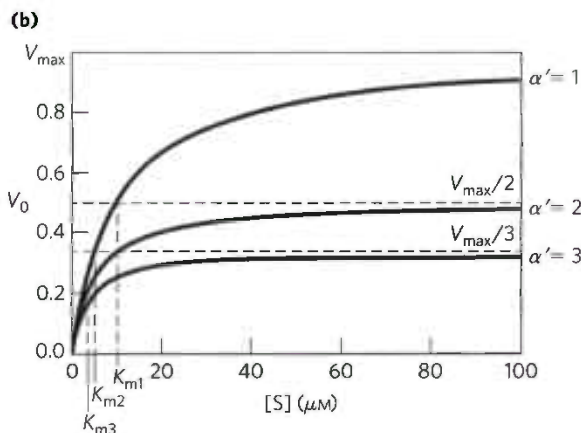
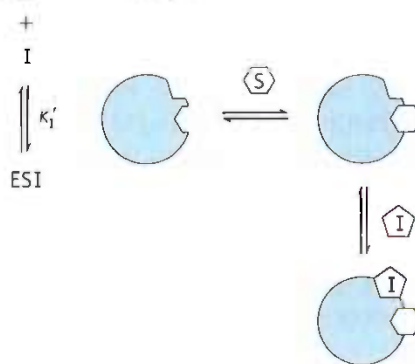


FIGURE 6-20 Uncompetitive inhibition. (a) Uncompetitive inhibitors bind at a separate site, but bind only to the ES complex; K'_I is the equilibrium constant for an inhibitor binding to ES. (b) In the presence of an uncompetitive inhibitor, both the K_m and the V_{max} decline, and by equivalent factors. In the V_0 versus $[S]$ plot, the decline in V_{max} is somewhat easier to discern than the decline in K_m . (c) The Lineweaver-Burk plot for an uncompetitive inhibitor is quite diagnostic, as the lines generated in the presence and absence of an inhibitor are parallel. Note that the lines seen in the presence of an inhibitor are always above the line generated in the absence of an inhibitor, reflecting the decline in both K_m and V_{max} brought about by the inhibitor (i.e., the intercepts move up on the y axis and to the left on the x axis).

As described by Equation 6-32, at high concentrations of substrate, V_0 approaches V_{max}/α' . Thus, an uncompetitive inhibitor lowers the measured V_{max} . Apparent K_m also decreases, because the $[S]$ required to reach

one-half $V_{\max}$ decreases by the factor α' (Fig. 6-20b, c). This behavior can be explained as follows. Because the enzyme is inactive when the uncompetitive inhibitor is bound, but the inhibitor is not competing with substrate for binding, the inhibitor effectively removes some fraction of the enzyme molecules from the reaction. Given that $V_{\max}$ depends on $[E]$, the observed $V_{\max}$ decreases. Given that the inhibitor binds only to the ES complex, only ES (not free enzyme) is deleted from the reaction, so the $[S]$ needed to reach $\frac{1}{2}V_{\max}$ —that is, K_m —declines by the same amount.

A **mixed inhibitor** (Fig. 6-21a) also binds at a site distinct from the substrate active site, but it binds to either E or ES. The rate equation describing mixed inhibition is

$$V_0 = \frac{V_{\max}[S]}{\alpha K_m + \alpha'[S]} \quad (6-33)$$

where α and α' are defined as above. A mixed inhibitor usually affects both K_m and $V_{\max}$ (Fig. 6-21b, c). $V_{\max}$ is affected because the inhibitor renders some fraction of the available enzyme molecules inactive, lowering the effective $[E]$ on which $V_{\max}$ depends. The K_m may increase or decrease, depending on which enzyme form, E or ES, the inhibitor binds to most strongly. The special case of $\alpha = \alpha'$, rarely encountered in experiments, historically has been defined as **noncompetitive inhibition**. Examine Equation 6-33 to see why a noncompetitive inhibitor would affect the $V_{\max}$ but not the K_m .

Equation 6-33 is a general expression for the effects of reversible inhibitors, simplifying to the expressions for competitive inhibition and uncompetitive inhibition when $\alpha' = 1.0$ or $\alpha = 1.0$, respectively. From this expression we can summarize the effects of inhibitors on individual kinetic parameters. For all reversible inhibitors, the apparent $V_{\max} = V_{\max}/\alpha'$, because the right side of Equation 6-33 always simplifies to $V_{\max}/\alpha'$ at sufficiently high substrate concentrations. For competitive inhibitors, $\alpha' = 1.0$ and can thus be ignored. Taking this expression for apparent $V_{\max}$, we can also derive a general expression for apparent K_m to show how this parameter changes in the presence of reversible inhibitors. Apparent K_m , as always, equals the $[S]$ at which V_0 is one-half apparent $V_{\max}$ or, more generally, when $V_0 = V_{\max}/2\alpha'$. This condition is met when $[S] = \alpha K_m/\alpha'$. Thus, apparent $K_m = \alpha K_m/\alpha'$. The terms α and α' reflect the binding of inhibitor to E and ES, respectively. Thus, the term $\alpha K_m/\alpha'$ is a mathematical expression of the relative affinity of inhibitor for the two enzyme forms. This expression is simpler when either α or α' is 1.0 (for uncompetitive or competitive inhibitors), as summarized in Table 6-9.

In practice, uncompetitive inhibition and mixed inhibition are observed only for enzymes with two or more substrates—say, S_1 and S_2 —and are very important in the experimental analysis of such enzymes. If an inhibitor binds to the site normally occupied by S_1 ,

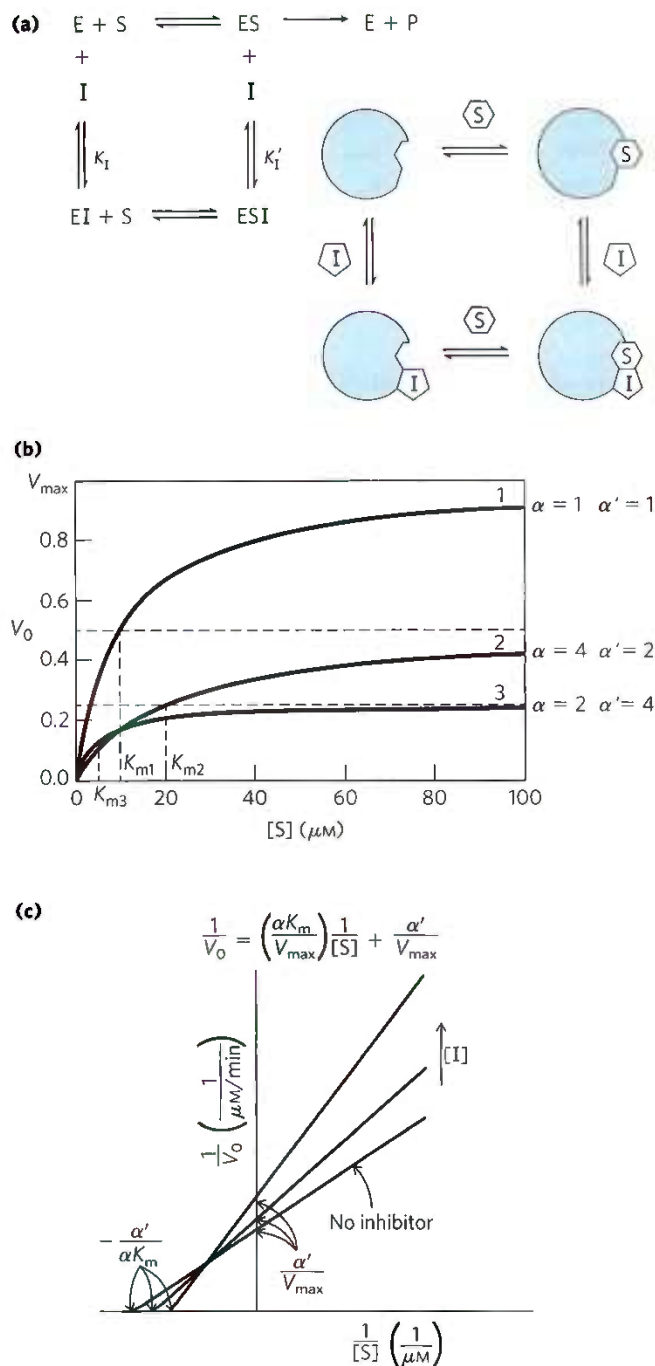


FIGURE 6-21 Mixed inhibition. (a) Mixed inhibitors bind at a separate site, but they may bind to either E or ES. (b) The kinetic patterns brought about by a mixed inhibitor are complex. The $V_{\max}$ always declines. The K_m may increase or decrease, depending upon the relative values of α and α' . (c) In a Lineweaver-Burk plot, the lines generated + and - inhibitor always intersect, but not on an axis. The intercepts on the y axis always move up, as the $V_{\max}$ declines. The lines may intersect either above or below the x axis. When they intersect above, as shown here, $\alpha > \alpha'$ and the observed K_m is increasing. When the lines intersect below the x axis (not shown), $\alpha < \alpha'$ and the observed K_m is decreasing.

it may act as a competitive inhibitor in experiments in which $[S_1]$ is varied. If an inhibitor binds to the site normally occupied by S_2 , it may act as a mixed or uncompetitive inhibitor of S_1 . The actual inhibition patterns

TABLE 6-9 Effects of Reversible Inhibitors on Apparent $V_{\max}$ and Apparent K_m

Inhibitor type	Apparent $V_{\max}$	Apparent K_m
None	$V_{\max}$	K_m
Competitive	$V_{\max}$	αK_m
Uncompetitive	$V_{\max}/\alpha'$	K_m/α'
Mixed	$V_{\max}/\alpha'$	$\alpha K_m/\alpha'$

observed depend on whether the S_1 - and S_2 -binding events are ordered or random, and thus the order in which substrates bind and products leave the active site can be determined. Product inhibition experiments in which one of the reaction products is provided as an inhibitor are often particularly informative. If only one of two reaction products is present, no reverse reaction can take place. However, a product generally binds to some part of the active site and can thus serve as an effective inhibitor. Enzymologists can combine steady-state kinetic studies involving different combinations and amounts of products and inhibitors with pre-steady state analysis to develop a detailed picture of the mechanism of a bisubstrate reaction.

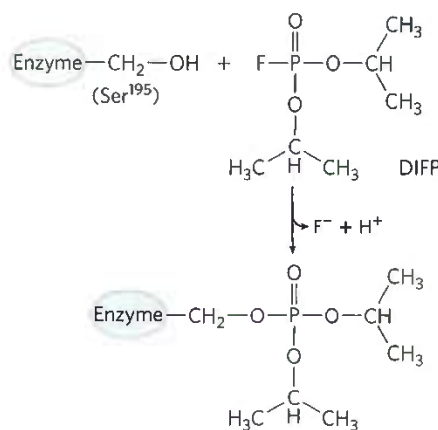
WORKED EXAMPLE 6-3 Effect of Inhibitor on K_m

The researchers working on happyase (see Worked Examples 6-1 and 6-2) discover that the compound STRESS is a potent competitive inhibitor of happyase. Addition of 1 nM STRESS increases the measured K_m for SAD by a factor of 2. What are the values for α and α' under these conditions?

SOLUTION: Recall that the apparent K_m , the K_m measured in the presence of a competitive inhibitor, is defined as αK_m . Because K_m for SAD increases by a factor of 2 in the presence of 1 nM STRESS, the value of α must be 2. The value of α' for a competitive inhibitor is 1, by definition.

Irreversible Inhibition The **irreversible inhibitors** bind covalently with or destroy a functional group on an enzyme that is essential for the enzyme's activity, or they form a highly stable noncovalent association. Formation of a covalent link between an irreversible inhibitor and an enzyme is a particularly effective way to inactivate an enzyme. Irreversible inhibitors are another useful tool for studying reaction mechanisms. Amino acids with key catalytic functions in the active site can sometimes be identified by determining which residue is covalently linked to an inhibitor after the enzyme is inactivated. An example is shown in Figure 6-22.

A special class of irreversible inhibitors is the **suicide inactivators**. These compounds are relatively unreactive until they bind to the active site of a specific enzyme. A suicide inactivator undergoes the first few

**FIGURE 6-22 Irreversible inhibition.** Reaction of chymotrypsin with diisopropylfluorophosphate (DIFP) modifies Ser¹⁹⁵ and irreversibly inhibits the enzyme. This has led to the conclusion that Ser¹⁹⁵ is the key active-site Ser residue in chymotrypsin.

chemical steps of the normal enzymatic reaction, but instead of being transformed into the normal product, the inactivator is converted to a very reactive compound that combines irreversibly with the enzyme. These compounds are also called **mechanism-based inactivators**, because they hijack the normal enzyme reaction mechanism to inactivate the enzyme. Suicide inactivators play a significant role in *rational drug design*, an approach to obtaining new pharmaceutical agents in which chemists synthesize novel substrates based on knowledge of substrates and reaction mechanisms. A well-designed suicide inactivator is specific for a single enzyme and is unreactive until it is within that enzyme's active site, so drugs based on this approach can offer the important advantage of few side effects (**Box 6-1**).

An irreversible inhibitor need not bind covalently to the enzyme. Noncovalent binding is enough, if that binding is so tight that the inhibitor dissociates only rarely. How does a chemist develop a tight-binding inhibitor?

P4 Recall that enzymes evolve to bind most tightly to the transition states of the reactions that they catalyze. In principle, if one can design a molecule that looks like that reaction transition state, it should bind tightly to the enzyme. Even though transition states cannot be observed directly, chemists can often predict the approximate structure of a transition state based on accumulated knowledge about reaction mechanisms. Although the transition state is by definition transient and thus unstable, in some cases stable molecules can be designed that resemble transition states. These are called **transition-state analogs**. They bind to an enzyme more tightly than does the substrate in the ES complex, because they fit into the active site better (that is, they form a greater number of weak interactions) than the substrate itself.



Curing African Sleeping Sickness with a Biochemical Trojan Horse

African sleeping sickness, or African trypanosomiasis, is caused by protists (single-celled eukaryotes) called trypanosomes (Fig. 1). This disease (and related trypanosome-caused diseases) is medically and economically significant in many developing nations. Until the late twentieth century, the disease was virtually incurable. Vaccines are ineffective because the parasite has a novel mechanism to evade the host immune system.

The cell coat of trypanosomes is covered with a single protein, which is the antigen to which the human immune system responds. Every so often, however, by a process of genetic recombination (see Table 28-1), a few cells in the population of infecting trypanosomes switch to a new protein coat, not recognized by the immune system. This process of “changing coats” can occur hundreds of times. The result is a chronic cyclic infection: the human host develops a fever, which subsides as the immune system beats back the first infection; trypanosomes with changed coats then become the seed for a second infection, and the fever recurs. This cycle can repeat for weeks, and the weakened person eventually dies.

One approach to treating African sleeping sickness uses pharmaceutical agents designed as mechanism-based enzyme inactivators (suicide inactivators). A vulnerable point in trypanosome metabolism is the pathway of polyamine biosynthesis. The polyamines spermine and spermidine, involved in DNA packaging, are required in large amounts in rapidly dividing cells. The first step in their synthesis is catalyzed by ornithine decarboxylase, an enzyme that requires for its function a coenzyme called pyridoxal phosphate. Pyridoxal phosphate (PLP), derived from vitamin B₆, forms a covalent bond

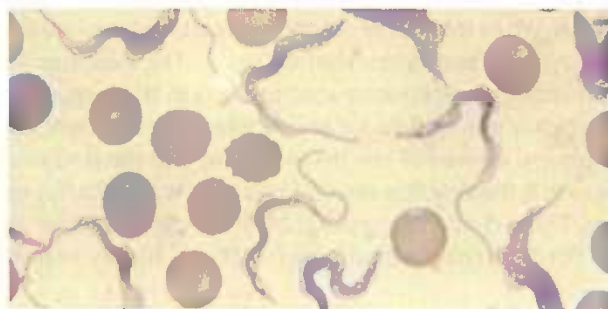
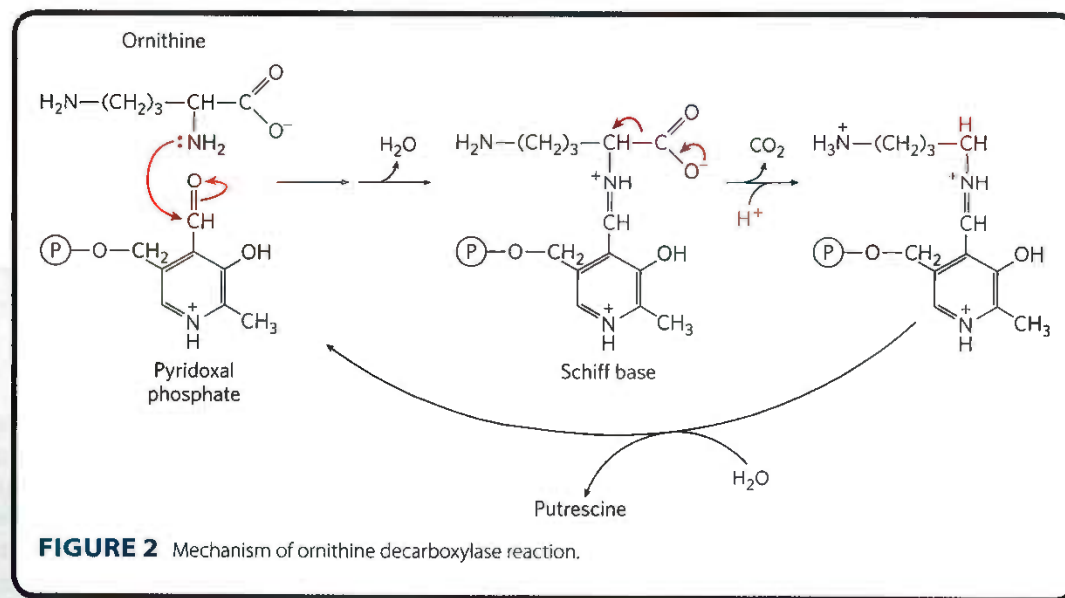


FIGURE 1 *Trypanosoma brucei rhodesiense*, one of several trypanosomes known to cause African sleeping sickness. [John Mansfield, University of Wisconsin–Madison, Department of Bacteriology.]

with the amino acid substrates of the reactions it is involved in and acts as an electron sink to facilitate a variety of reactions (see Fig. 22-32). In mammalian cells, ornithine decarboxylase undergoes rapid turnover—that is, a rapid, constant round of enzyme degradation and synthesis. In some trypanosomes, however, the enzyme (for reasons not well understood) is stable, not readily replaced by newly synthesized enzyme. An inhibitor of ornithine decarboxylase that binds permanently to the enzyme thus adversely affects the parasite, but has little effect on human cells, which can rapidly replace inactivated enzyme.

The first few steps of the normal reaction catalyzed by ornithine decarboxylase are shown in Figure 2. Once CO₂ is released, the electron movement is reversed and putrescine is produced (see Fig. 22-32). Based on this mechanism, several suicide inactivators have been designed, one of which is



(Continued on next page)

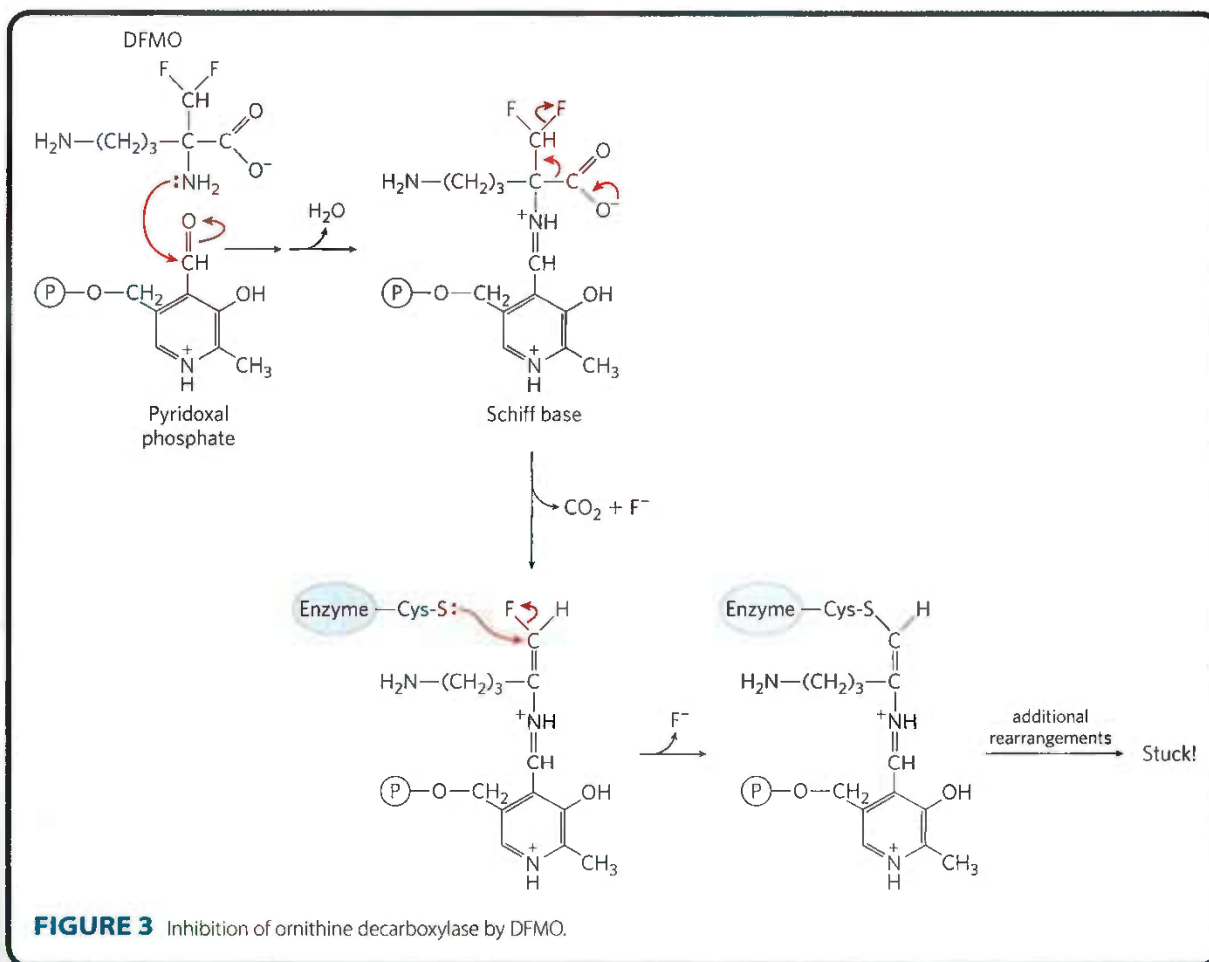


Curing African Sleeping Sickness with a Biochemical Trojan Horse (Continued)

difluoromethylornithine (DFMO). DFMO is relatively inert in solution. When it binds to ornithine decarboxylase, however, the enzyme is quickly inactivated (Fig. 3). The inhibitor acts by providing an alternative electron sink in the form of two strategically placed fluorine atoms, which are excellent leaving groups. Instead of electrons moving into the ring structure of PLP, the reaction results in displacement of a fluorine atom. The —S of a Cys residue at the enzyme's active site then forms a covalent complex with the highly reactive

PLP-inhibitor adduct, in an essentially irreversible reaction. In this way, the inhibitor makes use of the enzyme's own reaction mechanisms to kill it.

DFMO has proved highly effective in treating African sleeping sickness caused by *Trypanosoma brucei gambiense*. Approaches such as this show great promise for treating a wide range of diseases. The design of drugs based on enzyme mechanism and structure can complement the more traditional trial-and-error methods of developing pharmaceuticals.



The idea of transition-state analogs was suggested by Linus Pauling in the 1940s, and it has been explored using a variety of enzymes. For example, transition-state analogs designed to inhibit the glycolytic enzyme aldolase bind to that enzyme more than four orders of magnitude more tightly than do its substrates (Fig. 6-23). A transition-state analog cannot perfectly mimic a transition state. Some analogs, however, bind to a target

enzyme 10^2 to 10^8 times more tightly than does the normal substrate, providing good evidence that enzyme active sites are indeed complementary to transition states. The concept of transition-state analogs is important to the design of new pharmaceutical agents. As we shall see in Section 6.4, the powerful anti-HIV drugs called protease inhibitors were designed in part as tight-binding transition-state analogs.

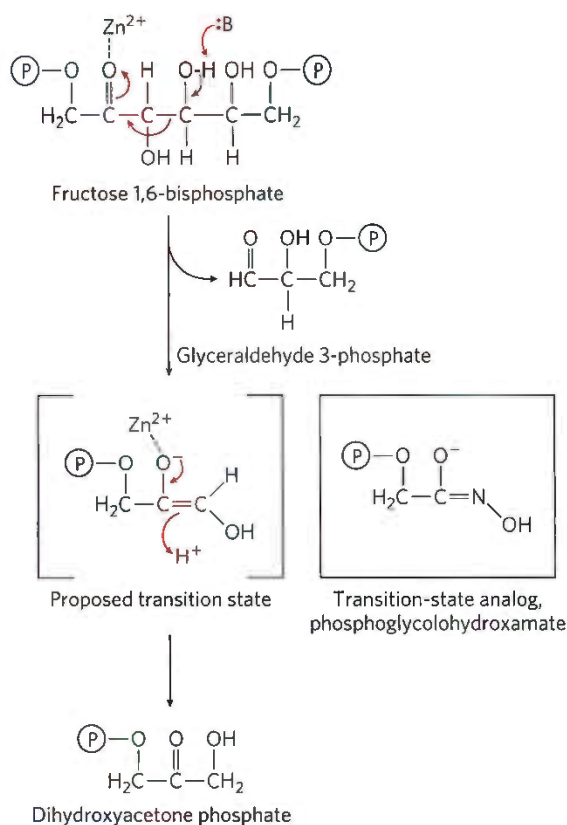


FIGURE 6-23 A transition-state analog. In glycolysis, a class II aldolase (found in bacteria and fungi) catalyzes the cleavage of fructose 1,6-bisphosphate to form glyceraldehyde 3-phosphate and dihydroxyacetone phosphate. The reaction proceeds via a reverse aldol-like mechanism. The compound phosphoglycolohydroxamate, which resembles the proposed enediolate transition state, binds to the enzyme nearly 10,000 times better than does the dihydroxyacetone phosphate product.

SUMMARY 6.3 Enzyme Kinetics as an Approach to Understanding Mechanism

■ Most enzymes have certain kinetic properties in common. When substrate is added to an enzyme, the reaction rapidly achieves a steady state in which the rate at which the ES complex forms balances the rate at which it breaks down. As $[S]$ increases, the steady-state activity of a fixed concentration of enzyme increases in a hyperbolic fashion to approach a characteristic maximum rate, $V_{\max}$, at which essentially all the enzyme has formed a complex with substrate.

■ The Michaelis-Menten equation

$$V_0 = \frac{V_{\max}[S]}{K_m + [S]}$$

relates initial velocity to $[S]$ and $V_{\max}$ through the Michaelis constant, K_m . Michaelis-Menten kinetics is also called steady-state kinetics. K_m and $V_{\max}$ have different meanings for different enzymes. However, the K_m is always

equal to the substrate concentration that results in a reaction rate equal to one-half $V_{\max}$.

■ The values of $V_{\max}$ and K_m can be determined by using transformations of the Michaelis-Menten equation that allow linear plotting of data and extrapolation (for example, Lineweaver-Burk), or (more accurately) by using non-linear regression.

■ The limiting rate of an enzyme-catalyzed reaction at saturation is described by the constant k_{cat} , the turnover number. The ratio k_{cat}/K_m provides a good measure of catalytic efficiency.

■ Most enzymes catalyze reactions involving multiple substrates, with the majority having two substrates and two products. The Michaelis-Menten equation is applicable to bisubstrate reactions, which occur by ternary complex or Ping-Pong (double-displacement) pathways.

■ Every enzyme has an optimum pH (or pH range) at which it has maximal activity. The pH-rate profile can provide mechanistic clues.

■ Pre-steady state kinetics can provide added insight into enzymatic reaction mechanisms.

■ Reversible inhibition of an enzyme may be competitive, uncompetitive, or mixed. Competitive inhibitors compete with substrate by binding reversibly to the active site, but they are not transformed by the enzyme. Uncompetitive inhibitors bind only to the ES complex, at a site distinct from the active site. Mixed inhibitors bind to either E or ES, again at a site distinct from the active site. In irreversible inhibition, an inhibitor binds permanently to an active site by forming a covalent bond or a very stable noncovalent interaction. Inhibition patterns can elucidate mechanism.

6.4 Examples of Enzymatic Reactions

Thus far we have focused on the general principles of catalysis and on introducing some of the kinetic parameters used to describe enzyme action. We now turn to several examples of specific enzyme reaction mechanisms.

To understand the complete mechanism of action of a purified enzyme, we need to identify all substrates, cofactors, products, and regulators. We also need to know (1) the temporal sequence in which enzyme-bound reaction intermediates form, (2) the structure of each intermediate and each transition state, (3) the rates of interconversion between intermediates, (4) the structural relationship of the enzyme to each intermediate, and (5) the energy that all reacting and interacting groups contribute to the intermediate complexes and transition states. There are still only a few enzymes for which we have an understanding that meets all these requirements.

Here we present the mechanisms for three enzymes: chymotrypsin, hexokinase, and enolase. These examples are not intended to cover all possible classes of enzyme

chemistry. They have been chosen in part because they are among the best-understood enzymes and in part because they clearly illustrate some general principles outlined in this chapter. We present the chymotrypsin example in order to review some of the conventions used to depict enzyme mechanisms. Much mechanistic detail and experimental evidence is necessarily omitted; no one book could completely document the rich experimental history of these enzymes. In addition, we consider only briefly the special contribution of coenzymes to the catalytic activity of many enzymes. The function of coenzymes is chemically varied, and we describe each coenzyme in detail as it is encountered in our discussion of metabolism in Part II of this book.

The Chymotrypsin Mechanism Involves Acylation and Deacylation of a Ser Residue

Bovine pancreatic chymotrypsin (M_r 25,191) is a **protease**, an enzyme that catalyzes the hydrolytic cleavage of peptide bonds. This protease is specific for peptide bonds adjacent to aromatic amino acid residues (Trp, Phe, Tyr). The three-dimensional structure of chymotrypsin is shown in **Figure 6-24**, with functional groups in the active site emphasized. The reaction catalyzed by this enzyme illustrates the principle of transition-state stabilization and also provides a classic example of general acid-base catalysis and covalent catalysis.

Chymotrypsin enhances the rate of peptide bond hydrolysis by a factor of at least 10^9 . It does not catalyze a direct attack of water on the peptide bond; instead, a transient covalent acyl-enzyme intermediate is formed. The reaction thus has two distinct phases. In the acylation phase, the peptide bond is cleaved and an ester linkage is formed between the peptide carbonyl carbon and the enzyme. In the deacylation phase, the ester linkage is hydrolyzed and the nonacylated enzyme is regenerated.

The first evidence for a covalent acyl-enzyme intermediate came from a classic application of pre-steady state kinetics. In addition to its action on polypeptides, chymotrypsin catalyzes the hydrolysis of small esters and amides. These reactions are much slower than hydrolysis of peptides because less binding energy is available with smaller substrates (the pre-steady state is also correspondingly longer), thus simplifying the analysis of the resulting reactions. In their investigations in 1954, B. S. Hartley and B. A. Kilby found that chymotrypsin hydrolysis of the ester *p*-nitrophenylacetate, as measured by release of *p*-nitrophenol, proceeds with a rapid burst before leveling off to a slower rate (**Fig. 6-25**). By extrapolating back to zero time, they concluded that the burst phase corresponded to the release of just under one molecule of *p*-nitrophenol for every enzyme molecule present (a small fraction of their enzyme molecules were inactive). Recall from Figure 6-18 that a burst implies that the rate-limiting step of catalysis occurs after release of the product being monitored. Hartley and Kilby interpreted the burst to reflect a rapid release of *p*-nitrophenol during a rapid acylation of all the enzyme

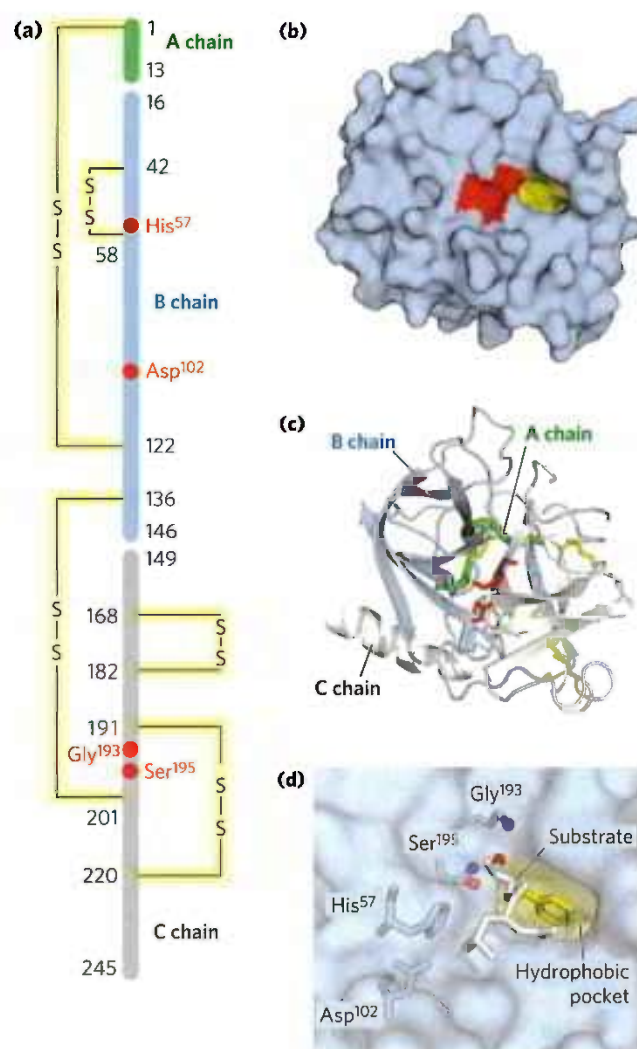


FIGURE 6-24 Structure of chymotrypsin. (a) A representation of primary structure, showing disulfide bonds and the amino acid residues crucial to catalysis. The protein consists of three polypeptide chains linked by disulfide bonds. (The numbering of residues in chymotrypsin, with "missing" residues 14, 15, 147, and 148, is explained in Fig. 6-42.) The active-site amino acid residues are grouped together in the three-dimensional structure. (b) A depiction of the enzyme emphasizing its surface. The hydrophobic pocket in which the aromatic amino acid side chain of the substrate is bound is shown in yellow. Key active-site residues, including Ser¹⁹⁵, His⁵⁷, and Asp¹⁰², are red. The roles of these residues in catalysis are illustrated in Figure 6-27. (c) The polypeptide backbone as a ribbon structure. Disulfide bonds are yellow; the three chains are colored as in part (a). (d) A close-up of the active site with a substrate (white and yellow) bound. The hydroxyl of Ser¹⁹⁵ attacks the carbonyl group of the substrate (the oxygens are red); the developing negative charge on the oxygen is stabilized by the oxyanion hole (amide nitrogens from Ser¹⁹⁵ and Gly¹⁹³, in blue), as explained in Figure 6-27. The aromatic amino acid side chain of the substrate (yellow) sits in the hydrophobic pocket. The amide nitrogen of the peptide bond to be cleaved (protruding toward the viewer and projecting the path of the rest of the substrate polypeptide chain) is shown in white. [(b, c, d) Data from PDB ID 7GCH, K. Brady et al., *Biochemistry* 29:7600, 1990]

molecules. They suggested that turnover of the enzyme was limited by a subsequent, slower deacylation step. Later work substantiated their hypothesis. The observation of a burst phase provides yet another example of the use of kinetics to break down a reaction into its constituent steps.

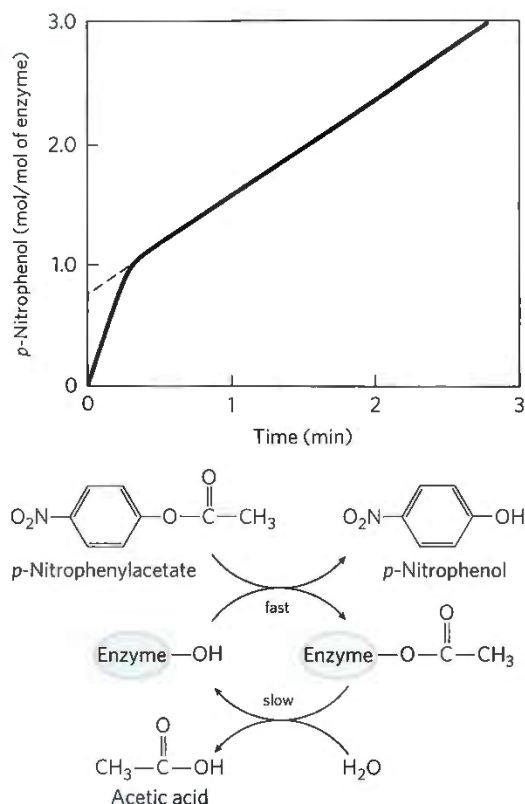


FIGURE 6-25 Pre-steady state kinetic evidence for an acyl-enzyme intermediate. The hydrolysis of *p*-nitrophenylacetate by chymotrypsin is measured by release of *p*-nitrophenol (a colored product). Initially, there is a rapid burst of *p*-nitrophenol release nearly stoichiometric with the amount of enzyme present. This reflects the fast acylation phase of the reaction. The subsequent rate is slower, because enzyme turnover is limited by the rate of the slower deacylation phase.

Researchers have discovered additional features of the chymotrypsin mechanism by analyzing the dependence of the reaction on pH. The rate of chymotrypsin-catalyzed cleavage generally exhibits a bell-shaped pH-rate profile (Fig. 6-26). The rates plotted in Figure 6-26a are obtained at low (subsaturating) substrate concentrations and therefore represent k_{cat}/K_m (see Eqn 6-30, p. 193). A more complete analysis of the rates at different substrate concentrations at each pH allows researchers to determine the individual contributions of the k_{cat} and K_m terms. After obtaining the maximum rates at each pH, one can plot the k_{cat} alone versus pH (Fig. 6-26b); after obtaining the K_m at each pH, researchers can then plot $1/K_m$ versus pH (Fig. 6-26c). Kinetic and structural analyses have revealed that the change in k_{cat} reflects the ionization state of His⁵⁷. The decline in k_{cat} at low pH results from protonation of His⁵⁷ (so that it can no longer extract a proton from Ser¹⁹⁵ in the first chemical step of the reaction). This rate reduction illustrates the importance of general acid and general base catalysis in the mechanism for chymotrypsin. The changes in the $1/K_m$ term reflect the ionization of the α -amino group of Ile¹⁶ (at the amino-terminal end of one of the enzyme's three polypeptide chains). This group forms a salt bridge to Asp¹⁹⁴, stabilizing the active conformation of the enzyme. When this group loses its proton at

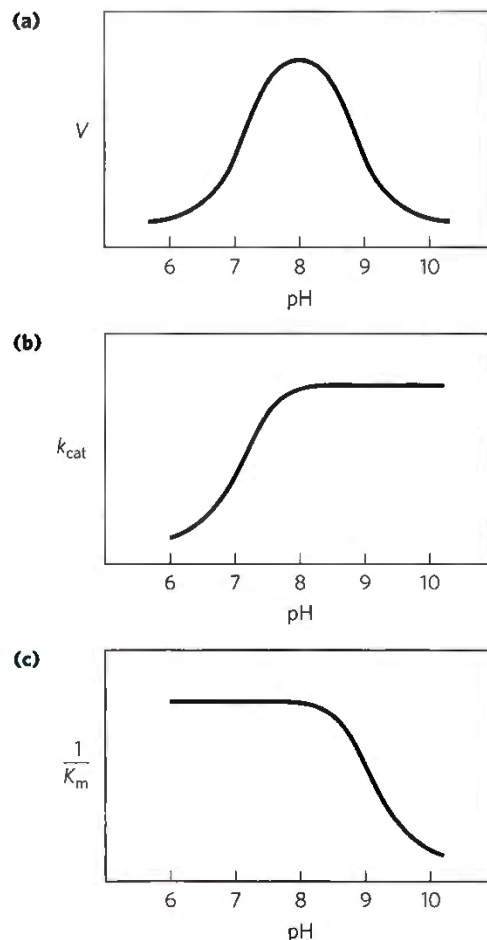
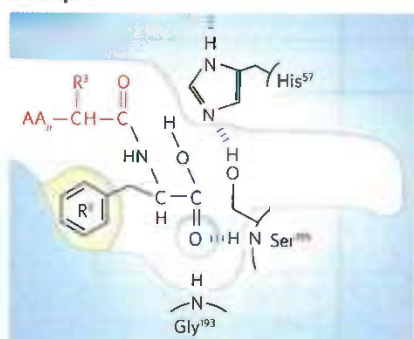
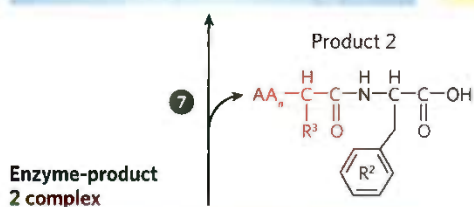
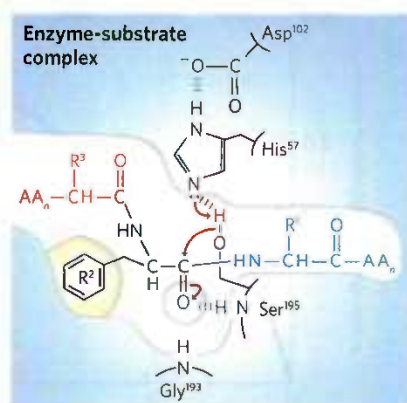
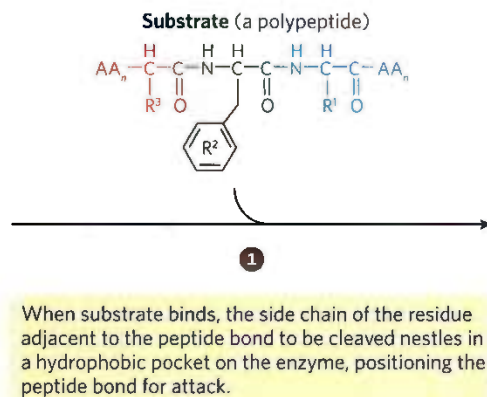
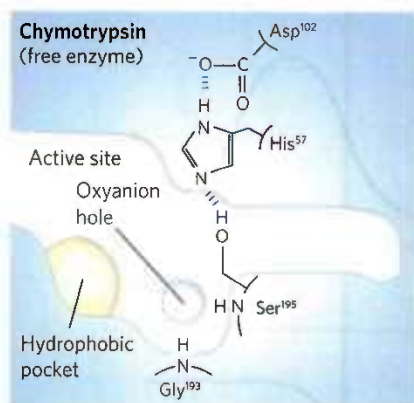


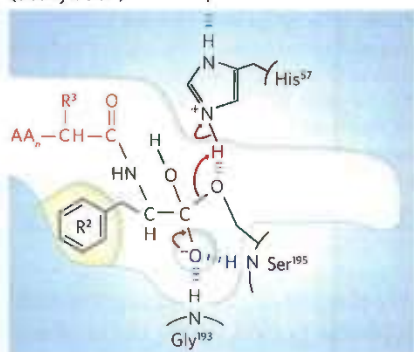
FIGURE 6-26 The pH dependence of chymotrypsin-catalyzed reactions. (a) The rates of chymotrypsin-mediated cleavage produce a bell-shaped pH-rate profile with an optimum at pH 8.0. The rate (V) plotted here is that at low substrate concentrations and thus reflects the term k_{cat}/K_m . The plot can be broken down to its components by using kinetic methods to determine the terms k_{cat} and K_m separately at each pH. When this is done (b, c), it becomes clear that the transition just above pH 7 is due to changes in k_{cat} , whereas the transition above pH 8.5 is due to changes in $1/K_m$. Kinetic and structural studies have shown that the transitions illustrated in (b) and (c) reflect the ionization states of the His⁵⁷ side chain (when substrate is not bound) and the α -amino group of Ile¹⁶ (at the amino terminus of the B chain), respectively. For optimal activity, His⁵⁷ must be unprotonated and Ile¹⁶ must be protonated.

high pH, the salt bridge is eliminated, and a conformational change closes the hydrophobic pocket where the aromatic amino acid side chain of the substrate inserts (Fig. 6-24). Substrates can no longer bind properly, which is measured kinetically as an increase in K_m .

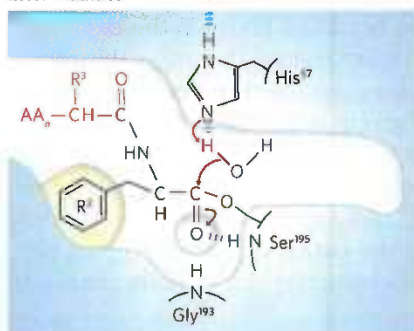
The chymotrypsin reaction is detailed in Figure 6-27. The nucleophile in the acylation phase is the oxygen of Ser¹⁹⁵. (Proteases with a Ser residue that plays this role in reaction mechanisms are called **serine proteases**.) The pK_a of a Ser hydroxyl group is generally too high for the unprotonated form to be present in significant concentrations at physiological pH. However, in chymotrypsin, Ser¹⁹⁵ is linked to His⁵⁷ and Asp¹⁰² in a hydrogen-bonding network referred to as the **catalytic triad**. When a peptide substrate binds to chymotrypsin, a subtle change in



Short-lived intermediate* (deacylation)

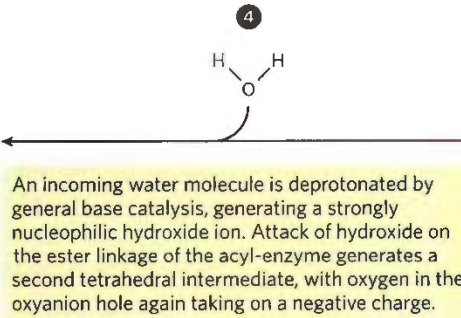


Acyl-enzyme intermediate



Dissociation of the second product from the active site regenerates free enzyme.

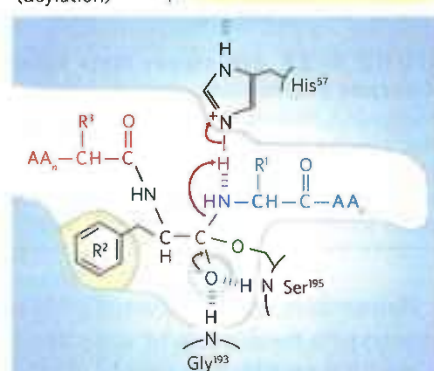
Collapse of the tetrahedral intermediate forms the second product, a carboxylate anion, and displaces Ser195.



Interaction of Ser195 and His57 generates a strongly nucleophilic alkoxide ion on Ser195; the ion attacks the peptide carbonyl group, forming a tetrahedral acyl-enzyme. This is accompanied by formation of a short-lived negative charge on the carbonyl oxygen of the substrate, which is stabilized by hydrogen bonding in the oxyanion hole.

2

Short-lived intermediate* (acylation)



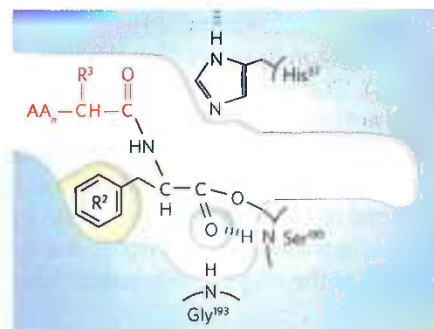
Product 1

$$\text{HN} - \overset{\text{R}^1}{\text{CH}} - \overset{\text{O}}{\parallel} \text{C} - AA_n$$

Instability of the negative charge on the substrate carbonyl oxygen leads to collapse of the tetrahedral intermediate; re-formation of a double bond with carbon displaces the bond between carbon and the amino group of the peptide linkage, breaking the peptide bond. The amino leaving group is protonated by His57, facilitating its displacement.

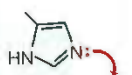
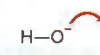
3

Acyl-enzyme intermediate



MECHANISM FIGURE 6-27 Hydrolytic cleavage of a peptide bond by chymotrypsin. The reaction has two phases. In the acylation phase (steps 1 to 4), formation of a covalent acyl-enzyme intermediate is coupled to cleavage of the peptide bond. In the deacylation phase (steps 5 to 7), deacylation regenerates the free enzyme; this is essentially the reverse of the acylation phase, with water mirroring, in reverse, the role of the amine component of the substrate.

*The short-lived tetrahedral intermediate following step 2 and the second tetrahedral intermediate that forms later, following step 5, are sometimes referred to as transition states, but this terminology can cause confusion. An *intermediate* is any chemical species with a finite lifetime, "finite" being defined as longer than the time required for a molecular vibration ($\sim 10^{-13}$ second). A *transition state* is simply the maximum-energy species formed on the reaction coordinate, and it does not have a finite lifetime. The tetrahedral intermediates formed in the chymotrypsin reaction closely resemble, both energetically and structurally, the transition states leading to their formation and breakdown. However, the intermediate represents a committed stage of completed bond formation, whereas the transition state is part of the process of reaction. In the case of chymotrypsin, given the close relationship between the intermediate and the actual transition state, the distinction between them is routinely glossed over. Furthermore, the interaction of the negatively charged oxygen with the amide nitrogens in the oxyanion hole, often referred to as transition-state stabilization, also serves to stabilize the intermediate in this case. Not all intermediates are so short-lived that they resemble transition states. The chymotrypsin acyl-enzyme intermediate is much more stable and more readily detected and studied, and it is never confused with a transition state.

Nucleophiles	Electrophiles
 Negatively charged oxygen (as in an unprotonated hydroxyl group or an ionized carboxylic acid)	 Carbon atom of a carbonyl group (the more electronegative oxygen of the carbonyl group pulls electrons away from the carbon)
 Negatively charged sulfhydryl	 Protonated imine group (activated for nucleophilic attack at the carbon by protonation of the imine)
 Carbanion	 Phosphorus of a phosphate group
 Uncharged amine group	 Proton
 Imidazole	
 Hydroxide ion	

How to Read Reaction Mechanisms—A Refresher

Chemical reaction mechanisms, which trace the formation and breakage of covalent bonds, are communicated with dots and curved arrows, a convention known informally as "electron pushing." A covalent bond consists of a shared pair of electrons. Nonbonded electrons important to the reaction mechanism are designated by dots ($\cdot\text{OH}$). Curved arrows ($\curvearrowright$) represent the movement of electron pairs. For movement of a single electron (as in a free radical reaction), a single-headed (fishhook-type) arrow is used ($\frown$). Most reaction steps involve an unshared electron pair (as in the chymotrypsin mechanism).

Some atoms are more electronegative than others; that is, they more strongly attract electrons. The relative electronegativities of atoms encountered in this text are $\text{F} > \text{O} > \text{N} > \text{C} \approx \text{S} > \text{P} \approx \text{H}$. For example, the two electron pairs making up a $\text{C}=\text{O}$ (carbonyl) bond are not shared equally; the carbon is relatively electron-deficient as the oxygen draws away the electrons. Many reactions involve an electron-rich atom (a nucleophile) reacting with an electron-deficient atom (an electrophile). Some common nucleophiles and electrophiles in biochemistry are shown at right.

In general, a reaction mechanism is initiated at an unshared electron pair of a nucleophile. In mechanism diagrams, the base of the electron-pushing arrow originates near the electron-pair dots, and the head of the arrow points directly at the electrophilic center being attacked. Where the unshared electron pair confers a formal negative charge on the nucleophile, the negative charge symbol itself can represent the unshared electron pair and serves as the base of the arrow. In the chymotrypsin mechanism, the nucleophilic electron pair in the ES complex between steps 1 and 2 is provided by the oxygen of the Ser¹⁹⁵ hydroxyl group. This electron pair (2 of the 8 valence electrons of the hydroxyl oxygen) provides the base of the curved arrow. The electrophilic center under attack is the carbonyl carbon of the peptide bond to be cleaved. The C, O, and N atoms have a maximum of 8 valence electrons, and H has a maximum of 2. These atoms are occasionally found in unstable states with less than their maximum allotment of electrons, but C, O, and N cannot have more than 8. Thus, when the electron pair from chymotrypsin's Ser¹⁹⁵ attacks the substrate's carbonyl carbon, an electron pair is displaced from the carbon valence shell (you cannot have 5 bonds to carbon!). These electrons move toward the more electronegative carbonyl oxygen. The oxygen has 8 valence electrons both before and after this chemical process, but the number shared with the carbon is reduced from 4 to 2, and the carbonyl oxygen acquires a negative charge. In step 3, the electron pair conferring the negative charge on the oxygen moves back to re-form a bond with carbon and reestablish the carbonyl linkage. Again, an electron pair must be displaced from the carbon, and this time it is the electron pair shared with the amino group of the peptide linkage. This breaks the peptide bond. The remaining steps follow a similar pattern.

conformation compresses the hydrogen bond between His⁵⁷ and Asp¹⁰², resulting in a stronger interaction, called a low-barrier hydrogen bond. This enhanced interaction increases the pK_a of His⁵⁷ from ~7 (for free histidine) to >12, allowing the His residue to act as an enhanced general base that can remove the proton from the Ser¹⁹⁵ hydroxyl group. Deprotonation prevents development of a highly unstable positive charge on the Ser¹⁹⁵ hydroxyl and makes the Ser side chain a stronger nucleophile. At later reaction stages, His⁵⁷ also acts as a proton donor, protonating the amino group in the displaced portion of the substrate (the leaving group).

As the Ser¹⁹⁵ oxygen attacks the carbonyl group of the substrate (Fig. 6-27, step 2), a very short-lived tetrahedral intermediate is formed in which the carbonyl oxygen acquires a negative charge. This charge, forming within a pocket on the enzyme called the oxyanion hole, is stabilized by hydrogen bonds contributed by the amide groups of two peptide bonds in the chymotrypsin backbone. One of these hydrogen bonds (contributed by Gly¹⁹³) is present only in this intermediate and in the transition states for its formation and breakdown; it reduces the energy required to reach these states. **P4** This is an example of the use of binding energy in catalysis through enzyme–transition state complementarity. The intermediate collapses in step 3, breaking the peptide bond. The amino group of the first product is protonated by His⁵⁷, now acting as a general acid catalyst. Water is the second substrate, entering the active site in step 4. As water attacks the carbon in the ester linkage in step 5, and the resulting intermediate collapses to break the ester linkage and generate the second product in step 6, His⁵⁷ again acts—first as a general base to deprotonate the water, and then as a general acid to protonate the Ser oxygen as it leaves. Dissociation of the second product (step 7) completes the reaction cycle.

An Understanding of Protease Mechanisms Leads to New Treatments for HIV Infection

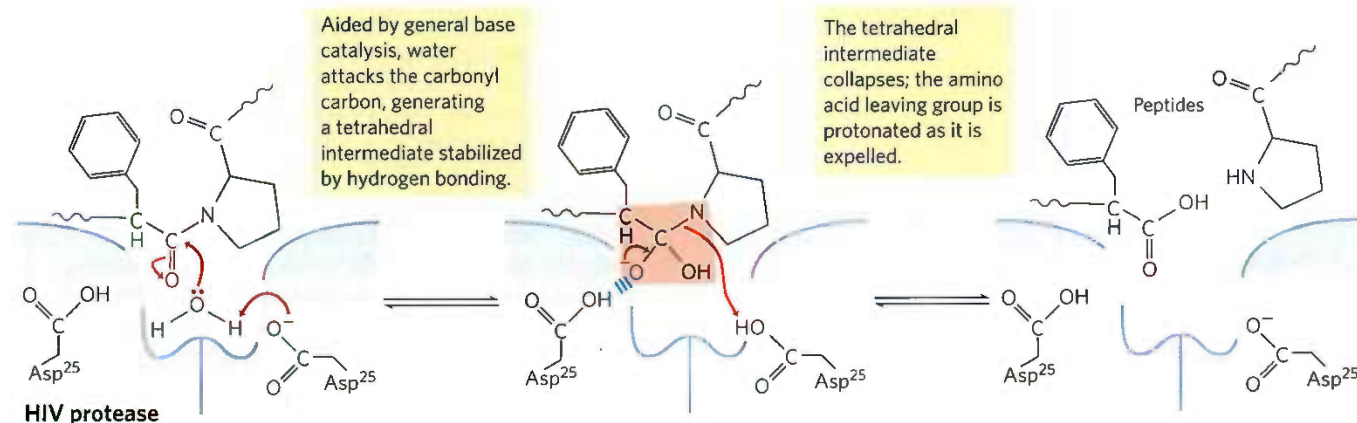


New pharmaceutical agents are almost always designed to inhibit an enzyme. The extremely successful therapies developed to treat HIV infection

provide a case in point. The human immunodeficiency virus (HIV) is the agent that causes acquired immune deficiency syndrome (AIDS). In 2018, 38 million people worldwide were living with HIV infection, with about 1.7 million new infections that year and approximately 770,000 fatalities. AIDS first surfaced as a worldwide epidemic in the 1980s; HIV was discovered soon after and was identified as a **retrovirus**. Retroviruses possess (1) an RNA genome and (2) an enzyme, reverse transcriptase, that is capable of using RNA to direct the synthesis of a complementary DNA. Efforts to understand HIV and develop therapies for HIV infection benefited from decades of basic research, both on enzyme mechanisms and on the properties of other retroviruses.

A retrovirus such as HIV has a relatively simple life cycle (see Fig. 26-29). Its RNA genome is converted to duplex DNA in several steps catalyzed by the reverse transcriptase (described in Chapter 26). The duplex DNA is then inserted into a chromosome in the nucleus of the host cell by the enzyme integrase (described in Chapter 25). The integrated copy of the viral genome can remain dormant indefinitely. Alternatively, it can be transcribed back into RNA, which can then be translated into proteins to construct new virus particles. Most of the viral genes are translated into large polypeptides, which are cut by an HIV protease into the individual proteins needed to make the virus (see Fig. 26-30). Only three key enzymes operate in this cycle: the reverse transcriptase, the integrase, and the protease. These enzymes thus represent the most promising drug targets.

There are four major subclasses of proteases. The serine proteases, such as chymotrypsin and trypsin, and the cysteine proteases (in which a Cys residue serves a catalytic role similar to that of Ser in the active site) form covalent enzyme–substrate complexes; the aspartyl proteases and metalloproteases do not. The HIV protease is an aspartyl protease. Two active-site Asp residues facilitate the direct attack of a water molecule on the carbonyl group of the peptide bond to be cleaved (Fig. 6-28). The initial product of this attack is an unstable tetrahedral intermediate, much like that in the chymotrypsin reaction.



MECHANISM FIGURE 6-28 Mechanism of action of HIV protease. Two active-site Asp residues (from different subunits) act as general

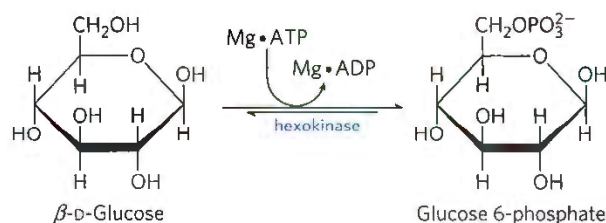
acid-base catalysts, facilitating the attack of water on the peptide bond. The unstable tetrahedral intermediate in the reaction pathway is shaded light red.

P4 This intermediate is close in structure and energy to the reaction transition state. The drugs that have been developed as HIV protease inhibitors form non-covalent complexes with the enzyme, but they bind to it so tightly that they can be considered irreversible inhibitors. The tight binding is derived in part from their design as transition-state analogs. The success of these drugs makes a point worth emphasizing: the catalytic principles we have studied in this chapter are not simply abstruse ideas to be memorized—their application saves lives.

The HIV protease is most efficient at cleaving peptide bonds between Phe and Pro residues. The active site has a pocket that binds an aromatic group next to the bond to be cleaved. Several HIV protease inhibitors are shown in **Figure 6-29**. Although the structures appear varied, they all share a core structure: a main chain with a hydroxyl group positioned next to a branch containing a benzyl group. This arrangement targets the benzyl group to an aromatic (hydrophobic) binding pocket. The adjacent hydroxyl group mimics the negatively charged oxygen in the tetrahedral intermediate in the normal reaction, providing a transition-state analog that facilitates very tight binding. The remainder of each inhibitor structure was designed to fit into and bind to various crevices along the surface of the enzyme, enhancing overall binding. The availability of these effective drugs has vastly increased the life span and quality of life of millions of people with HIV and AIDS. In 2018, 23.3 million of the 38 million people living with HIV infection were receiving antiretroviral therapy. ■

Hexokinase Undergoes Induced Fit on Substrate Binding

Yeast hexokinase (M_r 107,862) is a bisubstrate enzyme that catalyzes this reversible reaction:



ATP and ADP always bind to enzymes as a complex with the metal ion Mg^{2+} .

In the hexokinase reaction, the γ -phosphoryl of ATP is transferred to the hydroxyl at C-6 of glucose. This hydroxyl is similar in chemical reactivity to water, and water freely enters the enzyme active site. Yet hexokinase favors the reaction with glucose by a factor of 10^6 . The enzyme can discriminate between glucose and water because of a conformational change in the enzyme when the correct substrate binds (**Fig. 6-30**). Hexokinase thus provides a good example of induced fit. When glucose is not present, the enzyme is in an inactive conformation, with the active-site amino acid side chains out of position for reaction. When glucose (but not water) and

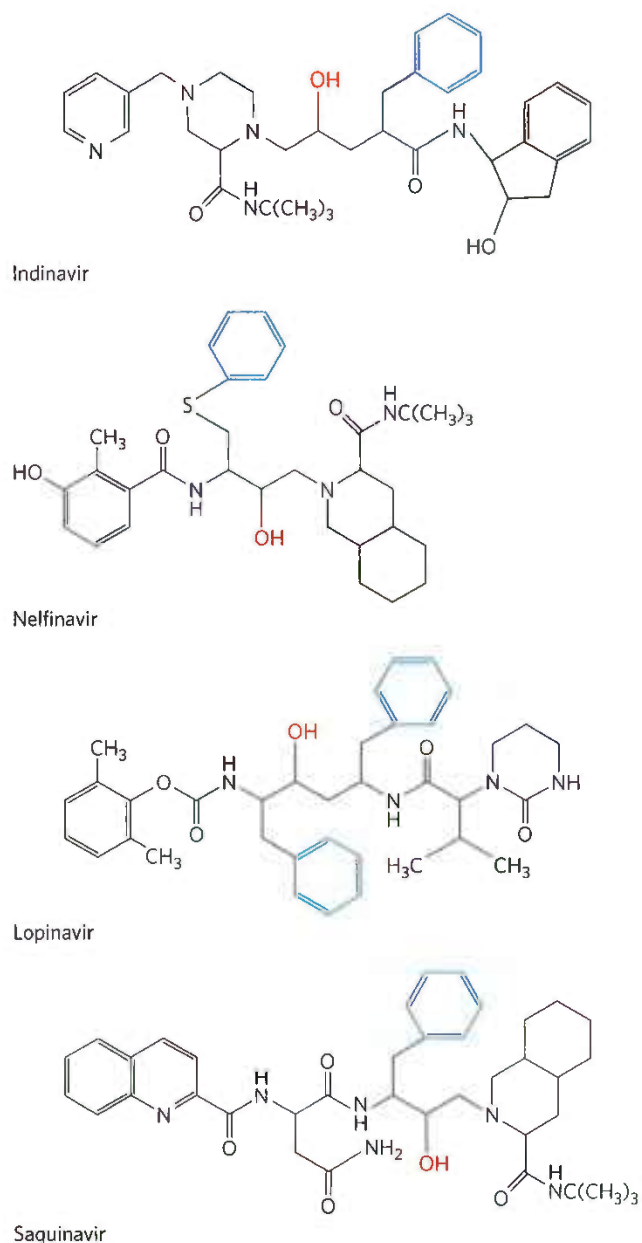


FIGURE 6-29 HIV protease inhibitors. The hydroxyl group (red) acts as a transition-state analog, mimicking the oxygen of the tetrahedral intermediate. The adjacent benzyl group (blue) helps to properly position the drug in the active site.

$\text{Mg} \cdot \text{ATP}$ bind, the binding energy derived from this interaction induces a conformational change in hexokinase to the catalytically active form.

This model has been reinforced by kinetic studies. The five-carbon sugar xylose, stereochemically similar to glucose but one carbon shorter, binds to hexokinase but in a position where it cannot be phosphorylated. Nevertheless, addition of xylose to the reaction mixture increases the rate of ATP hydrolysis. Evidently, the binding of xylose is sufficient to induce a change in hexokinase to its active conformation, and the enzyme is thereby “tricked” into phosphorylating water. The hexokinase reaction also illustrates that

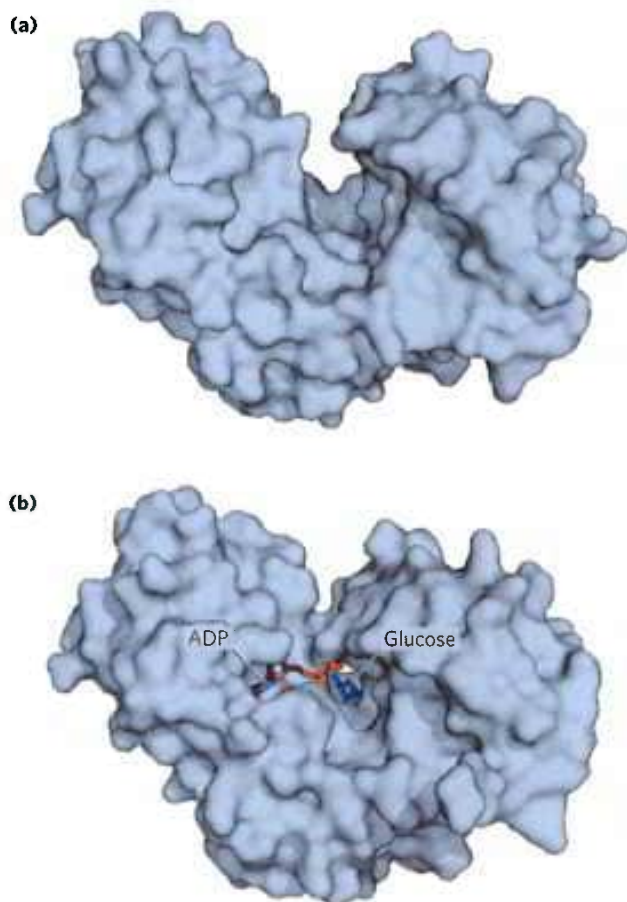
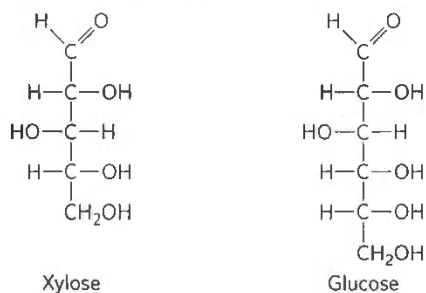


FIGURE 6-30 Induced fit in hexokinase. (a) Hexokinase has a U-shaped structure. (b) The ends pinch toward each other in a conformational change induced by binding D-glucose. [Data from (a) PDB ID 2YHX, C. M. Anderson et al., *J. Mol. Biol.* 123:15, 1978. (b) PDB ID 2E2O, modeled with ADP derived from PDB ID 2E2O, H. Nishimasu et al., *J. Biol. Chem.* 282:9923, 2007.]

enzyme specificity is not always a simple matter of binding one compound but not another. In the case of hexokinase, specificity is observed not in the formation of the ES complex but in the relative rates of subsequent catalytic steps. Reaction rates increase greatly in the presence of a substrate, glucose, that is able to accept a phosphoryl group.

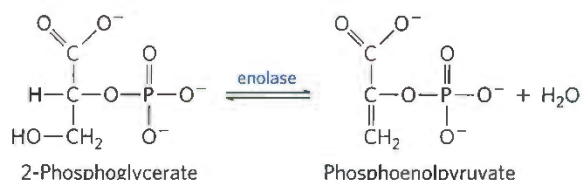


P4 Induced fit is only one aspect of the catalytic mechanism of hexokinase — like chymotrypsin,

hexokinase uses several catalytic mechanisms. For example, the active-site amino acid residues (those brought into position by the conformational change that follows substrate binding) participate in general acid-base catalysis and transition-state stabilization.

The Enolase Reaction Mechanism Requires Metal Ions

Another glycolytic enzyme, enolase, catalyzes the reversible dehydration of 2-phosphoglycerate to phosphoenolpyruvate:

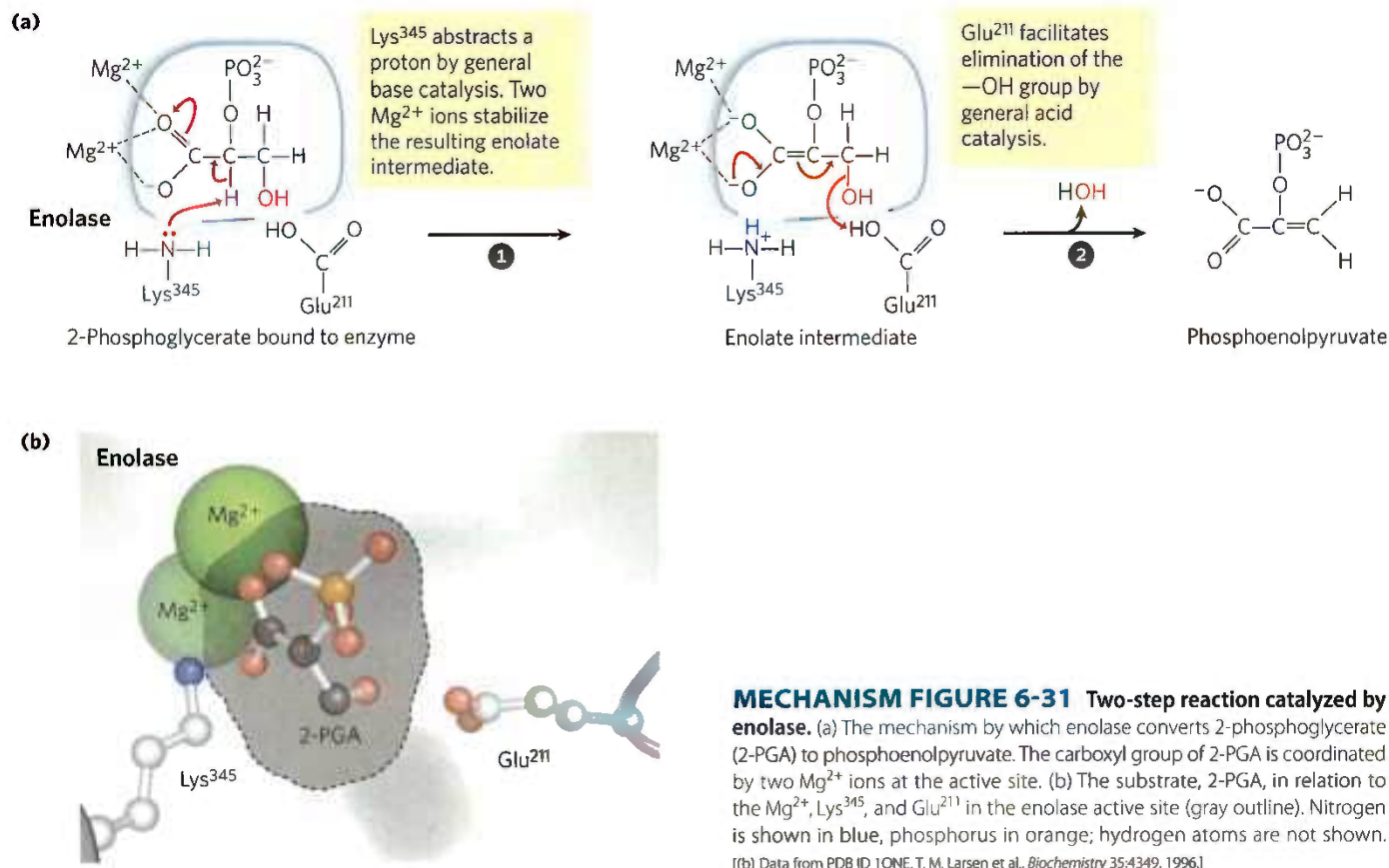


The reaction provides an example of the use of an enzymatic cofactor, in this case a metal ion (another example of coenzyme function is provided in Box 6-1). Yeast enolase (M_r 93,316) is a dimer with 436 amino acid residues per subunit. The enolase reaction illustrates one type of metal ion catalysis and provides an additional example of general acid-base catalysis and transition-state stabilization. The reaction occurs in two steps (**Fig. 6-31a**). First, Lys³⁴⁵ acts as a general base catalyst, abstracting a proton from C-2 of 2-phosphoglycerate; then Glu²¹¹ acts as a general acid catalyst, donating a proton to the —OH leaving group. The proton at C-2 of 2-phosphoglycerate is not acidic and thus is quite resistant to its removal by Lys³⁴⁵. However, the electronegative oxygen atoms of the adjacent carboxyl group pull electrons away from C-2, making the attached protons somewhat more labile. In the active site, the carboxyl group of 2-phosphoglycerate undergoes strong ionic interactions with two bound Mg²⁺ ions (**Fig. 6-31b**), greatly enhancing the electron withdrawal by the carboxyl. Together, these effects render the C-2 protons sufficiently acidic (lowering the pK_a) that one proton can be abstracted to initiate the reaction. As the unstable enolate intermediate is formed, the metal ions further act to shield the two negative charges (on the carboxyl oxygen atoms) that transiently exist in close proximity to each other. Hydrogen bonding to other active-site amino acid residues also contributes to the overall mechanism. The various interactions effectively stabilize both the enolate intermediate and the transition state preceding its formation.

An Understanding of Enzyme Mechanism Produces Useful Antibiotics



Penicillin was discovered in 1928 by Alexander Fleming, but another 15 years passed before this



relatively unstable compound was understood well enough for it to be used as a pharmaceutical agent to treat bacterial infections. Penicillin interferes with the synthesis of peptidoglycan, the major component of the rigid cell wall that protects bacteria from osmotic lysis. Peptidoglycan consists of polysaccharides and peptides cross-linked in several steps that include a transpeptidase reaction (Fig. 6-32). It is this reaction that is inhibited by penicillin and related compounds (Fig. 6-33a), all of which are irreversible inhibitors of transpeptidase. They bind to the active site of transpeptidase through a segment that mimics one conformation of the D-Ala-D-Ala segment of the peptidoglycan precursor. The peptide bond in the precursor is replaced by a highly reactive β -lactam ring in the antibiotic. When penicillin binds to the transpeptidase, an active-site Ser attacks the carbonyl of the β -lactam ring and generates a covalent adduct between penicillin and the enzyme. The leaving group remains attached, however, because it is linked by the remnant of the β -lactam ring (Fig. 6-33b). The covalent complex irreversibly inactivates the enzyme. This, in turn, blocks synthesis of the bacterial cell wall, and most bacteria die as the fragile inner membrane bursts under osmotic pressure.

Human use of penicillin and its derivatives has led to the evolution of strains of pathogenic bacteria that

express β -lactamases (Fig. 6-34a), enzymes that cleave β -lactam antibiotics, rendering them inactive. The bacteria thereby become resistant to the antibiotics. The genes for these enzymes have spread rapidly through bacterial populations under the selective pressure imposed by the use (and often overuse) of β -lactam antibiotics. Human medicine responded with the development of compounds such as clavulanic acid, a suicide inactivator, which irreversibly inactivates the β -lactamases (Fig. 6-34b). Clavulanic acid mimics the structure of a β -lactam antibiotic and forms a covalent adduct with a Ser in the β -lactamase active site. This leads to a rearrangement that creates a much more reactive derivative, which is subsequently attacked by another nucleophile in the active site to irreversibly acylate the enzyme and inactivate it. Amoxicillin and clavulanic acid are combined in a widely used pharmaceutical formulation with the trade name Augmentin. The cycle of chemical warfare between humans and bacteria continues unabated. Strains of disease-causing bacteria that are resistant to both amoxicillin and clavulanic acid have been discovered. Mutations in β -lactamase within these strains render it unreactive to clavulanic acid. The development of new antibiotics promises to be a growth industry for the foreseeable future. ■

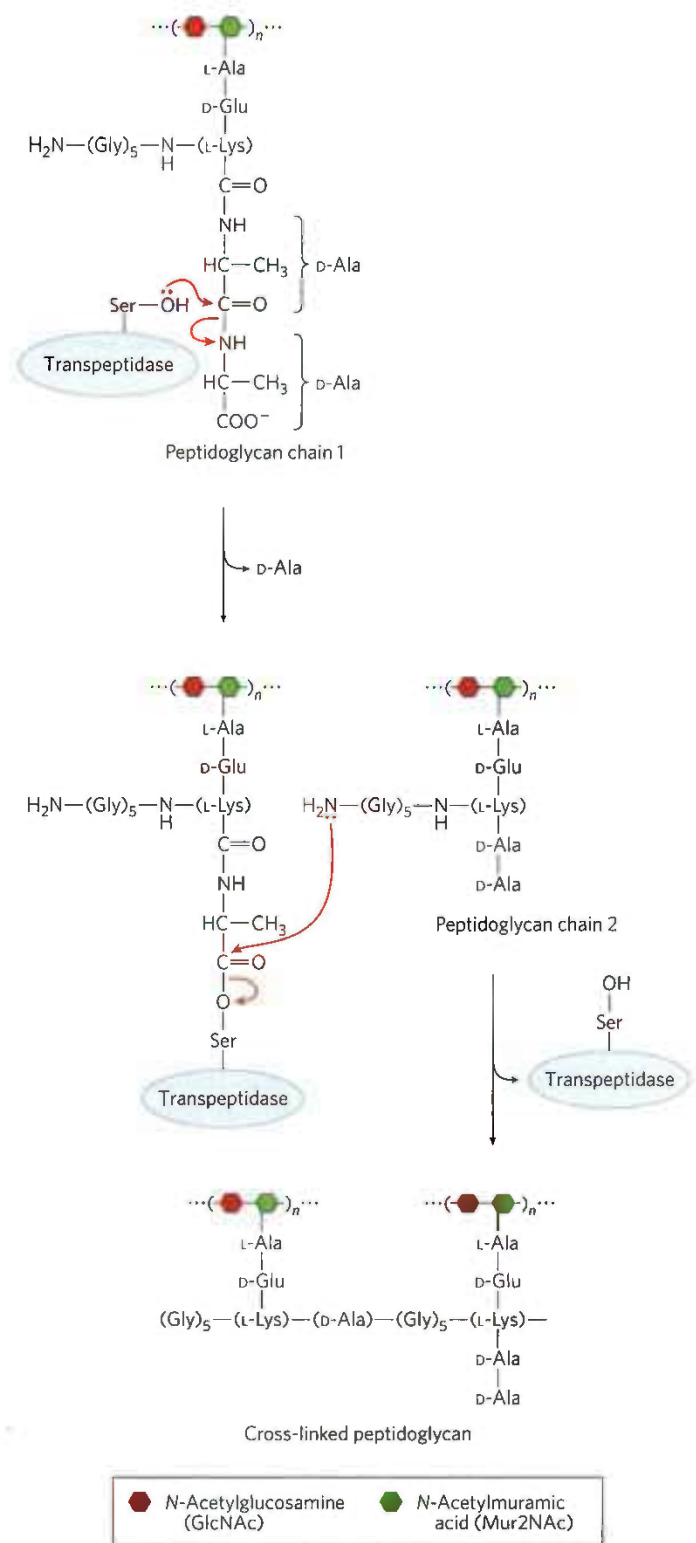


FIGURE 6-32 The transpeptidase reaction. This reaction, which links two peptidoglycan precursors into a larger polymer, is facilitated by an active-site Ser and a covalent catalytic mechanism similar to that of chymotrypsin. Note that peptidoglycan is one of the few places in nature where D-amino acid residues are found. The active-site Ser attacks the carbonyl of the peptide bond between the two D-Ala residues, creating a covalent ester linkage between the substrate and the enzyme, with release of the terminal D-Ala residue. An amino group from the second peptidoglycan precursor then attacks the ester linkage, displacing the enzyme and cross-linking the two precursors.

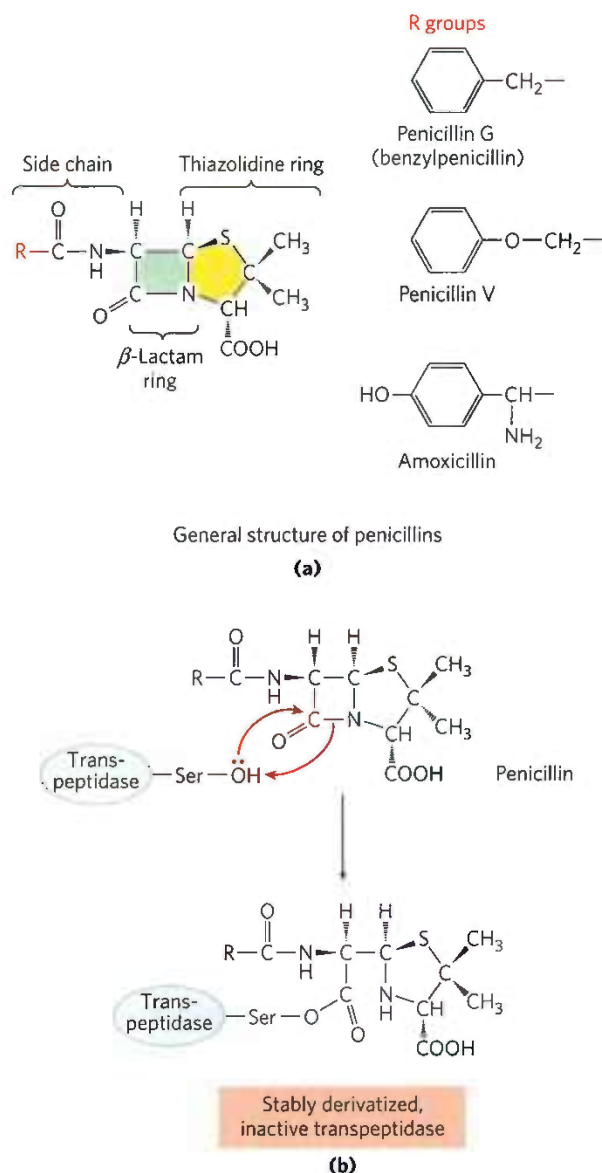


FIGURE 6-33 Transpeptidase inhibition by β -lactam antibiotics.

(a) β -Lactam antibiotics have a five-membered thiazolidine ring fused to a four-membered β -lactam ring. The latter ring is strained and includes an amide moiety that plays a critical role in the inactivation of peptidoglycan synthesis. The R group differs with the type of penicillin. Penicillin G was the first to be isolated and remains one of the most effective, but it is degraded by stomach acid and must be administered by injection. Penicillin V is nearly as effective and is acid stable, so it can be administered orally. Amoxicillin has a broad range of effectiveness, is readily administered orally, and thus is the most widely prescribed β -lactam antibiotic. (b) Attack on the amide moiety of the β -lactam ring by a transpeptidase active-site Ser results in a covalent acyl-enzyme product. This is hydrolyzed so slowly that adduct formation is practically irreversible, and the transpeptidase is inactivated.

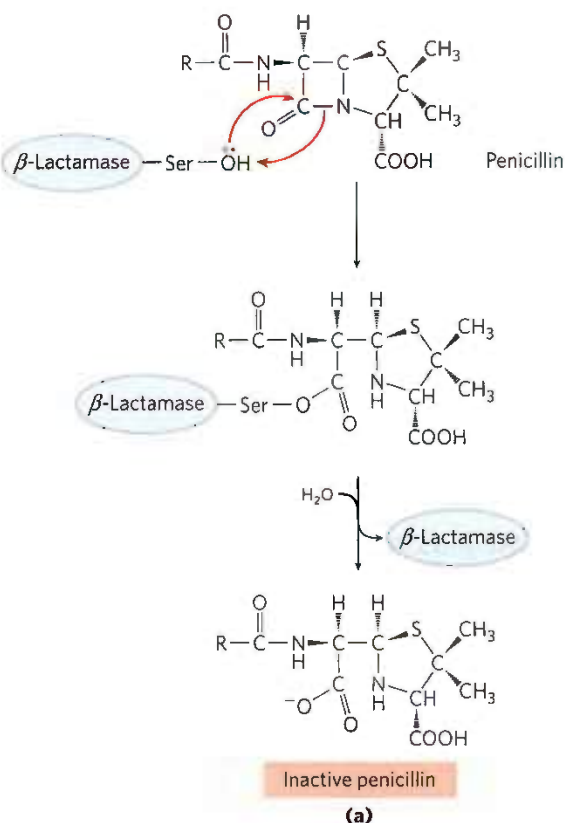
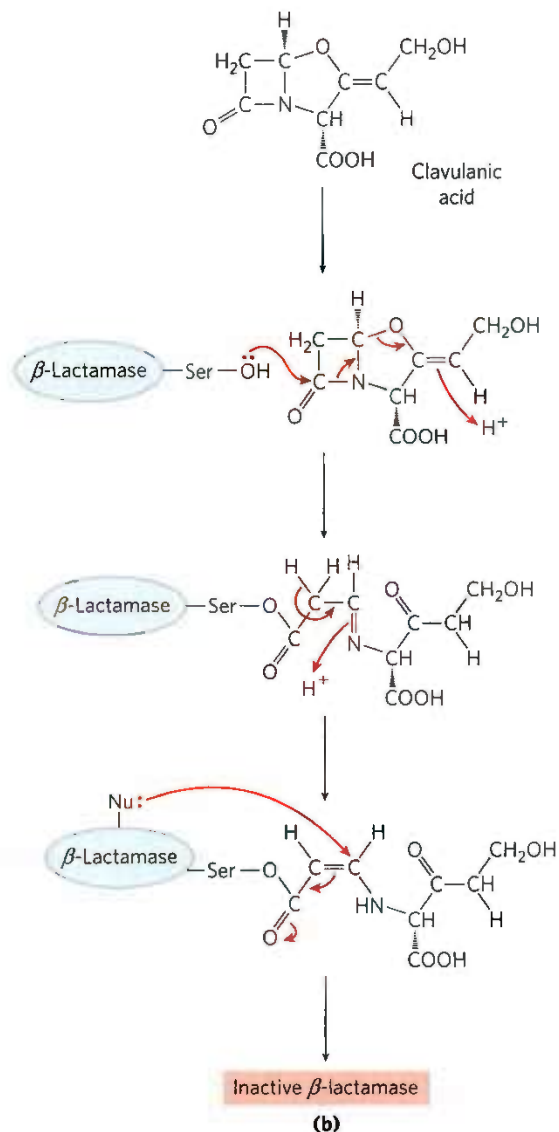


FIGURE 6-34 β -Lactamases and β -lactamase inhibition.

(a) β -Lactamases promote cleavage of the β -lactam ring in β -lactam antibiotics, thus inactivating them. (b) Clavulanic acid is a suicide inhibitor, making use of the normal chemical mechanism of β -lactamases to create a reactive species at the active site. This reactive species is attacked by a nucleophilic group (Nu:) in the active site to irreversibly acylate the enzyme.



SUMMARY 6.4 Examples of Enzymatic Reactions

- Chymotrypsin is a serine protease with a well-understood mechanism, featuring general acid-base catalysis, covalent catalysis, and transition-state stabilization.
- Hexokinase provides an excellent example of induced fit as a means of using substrate binding energy.
- The enolase reaction proceeds via metal ion catalysis.
- Understanding enzyme mechanism allows for the development of drugs to inhibit enzyme action.

6.5 Regulatory Enzymes

In cellular metabolism, groups of enzymes work together in sequential pathways to carry out a given metabolic process, such as the multireaction breakdown of glucose to lactate or the multireaction synthesis of an amino acid from simpler precursors. Each separate reaction is catalyzed by a different enzyme. In such enzyme systems, the

reaction product of one enzyme becomes the substrate of the next. This functional compartmentalization of cellular chemistry does more than accelerate individual reactions; it provides opportunities for the exquisitely precise regulation of all cellular processes.

Most of the enzymes in each metabolic pathway follow the kinetic patterns we have already described. Each pathway, however, includes one or more enzymes that have a greater effect on the rate of the overall sequence. The catalytic activity of these **regulatory enzymes** increases or decreases in response to certain signals. **P5** Adjustments in the rate of reactions catalyzed by regulatory enzymes, and therefore in the rate of entire metabolic sequences, allow the cell to meet changing needs for energy and for biomolecules required in growth and repair.

The activities of regulatory enzymes are modulated in a variety of ways. **Allosteric enzymes** function through reversible, noncovalent binding of regulatory compounds called **allosteric modulators** or **allosteric effectors**, which are generally small metabolites or

cofactors. Other enzymes are regulated by reversible **covalent modification**. Both classes of regulatory enzymes tend to be multisubunit proteins, and in some cases the regulatory site(s) and the active site are on separate subunits. Metabolic systems have at least two other mechanisms of enzyme regulation. Some enzymes are stimulated or inhibited when they are bound by separate **regulatory proteins**. Others are activated when peptide segments are removed by **proteolytic cleavage**; unlike effector-mediated regulation, regulation by proteolytic cleavage is irreversible. Important examples of both mechanisms are found in physiological processes such as digestion, blood clotting, hormone action, and vision.

Cell growth and survival depend on efficient use of resources, and this efficiency is made possible by regulatory enzymes. No single rule governs which of the various types of regulation occur in different systems. Several types of regulation may occur in a single regulatory enzyme. The remainder of this chapter is devoted to a discussion of these major mechanisms of enzyme regulation.

Allosteric Enzymes Undergo Conformational Changes in Response to Modulator Binding

As we saw in Chapter 5, allosteric proteins are those having “other shapes” or conformations induced by the binding of modulators. The same concept applies to certain regulatory enzymes, as conformational changes induced by one or more modulators interconvert more-active and less-active forms of the enzyme. The modulators for allosteric enzymes may be inhibitory or stimulatory. The modulator can be the substrate itself; regulation in which substrate and modulator are identical is referred to as **homotropic**. The effect is similar to that of O_2 binding to hemoglobin (Chapter 5): binding of the ligand—or substrate, in the case of enzymes—causes conformational changes that affect the subsequent activity of other sites on the protein. In most cases, the conformational change converts a relatively inactive conformation (often referred to as a T state—a convention based on the early hemoglobin literature) to a more active conformation (an R state). When the modulator is a molecule other than the substrate, the enzyme is said to be **heterotropic**. Note that heterotropic modulators should not be confused with uncompetitive and mixed inhibitors. Although the latter bind at a second site on the enzyme, they do not necessarily mediate conformational changes between active and inactive forms, and the kinetic effects are distinct.

The properties of allosteric enzymes are significantly different from those of simple nonregulatory enzymes. Some of the differences are structural. In addition to active sites, allosteric enzymes often have one or more regulatory, or allosteric, sites for binding to each heterotropic modulator (Fig. 6-35). Just as an enzyme's active site is specific for its substrate, each regulatory site is

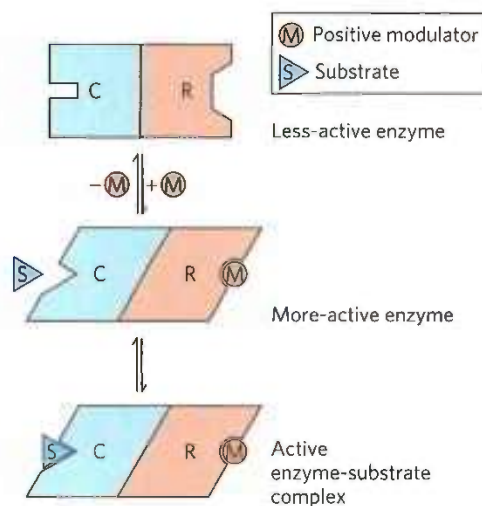
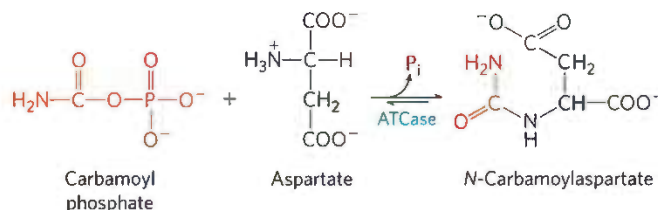


FIGURE 6-35 Subunit interactions in an allosteric enzyme, and interactions with inhibitors and activators. In many allosteric enzymes, the substrate-binding site and the modulator-binding site(s) are on different subunits: the catalytic (C) and regulatory (R) subunits, respectively. Binding of the positive (stimulatory) modulator (M) to its specific site on the regulatory subunit is communicated to the catalytic subunit through a conformational change. This change renders the catalytic subunit active and capable of binding the substrate (S) with higher affinity. Upon dissociation of the modulator from the regulatory subunit, the enzyme reverts to its inactive or less-active form.

specific for its modulator. Enzymes with several modulators generally have different specific binding sites for each. In homotropic enzymes, the active site and regulatory site are the same.

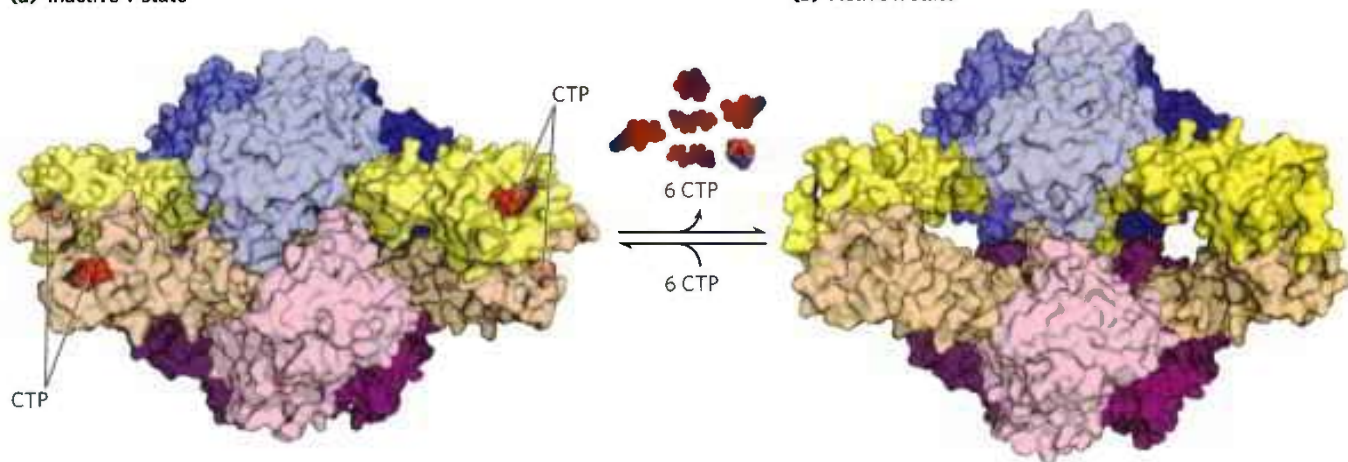
Allosteric enzymes are typically larger and more complex than nonallosteric enzymes, with two or more subunits. A classic example is aspartate transcarbamoylase (often abbreviated ATCase), which catalyzes an early step in the biosynthesis of pyrimidine nucleotides, the reaction of carbamoyl phosphate and aspartate to form carbamoyl aspartate:



ATCase has 12 polypeptide chains organized into 6 catalytic subunits (organized as 2 trimeric complexes) and 6 regulatory subunits (organized as 3 dimeric complexes). **Figure 6-36** shows the quaternary structure of this enzyme, deduced from x-ray analysis. The enzyme exhibits allosteric behavior as detailed below, as the catalytic subunits function cooperatively. The regulatory subunits have binding sites for ATP and CTP, which function as positive and negative regulators, respectively. CTP is one of the end products of the pathway, and negative

(a) Inactive T state

(b) Active R state

**FIGURE 6-36** The regulatory enzyme aspartate transcarbamoylase.

(a) The inactive T state and (b) the active R state of the enzyme are shown. This allosteric regulatory enzyme has two stacked catalytic clusters, each with three catalytic polypeptide chains (in shades of blue and purple), and three regulatory clusters, each with two regulatory polypeptide chains (in beige and yellow). The regulatory clusters form the points of a triangle (not evident

in this side view) surrounding the catalytic subunits. Binding sites for allosteric modulators (including CTP) are on the regulatory subunits. Modulator binding produces large changes in enzyme conformation and activity. The role of this enzyme in nucleotide synthesis, and details of its regulation, are discussed in Chapter 22. [Data from (a) PDB ID 1RAB, R. P. Kosman et al., *Proteins* 15:147, 1993; (b) PDB ID 1F1B, L. Jin et al., *Biochemistry* 39:8058, 2000.]

regulation by CTP serves to limit ATCase action under conditions when CTP is abundant. On the other hand, high concentrations of ATP indicate that cellular metabolism is robust, that the cell is growing, and that additional pyrimidine nucleotides may be needed to support RNA transcription and DNA replication.

The Kinetic Properties of Allosteric Enzymes Diverge from Michaelis-Menten Behavior

Allosteric enzymes show relationships between V_0 and $[S]$ that differ from Michaelis-Menten kinetics. They do exhibit saturation with the substrate when $[S]$ is sufficiently high, but for allosteric enzymes, plots of V_0 versus $[S]$ (Fig. 6-37) usually produce a sigmoid saturation curve, rather than the hyperbolic curve typical of nonregulatory enzymes. On the sigmoid saturation curve we can find a value of $[S]$ at which V_0 is half-maximal, but we cannot refer to it with the designation K_m , because the enzyme does not follow the hyperbolic Michaelis-Menten relationship. Instead, the symbol $[S]_{0.5}$ or $K_{0.5}$ is often used to represent the substrate concentration giving half-maximal velocity of the reaction catalyzed by an allosteric enzyme (Fig. 6-37).

Sigmoid kinetic behavior reflects cooperative interactions between multiple protein subunits. In other words, changes in the structure of one subunit are translated into structural changes in adjacent subunits, an effect mediated by noncovalent interactions at the interface between subunits. Sigmoid kinetic behavior for enzymes is well explained by the concerted and sequential models for subunit interactions we previously encountered when considering O_2 binding to hemoglobin (see Fig. 5-14).

ATCase effectively illustrates both homotropic and heterotropic allosteric kinetic behavior. Binding of the substrates, aspartate and carbamoyl phosphate, to the enzyme gradually brings about a transition from the relatively inactive T state to the more active R state. This accounts for the sigmoid rather than hyperbolic change in V_0 with increasing $[S]$. One characteristic of sigmoid kinetics is that small changes in the concentration of a modulator can be associated with large changes in activity. As exemplified in Figure 6-37a, a relatively small increase in $[S]$ in the steep part of the curve causes a comparatively large increase in V_0 .

The heterotropic allosteric regulation of ATCase is brought about by its interactions with ATP and CTP. For heterotropic allosteric enzymes, an activator may cause the curve to become more nearly hyperbolic, with a decrease in $K_{0.5}$ but no change in V_{max} , resulting in an increased reaction velocity at a fixed substrate concentration. For ATCase, the interaction with ATP brings this about, and the enzyme exhibits a V_0 versus $[S]$ curve that is characteristic of the active R state at sufficiently high ATP concentrations (V_0 is higher for any value of $[S]$; Fig. 6-37b). A negative modulator (an inhibitor) may produce a *more* sigmoid substrate-saturation curve, with an increase in $K_{0.5}$, as illustrated by the effects of CTP on ATCase kinetics (see curves for a negative modulator in Fig. 6-37b). Other heterotropic allosteric enzymes respond to an activator by an increase in V_{max} with little change in $K_{0.5}$ (Fig. 6-37c). Heterotropic allosteric enzymes therefore show different kinds of responses in their substrate-activity curves, because some have inhibitory modulators, some have activating modulators, and some (like ATCase) have both.

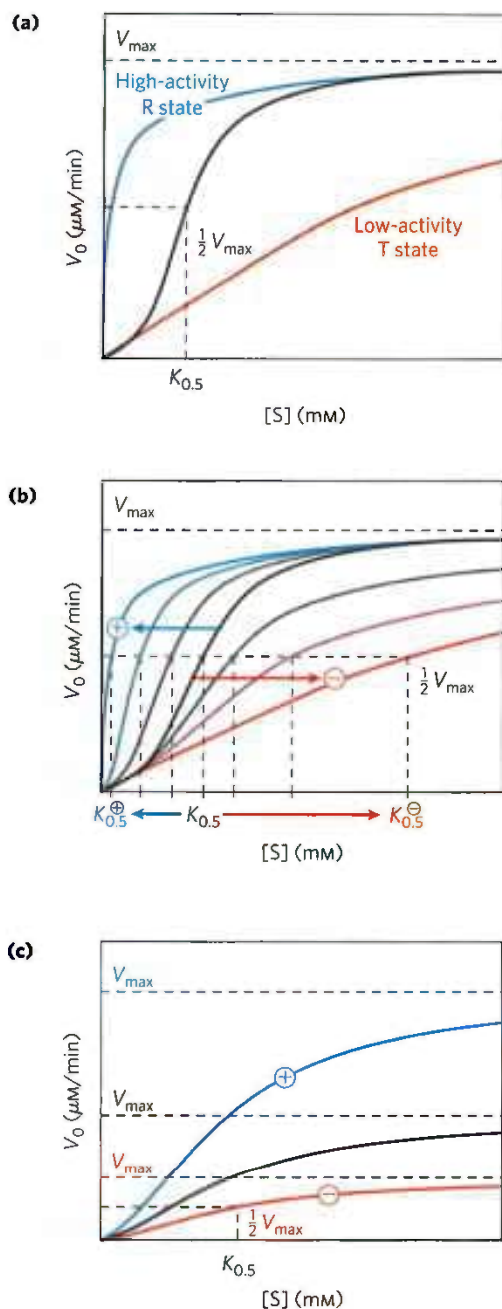


FIGURE 6-37 Substrate-activity curves for representative allosteric enzymes. Three examples of complex responses of allosteric enzymes to their modulators. (a) The sigmoid curve of a homotropic enzyme, in which the substrate also serves as a positive (stimulatory) modulator, or activator. Notice the resemblance to the oxygen-saturation curve of hemoglobin (see Fig. 5-12). The sigmoid curve is a hybrid curve in which the enzyme is present primarily in the relatively inactive T state at low substrate concentration, and primarily in the more active R state at high substrate concentration. The curves for the pure T and R states are plotted separately in color. ATCase exhibits a kinetic pattern similar to this. (b) The effects of several different concentrations of a positive modulator (+) or a negative modulator (−) on an allosteric enzyme in which $K_{0.5}$ is altered without a change in V_{max} . For ATCase, CTP is a negative modulator and ATP is a positive modulator. (c) A less common type of modulation, in which V_{max} is altered and $K_{0.5}$ is nearly constant.

Some Enzymes Are Regulated by Reversible Covalent Modification

In another important class of regulatory enzymes, activity is modulated by covalent modification of one or more of the amino acid residues in the enzyme molecule. Over 500 different types of covalent modification have been found in proteins. Common modifying groups include phosphoryl, acetyl, adenylyl, uridylyl, methyl, amide, carboxyl, myristoyl, palmitoyl, prenyl, hydroxyl, sulfate, and adenosine diphosphate ribosyl groups (Fig. 6-38). There are even entire proteins that function as specialized modifying groups. These include ubiquitin and SUMO (small ubiquitin-like modifier), which are attached and detached from other proteins to regulate their activity in some way. All of these groups, small and large, are linked to and removed from a regulated enzyme by separate enzymes. When an amino acid residue in an enzyme is modified, a novel amino acid with altered properties has effectively been introduced into the enzyme. Introduction of a charge can alter the local properties of the enzyme and induce a change in conformation. Introduction of a hydrophobic group can trigger association with a membrane. The changes are often substantial and can be critical to the function of the altered enzyme.

Although there are too many examples of covalent modification of proteins to cover in detail, a few examples are instructive. One enzyme that is regulated by methylation is the methyl-accepting chemotaxis protein of bacteria. This protein is part of a system that permits a bacterium to swim toward an attractant (such as a sugar) in solution and away from repellent chemicals. The methylating agent is *S*-adenosylmethionine (adoMet) (see Fig. 18-18). Acetylation is another common modification, with approximately 80% of the soluble proteins in eukaryotes, including many enzymes, acetylated at their amino terminus. Ubiquitin is added to proteins as a tag that destines them for proteolytic degradation (see Fig. 27-47). Ubiquitination can also have a regulatory function. SUMO is found attached to many eukaryotic nuclear proteins and has roles in the regulation of transcription, chromatin structure, and DNA repair.

ADP-ribosylation is an especially interesting reaction; the ADP-ribose is derived from nicotinamide adenine dinucleotide (NAD) (see Fig. 8-41). This type of modification occurs for the bacterial enzyme dinitrogenase reductase, resulting in regulation of the important process of biological nitrogen fixation. Diphtheria toxin and cholera toxin are enzymes that catalyze the ADP-ribosylation (and inactivation) of key cellular enzymes or other proteins.

By far the most common type of regulatory modification is phosphorylation. About one-third of all proteins in a eukaryotic cell are phosphorylated, and one or (often) many phosphorylation events are part of virtually every regulatory process. Some proteins have only one

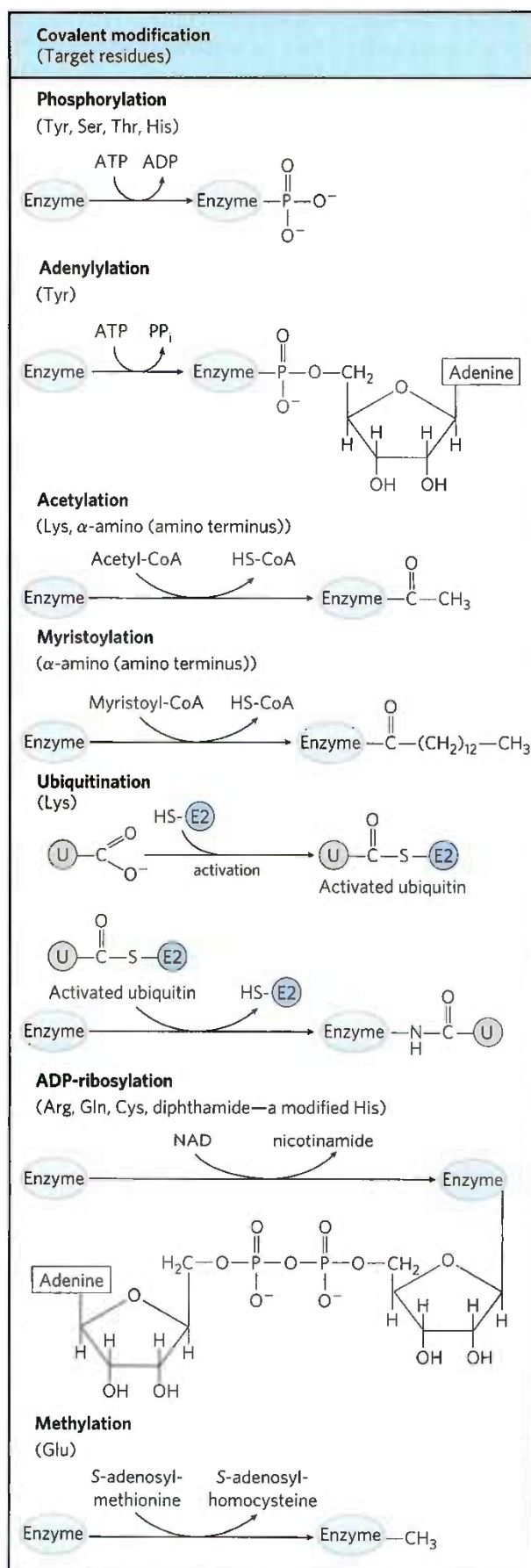


FIGURE 6-38 Some enzyme modification reactions.

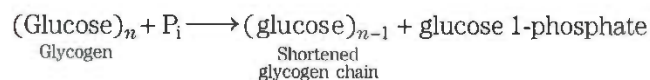
phosphorylated residue, others have several, and a few have dozens of sites for phosphorylation. This mode of covalent modification is central to a large number of regulatory pathways. We discuss it in some detail here, and again in Chapter 12.

We will encounter all of these types of enzyme modification again in later chapters.

Phosphoryl Groups Affect the Structure and Catalytic Activity of Enzymes

The attachment of phosphoryl groups to specific amino acid residues of a protein is catalyzed by **protein kinases**. More than 500 genes encoding these critical enzymes are found in the human genome. In the reactions they catalyze, the γ -phosphoryl group derived from a nucleoside triphosphate (usually ATP) is transferred to a particular Ser, Thr, or Tyr residue (occasionally His as well) on the target protein. This introduces a bulky, charged group into a region of the target protein that was only moderately polar prior to modification. The oxygen atoms of a phosphoryl group can hydrogen-bond with one or several groups in a protein, commonly the amide groups of the peptide backbone at the start of an α helix or the charged guanidinium group of an Arg residue. The two negative charges on a phosphorylated side chain can also repel neighboring negatively charged (Asp or Glu) residues. When the modified side chain is located in a region of an enzyme critical to its three-dimensional structure, phosphorylation can have dramatic effects on enzyme conformation and thus on substrate binding and catalysis. Removal of phosphoryl groups from these same target proteins is catalyzed by **phosphoprotein phosphatases**, also called simply **protein phosphatases**.

An important example of enzyme regulation by phosphorylation is the case of glycogen phosphorylase (M_r 94,500) of muscle and liver (Chapter 15), which catalyzes the reaction



The glucose 1-phosphate so formed can be used for ATP synthesis in muscle or converted to free glucose in the liver. Note that glycogen phosphorylase, though it adds a phosphate to a substrate, is not itself a kinase, because it does not utilize ATP or any other nucleotide triphosphate as a phosphoryl donor in its catalyzed reaction. It is, however, the substrate for a protein kinase that phosphorylates it. In the discussion below, the phosphoryl groups we are concerned with are those involved in regulation of the enzyme, as distinguished from its catalytic function.

Glycogen phosphorylase occurs in two forms: the more active phosphorylase *a* and the less active phosphorylase *b* (Fig. 6-39). Phosphorylase *a* has two subunits, each with a specific Ser residue (Ser¹⁴) that is phosphorylated at its hydroxyl group. Phosphorylase *b* is covalently transformed into active phosphorylase *a* by

TABLE 6-10 Consensus Recognition Sequences for a Few Protein Kinases

Protein kinase	Consensus sequence and phosphorylated residue
Protein kinase A	-x-R-[RK]-x-[ST]-B-
Protein kinase G	-x-R-[RK]-x-[ST]-X-
Protein kinase C	-[RK](2)-x-[ST]-B-[RK](2)-
Protein kinase B	R-x-x-R-x-[ST]-x-ψ ^a -
Ca ²⁺ /calmodulin kinase I	-B-x-R-x(2)-[ST]-x(3)-B-
Ca ²⁺ /calmodulin kinase II	-B-x-[RK]-x(2)-[ST]-x(2)-
Myosin light chain kinase (smooth muscle)	-K(2)-R-x(2)- S -x-B(2)-
Phosphorylase b kinase	- K-R-K-Q-I-S-V-R -
Extracellular signal-regulated kinase (ERK)	-P-x-[ST]-P(2)-
Cyclin-dependent protein kinase (cdc2)	-x-[ST]-P-x-[KR]-
Casein kinase I	-[SpTp]-x(2)-[ST]-B ^b
Casein kinase II	-x-[ST]-x(2)-[ED]-x-
β-Adrenergic receptor kinase	-[DE](n)-[ST]-x(3)
Rhodopsin kinase	-x(2)-[ST]-E(n)-vABL-[YLV]- Y -X ₁₋₃ -[PF]-
Epidermal growth factor (EGF) receptor kinase	-E(4)- Y-F-E-L-V -

Information from L. A. Pinna and M. H. Ruzzene, *Biochim. Biophys. Acta* 1314:191, 1996; B. E. Kemp and R. B. Pearson, *Trends Biochem. Sci.* 15:342, 1990; P. J. Kennelly and E. G. Krebs, *J. Biol. Chem.* 266:15,555, 1991; T. P. Cujec, P. F. Madelos, P. Hammond, C. Rise, and B. L. Kreider, *Chem. Biol.* 9:253, 2002.

Note: Shown here are deduced consensus sequences (in roman type) and actual sequences from known substrates (italic). The Ser (S), Thr (T), or Tyr (Y) residue that undergoes phosphorylation is in bold red; all amino acid residues are shown as their one-letter abbreviations (see Table 3-1). An x represents any amino acid; B, any hydrophobic amino acid; Sp and Tp are Ser and Thr residues that must already be phosphorylated for the kinase to recognize the site.

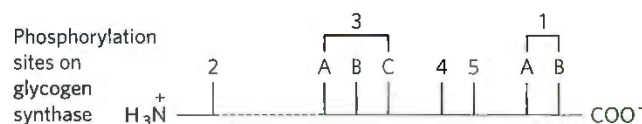
^aψ denotes any amino acid with a bulky hydrophobic side chain.

^bThe best target site has two amino acid residues separating the phosphorylated and target Ser/Thr residues; target sites with one or three intervening residues function at a reduced level.

Multiple Phosphorylations Allow Exquisite Regulatory Control

The Ser, Thr, or Tyr residues that are typically phosphorylated in regulated proteins occur within common structural motifs, called consensus sequences, that are recognized by specific protein kinases (Table 6-10). Some kinases are basophilic, preferentially phosphorylating a residue that has basic neighbors; others have different substrate preferences, such as for a residue near a Pro residue. However, amino acid sequence is not the only important factor in determining whether a given residue will be phosphorylated. Protein folding brings together residues that are distant in the primary sequence; the resulting three-dimensional structure can determine whether a protein kinase has access to a given residue and can recognize it as a substrate.

Regulation by phosphorylation is often complicated. Some proteins have consensus sequences recognized by several different protein kinases, each of which can phosphorylate the protein and alter its enzymatic activity. In some cases, phosphorylation is hierarchical: a certain residue can be phosphorylated only if a neighboring residue has already been phosphorylated. For example, glycogen synthase, the enzyme that catalyzes the condensation of glucose monomers to form glycogen (Chapter 15), is inactivated by phosphorylation of specific Ser residues and is also modulated by at least four other protein kinases that phosphorylate four other sites in the enzyme (Fig. 6-41). The enzyme does not become



Kinase	Phosphorylation sites	Degree of synthase inactivation
Protein kinase A	1A, 1B, 2, 4	+
Protein kinase G	1A, 1B, 2	+
Protein kinase C	1A	+
Ca ²⁺ /calmodulin kinase	1B, 2	+
Phosphorylase b kinase	2	+
Casein kinase I	At least nine	++++
Casein kinase II	5	0
Glycogen synthase kinase 3	3A, 3B, 3C	+++
Glycogen synthase kinase 4	2	+

FIGURE 6-41 Multiple regulatory phosphorylations. The enzyme glycogen synthase has at least nine separate sites in five designated regions that are susceptible to phosphorylation by one of the cellular protein kinases. Thus, regulation of this enzyme is a matter not of binary (on/off) switching but of finely tuned modulation of activity over a wide range in response to a variety of signals.

a substrate for glycogen synthase kinase 3 until one site has been phosphorylated by casein kinase II. Some phosphorylations inhibit glycogen synthase more than others, and some combinations of phosphorylations are cumulative. These multiple regulatory phosphorylations provide the potential for extremely subtle modulation of enzyme activity.

As is the case for kinases, there are many different enzymes that remove phosphoryl groups from proteins. Cells contain a family of phosphoprotein phosphatases that each hydrolyze specific P-Ser , P-Thr , and P-Tyr esters, releasing P_i . The phosphoprotein phosphatases generally act on only a subset of phosphorylated proteins, but they show less substrate specificity than protein kinases.

Some Enzymes and Other Proteins Are Regulated by Proteolytic Cleavage of an Enzyme Precursor

For some enzymes, an inactive precursor called a **zymogen** is cleaved to form the active enzyme. Many proteolytic enzymes (proteases) of the stomach and pancreas are regulated in this way. Chymotrypsin and trypsin are initially synthesized as chymotrypsinogen and trypsinogen (Fig. 6-42). Specific cleavage causes conformational changes that expose the enzyme active site. This type of activation is irreversible, and additional regulation requires a different mechanism. When necessary, a previously activated protease is inactivated by inhibitor proteins that bind very tightly to the enzyme active site. For example, pancreatic trypsin inhibitor (M_r 6,000) binds to and inhibits trypsin. α_1 -Antitrypsin (M_r 53,000) primarily inhibits neutrophil elastase (neutrophils are a type of leukocyte, or white blood cell; elastase is a protease that acts on elastin, a component of some connective tissues).

An insufficiency of α_1 -antitrypsin, which can be caused by exposure to cigarette smoke, has been associated with lung damage, including emphysema.

Proteases are not the only proteins activated by proteolysis. In other cases, however, the precursors are called not zymogens but, more generally, **proproteins** or **proenzymes**, as appropriate. For example, the connective tissue protein collagen is initially synthesized as the soluble precursor procollagen.

A Cascade of Proteolytically Activated Zymogens Leads to Blood Coagulation

The formation of a blood clot provides a well-studied example of a **regulatory cascade**, a mechanism that allows a very sensitive response to—and amplification of—a molecular signal. The blood coagulation pathways also bring together several other types of regulation, including proteolytic activation and regulatory proteins. We have not yet encountered the latter type of regulation. It employs proteins whose only function is to regulate the activity of other proteins by interacting with them noncovalently.

In a regulatory cascade, a signal leads to the activation of protein X. Protein X catalyzes the activation of protein Y. Protein Y catalyzes the activation of protein Z, and so on. Since proteins X, Y, and Z are catalysts and activate multiple copies of the next protein in the chain, the signal is amplified in each step. In some cases, the activation steps involve proteolytic cleavage and are thus effectively irreversible. In other cases, activation entails readily reversible protein modification steps such as phosphorylation. Regulatory cascades govern a wide range of biological processes, including, besides blood coagulation, some aspects

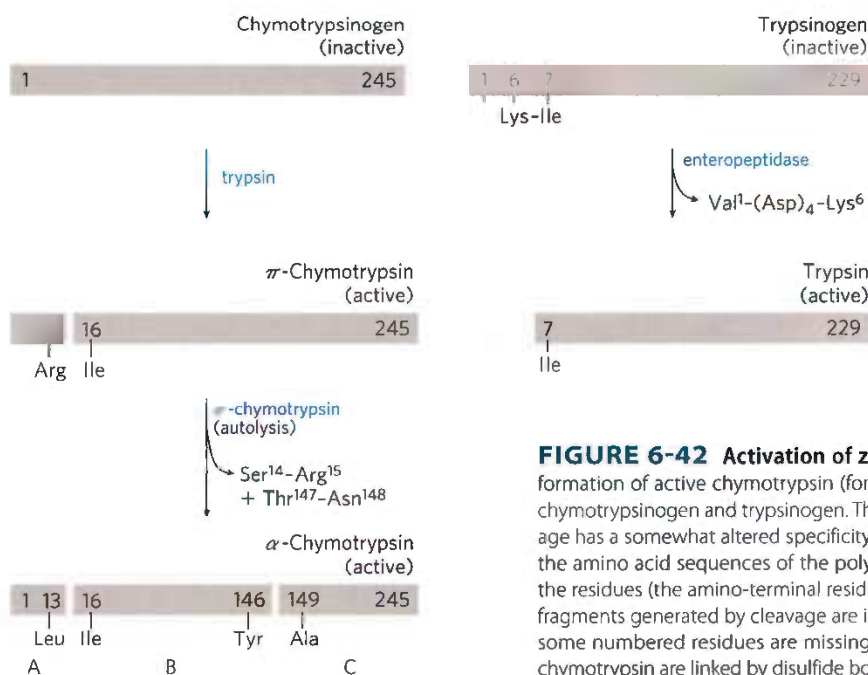


FIGURE 6-42 Activation of zymogens by proteolytic cleavage. Shown here is the formation of active chymotrypsin (formally, α -chymotrypsin) and trypsin from their zymogens, chymotrypsinogen and trypsinogen. The π -chymotrypsin intermediate generated by trypsin cleavage has a somewhat altered specificity relative to the mature α -chymotrypsin. The bars represent the amino acid sequences of the polypeptide chains, with numbers indicating the positions of the residues (the amino-terminal residue is number 1). Residues at the termini of the polypeptide fragments generated by cleavage are indicated below the bars. Note that in the final active forms, some numbered residues are missing. Recall that the three polypeptide chains (A, B, and C) of chymotrypsin are linked by disulfide bonds (see Fig. 6-24).

of cell fate determination during development, the detection of light by retinal rods, and programmed cell death (apoptosis). A regulatory cascade is also one of the strategies governing the overall activity of glycogen phosphorylase (see Fig. 15-13).

A blood clot is an aggregate of specialized cell fragments that lack nuclei, called **platelets**. They are cross-linked and stabilized by proteinaceous fibers consisting mainly of the protein **fibrin** (Fig. 6-43a). Fibrin is derived from a soluble zymogen called **fibrinogen**. After albumins and globulins, fibrinogen is usually the third most abundant type of protein in blood plasma. Blood clotting begins with the activation of circulating platelets at the site of a wound. Tissue damage causes collagen molecules present beneath the epithelial cell layer that lines each blood vessel to become exposed to the blood. Platelet activation is primarily triggered by interaction with this collagen. Activation leads to the presentation of anionic phospholipids on the surface of each platelet and the release of signaling molecules such as **thromboxanes** (p. 355) that help stimulate the activation of additional platelets. The activated platelets aggregate at the site of

a wound, forming a loose clot. Stabilization of the clot requires fibrin, generated by fibrinogen cleavage as the endpoint of regulatory cascades.

Fibrinogen is a dimer of heterotrimers ($A\alpha_2B\beta_2\gamma_2$) with three different but evolutionarily related types of subunits (Fig. 6-43b). Fibrinogen is converted to fibrin ($\alpha_2\beta_2\gamma_2$), and thereby activated for blood clotting, by the proteolytic removal of 16 amino acid residues from the amino-terminal end (the A peptide) of each α subunit and 14 amino acid residues from the amino-terminal end (the B peptide) of each β subunit. Peptide removal is catalyzed by the serine protease **thrombin**. The newly exposed amino termini of the α and β subunits fit neatly into binding sites in the carboxyl-terminal globular portions of the γ and β subunits, respectively, of another fibrin protein. Fibrin thus polymerizes into a gel-like matrix to generate a soft clot. Covalent cross-links between the associated fibrins are generated by the condensation of particular Lys residues in one fibrin heterotrimer with Gln residues in another, catalyzed by a transglutaminase, **factor XIIIa**. The covalent cross-links convert the soft clot into a hard clot.

Fibrinogen activation to produce fibrin is the end point of not one but two parallel but intertwined regulatory cascades (Fig. 6-44). One of these is referred to as

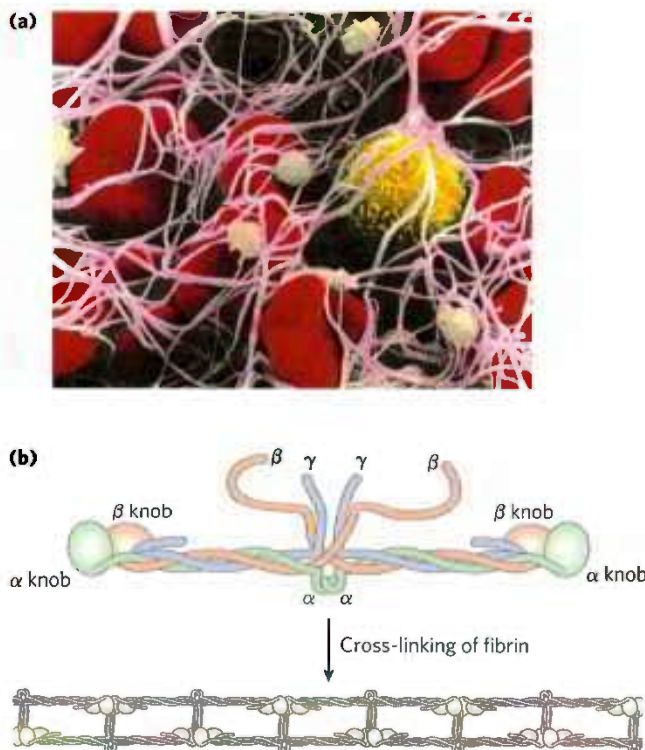


FIGURE 6-43 The function of fibrin in blood clots. (a) A blood clot consists of aggregated platelets (small, light-colored cells) tied together with strands of cross-linked fibrin. Erythrocytes (red in this colorized scanning electron micrograph) are also trapped in the matrix. (b) The soluble plasma protein fibrinogen consists of two complexes of α , β , and γ subunits ($\alpha_2\beta_2\gamma_2$). The removal of amino-terminal peptides from the α and β subunits (not shown) leads to the formation of higher-order complexes and eventual covalent cross-linking that results in the formation of fibrin fibers. The "knobs" are globular domains at the ends of the proteolyzed subunits. [(a) CNRI/Science Source.]

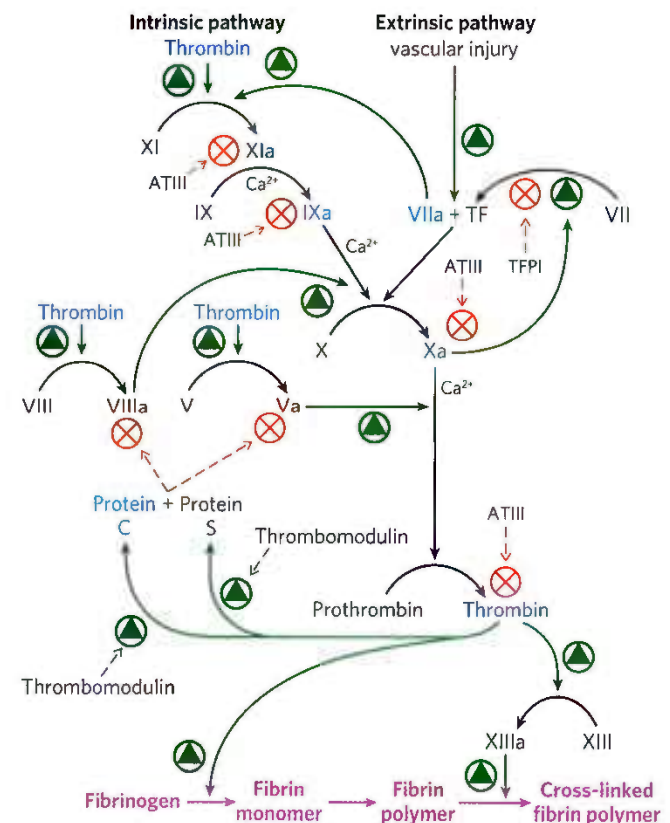


FIGURE 6-44 The coagulation cascades. The interlinked intrinsic and extrinsic pathways leading to the cleavage of fibrinogen to form active fibrin are shown. Active serine proteases in the pathways are shown in blue. Green arrows denote activating steps, and red arrows indicate inhibitory processes.

the contact activation pathway ("contact" refers to interaction of key components of this system with anionic phospholipids presented on the surface of platelets at the site of a wound). As all components of this pathway are found in the blood plasma, it is also called the **intrinsic pathway**. The second path is the tissue factor or **extrinsic pathway**. A major component of this pathway, the protein **tissue factor (TF)**, is not present in the blood. Most of the protein factors in both pathways are designated by roman numerals. Many of these factors are either chymotrypsin-like serine proteases or regulatory proteins, most with zymogen precursors that are synthesized in the liver and exported to the blood. For those with both inactive and active forms, the roman numeral designates the inactive zymogen (VII, X, for example). An "a" is added to designate the cleaved and active form (VIIa, Xa). The regulatory proteins bind to particular serine proteases and help to activate them.

The extrinsic pathway comes into play first. Tissue damage exposes the blood plasma to TF embedded largely in the membranes of fibroblasts and smooth muscle cells beneath the endothelial layer. An initiating complex is formed between TF and factor VII, present in the blood plasma. **Factor VII** is a zymogen of a serine protease, and TF is a regulatory protein required for its function. Factor VII is converted to its active form, **factor VIIa**, by proteolytic cleavage catalyzed by **factor Xa** (another serine protease). The TF-VIIa complex then cleaves **factor X**, creating the active form, factor Xa.

If TF-VIIa is needed to cleave X, and Xa is needed to cleave TF-VII, how does the process ever get started? A very small amount of factor VIIa is present in the blood at all times, enough to form a small amount of the active TF-VIIa complex immediately after tissue is damaged. This allows formation of factor Xa and establishes the initiating feedback loop. Once levels of factor Xa begin to build up, Xa (in a complex with another regulatory protein, factor Va) cleaves prothrombin to form active thrombin, and thrombin cleaves fibrinogen.

The extrinsic pathway thus provides a burst of thrombin. However, the TF-VIIa complex is quickly shut down by the protein **tissue factor pathway inhibitor (TFPI)**. Clot formation is sustained by the activation of components of the second cascade, the intrinsic pathway. **Factor IX** is converted to the active serine protease **factor IXa** by the TF-VIIa protease during initiation of the clotting sequence. Factor IXa, in a complex with the regulatory protein **VIIIa**, is relatively stable and provides an alternative enzyme for the proteolytic conversion of factor X to Xa. Activated IXa can also be produced by the serine protease factor XIa. Most of the XIa is generated by cleavage of **factor XI** zymogen by thrombin in a feedback loop.

Left uncontrolled, blood coagulation could eventually lead to blockage of blood vessels, causing heart attacks or strokes. More regulation is thus needed. As a hard clot forms, regulatory pathways are already acting to limit the time during which the coagulation cascade is active. In

addition to cleaving fibrinogen, thrombin forms a complex with a protein embedded in the vascular surface of endothelial cells, **thrombomodulin**. The thrombin-thrombomodulin complex cleaves the serine protease zymogen **protein C**. Activated protein C, in a complex with the regulatory **protein S**, cleaves and inactivates factors Va and VIIIa, leading to suppression of the overall cascade. Another protein, **antithrombin III (ATIII)**, is a serine protease inhibitor. ATIII makes a covalent 1:1 complex between an Arg residue on ATIII and the active-site Ser residue of serine proteases, particularly thrombin and factor Xa. These two regulatory systems, in concert with TFPI, help to establish a threshold or level of exposure to TF that is needed to activate the coagulation cascade. Individuals with genetic defects that eliminate or decrease levels of protein C or ATIII in the blood have a greatly elevated risk of thrombosis (inappropriate formation of blood clots).



The control of blood coagulation has important roles in medicine, particularly in the prevention of blood clotting during surgery and in patients at risk for heart attacks or strokes. Several different medical approaches to anticoagulation are available. The first takes advantage of another feature of several proteins in the coagulation cascade that we have not yet considered. The factors VII, IX, X, and prothrombin, along with proteins C and S, have calcium-binding sites that are critical to their function. In each case, the calcium-binding sites are formed by modification of multiple Glu residues near the amino terminus of each protein to **γ -carboxyglutamate** residues (abbreviated **Gla**; p. 76). The Glu-to-Gla modifications are carried out by enzymes that depend on the function of the fat-soluble vitamin K (p. 359). Bound calcium functions to adhere these proteins to the anionic phospholipids that appear on the surface of activated platelets, effectively localizing the coagulation factors to the area where the clot is to form. Vitamin K antagonists such as **warfarin** (Coumadin) have proven highly effective as anticoagulants. A second approach to anticoagulation is the administration of heparins. **Heparins** are highly sulfated polysaccharides (see Fig. 7-19). They act as anticoagulants by increasing the affinity of ATIII for factor Xa and thrombin, thus facilitating the inactivation of key cascade elements (see Figs 7-23 and 7-24). Finally, **aspirin** (acetylsalicylate; Fig. 21-15b) is effective as an anticoagulant. Aspirin inhibits the enzyme cyclooxygenase, required for the production of thromboxanes. As aspirin reduces thromboxane release from platelets, the capacity of the platelets to aggregate declines.

Humans born with a deficiency in any one of most components of the clotting cascade have an increased tendency to bleed that varies from mild to essentially uncontrollable, a fatal condition. Genetic defects in the genes encoding the proteins required for blood clotting result in diseases referred to as hemophilias. Hemophilia B is a sex-linked trait resulting from a deficiency in factor IX. This type of hemophilia affects about one in 25,000 males worldwide. The most famous example of the inheritance

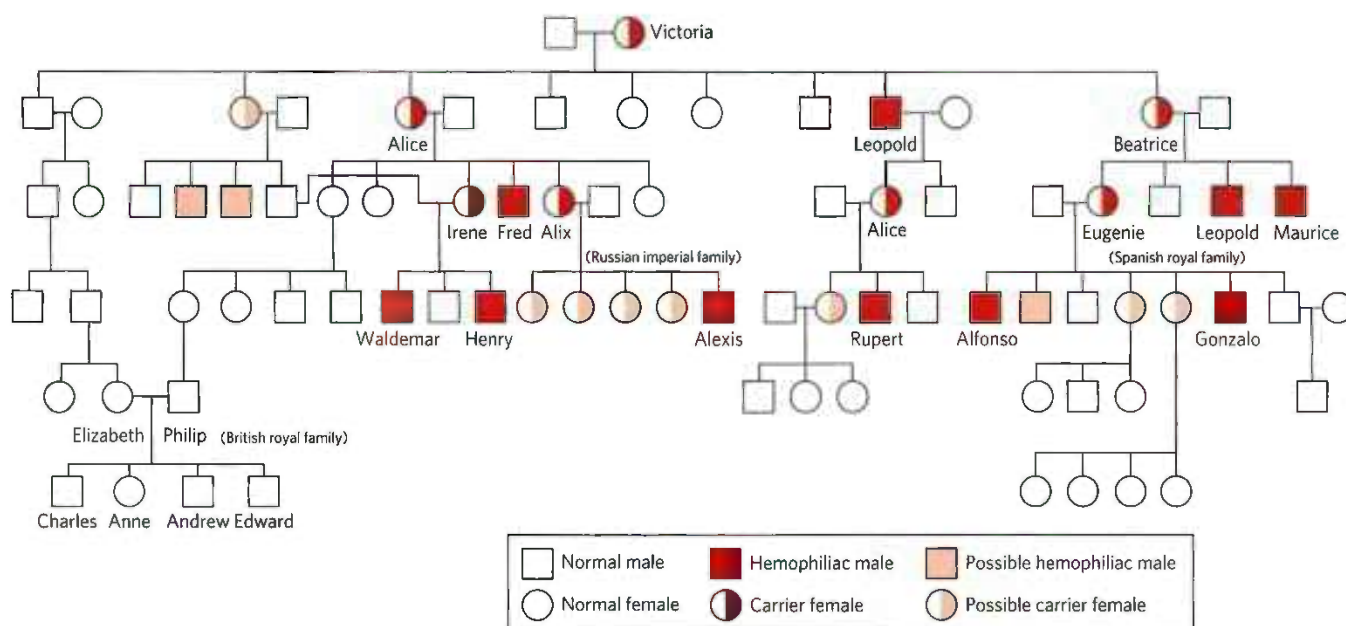


FIGURE 6-45 The royal families of Europe, and inheritance of hemophilia B. Males are indicated by squares and females by circles. Males

who suffered from hemophilia are represented by red squares, presumed female carriers by half-red circles.

of hemophilia B occurred among European royalty. Queen Victoria (1819–1901) was evidently a carrier. Prince Leopold, her eighth child, suffered from hemophilia B and died at the age of 31 after a minor fall. At least two of her daughters were carriers and passed the defective gene to other royal families of Europe (Fig. 6-45). ■

Some Regulatory Enzymes Use Several Regulatory Mechanisms

Glycogen phosphorylase and the blood clotting cascade are not the only examples of complex regulatory patterns. Additional examples of regulatory complexity are found at key metabolic crossroads. Bacterial glutamine synthetase, which catalyzes a reaction that introduces reduced nitrogen into cellular metabolism (Chapter 22), is among the most complex regulatory enzymes known. It is regulated allosterically, with at least eight different modulators; by reversible covalent modification; and by the association of other regulatory proteins, a mechanism examined in detail when we consider the regulation of specific metabolic pathways in Part II of this book.

What is the advantage of such complexity in the regulation of enzymatic activity? We began this chapter by stressing the central importance of catalysis to the existence of life. **P5** The control of catalysis is also critical to life. If all possible reactions in a cell were catalyzed simultaneously, macromolecules and metabolites would quickly be broken down to much simpler chemical forms. Instead, cells catalyze only the reactions they need at a given moment. When chemical resources are plentiful, cells synthesize and store glucose and other metabolites. When chemical resources are scarce, cells use these stores to fuel cellular metabolism. Chemical

energy is used economically, parceled out to various metabolic pathways as cellular needs dictate. The availability of powerful catalysts, each specific for a given reaction, makes the regulation of these reactions possible. This, in turn, gives rise to the complex, highly regulated symphony we call life.

SUMMARY 6.5 Regulatory Enzymes

- The activities of metabolic pathways in cells are regulated by control of the activities of certain enzymes.
- The activity of an allosteric enzyme is adjusted by reversible binding of a specific modulator to a regulatory site. A modulator may be the substrate itself or some other metabolite, and the effect of the modulator may be inhibitory or stimulatory.
- The kinetic behavior of allosteric enzymes reflects cooperative interactions among enzyme subunits.
- Other regulatory enzymes are modulated by covalent modification of a specific functional group necessary for activity.
- The phosphorylation of specific amino acid residues is a particularly common way to regulate enzyme activity.
- Many proteolytic enzymes are synthesized as inactive precursors called zymogens, which are activated by cleavage to release small peptide fragments.
- Blood clotting is mediated by two interlinked regulatory cascades of proteolytically activated zymogens.
- Enzymes at important metabolic intersections may be regulated by complex combinations of effectors, allowing coordination of the activities of interconnected pathways.

KEY TERMS

Terms in bold are defined in the glossary.

enzyme 178	Michaelis constant
cofactor 178	(K_m) 190
coenzyme 178	Lineweaver-Burk
prosthetic group 179	equation 191
holoenzyme 179	Michaelis-Menten
apoenzyme 179	kinetics 192
apoprotein 179	dissociation constant
active site 179	(K_d) 192
substrate 180	k_{cat} 192
ground state 180	turnover number 193
standard free-energy	Cleland nomenclature 195
change, ΔG° 180	reversible inhibition 197
biochemical standard	competitive inhibition 197
free-energy change,	uncompetitive
$\Delta G'^\circ$ 180	inhibition 198
transition state 180	mixed inhibition 199
activation energy	noncompetitive
($\Delta G^\ddagger$) 181	inhibition 199
reaction intermediate 181	irreversible inhibitors 200
rate-limiting step 181	suicide inactivator 200
equilibrium constant	transition-state analog 200
(K_{eq}) 182	serine proteases 205
rate constant 182	retrovirus 208
binding energy	regulatory enzyme 213
(ΔG_b) 183	allosteric enzyme 213
specificity 185	allosteric modulator (allosteric
desolvation 186	effector) 213
induced fit 186	regulatory protein 214
specific acid-base	homotropic 214
catalysis 187	heterotropic 214
general acid-base	protein kinases 217
catalysis 187	protein phosphatases 217
covalent catalysis 187	zymogen 220
enzyme kinetics 188	proteins
pre-steady state 188	(proenzymes) 220
steady state 188	regulatory cascade 220
steady-state kinetics 188	platelets 221
V_0 189	fibrin 221
V_{max} 189	fibrinogen 221
Michaelis-Menten	thromboxane 221
equation 190	thrombin 221
steady-state	intrinsic pathway 222
assumption 190	extrinsic pathway 222

PROBLEMS

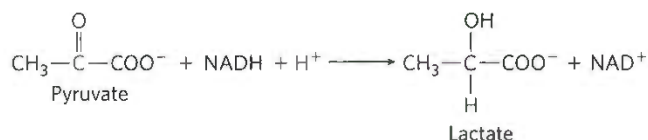
1. Keeping the Sweet Taste of Corn The sweet taste of freshly picked corn (maize) is due to the high level of sugar in the kernels. Store-bought corn (several days after picking) is not as sweet, because about 50% of the free sugar is converted to starch within one day of picking. To preserve the sweetness of fresh corn, the husked ears can be immersed in boiling water for a few minutes ("blanched"), then cooled in cold water. Corn processed in this way and stored in a freezer maintains its sweetness. What is the biochemical basis for this procedure?

2. Intracellular Concentration of Enzymes To approximate the concentration of enzymes in a bacterial cell, assume that the cell contains equal concentrations of 1,000 different enzymes in solution in the cytosol and that each protein has a molecular weight of 100,000. Assume also that the bacterial cell is a cylinder (diameter 1.0 μm , height 2.0 μm), that the cytosol (specific gravity 1.20) is 20% soluble protein by weight, and that the soluble protein consists entirely of enzymes. Calculate the *average* molar concentration of each enzyme in this hypothetical cell.

3. Rate Enhancement by Urease The enzyme urease enhances the rate of urea hydrolysis at pH 8.0 and 20 °C by a factor of 10^{14} . Suppose that a given quantity of urease can completely hydrolyze a given quantity of urea in 5.0 min at 20 °C and pH 8.0. How long would it take for this amount of urea to be hydrolyzed under the same conditions in the absence of urease? Assume that both reactions take place in sterile systems so that bacteria cannot attack the urea.

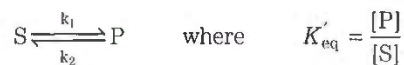
4. Protection of an Enzyme against Denaturation by Heat When enzyme solutions are heated, there is a progressive loss of catalytic activity over time due to denaturation of the enzyme. A solution of the enzyme hexokinase incubated at 45 °C lost 50% of its activity in 12 min, but when incubated at 45 °C in the presence of a very large concentration of one of its substrates, it lost only 3% of its activity in 12 min. Suggest why thermal denaturation of hexokinase was retarded in the presence of one of its substrates.

5. Quantitative Assay for Lactate Dehydrogenase The muscle enzyme lactate dehydrogenase catalyzes the reaction



NADH and NAD⁺ are the reduced and oxidized forms, respectively, of the coenzyme NAD. Solutions of NADH, but *not* NAD⁺, absorb light at 340 nm. This property is used to determine the concentration of NADH in solution by measuring spectrophotometrically the amount of light absorbed at 340 nm (A_{340}) by the solution. Explain how these properties of NADH can be used to design a quantitative assay for lactate dehydrogenase.

6. Effect of Enzymes on Reactions Consider this simple reaction:



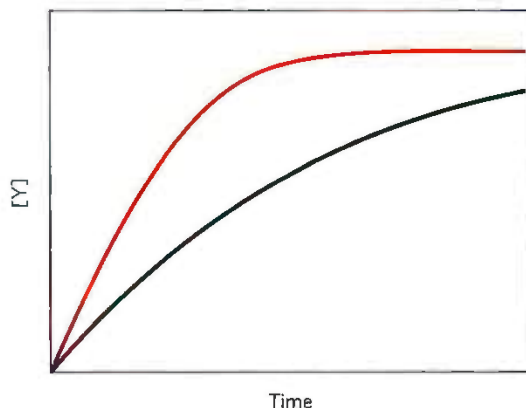
Which of the listed effects would be brought about by an enzyme catalyzing the simple reaction? (a) increased k_1 (b) increased K'_{eq} (c) decreased $\Delta G^\ddagger$ (d) more negative $\Delta G'^\circ$ (e) increased k_2

7. Relation between Reaction Velocity and Substrate Concentration: Michaelis-Menten Equation The K_m of an enzyme is 5.0 mM.

(a) Calculate the substrate concentration when this enzyme operates at one-quarter of its maximum rate.

(b) Determine the fraction of V_{max} that would be obtained when the substrate concentration, [S], is 0.5 K_m , 2 K_m , and 10 K_m .

(c) An enzyme that catalyzes the reaction $X \rightleftharpoons Y$ is isolated from two bacterial species. The enzymes have the same $V_{\max}$ but different K_m values for the substrate X. Enzyme A has a K_m of $2.0 \mu\text{M}$, and enzyme B has a K_m of $0.5 \mu\text{M}$. Kinetic experiments used the same concentration of each enzyme and $1 \mu\text{M}$ substrate X. The graph plots the concentration of product Y formed over time. Which curve corresponds to which enzyme?



8. Applying the Michaelis-Menten Equation I An enzyme has a $V_{\max}$ of $1.2 \mu\text{M s}^{-1}$. The K_m for its substrate is $10 \mu\text{M}$. Calculate the initial velocity of the reaction, V_0 , when the substrate concentration is (a) $2 \mu\text{M}$ (b) $10 \mu\text{M}$ (c) $30 \mu\text{M}$.

9. Applying the Michaelis-Menten Equation II An enzyme is present at a concentration of 1 nM and has a $V_{\max}$ of $2 \mu\text{M s}^{-1}$. The K_m for its primary substrate is $4 \mu\text{M}$.

(a) Calculate k_{cat} .

(b) Calculate the apparent (measured) $V_{\max}$ and apparent (measured) K_m of this enzyme in the presence of sufficient amounts of an uncompetitive inhibitor to generate an α' of 2. Assume that the enzyme concentration remains at 1 nM .

10. Applying the Michaelis-Menten Equation III A research group discovers a new version of happyase, which they call happyase*, that catalyzes the chemical reaction $\text{HAPPY} \rightleftharpoons \text{SAD}$. The researchers begin to characterize the enzyme.

(a) In the first experiment, with $[E_t]$ at 4 nM , they find that the $V_{\max}$ is $1.6 \mu\text{M s}^{-1}$. Based on this experiment, what is the k_{cat} for happyase*? (Include appropriate units.)

(b) In the second experiment, with $[E_t]$ at 1 nM and $[\text{HAPPY}]$ at $30 \mu\text{M}$, the researchers find that $V_0 = 300 \text{ nM s}^{-1}$. What is the measured K_m of happyase* for its substrate HAPPY? (Include appropriate units.)

(c) Further research shows that the purified happyase* used in the first two experiments was actually contaminated with a reversible inhibitor called ANGER. When ANGER is carefully removed from the happyase* preparation and the two experiments are repeated, the measured $V_{\max}$ in (a) is increased to $4.8 \mu\text{M s}^{-1}$, and the measured K_m in (b) is now $15 \mu\text{M}$. Calculate the values of α and α' for ANGER.

(d) Based on the information given, what type of inhibitor is ANGER?

11. Applying the Michaelis-Menten Equation IV Researchers discover an enzyme that catalyzes the reaction $X \rightleftharpoons Y$. They find that the K_m for the substrate X is $4 \mu\text{M}$, and the k_{cat} is 20 min^{-1} .

(a) In an experiment, $[X] = 6 \text{ mM}$, and $V_0 = 480 \text{ nM min}^{-1}$. What was the $[E_t]$ used in the experiment?

(b) In another experiment, $[E_t] = 0.5 \mu\text{M}$, and the measured $V_0 = 5 \mu\text{M min}^{-1}$. What was the $[X]$ used in the experiment?

(c) The researchers discover that compound Z is a very strong competitive inhibitor of the enzyme. In an experiment with the same $[E_t]$ as in (a), but a different $[X]$, they add an amount of Z that produces an α of 10 and reduces V_0 to 240 nM min^{-1} . What is the $[X]$ in this experiment?

(d) Based on the kinetic parameters given, has this enzyme evolved to achieve catalytic perfection? Explain your answer briefly, using the kinetic parameter(s) that define catalytic perfection.

12. Estimation of $V_{\max}$ and K_m by Inspection Graphical methods are available for accurate determination of the $V_{\max}$ and K_m of an enzyme-catalyzed reaction. However, these quantities can sometimes be estimated by inspecting values of V_0 at increasing $[S]$. Estimate the $V_{\max}$ and K_m of the enzyme-catalyzed reaction for which the data in the table were obtained.

$[S] \text{ (M)}$	$V_0 \text{ (}\mu\text{M/min)}$
2.5×10^{-6}	28
4.0×10^{-6}	40
1×10^{-5}	70
2×10^{-5}	95
4×10^{-5}	112
1×10^{-4}	128
2×10^{-3}	139
1×10^{-2}	140

13. Properties of an Enzyme of Prostaglandin Synthesis Prostaglandins are one class of the fatty acid derivatives called eicosanoids. Prostaglandins produce fever and inflammation, as well as the pain associated with inflammation. The enzyme prostaglandin endoperoxide synthase, a cyclooxygenase, uses oxygen to convert arachidonic acid to PGG_2 , the immediate precursor of many different prostaglandins (prostaglandin synthesis is described in Chapter 21).

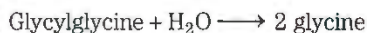
Ibuprofen inhibits prostaglandin endoperoxide synthase, thereby reducing inflammation and pain. The kinetic data given in the table are for the reaction catalyzed by prostaglandin endoperoxide synthase in the absence and presence of ibuprofen.

(a) Based on the data, determine the $V_{\max}$ and K_m of the enzyme.

[Arachidonic acid] (mM)	Rate of formation of PGG_2 (mM min ⁻¹)	Rate of formation of PGG_2 with 10 mg/mL ibuprofen (mM min ⁻¹)
0.5	23.5	16.67
1.0	32.2	25.25
1.5	36.9	30.49
2.5	41.8	37.04
3.5	44.0	38.91

(b) Based on the data, determine the type of inhibition that ibuprofen exerts on prostaglandin endoperoxide synthase.

14. Graphical Analysis of $V_{\max}$ and K_m A kinetic study of an intestinal peptidase using glycylglycine as the substrate produced the experimental data shown in the table. The peptidase catalyzes this reaction:



[S] (mM)	Product formed ($\mu\text{mol/min}$)
1.5	0.21
2.0	0.24
3.0	0.28
4.0	0.33
8.0	0.40
16.0	0.45

Use the Lineweaver-Burk equation to determine the $V_{\max}$ and K_m for this enzyme preparation and substrate.

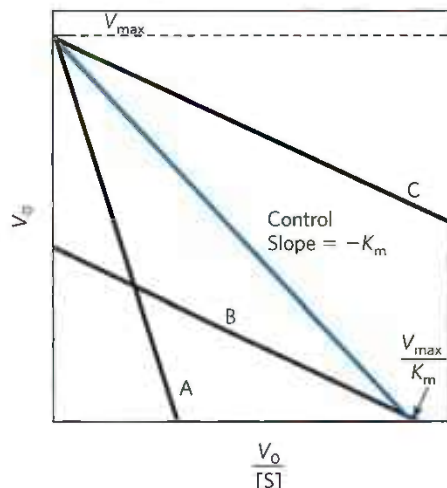
15. The Eadie-Hofstee Equation There are several ways to transform the Michaelis-Menten equation so as to plot data and derive kinetic parameters, each with different advantages depending on the data set being analyzed. One transformation of the Michaelis-Menten equation is the Lineweaver-Burk, or double-reciprocal, equation. Multiplying both sides of the Lineweaver-Burk equation by $V_{\max}$ and rearranging gives the Eadie-Hofstee equation:

$$V_0 = (-K_m) \frac{V_0}{[S]} + V_{\max}$$

Consider the plot of V_0 versus $V_0/[S]$ for an enzyme-catalyzed reaction. The slope of the line is $-K_m$. The x intercept is $V_{\max}/K_m$. The control reactions (the blue line in the plot) did not contain any inhibitor.

(a) Which of the other lines (A, B, or C) depicts this enzyme's activity in the presence of a competitive inhibitor? Hint: See Equation 6-33.

(b) Which line (A, B, or C) depicts this enzyme's activity in the presence of an uncompetitive inhibitor?



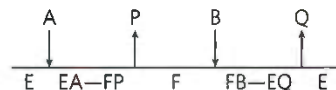
16. The Turnover Number of Carbonic Anhydrase Carbonic anhydrase of erythrocytes (M_r 30,000) has one of the highest turnover numbers known. It catalyzes the reversible hydration of CO_2 :



This is an important process in the transport of CO_2 from the tissues to the lungs. If $10.0 \mu\text{g}$ of pure carbonic anhydrase catalyzes the hydration of 0.30 g of CO_2 in 1 min at 37°C at $V_{\max}$, what is the turnover number (k_{cat}) of carbonic anhydrase (in units of min^{-1})?

17. Describing Reactions with the Cleland Shorthand The chymotrypsin-catalyzed reaction is diagrammed using the Cleland shorthand. Match the letters in the drawing with each description:

- The product that includes the amino group from the cleaved peptide bond
- The product that includes the carbonyl group from the cleaved peptide bond
- Free chymotrypsin (nothing bound to it)
- Water
- The peptide substrate
- The acyl-enzyme intermediate



18. Kinetic Inhibition Patterns Indicate how the observed K_m of an enzyme would change in the presence of inhibitors having the given effect on α and α' :

- $\alpha > \alpha'$; $\alpha' = 1.0$
- $\alpha' > \alpha$
- $\alpha = \alpha'$; $\alpha' > 1.0$
- $\alpha = \alpha'$; $\alpha' = 1.0$

19. Deriving a Rate Equation for Competitive Inhibition The Michaelis-Menten rate equation for an enzyme subject to competitive inhibition is

$$V_0 = \frac{V_{\max} [S]}{\alpha K_m + [S]}$$

Beginning with a new definition of total enzyme as

$$[E_t] = [E] + [ES] + [EI]$$

and the definitions of α and K_I provided in the text, derive the first rate equation. Use the derivation of the Michaelis-Menten equation as a guide.

20. Irreversible Inhibition of an Enzyme Many enzymes are inhibited irreversibly by heavy metal ions such as Hg^{2+} , Cu^{2+} , or Ag^+ , which can react with essential sulfhydryl groups to form mercaptides:

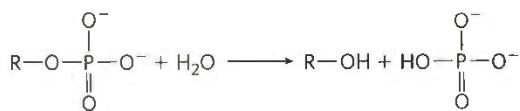


The affinity of Ag^+ for sulfhydryl groups is so great that Ag^+ can be used to titrate $-\text{SH}$ groups quantitatively. An investigator added just enough AgNO_3 to completely inactivate a 10.0 mL solution containing 1.0 mg/mL enzyme. A total of $0.342 \mu\text{mol}$ of AgNO_3 was required. Calculate the minimum molecular weight of the enzyme. Why does the value obtained in this way give only the *minimum* molecular weight?



21. Clinical Application of Differential Enzyme

Inhibition Human blood serum contains a class of enzymes known as acid phosphatases, which hydrolyze biological phosphate esters under slightly acidic conditions (pH 5.0):

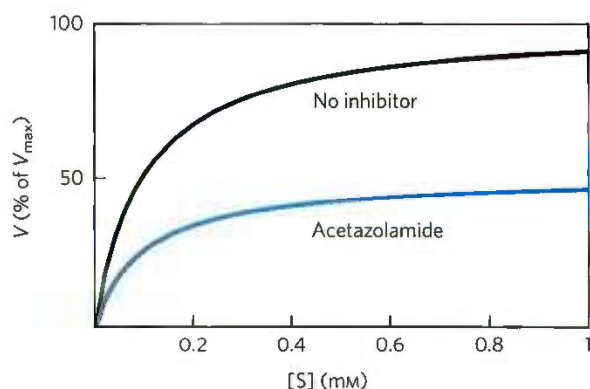


Acid phosphatases are produced by erythrocytes and by the liver, kidney, spleen, and prostate gland. The enzyme of the prostate gland is clinically important, because its increased activity in the blood can be an indication of prostate cancer. The phosphatase from the prostate gland is strongly inhibited by tartrate ion, but acid phosphatases from other tissues are not. How can this information be used to develop a specific procedure for measuring the activity of prostatic acid phosphatase in human blood serum?



22. Inhibition of Carbonic Anhydrase by Acetazolamide

Carbonic anhydrase is strongly inhibited by the drug acetazolamide, which is used as a diuretic (i.e., to increase the production of urine) and to lower excessively high pressure in the eye (due to accumulation of intraocular fluid) in glaucoma. Carbonic anhydrase plays an important role in these and other secretory processes because it participates in regulating the pH and bicarbonate content of several body fluids. Carbonic anhydrase activity can be analyzed using the initial reaction velocity (as percentage of $V_{\max}$) versus $[S]$. The black curve of the graph shows the uninhibited activity; the blue curve shows activity in the presence of acetazolamide. Based on the data provided, determine the nature of the inhibition by acetazolamide. Explain your reasoning.



23. The Effects of Reversible Inhibitors The Michaelis-Menten rate equation for reversible mixed inhibition is written as

$$V_0 = \frac{V_{\max}[S]}{\alpha K_m + \alpha'[S]}$$

Apparent, or observed, K_m is equivalent to the $[S]$ at which

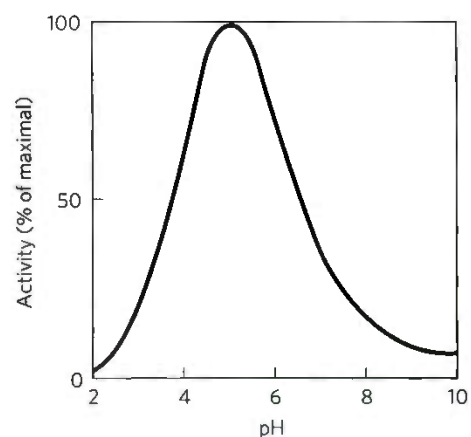
$$V_0 = \frac{V_{\max}}{2\alpha'}$$

Derive an expression for the effect of a reversible inhibitor on apparent K_m from the previous equation.

24. Perturbed pK_a Values in Enzyme Active Sites Alanine racemase is a bacterial enzyme that converts L-alanine to D-alanine,

which is needed in small amounts to synthesize the bacterial cell wall. The active site of alanine racemase includes a Tyr residue with a pK_a value of 7.2. The pK_a of free tyrosine is 10. The altered pK_a of this residue is due largely to the presence of a nearby charged amino acid residue. Which amino acid(s) could lower the pK_a of the neighboring Tyr residue? Explain your reasoning.

25. pH Optimum of Lysozyme The enzyme lysozyme hydrolyzes glycosidic bonds in peptidoglycan, an oligosaccharide found in bacterial cell walls. The active site of lysozyme contains two amino acid residues essential for catalysis: Glu³⁵ and Asp⁵². The pK_a values of the carboxyl side chains of these residues are 5.9 and 4.5, respectively. What is the ionization state (protonated or deprotonated) of each residue at pH 5.2, the pH optimum of lysozyme? How can the ionization states of these residues explain the pH-activity profile of lysozyme shown?

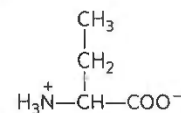


DATA ANALYSIS PROBLEM

26. Mirror-Image Enzymes As noted in Chapter 3, "The amino acid residues in protein molecules are almost all L stereoisomers." It is not clear whether this selectivity is necessary for proper protein function or is an accident of evolution. To explore this question, Milton and colleagues (1992) studied an enzyme made entirely of D stereoisomers. The enzyme they chose was HIV protease, the proteolytic enzyme made by HIV that converts inactive viral pre-proteins to their active forms as described earlier in Figure 6-28.

Previously, Wlodawer and coworkers (1989) had reported the complete chemical synthesis of HIV protease from L-amino acids (the L-enzyme), using the process shown in Figure 3-30. Normal HIV protease contains two Cys residues at positions 67 and 95. Because chemical synthesis of proteins containing Cys is technically difficult, Wlodawer and colleagues substituted the synthetic amino acid L- α -amino-n-butyric acid (Aba) for the two Cys residues in the protein. They did this to "reduce synthetic difficulties associated with Cys deprotection and ease product handling."

(a) The structure of Aba is shown below.



Why was this a suitable substitution for a Cys residue?

In their study, Milton and coworkers synthesized HIV protease from D-amino acids, using the same protocol as the earlier study (Wlodawer et al.). Formally, there are three possible outcomes for the folding of the D-protease: (1) the same shape as the L-protease; (2) the mirror image of the L-protease, or (3) something else, possibly inactive.

(b) For each possibility, decide whether or not it is a likely outcome and defend your position.

In fact, the D-protease was active: it cleaved a particular synthetic substrate and was inhibited by specific inhibitors. To examine the structure of the D- and L-enzymes, Milton and coworkers tested both forms for activity with D and L forms of a chiral peptide substrate and for inhibition by D and L forms of a chiral peptide-analog inhibitor. Both forms were also tested for inhibition by the achiral inhibitor Evans blue. The findings are given in the table.

HIV protease	Substrate hydrolysis		Inhibition	
	D form	L form	Peptide inhibitor D form	L form
L form	—	+	—	+
D form	+	—	+	—

(c) Which of the three models proposed above is supported by these data? Explain your reasoning.

(d) Would you expect chymotrypsin to digest the D-protease? Explain your reasoning.

(e) Would you expect total synthesis from D-amino acids followed by renaturation to yield active enzyme for any enzyme? Explain your reasoning.

References

Milton, R.C., Milton, S.C., and Kent, S.B. 1992. Total chemical synthesis of a D-enzyme: the enantiomers of HIV-1 protease show demonstration of reciprocal chiral substrate specificity. *Science* 256, 1445–1448.

Wlodawer, A., Miller, M., Jaskólski, M., Sathyanarayana, B.K., Baldwin, E., Weber, I.T., Selk, L.M., Clawson, L., Schneider, J., and Kent, S.B. 1989. Conserved folding in retroviral proteases: crystal structure of a synthetic HIV-1 protease. *Science* 245, 616–621.

CARBOHYDRATES AND GLYCOBIOLOGY

- 7.1 Monosaccharides and Disaccharides 230
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Carbohydrates are the most abundant biomolecules on Earth. Each year, photosynthesis converts more than 100 billion metric tons of CO_2 and H_2O into cellulose and other plant products. The carbohydrates in these plant products are a dietary staple in most parts of the world, and the oxidation of carbohydrates is the central energy-yielding pathway in most nonphotosynthetic cells. Carbohydrate polymers called glycans serve as structural and protective elements in the cell walls of bacteria, fungi, and plants, and in the connective tissues of animals. Other carbohydrate polymers lubricate skeletal joints and participate in cell-cell recognition and adhesion. In addition, some complex carbohydrate polymers covalently attached to proteins or lipids act as signals that determine the intracellular destination or metabolic fate of these hybrid molecules, called glycoconjugates.

Carbohydrates are aldehydes or ketones with at least two hydroxyl groups, or substances that yield such compounds on hydrolysis. Many, but not all, carbohydrates have the empirical formula $(\text{CH}_2\text{O})_n$; some also contain nitrogen, phosphorus, or sulfur. There are three major size classes of carbohydrates: monosaccharides, oligosaccharides, and polysaccharides (the word “saccharide” is derived from the Greek *sakcharon*, meaning “sugar”). **Monosaccharides**, or simple sugars, consist of a single polyhydroxy aldehyde or ketone unit. The most abundant monosaccharide in nature is the six-carbon sugar D-glucose, sometimes referred to as dextrose.

Oligosaccharides consist of short chains of monosaccharide units, or residues, joined by characteristic

linkages called glycosidic bonds. The most abundant are the **disaccharides**, with two monosaccharide units. Sucrose (table sugar), for example, consists of the six-carbon sugars D-glucose and D-fructose. All common monosaccharides and disaccharides have names ending with the suffix “-ose.” In cells, most oligosaccharides consisting of three or more units do not occur as free entities but are joined to nonsugar molecules (lipids or proteins) in glycoconjugates.

The **polysaccharides** are sugar polymers containing more than 10 monosaccharide units; some have hundreds or thousands of units. Some polysaccharides, such as cellulose, are linear chains; others, such as glycogen, are branched. Both cellulose and glycogen consist of recurring units of D-glucose, but they differ in the type of glycosidic linkage and consequently have strikingly different properties and biological roles.

This chapter introduces the major classes of carbohydrates and glycoconjugates and provides examples of their many structural and functional roles. Learning the structures and chemical properties of biomolecules is essential, because they are the vocabulary and the grammar of biochemistry. As you read about carbohydrates, note how specific cases illustrate these general principles that underlie all of biochemistry.

P1 Carbohydrates can have multiple chiral carbons; the configuration of groups around each carbon atom determines how the compound interacts with other biomolecules.

As we saw for L-amino acids in proteins, with rare exceptions, biological evolution selected one stereochemical series (D-series) for sugars.

P2 Monomeric subunits, monosaccharides, serve as the building blocks of large carbohydrate polymers. The specific sugar, the way the units are linked, and whether the polymer is branched determine its properties and thus its function.

P3 Storage of low molecular weight metabolites in polymeric form avoids the very high osmolarity that would result from storing them as individual monomers. If the glucose in liver glycogen were monomeric, the glucose

concentration in liver would be so high that cells would swell and lyse from the entry of water by osmosis.

P4 The sequences of complex polysaccharides are determined by the intrinsic properties of the biosynthetic enzymes that add each monomeric unit to the growing polymer. This is in contrast with DNA, RNA, and proteins, which are synthesized on templates that direct their sequence.

P5 Polysaccharides assume three-dimensional structures with the lowest-energy conformations, determined by covalent bonds, hydrogen bonds, charge interactions, and steric factors. Starch folds into a helical structure stabilized by internal hydrogen bonds; cellulose assumes an extended structure in which intermolecular hydrogen bonds are more important.

P6 Molecular complementarity is central to function. The recognition of oligosaccharides by sugar-binding proteins (lectins) results from a perfect fit between lectin and ligand.

P7 An almost infinite variety of discrete structures can be built from a small number of monomeric subunits. Even short polymers, when arranged in different sequences, joined through different linkages, and branched to specific degrees, present unique faces recognized by their molecular partners.

7.1 Monosaccharides and Disaccharides

The simplest of the carbohydrates, the monosaccharides, are either aldehydes or ketones with two or more hydroxyl groups; the six-carbon monosaccharides glucose and fructose have five hydroxyl groups. Many of the carbon atoms to which the hydroxyl groups are attached are chiral centers, which give rise to the many sugar stereoisomers found in nature. Stereoisomerism in sugars is biologically significant because the enzymes that act on sugars are strictly

stereospecific, typically preferring one stereoisomer to another by three or more orders of magnitude, as reflected in K_m values or binding constants. It is as difficult to fit the wrong sugar stereoisomer into an enzyme's binding site as it is to put your left glove on your right hand.

We begin by describing the families of monosaccharides with backbones of three to seven carbons—their structure, their stereoisomeric forms, and the means of representing their three-dimensional structures on paper. We then discuss several chemical reactions of the carbonyl groups of monosaccharides. One such reaction, the addition of a hydroxyl group from within the same molecule, generates cyclic forms with four or more backbone carbons (the forms that predominate in aqueous solution).

P1 This ring closure creates a new chiral center, adding further stereochemical complexity to this class of compounds. The nomenclature for unambiguously specifying the configuration about each carbon atom in a cyclic form and the means of representing these structures on paper are described in some detail; this information will be useful as we discuss the metabolism of monosaccharides in Part II. We also introduce here some important monosaccharide derivatives that will be examined closely in later chapters.

The Two Families of Monosaccharides Are Aldoses and Ketoses

Monosaccharides are colorless, crystalline solids that are freely soluble in water but insoluble in nonpolar solvents. Most have a sweet taste (Box 7-1). The backbones of common monosaccharides are unbranched carbon chains in which all the carbon atoms are linked by single bonds. In this open-chain form, one of the carbon atoms is double-bonded to an oxygen atom to form a carbonyl group; each of the other carbon atoms has a hydroxyl group. If the carbonyl group is at an end of the carbon chain (that is, in an aldehyde group), the monosaccharide is an **aldose**; if the carbonyl group is at any other position (in a ketone group), the monosaccharide is a **ketose**. The simplest monosaccharides are the two three-carbon trioses: glyceraldehyde, an aldotriose, and dihydroxyacetone, a ketotriose (Fig. 7-1a).

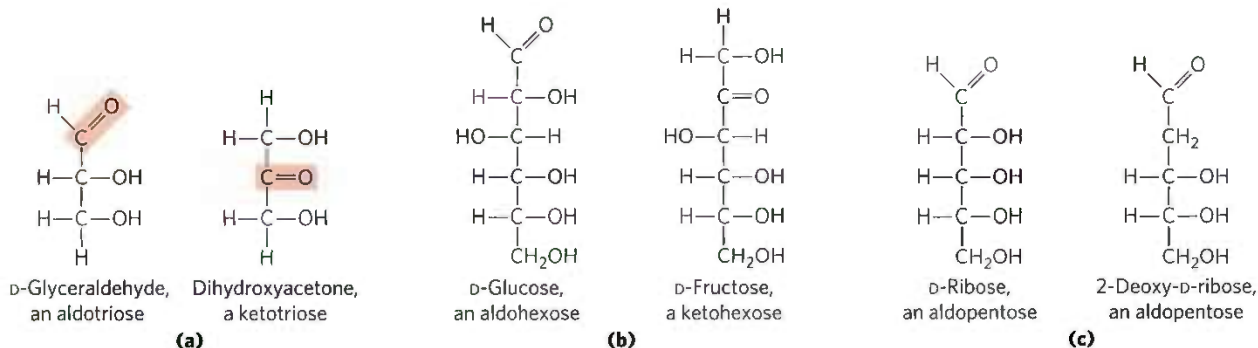


FIGURE 7-1 Representative monosaccharides. (a) Two trioses: an aldose and a ketose. The carbonyl group in each is shaded. (b) Two common hexoses. (c) The pentose components of nucleic acids. D-Ribose is a

component of ribonucleic acid (RNA), and 2-deoxy-D-ribose is a component of deoxyribonucleic acid (DNA).



What Makes Sugar Sweet?

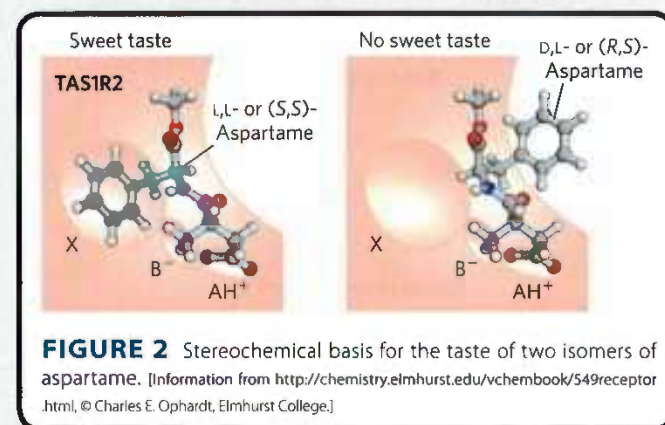
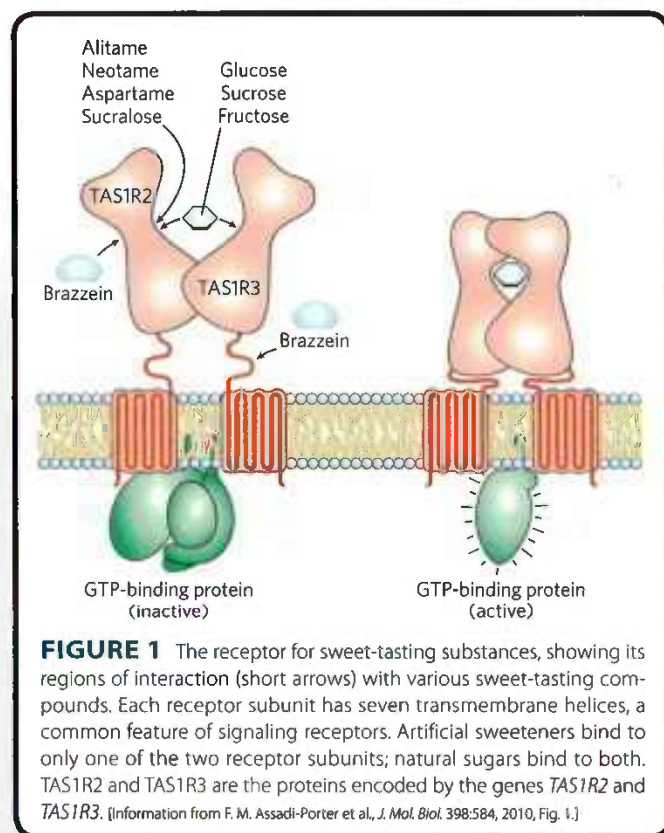
Sweetness is one of the five basic flavors that humans can taste; the others are sour, bitter, salty, and umami. Sweetness is detected by protein receptors in the plasma membranes of cells in the taste buds on the surface of the tongue. In humans, two closely related genes (*TAS1R2* and *TAS1R3*) encode sweet-taste receptors (Fig. 1). When a molecule with a compatible structure binds a sweet-taste receptor's extracellular domain, it triggers a series of events in the cell (including activation of a GTP-binding protein; see Fig. 12-20) that generates an electrical signal to the brain that is interpreted as "sweet."

During evolution, there has probably been selection for the ability to taste compounds found in foods containing important nutrients, such as the carbohydrates that are major fuels for most organisms. Most sugars, including sucrose, glucose, lactose, and fructose, taste sweet, but other classes of compounds also bind sweet-taste receptors. The amino acids glycine, alanine, and serine are mildly sweet and harmless; nitrobenzene and ethylene glycol have a strong sweet taste

but are toxic. (See Box 18-2 for a remarkable medical mystery involving ethylene glycol poisoning.) Several natural products are extraordinarily sweet. Stevioside, a sugar derivative isolated from the leaves of the stevia plant (*Stevia rebaudiana* Bertoni), is several hundred times sweeter than an equivalent amount of sucrose. The small protein brazzein (54 amino acids), isolated from berries of the Oubli vine (*Pentadiplandra brazzeana* Baillon) in Gabon and Cameroon, is 17,000 times sweeter than sucrose on a molar basis. Presumably, the sweetness of the berries encourages their consumption by animals that then disperse the seeds so that new plants are established.

In societies where obesity is a major health problem, compounds that give foods a sweet taste without adding the calories found in sugars are common food additives. The artificial sweetener aspartame demonstrates the importance of stereochemistry in biology (Fig. 2). One simple model of binding to the sweet-taste receptor involves three sites: AH^+ , B^- , and X. Site AH^+ has a group (an alcohol or an amine) that binds the partially negative oxygen of the carboxylic acid of (*S,S*)-aspartame. Site B^- has a partially negative oxygen available to hydrogen-bond with the amine nitrogen of (*S,S*)-aspartame. Site X is oriented perpendicular to the other two groups and can accommodate the hydrophobic benzene ring of (*S,S*)-aspartame.

When the steric match is correct, as for (*S,S*)-aspartame, on the left in Figure 2, the sweet receptor is stimulated and the signal "sweet" is conducted to the brain. When the match is not correct, as for (*R,S*)-aspartame, the sweet receptor is not stimulated; in fact, (*R,S*)-aspartame stimulates a separate receptor specifically for bitterness. Stereoisomerism really matters!



Monosaccharides with four, five, six, and seven carbon atoms in their backbones are called, respectively, tetroses, pentoses, hexoses, and heptoses. There are aldoses and ketoses of each of these chain lengths: aldotetroses and ketotetroses, aldopentoses and ketopentoses, and so on. The hexoses, which include the aldohexose D-glucose

and the ketohexose D-fructose (Fig. 7-1b), are the most common monosaccharides in nature—the products of photosynthesis and key intermediates in the central energy-yielding reaction sequence in most organisms. The aldopentoses D-ribose and 2-deoxy-D-ribose (Fig. 7-1c) are components of nucleotides and nucleic acids (Chapter 8).

Monosaccharides Have Asymmetric Centers

P1 All the monosaccharides except dihydroxyacetone contain one or more asymmetric (chiral) carbon atoms and thus occur in optically active isomeric forms (pp. 16–17). The simplest aldose, glyceraldehyde, contains one chiral center (the middle carbon atom) and therefore has two different optical isomers, or **enantiomers** (Fig. 7-2).

KEY CONVENTION One of the two enantiomers of glyceraldehyde is, by convention, designated the **D** isomer; the other is the **L** isomer. As for other biomolecules with chiral centers, the absolute configurations of sugars are known from x-ray crystallography. To represent three-dimensional sugar structures on paper, we often use **Fischer projection formulas** (Fig. 7-2). In these projections, bonds drawn horizontally indicate bonds that project out of the plane of the paper, toward the reader; bonds drawn vertically project behind the plane of the paper, away from the reader. ■

In general, a molecule with n chiral centers can have 2^n stereoisomers. Glyceraldehyde has $2^1 = 2$; the aldohexoses, with four chiral centers, have $2^4 = 16$. The

stereoisomers of monosaccharides of each carbon-chain length can be divided into two groups that differ in the configuration about the chiral center *most distant* from the carbonyl carbon. Those in which the configuration at this reference carbon is the same as that of **D**-glyceraldehyde are designated **D** isomers, and those with the same configuration as **L**-glyceraldehyde are **L** isomers. In other words, when the hydroxyl group on the reference carbon is on the right (*dextro*) in a projection formula that has the carbonyl carbon at the top, the sugar is the **D** isomer; when the hydroxyl group is on the left (*levo*), it is the **L** isomer. Of the 16 possible aldohexoses, eight are **D** forms and eight are **L**. **P1** Most of the hexoses of living organisms are **D** isomers. Why **D** isomers? An interesting and unanswered question. Recall that all of the amino acids found in proteins are exclusively one of two possible stereoisomers, **L** (p. 73). The basis for this initial preference for one isomer during evolution is unknown; however, once one isomer became prevalent, evolving enzymes able to use that isomer efficiently would have a selective advantage.

Figure 7-3 shows the structures of the **D** stereoisomers of all the aldoses and ketoses having three to six carbon atoms. The carbons of a sugar are numbered beginning at the end of the chain nearest the carbonyl group. Each of the eight **D**-aldohexoses, which differ in the stereochemistry at C-2, C-3, or C-4, has its own name: **D**-glucose, **D**-galactose, **D**-mannose, and so forth (Fig. 7-3a). The four- and five-carbon ketoses are designated by inserting “ul” into the name of a corresponding aldose; for example, **D**-ribulose is the ketopentose corresponding to the aldopentose **D**-ribose. (The importance of ribulose will become clear when we discuss the fixation of atmospheric CO_2 by green plants, in Chapter 20.) The ketohexoses are named otherwise: for example, fructose (from the Latin *fructus*, “fruit”; fruits are one source of this sugar) and sorbose (from *Sorbus*, the genus of mountain ash, which has berries rich in the related sugar alcohol sorbitol). Two sugars that differ only in the configuration around one carbon atom are called **epimers**; **D**-glucose and **D**-mannose, which differ only in the stereochemistry at C-2, are epimers, as are **D**-glucose and **D**-galactose (which differ at C-4) (Fig. 7-4).

Some sugars occur naturally in their **L** form; examples are **L**-arabinose and the **L** isomers of some sugar derivatives that are common components of glycoconjugates (Section 7.3).

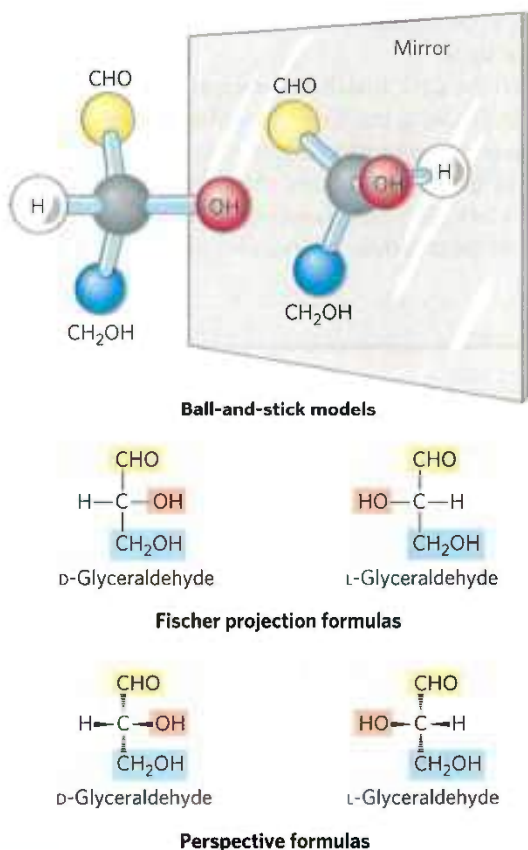
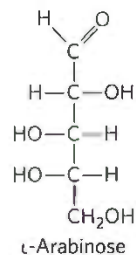


FIGURE 7-2 Three ways to represent the two enantiomers of glyceraldehyde. The enantiomers are mirror images of each other. Ball-and-stick models show the actual configuration of molecules. In Fischer projections, vertical lines point behind the plane of the page, and horizontal lines project above the page. Recall (see Fig. 1-17) that in perspective formulas, the wide end of a solid wedge projects out of the plane of the paper, toward the reader; a dashed wedge extends behind.



(a) D-Aldoses

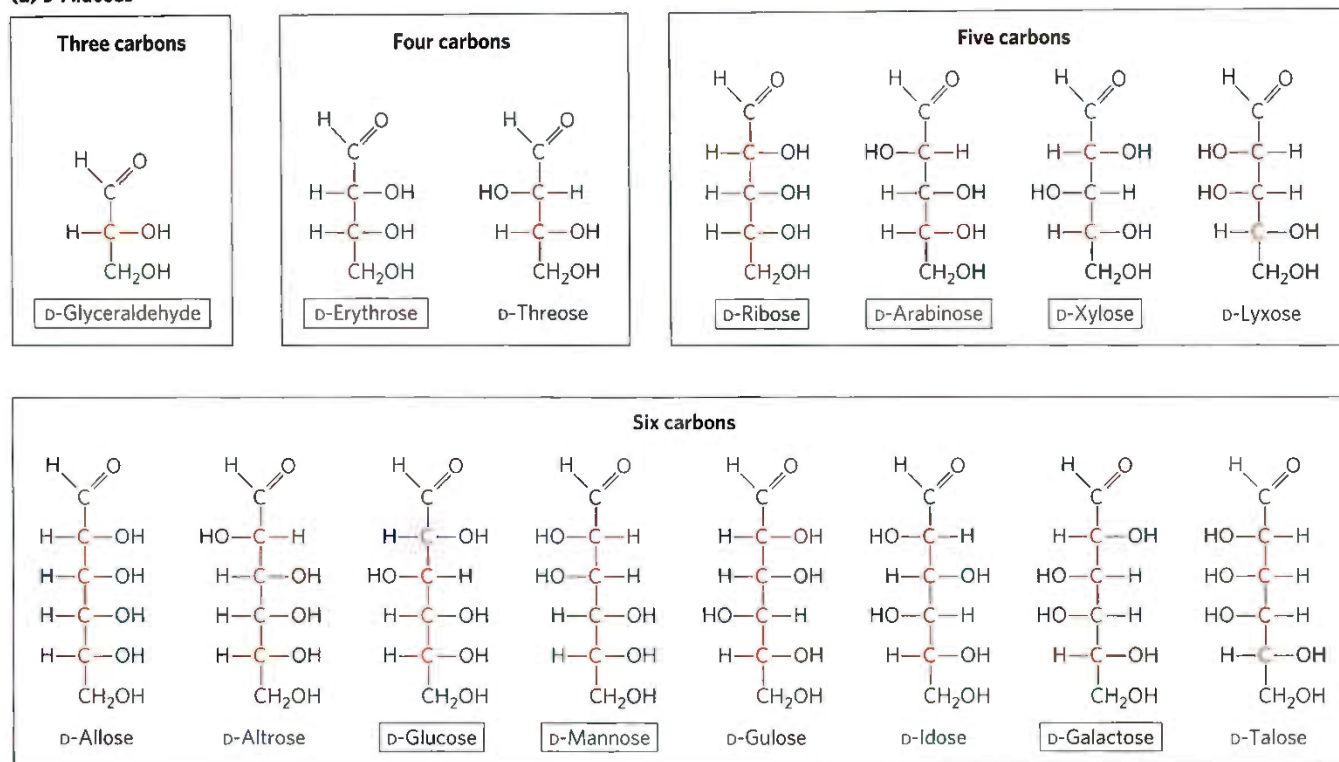
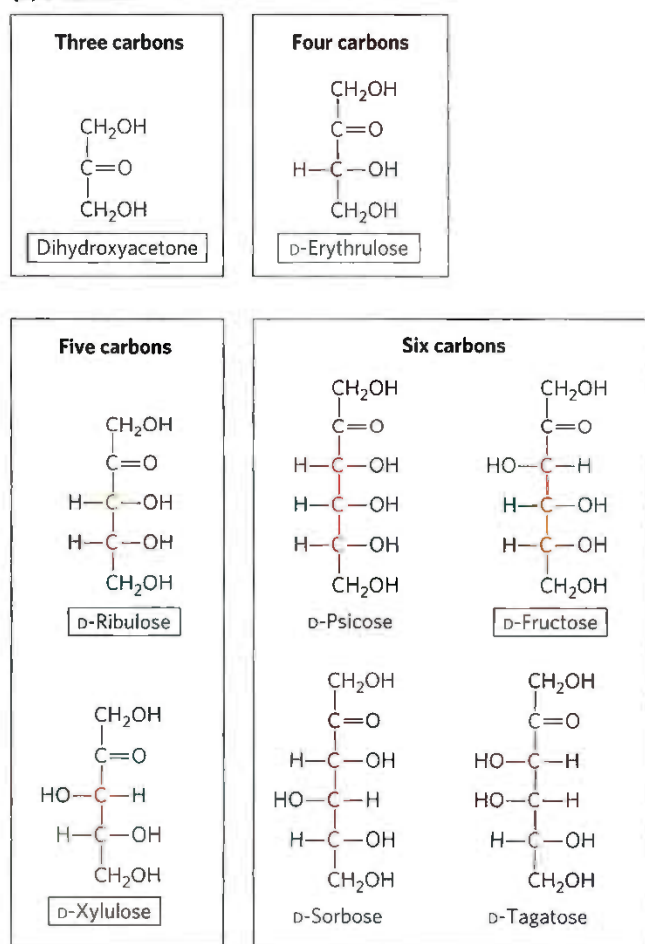


FIGURE 7-3 Aldoses and ketoses. The series of (a) D-aldoses and (b) D-ketoses having from three to six carbon atoms, shown as projection formulas. The carbon atoms in red are chiral centers. In all of these D isomers, the chiral carbon *most distant from the carbonyl carbon* has the same configuration as the chiral carbon in D-glyceraldehyde. Shown are the most common sugars in nature; you will encounter these again in this and later chapters.

(b) D-Ketoses



The Common Monosaccharides Have Cyclic Structures

For simplicity, we have thus far represented the structures of aldoses and ketoses as straight-chain molecules (Figs 7-3, 7-4). In fact, in aqueous solution, aldotetroses and all monosaccharides with five or more carbon atoms in the backbone occur predominantly as cyclic (ring) structures in which the carbonyl group has formed a covalent bond with the oxygen of a hydroxyl group in the same sugar molecule. The formation of these ring

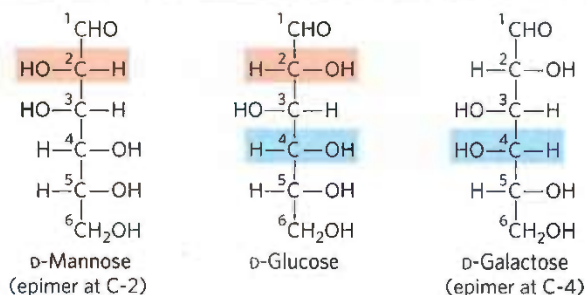


FIGURE 7-4 Epimers. D-Glucose and two of its epimers are shown as projection formulas. Each epimer differs from D-glucose in the configuration at one chiral center (shaded light red or blue).

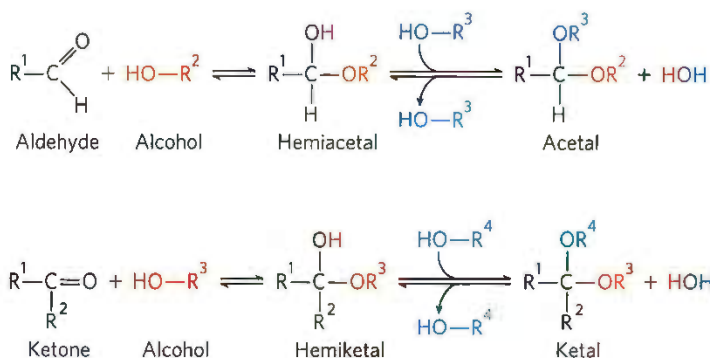


FIGURE 7-5 Formation of hemiacetals and hemiketals. An aldehyde or a ketone can react with an alcohol in a 1:1 ratio to yield a hemiacetal or a hemiketal, respectively, creating a new chiral center at the carbonyl carbon. Substitution of a second alcohol molecule produces an acetal or a ketal. When the second alcohol is part of another sugar molecule, the bond produced is a glycosidic bond.

structures is the result of a general reaction between alcohols and aldehydes or ketones to form derivatives called **hemiacetals** or **hemiketals**. Two molecules of an alcohol can add to a carbonyl carbon; the product of the first addition is a hemiacetal (for addition to an aldose) or a hemiketal (for addition to a ketose). If the —OH and carbonyl groups are on the same molecule, a five- or six-membered ring results. Addition of the second molecule of alcohol produces the full **acetal** or **ketal** (Fig. 7-5), and the linkage formed is a glycosidic bond. When the two molecules that react are monosaccharides, the acetal or ketal formed is a disaccharide.

The reaction with the first molecule of alcohol creates an additional chiral center (the carbonyl carbon). Because the alcohol can add in either of two ways, attacking either the “front” or the “back” of the carbonyl carbon, the reaction can produce either of two stereoisomeric configurations, denoted α and β . For example, D-glucose exists in solution as an intramolecular hemiacetal in which the free hydroxyl group at C-5 has reacted with the aldehydic C-1, rendering the latter carbon asymmetric and producing two possible stereoisomers, designated α and β (Fig. 7-6). Isomeric forms of monosaccharides that differ only in their configuration about the hemiacetal or hemiketal carbon atom are called **anomers**, and the carbonyl carbon atom is called the **anomeric carbon**.

Six-membered ring compounds are called **pyranoses** because they resemble the six-membered ring compound pyran (shown in Fig. 7-7). The systematic names for the two ring forms of D-glucose are therefore $\alpha\text{-D-glucopyranose}$ and $\beta\text{-D-glucopyranose}$. Ketohehexoses (such as fructose) also occur as cyclic compounds with α and β anomeric forms. In these compounds, the hydroxyl group at C-5 (or C-6) reacts with the keto group at C-2 to form a **furanose** (or pyranose) ring containing a hemiketal linkage (Fig. 7-5). D-Fructose readily forms the furanose ring (Fig. 7-7); the more common anomer of this sugar in combined forms or in derivatives is $\beta\text{-D-fructofuranose}$.

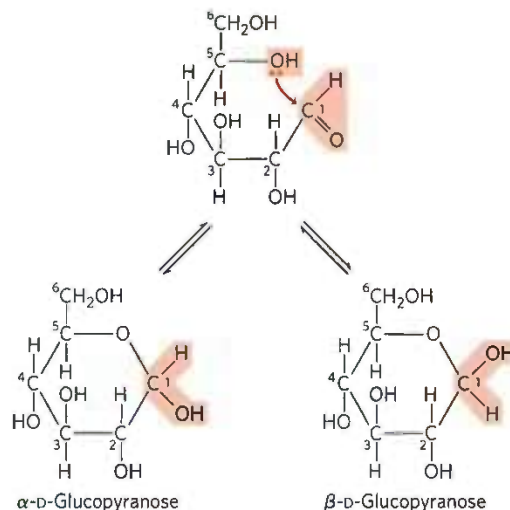
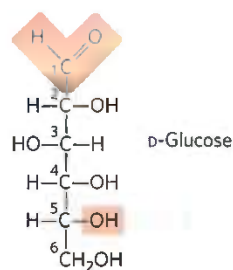


FIGURE 7-6 Formation of the two cyclic forms of D-glucose . Reaction between the aldehyde group at C-1 and the hydroxyl group at C-5 forms a hemiacetal linkage, producing either of two stereoisomers, the α and β anomers, which differ only in the stereochemistry around the hemiacetal carbon. This reaction is reversible. The interconversion of α and β anomers is called mutarotation.

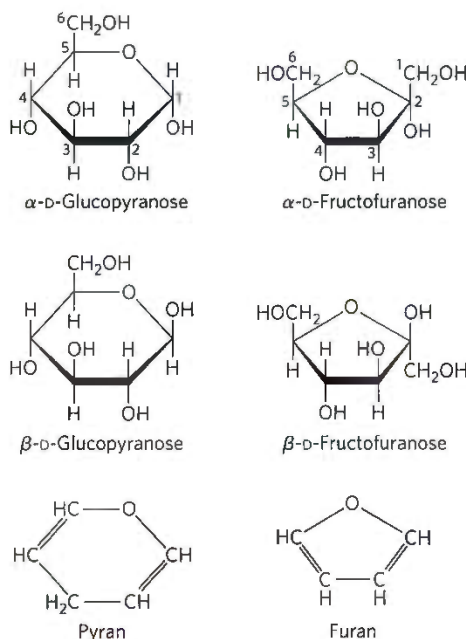
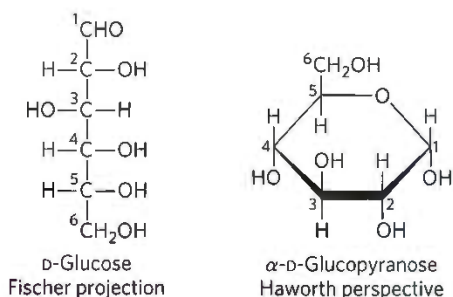


FIGURE 7-7 Pyranoses and furanoses. The pyranose forms of D-glucose and the furanose forms of D-fructose are shown here as Haworth perspective formulas. The edges of the ring nearest the reader are represented by bold lines. Hydroxyl groups below the plane of the ring in these Haworth perspectives would appear at the right side of a Fischer projection (compare with Fig. 7-6). Pyran and furan are shown for comparison.

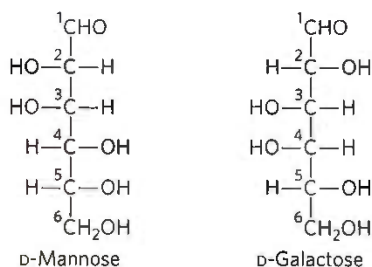
Cyclic sugar structures are more accurately represented in **Haworth perspective formulas** than in the Fischer projections commonly used for linear sugar structures. In Haworth perspectives, the six-membered ring is tilted to make its plane almost perpendicular to that of the paper, with the bonds closest to the reader drawn thicker than those farther away, as in Figure 7-7.

KEY CONVENTION To convert the Fischer projection formula of any linear D-hexose to a Haworth perspective formula showing the molecule's cyclic structure, draw the six-membered ring (five carbons, and one oxygen at the upper right), number the carbons in a clockwise direction beginning with the anomeric carbon, then place the hydroxyl groups. If a hydroxyl group is to the right in the Fischer projection, it is placed pointing down (i.e., below the plane of the ring) in the Haworth perspective; if it is to the left in the Fischer projection, it is placed pointing up (i.e., above the plane) in the Haworth perspective. The terminal $\text{—CH}_2\text{OH}$ group projects upward for the D enantiomer, downward for the L enantiomer. The hydroxyl on the anomeric carbon can point up or down. When the anomeric hydroxyl of a D-hexose is on the same side of the ring as C-6, the structure is by definition β ; when it is on the opposite side from C-6, the structure is α .



WORKED EXAMPLE 7-1 Conversion of Fischer Projection to Haworth Perspective Formulas

Draw the Haworth perspective formulas for D-mannose and D-galactose.



SOLUTION: Pyranoses are six-membered rings, so start with six-membered Haworth structures with the oxygen atom at the top right. Number the carbon atoms clockwise, starting with the aldose carbon. For mannose, place the hydroxyls on C-2, C-3, and C-4 above, above, and below the ring, respectively (because in the Fischer projection they are on the left, left, and right sides of the mannose backbone). For D-galactose, the hydroxyls are oriented

below, above, and above the ring for C-2, C-3, and C-4, respectively. The hydroxyl at C-1 can point either up or down; there are two possible configurations, α and β , at this carbon.

WORKED EXAMPLE 7-2 Drawing Haworth Perspective Formulas of Sugar Isomers

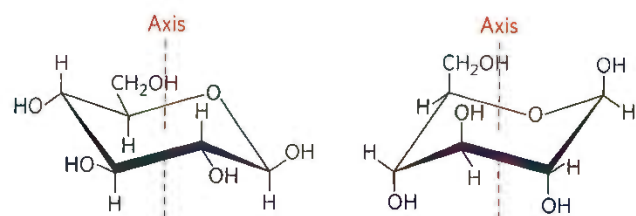
Draw the Haworth perspective formulas for α -D-mannose and β -L-galactose.

SOLUTION: The Haworth perspective formula of D-mannose from Worked Example 7-1 can have the hydroxyl group at C-1 pointing either up or down. According to the Key Convention, for the α form, the C-1 hydroxyl is pointing down when C-6 is up, as it is in D-mannose.

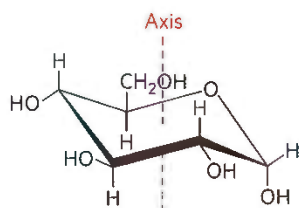
For β -L-galactose, use the Fischer representation of D-galactose (see Worked Example 7-1) to draw the correct Fischer representation of L-galactose, which is its mirror image: the hydroxyls at C-2, C-3, C-4, and C-5 are on the left, right, right, and left sides, respectively. Now draw the Haworth perspective, a six-membered ring in which the —OH groups on C-2, C-3, and C-4 are oriented up, down, and down, respectively, because in the Fischer representation they are on the left, right, and right sides. Because it is the β form, the —OH on the anomeric carbon points down (same side as C-5).

The α and β anomers of D-glucose interconvert in aqueous solution by a process called **mutarotation**, in which one ring form (say, the α anomer) opens briefly into the linear form, then closes again to produce the β anomer (Fig. 7-6). Thus, a solution of β -D-glucose and a solution of α -D-glucose eventually form identical equilibrium mixtures having identical optical properties. This mixture consists of about one-third α -D-glucose, two-thirds β -D-glucose, and very small amounts of the linear form and the five-membered ring (glucofuranose) form.

Haworth perspective formulas like those in Figure 7-7 are commonly used to show the stereochemistry of ring forms of monosaccharides. However, the six-membered pyranose ring is not planar, as Haworth perspectives suggest, but tends to assume either of two “chair” conformations (Fig. 7-8). Recall from Chapter 1 that **P5** two conformations of a molecule are interconvertible without the breakage of covalent bonds, whereas two configurations can be interconverted only by breaking a covalent bond. To interconvert α and β configurations, the bond involving the ring oxygen atom has to be broken, but interconversion of the two chair forms (which are conformers) does not require bond breakage and does not change configurations at any of the ring carbons. The specific three-dimensional structures of the monosaccharide units are important in determining the biological properties and functions of some polysaccharides, as we shall see.

Two possible chair forms of β -D-glucopyranose

(a)

 α -D-Glucopyranose

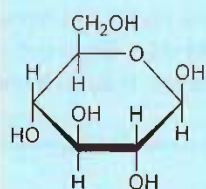
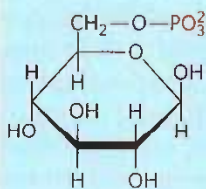
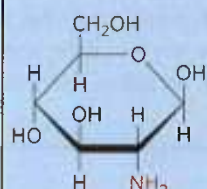
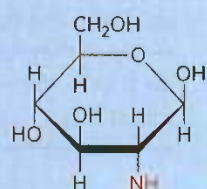
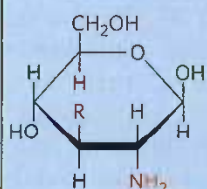
(b)

FIGURE 7-8 Conformational formulas of pyranoses. (a) Two chair forms of the pyranose ring of β -D-glucopyranose. Two conformers such as these are not readily interconvertible; an input of about 46 kJ of energy per mole of sugar is required to force the interconversion of chair forms. Another conformation, the "boat" (not shown), is seen only in derivatives with very bulky substituents. (b) The preferred chair conformation of α -D-glucopyranose.

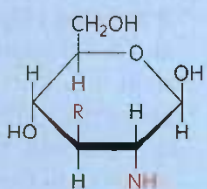
Organisms Contain a Variety of Hexose Derivatives

In addition to simple hexoses such as glucose, galactose, and mannose, there are many sugar derivatives in which a hydroxyl group in the parent compound is replaced with another substituent, or a carbon atom is oxidized to a carboxyl group (Fig. 7-9). In glucosamine, galactosamine, and mannosamine, the hydroxyl at C-2 of the parent compound is replaced with an amino group.

Glucose family

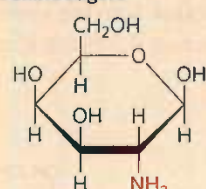
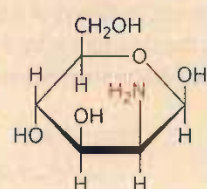
 β -D-Glucose β -D-Glucose 6-phosphate β -D-GlucosamineN-Acetyl- β -D-glucosamine

Muramic acid

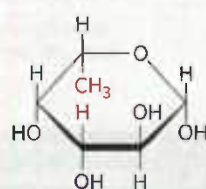
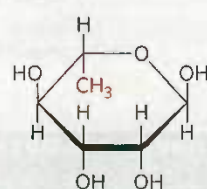


N-Acetylmuramic acid

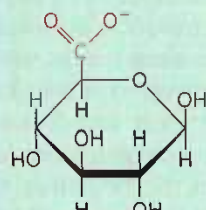
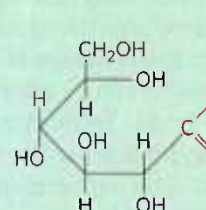
Amino sugars

 β -D-Galactosamine β -D-Mannosamine

Deoxy sugars

 β -L-Fucose α -L-Rhamnose

Acidic sugars

 β -D-Glucuronate

D-Gluconate

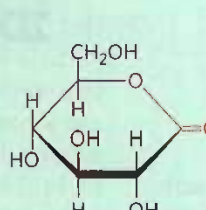
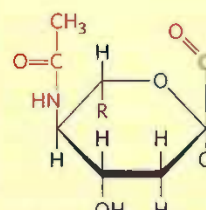
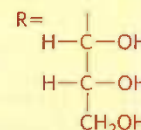
D-Glucono- δ -lactoneN-Acetylneuraminic acid
(a sialic acid)

FIGURE 7-9 Some hexose derivatives important in biology.

In amino sugars, an —NH_2 group replaces one of the —OH groups in the parent hexose. Substitution of —H for —OH produces a deoxy sugar; note that the deoxy sugars shown here occur in nature as the L isomers. The

acidic sugars contain a carboxylate group, which confers a negative charge at neutral pH. D-Glucono- δ -lactone results from formation of an ester linkage between the C-1 carboxylate group and the C-5 (also known as the δ carbon) hydroxyl group of D-gluconate.

The amino group is commonly condensed with acetic acid, as in *N*-acetylglucosamine. This glucosamine derivative is part of many structural polymers, including those of the bacterial cell wall. Substitution of a hydrogen for the hydroxyl group at C-6 of L-galactose or L-mannose produces L-fucose or L-rhamnose, respectively. L-Fucose is found in the complex oligosaccharide components of glycoproteins and glycolipids; L-rhamnose is found in plant polysaccharides.

Oxidation of the carbonyl (aldehyde) carbon of glucose to the carboxyl level produces gluconic acid, used in medicine as an innocuous counterion when administering positively charged drugs (such as quinine) or ions (such as Ca^{2+}). Other aldoses yield other **aldonic acids**. Oxidation of the carbon at the other end of the carbon chain—C-6 of glucose, galactose, or mannose—forms the corresponding **uronic acid**: glucuronic, galacturonic, or mannuronic acid. Both aldonic acids and uronic acids form stable intramolecular esters called lactones (Fig. 7-9, lower left). The sialic acids are a family of sugars with the same nine-carbon backbone. One of them, *N*-acetylneuraminic acid (often referred to simply as “sialic acid”), is a derivative of *N*-acetylmannosamine that occurs in many glycoproteins and glycolipids on animal cell surfaces, providing sites of recognition by other cells or extracellular carbohydrate-binding proteins. The carboxylic acid groups of the acidic sugar derivatives are ionized at pH 7, and the compounds are therefore correctly named as the carboxylates—glucuronate, galacturonate, and so forth.

In the synthesis and metabolism of carbohydrates, the intermediates are very often not the sugars themselves but their phosphorylated derivatives. Condensation of phosphoric acid with one of the hydroxyl groups of a sugar forms a phosphate ester, as in glucose 6-phosphate (Fig. 7-9), the first metabolite in the pathway by which most organisms oxidize glucose for energy. Sugar phosphates are relatively stable at neutral pH and bear a negative charge. One effect of sugar phosphorylation within cells is to trap the sugar inside the cell; most cells do not have plasma membrane transporters for phosphorylated sugars. Phosphorylation also activates sugars for subsequent chemical transformation. Several important phosphorylated derivatives of sugars are components of nucleotides (discussed in the next chapter).

Sugars That Are, or Can Form, Aldehydes Are Reducing Sugars

Free aldehyde groups in sugars undergo a characteristic redox reaction with Cu^{2+} under alkaline conditions. As the sugar is oxidized from aldehyde to carboxylic acid, Cu^{2+} is reduced to Cu^+ , which forms a brick-red precipitate. This reaction defines **reducing sugars**, which include, for example, glucose, galactose, mannose, ribose, and glyceraldehyde. The reaction occurs only with a *free* aldose; but because cyclic aldoses are in equilibrium with

their linear forms, which do have free aldehyde groups (Fig. 7-6), all aldose monosaccharides are reducing sugars. Ketoses that can rearrange (tautomerize) to form aldehydes are also reducing sugars; fructose and ribulose are reducing sugars, for example.

The metabolism of glucose in people with diabetes can be monitored by measuring urinary glucose with a simple qualitative assay for reducing sugar, or by measuring quantitatively the nonenzymatic reaction between glucose and the hemoglobin in blood (**Box 7-2**).

Disaccharides (such as maltose, lactose, and sucrose) consist of two monosaccharides joined covalently by an **O-glycosidic bond**, which is formed when a hydroxyl group of one sugar molecule, typically in its cyclic form, reacts with the anomeric carbon of the other (Fig. 7-10). This reaction represents the formation of an acetal from a hemiacetal (such as glucopyranose) and an alcohol (a hydroxyl group of the second sugar molecule) (Fig. 7-5), and the resulting compound is called a glycoside. Glycosidic bonds are readily hydrolyzed by acid but resist cleavage by base. Thus disaccharides can be hydrolyzed to yield their free monosaccharide components by boiling with dilute acid.

When the anomeric carbon is involved in a glycosidic bond, the easy interconversion of linear and cyclic forms shown in Figure 7-6 is prevented. Formation of a glycosidic bond therefore renders a sugar nonreducing. In describing disaccharides or polysaccharides, the end of a chain with a free anomeric carbon (one not involved in a glycosidic bond) is called the **reducing end**.

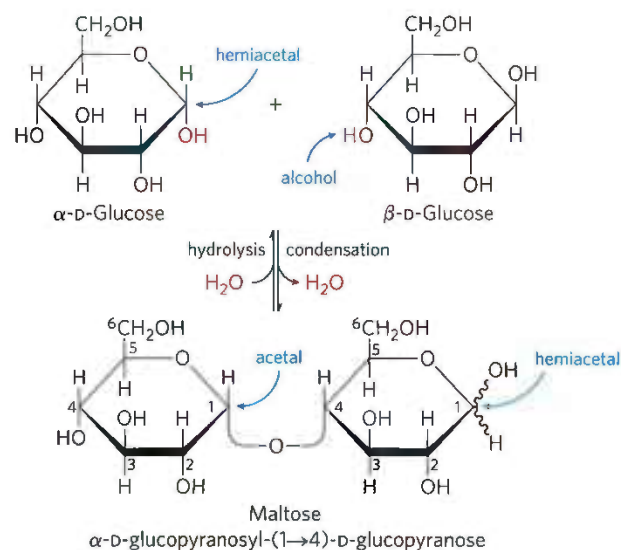


FIGURE 7-10 Formation of maltose. A disaccharide is formed from two monosaccharides (here, two molecules of D-glucose) when an —OH (alcohol) of one monosaccharide molecule (right) condenses with the intramolecular hemiacetal of the other (left), with elimination of H_2O and formation of a glycosidic bond. The reversal of this reaction is hydrolysis—attack by H_2O on the glycosidic bond. The maltose molecule, shown here, retains a reducing hemiacetal at the C-1 not involved in the glycosidic bond. Because mutarotation interconverts the α and β forms of the hemiacetal, the bonds at this position are sometimes depicted with wavy lines to indicate that the structure may be either α or β .



Blood Glucose Measurements in the Diagnosis and Treatment of Diabetes

Glucose is the principal fuel for the brain. When the amount of glucose reaching the brain is too low, the consequences can be dire: lethargy, coma, permanent brain damage, and death. Complex hormonal mechanisms have evolved to ensure that the concentration of glucose in the blood remains high enough (about 5 mM) to satisfy the brain's needs—but not too high, because elevated blood glucose can also have serious physiological consequences.

Individuals with insulin-dependent diabetes mellitus do not produce sufficient insulin, the hormone that normally serves to reduce blood glucose concentration. If the diabetes is untreated, blood glucose levels may rise to several-fold higher than normal. These high glucose levels are believed to be at least one cause of the serious long-term consequences of untreated diabetes—kidney failure, cardiovascular disease, blindness, and impaired wound healing—so one goal of therapy is to provide just enough insulin (by injection) to keep blood glucose levels near normal. To maintain the correct balance of exercise, diet, and insulin, individuals with diabetes must measure their blood glucose concentration several times a day, and adjust the amount of insulin injected appropriately.

The concentrations of glucose in blood can be determined by a simple assay for reducing sugar. A single drop of blood is added to a test strip containing the enzyme glucose oxidase, which catalyzes the reaction:



A second enzyme, a peroxidase, catalyzes reaction of the H_2O_2 with a colorless compound to create a colored product, which is quantified with a simple photometer that reads out the blood glucose concentration.

Because blood glucose levels change with the timing of meals and exercise, single-time measurements do not reflect the *average* blood glucose over hours and days, so dangerous increases may go undetected. The average glucose concentration can be assessed by looking at its effect on hemoglobin, the oxygen-carrying protein in erythrocytes (p. 153). Transporters in the erythrocyte membrane equilibrate intracellular and plasma glucose concentrations, so hemoglobin is constantly exposed to glucose at whatever concentration is present in the blood. A series of relatively slow *nonenzymatic* reactions (Fig. 1) occurs between glucose and primary amino groups in hemoglobin (either the amino-terminal Val or the ϵ -amino groups of Lys residues) (Fig. 2). The rate of this process is proportional to the concentration of glucose, so the reaction can be used to estimate the average blood glucose

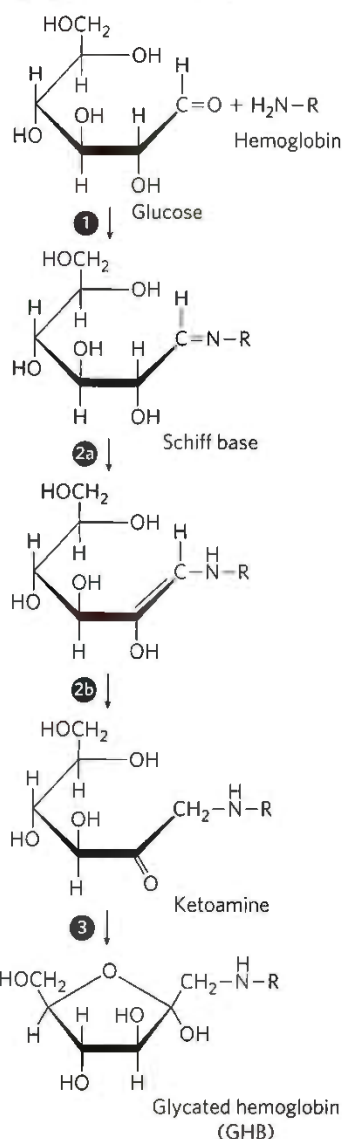


FIGURE 1 The nonenzymatic reaction of glucose with a primary amino group in hemoglobin begins with **1** formation of a Schiff base, which **2** undergoes a rearrangement to generate a stable product; **3** this ketoamine can further cyclize to yield GHB.

level over weeks. The amount of glycated hemoglobin (GHB) present at any time reflects the average blood glucose concentration over the circulating lifetime of the erythrocyte (about 120 days), although the concentration in the two weeks before the test is the most important in setting the level of GHB.

The disaccharide maltose (Fig. 7-10) contains two D-glucose residues joined by a glycosidic linkage between C-1 (the anomeric carbon) of one glucose residue and C-4 of the other. Because the disaccharide retains a free anomeric carbon (C-1 of the glucose

residue on the right in Fig. 7-10), maltose is a reducing sugar. The configuration of the anomeric carbon atom in the glycosidic linkage is α . The glucose residue with the free anomeric carbon is capable of existing in α - and β -pyranose forms.

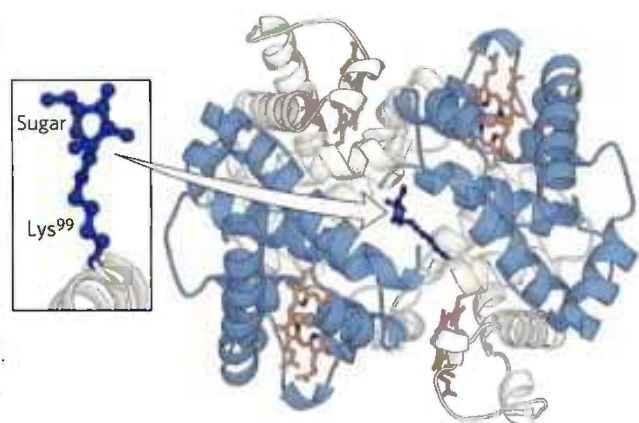


FIGURE 2 Glycated hemoglobin, showing the hemes (red) and the glycated Lys⁹⁹ residue (blue). [Data from PDB ID 3B75, N. T. Saraswathi et al., 2008.]

The extent of **hemoglobin glycation** (so named to distinguish it from glycosylation, the *enzymatic* transfer of glucose to a protein) is measured clinically by extracting hemoglobin from a small sample of blood and separating GHB from unmodified hemoglobin electrophoretically (Fig. 3), taking advantage of the charge difference resulting from modification of the amino group(s). Normal values of the monoglycated hemoglobin referred to as HbA1c are about 5% of total hemoglobin (corresponding to an average blood glucose level of 120 mg/100 mL). In people with untreated diabetes, however, this value may be as high as 13%, indicating an average blood glucose level of about 300 mg/100 mL — dangerously high. One criterion for success in an individual program of insulin therapy (the timing, frequency, and amount of insulin injected) is maintaining HbA1c values at about 7%.

Glycated hemoglobin undergoes a series of rearrangements, oxidations, and dehydrations of the carbohydrate moiety to produce a heterogeneous mixture of **advanced glycation end products (AGE)**, such as ϵ -N-carboxymethyllysine and methylglyoxal. AGE are believed to be responsible for at least some of the pathology of diabetes. AGE can leave the erythrocyte and form covalent cross-links between proteins, interfering with normal protein function (Fig. 4). The accumulation of relatively high concentrations of AGE in people with diabetes may, by cross-linking critical proteins, cause the damage to the kidneys, retinas, and cardiovascular system that characterizes the disease. Some AGE also can act

Analyte ID	Percent	Time (min)	Area
Injection	0.0	0.11	17,682
A1a	0.4	0.30	8,051
A1b	1.0	0.45	19,267
A1c	5.9	1.10	110,946
A0	92.6	1.56	1,727,669

Total area 1,883,615

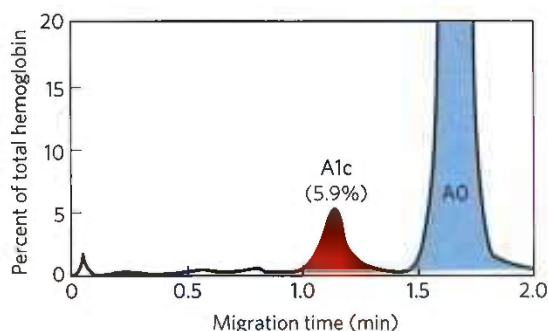


FIGURE 3 Pattern of hemoglobin (detected by its absorption at 415 nm) after electrophoretic separation of nonglycated (A0) and monoglycated (A1c) forms in a thin glass capillary. Integration of the area under the peaks allows calculation of the amount of GHB (HbA1c) as a percentage of total hemoglobin. Shown here is the profile of an individual with a normal level of HbA1c (5.9%).

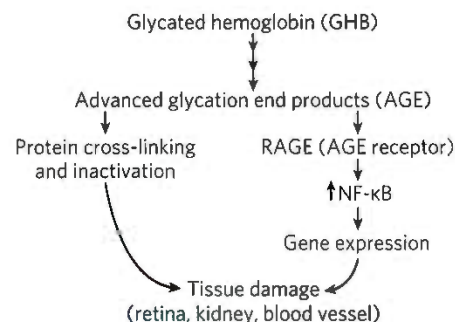


FIGURE 4 Pathways from glycated hemoglobin to the tissue damage associated with diabetes.

through membrane receptors for AGE (RAGE), triggering intracellular responses that include activation of a transcription factor (NF- κ B) and consequent changes in gene expression. This pathogenic process is a potential target for drug action.

KEY CONVENTION To name reducing disaccharides such as maltose unambiguously, and especially to name more complex oligosaccharides, several rules are followed. By convention, the name describes the compound written with its nonreducing end to the left, and we can “build up” the name in the following

order. (1) Give the configuration (α or β) at the anomeric carbon joining the first monosaccharide unit (on the left) to the second. (2) Name the nonreducing residue; to distinguish five- and six-membered ring structures, insert “furano” or “pyrano” into the name. (3) Indicate in parentheses the two carbon atoms joined by

TABLE 7-1 Symbols and Abbreviations for Common Monosaccharides and Some of Their Derivatives

Abequose	Abe	Glucuronic acid	◆ GlcA
Arabinose	Ara	Galactosamine	◻ GalN
Fructose	Fru	Glucosamine	◼ GlcN
Fucose	▲ Fuc	<i>N</i> -Acetylgalactosamine	◻ GalNAc
Galactose	◯ Gal	<i>N</i> -Acetylglucosamine	◼ GlcNAc
Glucose	● Glc	Iduronic acid	◊ IdoA
Mannose	● Man	Muramic acid	Mur
Rhamnose	Rha	<i>N</i> -Acetylmuramic acid	Mur2Ac
Ribose	Rib	<i>N</i> -Acetylneuraminic acid (a sialic acid)	◆ Neu5Ac
Xylose	★ Xyl		

Note: In a commonly used convention, hexoses are represented as circles, *N*-acetylhexosamines as squares, and hexosamines as squares divided diagonally. All sugars with the "gluco" configuration are blue, those with the "galacto" configuration are yellow, and "manno" sugars are green. Other substituents can be added as needed: sulfate (S), phosphate (P), *O*-acetyl (OAc), or *O*-methyl (OMe).

the glycosidic bond, with an arrow connecting the two numbers; for example, (1→4) shows that C-1 of the first-named sugar residue is joined to C-4 of the second. (4) Name the second residue. If there is a third residue, describe the second glycosidic bond by the same conventions. (To shorten the description of complex polysaccharides, three-letter abbreviations or colored symbols for the monosaccharides are often used, as given in Table 7-1.) Following this convention for naming oligosaccharides, maltose is α -D-glucopyranosyl-(1→4)-D-glucopyranose. Because most sugars encountered in this book are the D enantiomers and the pyranose form of hexoses predominates, we generally use a shortened version of the formal name of such compounds, giving the configuration of the anomeric carbon and naming the carbons joined by the glycosidic bond. In this abbreviated nomenclature, maltose is Glc(α 1→4)Glc. ■

The disaccharide lactose (Fig. 7-11), which yields D-galactose and D-glucose on hydrolysis, occurs naturally in milk and gives milk its sweetness. The anomeric carbon of the glucose residue is available for oxidation, and thus lactose is a reducing disaccharide. The reducing end of this disaccharide, which by convention is drawn on the right, is glucose, and the disaccharide is named as a derivative of glucose. Its abbreviated name is Gal(β 1→4)Glc, with the linkage shown in parentheses; the anomeric carbon of galactose is β , and C-1 of Gal is linked to C-4 of Glc.

The enzyme lactase—absent in lactose-intolerant individuals—begins the digestive process in the small intestine by splitting the (β 1→4) bond of lactose into monosaccharides, which can be absorbed from the small intestine. Lactose, like other disaccharides, is not absorbed from the small intestine, and in lactose-intolerant individuals the undigested lactose passes into the large intestine. Here, the increased osmolarity due

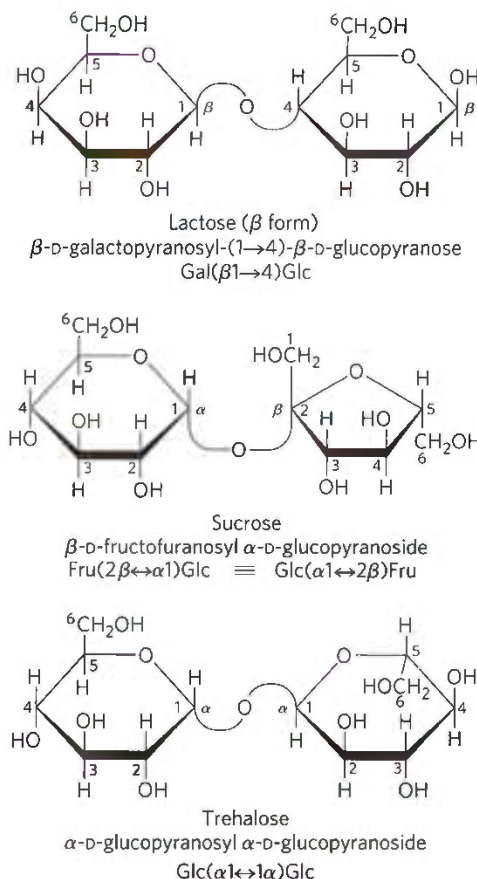


FIGURE 7-11 Three common disaccharides. Like maltose in Figure 7-10, these disaccharides are shown as Haworth perspectives. The common name, full systematic name, and abbreviation are given. Formal nomenclature for sucrose names glucose as the parent glycoside, although it is typically depicted as shown, with glucose on the left. The two abbreviated symbols shown for sucrose are equivalent ($\equiv$).

to dissolved lactose opposes the absorption of water from the intestine into the bloodstream, causing watery, loose stools. In addition, fermentation of the lactose by intestinal bacteria produces large volumes of CO₂, which leads to the bloating, cramps, and gas associated with lactose intolerance.

Sucrose (table sugar) is a disaccharide of glucose and fructose. It is formed by plants but not by animals. In contrast to maltose and lactose, sucrose contains no free anomeric carbon atom; the anomeric carbons of both monosaccharide units are involved in the glycosidic bond (Fig. 7-11). Sucrose is therefore a nonreducing sugar, and its stability—its resistance to oxidation—makes it a suitable molecule for the storage and transport of energy in plants. In the abbreviated nomenclature, a double-headed arrow connects the symbols specifying the anomeric carbons and their configurations. Thus the abbreviated name of sucrose is either Glc(α 1↔2 β)Fru or Fru(2 β ↔ α 1)Glc. Sucrose is a major intermediate product of photosynthesis; in many plants it is the principal form in which sugar is transported from the leaves to other parts of the plant body.

Trehalose, $\text{Glc}(\alpha 1 \leftrightarrow 1 \alpha) \text{Glc}$ (Fig. 7-11)—a disaccharide of D-glucose that, like sucrose, is a nonreducing sugar—is a major constituent of the circulating fluid (hemolymph) of insects. It serves as an energy-storage compound, and it serves as antifreeze in some invertebrate organisms.

SUMMARY 7.1 Monosaccharides and Disaccharides

- Sugars (saccharides) contain an aldehyde or ketone group and two or more hydroxyl groups.
- Monosaccharides generally contain several chiral carbons and therefore exist in a variety of stereochemical forms, which may be represented on paper as Fischer projections. Epimers are sugars that differ in configuration at only one carbon atom.
- Monosaccharides with at least six carbons commonly form cyclic structures, in which the aldehyde or ketone group is joined to a hydroxyl group of the same molecule. The cyclic structure can be represented as a Haworth perspective formula. The carbon atom originally found in the aldehyde or ketone group (the anomeric carbon) can assume either of two configurations, α and β , which are interconvertible by mutarotation. In the linear form of the monosaccharide, which is in equilibrium with the cyclic forms, the anomeric carbon is easily oxidized, making the compound a reducing sugar.
- Naturally occurring hexoses include some that have $-\text{NH}_2$ at C-2 (amino sugars), often with the amino group acetylated. Oxidation of the carbonyl carbon of glucose and other aldoses yields aldonic acids (gluconic acid); oxidation at C-6 produces uronic acids (glucuronate). Some sugar intermediates are phosphate esters (for example, glucose 6-phosphate).
- A hydroxyl group of one monosaccharide can add to the anomeric carbon of a second monosaccharide to form an acetal called a glycoside. Oligosaccharides are short polymers of several different monosaccharides joined by glycosidic bonds. At one end of the chain, the reducing end, is a monosaccharide residue with its anomeric carbon not involved in a glycosidic bond, and therefore available to be oxidized.
- Disaccharides and oligosaccharides are named as derivatives of the sugar at the reducing end. Their names provide the order of monosaccharide units, the configuration at each anomeric carbon, and the carbon atoms involved in the glycosidic linkage(s).

7.2 Polysaccharides

Most carbohydrates found in nature occur as polysaccharides, polymers of medium to high molecular weight ($M_r > 20,000$). **P2** Polysaccharides, also called **glycans**, differ from each other in the identity of their recurring monosaccharide units, in the length of their chains, in

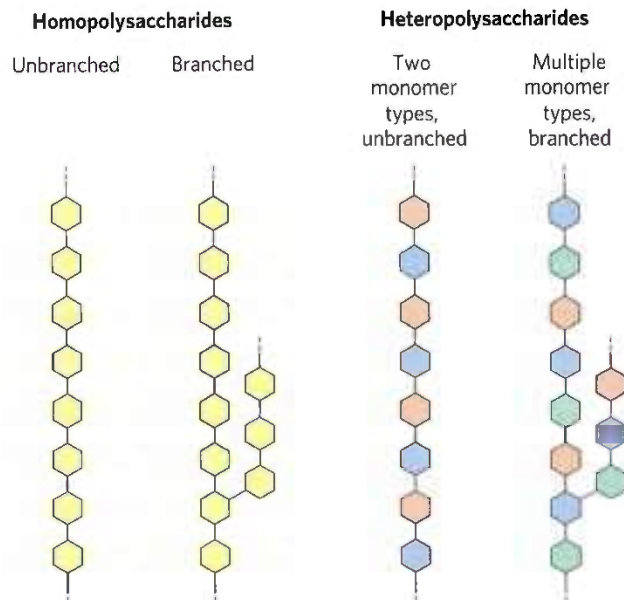


FIGURE 7-12 Homopolysaccharides and heteropolysaccharides. Polysaccharides may be composed of one, two, or several different monosaccharides, in straight or branched chains of varying length.

the types of bonds linking the units, and in the degree of branching. **Homopolysaccharides** contain only a single monomeric sugar species; **heteropolysaccharides** contain two or more kinds of monomers (Fig. 7-12). Some homopolysaccharides serve as storage forms of monosaccharides that are used as fuels; starch and glycogen are homopolysaccharides of this type. Other homopolysaccharides (cellulose and chitin, for example) serve as structural elements in plant cell walls and animal exoskeletons. Heteropolysaccharides provide extracellular support for organisms of all kingdoms. For example, the rigid layer of the bacterial cell envelope (the peptidoglycan) is composed in part of a heteropolysaccharide built from two alternating monosaccharide units (see Fig. 6-32). In animal tissues, the extracellular space is occupied by several types of heteropolysaccharides, which form a matrix that holds individual cells together and provides protection, shape, and support to cells, tissues, and organs.

P4 Unlike proteins, polysaccharides generally do not have defined lengths or molecular weights. This difference is a consequence of the mechanisms of assembly of the two types of polymer. As we shall see in Chapter 27, proteins are synthesized on a template (messenger RNA) of defined sequence and length, by enzymes that follow the template exactly. For polysaccharide synthesis there is no template; rather, the program for polysaccharide synthesis is intrinsic to the enzymes that catalyze the polymerization of the monomeric units, and there is no specific stopping point in the synthetic process; the products thus vary in length.

Some Homopolysaccharides Are Storage Forms of Fuel

The most important storage polysaccharides are starch in plant cells and glycogen in animal cells. Both

polysaccharides occur intracellularly as large clusters or granules. Starch and glycogen molecules are heavily hydrated, because they have many exposed hydroxyl groups available to hydrogen-bond with water. Most plant cells have the ability to form starch, and starch storage is especially abundant in tubers (underground stems), such as potatoes, and in seeds.

Starch contains two types of glucose polymer, amylose and amylopectin (Fig. 7-13). Amylose consists of long, unbranched chains of D-glucose residues connected by ($\alpha 1 \rightarrow 4$) linkages (as in maltose). Such chains vary in molecular weight from a few thousand to more than a million. Amylopectin is even larger (M_r up to 200 million) but unlike amylose is highly branched. The glycosidic linkages joining successive glucose residues in amylopectin chains are ($\alpha 1 \rightarrow 4$); the branch points (occurring every 24 to 30 residues) are ($\alpha 1 \rightarrow 6$) linkages.

Glycogen is the main storage polysaccharide of animal cells. Like amylopectin, glycogen is a polymer of ($\alpha 1 \rightarrow 4$)-linked glucose subunits, with ($\alpha 1 \rightarrow 6$)-linked branches, but glycogen is more extensively branched (on average, a branch every 8 to 12 residues) and more compact than starch. Glycogen is especially abundant in the liver, where it may constitute as much as 7% of the wet

weight; it is also present in skeletal muscle. In hepatocytes, glycogen is found in large α granules (see Fig. 15-1), which are clusters of smaller β granules, each of which is a single, highly branched glycogen molecule with an average molecular weight of several million. The large α glycogen granules also contain, in tightly bound form, the enzymes responsible for the synthesis and degradation of glycogen (see Fig. 15-16).

Because each branch of glycogen ends with a nonreducing sugar unit, a glycogen molecule with n branches has $n + 1$ nonreducing ends, but only one reducing end. When glycogen is used as an energy source, glucose units are removed one at a time from the nonreducing ends. Degradative enzymes that act only at nonreducing ends can work simultaneously on the many branches, speeding the conversion of the polymer to monosaccharides.

Why not store glucose in its monomeric form?

P3 Hepatocytes in the fed state store glycogen equivalent to a glucose concentration of 0.4 M. The actual concentration of glycogen, which contributes little to the osmolarity of the cytosol, is about 0.01 μM . If the cytosol contained 0.4 M glucose, the osmolarity would be threateningly elevated, leading to osmotic entry of water that might rupture the cell (see Fig. 2-12).

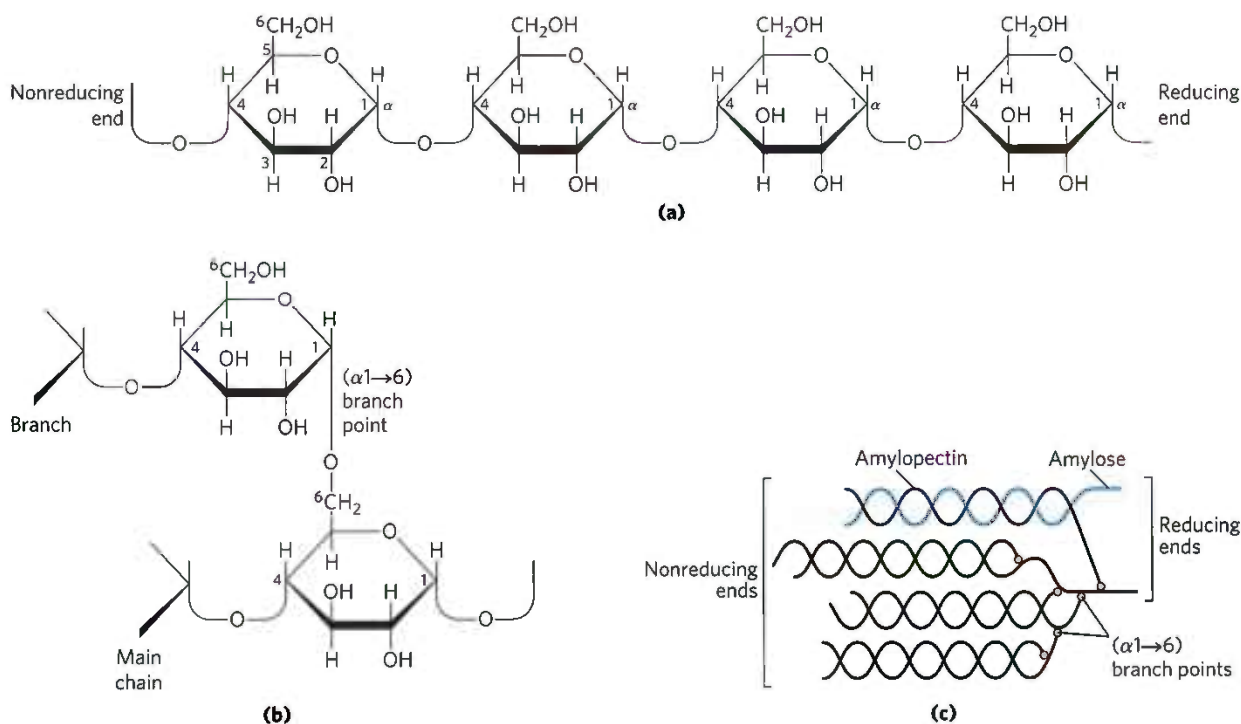


FIGURE 7-13 Starch and glycogen. (a) A short segment of a linear polymer of D-glucose residues in ($\alpha 1 \rightarrow 4$) linkage, shown here as Haworth perspectives. This is the basic structure of amylose of starch (in plants) and glycogen (in animals). A single chain of amylose can contain several thousand glucose residues. (b) An ($\alpha 1 \rightarrow 6$) branch point of amylopectin or glycogen. (c) A cluster of amylose and amylopectin like that believed to occur in starch

granules. Strands of amylopectin (black) form double-helical structures with each other or with amylose strands (blue). Amylopectin has frequent ($\alpha 1 \rightarrow 6$) branch points (red). Glucose residues at the nonreducing ends of the outer branches are removed enzymatically during the mobilization of starch for energy production. Glycogen has a similar structure but is more highly branched and more compact.

Some Homopolysaccharides Serve Structural Roles

Cellulose, a tough, fibrous, water-insoluble substance, is found in the cell walls of plants, particularly in stalks, stems, trunks, and all the woody portions of the plant body. Cellulose constitutes much of the mass of wood, and cotton is almost pure cellulose. Like amylose, the cellulose molecule is a linear, unbranched homopolysaccharide, consisting of 10,000 to 15,000 D-glucose units. But there is a very important difference: **P5** in cellulose the glucose residues have the β configuration (Fig. 7-14), whereas in amylose the glucose is in the α configuration. The glucose residues in cellulose are linked by (β 1 $\rightarrow$ 4) glycosidic bonds, in contrast to the (α 1 $\rightarrow$ 4) bonds of amylose. This difference causes individual molecules of cellulose and amylose to fold differently in space, giving them very different macroscopic structures and physical properties.

The tough, fibrous nature of cellulose makes it useful in such commercial products as cardboard and insulation material, and it is a major constituent of cotton and linen fabrics. Cellulose is also the starting material for the commercial production of cellophane, rayon, and lyocell.

Chitin is a linear homopolysaccharide composed of N-acetylglucosamine residues in (β 1 $\rightarrow$ 4) linkage (Fig. 7-15). The only chemical difference from cellulose is the replacement of the hydroxyl group at C-2 with an acetylated amino group, which makes chitin more hydrophobic and water-resistant than cellulose. Chitin forms

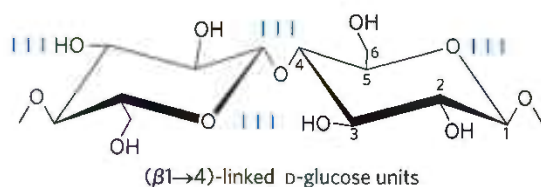


FIGURE 7-14 Cellulose. Two units of a cellulose chain; the D-glucose residues are in (β 1 $\rightarrow$ 4) linkage. The rigid chair structures can rotate relative to one another.

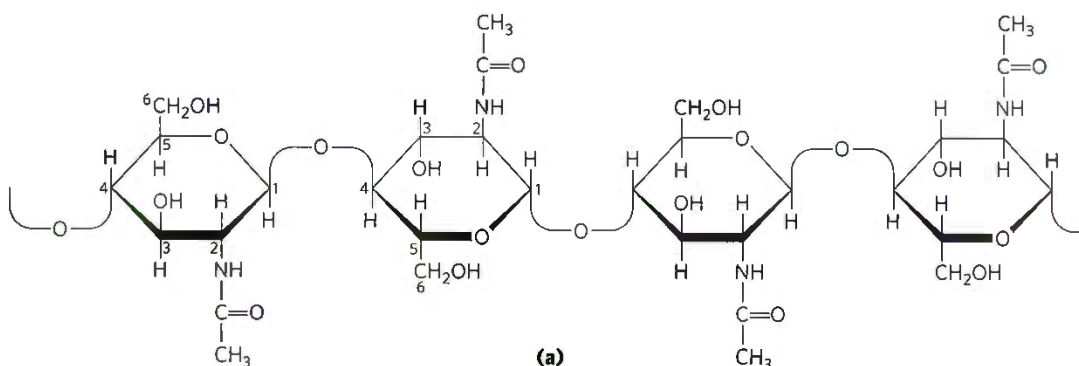


FIGURE 7-15 Chitin. (a) A short segment of chitin, a homopolymer of N-acetyl-D-glucosamine units in (β 1 $\rightarrow$ 4) linkage. (b) A spotted June bee-

extended fibers similar to those of cellulose, and like cellulose cannot be digested by vertebrates. Chitin is the principal component of the hard exoskeletons of nearly a million species of arthropods—insects, lobsters, and crabs, for example—and is probably the second most abundant polysaccharide, next to cellulose, in nature; an estimated 1 billion tons of chitin are produced in the biosphere each year.

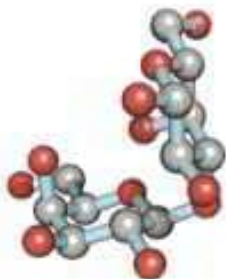
Steric Factors and Hydrogen Bonding Influence Homopolysaccharide Folding

P5 The folding of polysaccharides in three dimensions follows the same principles as those governing polypeptide structure: subunits with a more-or-less rigid structure dictated by covalent bonds form three-dimensional macromolecular structures that are stabilized by weak interactions within or between molecules, such as hydrogen bonds, interactions due to the hydrophobic effect, van der Waals interactions, and, for polymers with charged subunits, electrostatic interactions. Because polysaccharides have so many hydroxyl groups, hydrogen bonding has an especially important influence on their structure. Glycogen, starch, cellulose, and chitin are composed of pyranoside (six-membered ring) subunits, as are the oligosaccharides of glycoproteins and glycolipids, to be discussed later. Such molecules can be represented as a series of rigid pyranose rings connected by an oxygen atom bridging two carbon atoms (the glycosidic bond). There is, in principle, free rotation about both C—O bonds linking the residues, but as in polypeptides (see Figs 4-2, 4-8), rotation about each bond is limited by steric hindrance by substituents. The three-dimensional structures of these molecules can be described in terms of the dihedral angles, ϕ and ψ , about the glycosidic bond (Fig. 7-16). The bulkiness of the pyranose ring and its substituents, along with electronic effects at the anomeric carbon, place constraints on the angles ϕ and ψ ; thus, certain conformations are much more stable than others.

tle (*Pelidnota punctata*), showing its surface armor (exoskeleton) of chitin. [(b) Paul Whitten/Science Source.]



Low-energy conformation is extended and maximizes H-bonding.



High-energy conformation is sterically hindered.

FIGURE 7-16 Different energetic conformations of a disaccharide.

The torsion angles ϕ (phi) and ψ (psi), which define the spatial relationship between adjacent rings, can in principle have any value from 0° to 360° . In fact, some torsion angles give conformations that maximize hydrogen bonding and minimize energy, whereas others give conformations that are sterically hindered, as we can see in two conformers of the disaccharide Gal($\beta 1 \rightarrow 3$)Gal at opposite energetic extremes.

P5 The most stable three-dimensional structure for the ($\alpha 1 \rightarrow 4$)-linked chains of starch and glycogen is a tightly coiled helix (Fig. 7-17), stabilized by interchain hydrogen bonds. The average plane of each residue along the amylose chain forms a 60° angle with the average plane of the preceding residue, so the helical structure has six residues per turn. For amylose, the core of the helix is of precisely the right dimensions to accommodate iodine as complex ions (I_3^- and I_5^-). This interaction gives an intensely blue product, making it a common qualitative test for amylose.

P5 For cellulose, the most stable conformation is that in which each chair is turned 180° relative to its neighbors, yielding a straight, extended chain. All —OH groups are available for hydrogen bonding with neighboring chains. With several chains lying side by side, a stabilizing network of interchain and intrachain hydrogen bonds produces straight, stable supramolecular fibers of great tensile strength (Fig. 7-18). This property of cellulose has made it useful to civilizations for millennia. Many manufactured products, including papyrus, paper, cardboard, rayon, insulating tiles, and a variety of other useful materials, are derived from cellulose. The water content of these materials is low because extensive interchain hydrogen bonding between cellulose molecules satisfies their capacity for hydrogen-bond formation.

Peptidoglycan Reinforces the Bacterial Cell Wall

The rigid component of bacterial cell walls (peptidoglycan) is a heteropolymer of alternating ($\beta 1 \rightarrow 4$)-linked

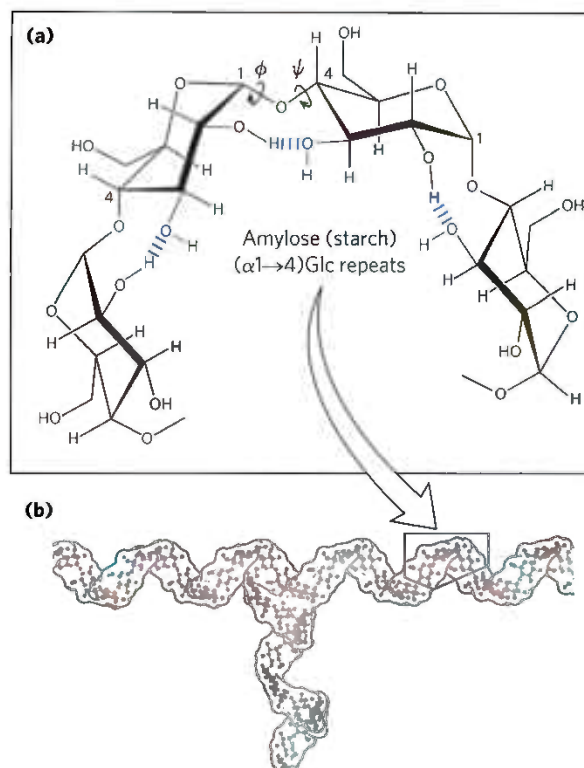


FIGURE 7-17 Helical structure of starch (amylose). (a) Four glucose units of starch (amylose), showing both the free rotation possible around the ψ and ϕ angles of the ($\alpha 1 \rightarrow 4$) glycosidic bonds and the hydrogen bonding possible between adjacent rings. In this most stable conformation, with adjacent rigid chairs at 60° to one another, the polysaccharide chain is curved. (b) A model of a segment of amylopectin (branched amylose). The conformation of ($\alpha 1 \rightarrow 4$) linkages in amylose, amylopectin, and glycogen causes these polymers to assume tightly coiled helical structures. [(b) Data from www.biotopics.co.uk/jsmol/amylopectin.html]

N-acetylglucosamine and *N*-acetylmuramic acid residues (see Fig. 6-32). The linear polymers lie side by side in the cell wall, cross-linked by short peptides, the exact structure of which depends on the bacterial species. The peptide cross-links weld the polysaccharide chains into a strong sheath (peptidoglycan) that envelops the entire cell and prevents cellular swelling and lysis due to the osmotic entry of water.

The enzyme lysozyme kills bacteria by hydrolyzing the ($\beta 1 \rightarrow 4$) glycosidic bond between *N*-acetylglucosamine and *N*-acetylmuramic acid. The enzyme is found in human tears, where it is presumably a defense against bacterial infections of the eye, and in hen's eggs, where it prevents bacterial infection of the developing embryo. Penicillin and related antibiotics kill bacteria by preventing synthesis of the peptidoglycan cross-links, leaving the cell wall too weak to resist osmotic lysis (p. 211).

Glycosaminoglycans Are Heteropolysaccharides of the Extracellular Matrix

The extracellular space in the tissues of multicellular animals is filled with a gel-like material, the **extracellular matrix (ECM)**, which holds the cells together and

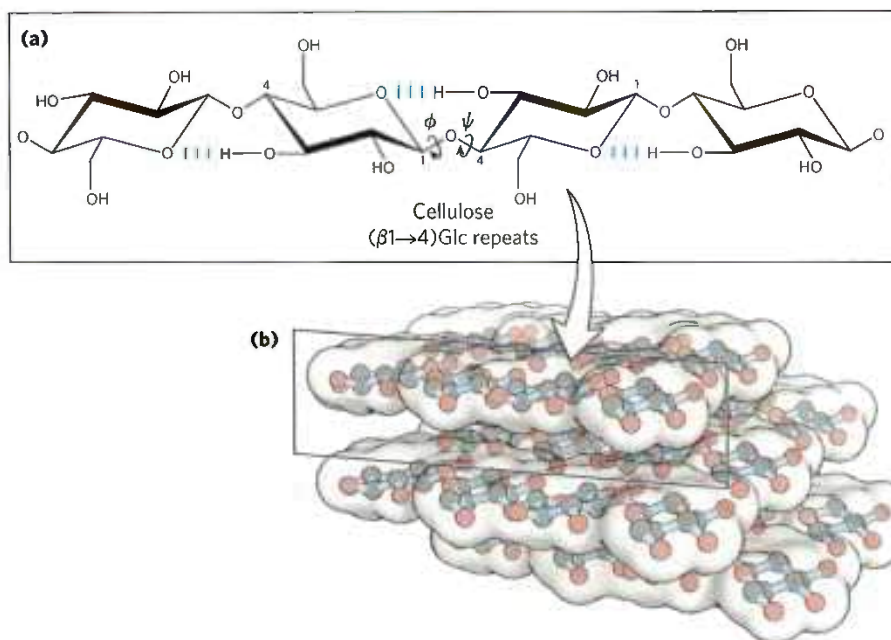


FIGURE 7-18 The linear structure of cellulose chains. (a) Four glucose units of cellulose. In the most stable conformation of cellulose, the (β1→4) glycosidic linkages put the adjacent rigid chairs at 180° relative to each other. (b) A model of straight, unbranched cellulose segments, held together in layers stabilized by interchain hydrogen bonds. [(b) Data from Cornell, B. 2016. Sugar Polymers. Available at: <http://ib.bioninja.com.au>. (Accessed 11 March 2020).]

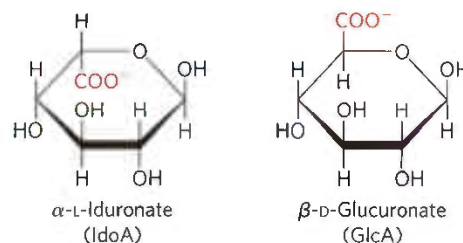
provides a porous pathway for the diffusion of nutrients and oxygen to individual cells. The ECM is composed of an interlocking meshwork of heteropolysaccharides (also called ground substance) and fibrous proteins such as fibrillar collagens, elastins, and fibronectins. The basement membrane is a specialized ECM that underlies epithelial cells; it consists of specialized collagens, laminins, and heteropolysaccharides.

These heteropolysaccharides, the **glycosaminoglycans**, are a family of linear polymers composed of repeating disaccharide units (Fig. 7-19). They are unique to animals and bacteria and are not found in plants. One of the two monosaccharides is always either *N*-acetylglucosamine or *N*-acetylgalactosamine; the other is in most cases a uronic acid, usually D-glucuronic or L-iduronic acid. Some glycosaminoglycans contain esterified sulfate groups. **P2** The combination of sulfate groups and the carboxylate groups of the uronic acid residues gives glycosaminoglycans a very high density of negative charge. To minimize the repulsive forces among neighboring charged groups, these molecules assume an extended conformation in solution, forming a rodlike helix in which the negatively charged carboxylate groups occur on alternate sides of the helix (as shown for heparin in Fig. 7-19). The extended rod form also provides maximum separation between the negatively charged sulfate groups. The specific patterns of sulfated and nonsulfated sugar residues in glycosaminoglycans allow specific recognition by a variety of protein ligands that bind electrostatically to these molecules. The sulfated glycosaminoglycans are attached to extracellular proteins to form proteoglycans (Section 7.3).

The glycosaminoglycan **hyaluronan** (hyaluronic acid) contains alternating residues of D-glucuronic acid and *N*-acetylglucosamine (Fig. 7-19). With up to 50,000 repeats of the basic disaccharide unit, hyaluronan has a

molecular weight of several million; it forms clear, highly viscous, noncompressible solutions that serve as lubricants in the synovial fluid of joints and give the vitreous humor of the vertebrate eye its jellylike consistency (the Greek *hyalos* means “glass”; hyaluronan can have a glassy or translucent appearance). Hyaluronan is also a component of the ECM of cartilage and tendons, to which it contributes tensile strength and elasticity as a result of its strong noncovalent interactions with other components of the matrix. Hyaluronidase, an enzyme secreted by some pathogenic bacteria, can hydrolyze the glycosidic linkages of hyaluronan, rendering tissues more susceptible to bacterial invasion. In many animal species, a similar enzyme in sperm hydrolyzes the outer glycosaminoglycan coat around an ovum, allowing sperm penetration.

Other glycosaminoglycans differ from hyaluronan in three respects: they are generally much shorter polymers, they are covalently linked to specific proteins (proteoglycans), and one or both monomeric units differ from those of hyaluronan. **Chondroitin sulfate** (Greek *chondros*, “cartilage”) contributes to the tensile strength of cartilage, tendons, ligaments, heart valves, and the walls of the aorta. **Dermatan sulfate** (Greek *derma*, “skin”) contributes to the pliability of skin and is also present in blood vessels and heart valves. In this polymer, many of the glucuronate residues present in chondroitin sulfate are replaced by their C-5 epimer, L-iduronate (IdoA).



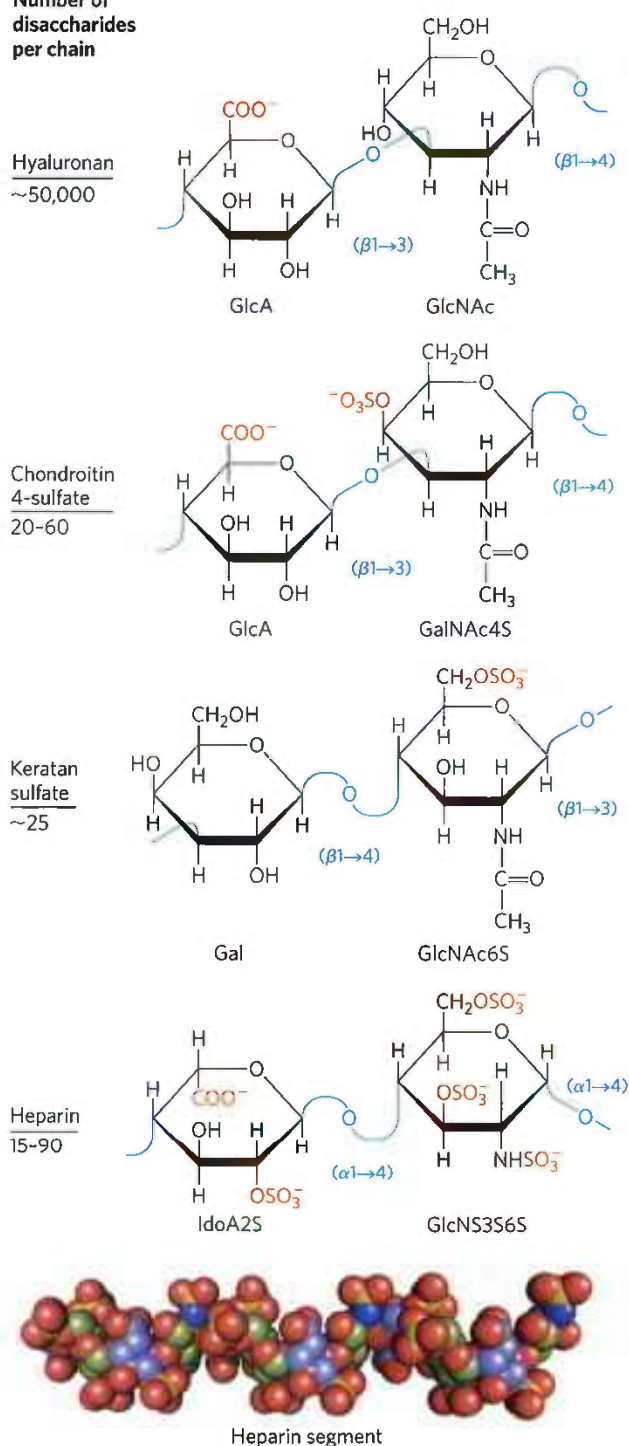
GlycosaminoglycanNumber of
disaccharides
per chain**Repeating disaccharide**

FIGURE 7-19 Repeating units of some common glycosaminoglycans of extracellular matrix. The molecules are copolymers of alternating uronic acid and amino sugar residues (keratan sulfate is the exception), with sulfate esters in any of several positions, except in hyaluronan. The ionized carboxylate and sulfate groups give these polymers their characteristic high negative charge. Therapeutic heparin contains primarily iduronic acid (IdoA) and a smaller proportion of glucuronic acid (not shown) and is generally highly sulfated and heterogeneous in length. The space-filling model shows a heparin segment as its structure in solution, as determined by NMR spectroscopy. Carbons in iduronic acid sulfate are colored blue; those in glucosamine sulfate are green. Oxygen and sulfur are shown in red and yellow, respectively. Hydrogen atoms are not shown (for clarity). [Data for molecular model from PDB ID 1HPN, B. Mulloy et al., *Biochem. J.* 293:849, 1993.]

Keratan sulfates (Greek *keras*, “horn”) have no uronic acid, and their sulfate content is variable. They are present in cornea, cartilage, bone, and a variety of horny structures formed from dead cells: horn, hair, hoofs, nails, and claws. **Heparan sulfate** (Greek *hēpar*, “liver”; it was originally isolated from dog liver) contains variable, nonrandom arrangements of sulfated and non-sulfated sugars. The exact sequence of sulfated residues gives the molecule the ability to interact specifically with a large number of proteins, including growth factors and ECM components, as well as various enzymes and factors present in plasma. Heparin is a highly sulfated, intracellular form of heparan sulfate produced primarily by mast cells (a type of leukocyte, or immune cell). Its physiological role is not yet clear, but purified heparin is used as a therapeutic agent to inhibit coagulation of blood through its capacity to bind the protease inhibitor antithrombin (see Fig. 7-24).

Table 7-2 summarizes the composition, properties, roles, and occurrence of the polysaccharides described in Section 7.2.

SUMMARY 7.2 Polysaccharides

- Homopolysaccharides contain only a single monomeric sugar species; heteropolysaccharides contain two or more kinds of monomers.
- The homopolysaccharides starch and glycogen are storage fuels in plant, animal, and bacterial cells. They consist of D-glucose units with ($\alpha 1 \rightarrow 4$) linkages, and both contain some branches.
- The homopolysaccharides cellulose, chitin, and dextran serve structural roles. Cellulose, composed of ($\beta 1 \rightarrow 4$)-linked D-glucose residues, lends strength and rigidity to plant cell walls. Chitin, a polymer of ($\beta 1 \rightarrow 4$)-linked N-acetylglucosamine, strengthens the exoskeletons of arthropods.
- Homopolysaccharides assume stable conformations dictated by weak interactions. The chair form of the pyranose ring is essentially rigid, so the conformation of the polymers is determined by rotation about the C—O bonds of the glycosidic linkage. Starch and glycogen form helical structures with intrachain hydrogen bonding; cellulose and chitin form long, straight strands that interact with neighboring strands.
- Bacterial cell walls are strengthened by peptidoglycan in which the repeating disaccharide is GlcNAc($\beta 1 \rightarrow 4$) Mur2Ac.
- Glycosaminoglycans are extracellular heteropolysaccharides in which one of the two monosaccharide units is a uronic acid (keratan sulfate is an exception) and the other is an N-acetylated amino sugar. The high density of negative charge on these molecules forces them to assume extended conformations. These polymers (hyaluronan, chondroitin sulfate, dermatan sulfate, and

TABLE 7-2 Structures and Roles of Some Polysaccharides

Polymer	Type ^a	Repeating unit ^b	Size (number of monosaccharide units)	Roles/significance
Starch				Energy storage: in plants
Amylose	Homo-	(α 1 $\rightarrow$ 4)Glc, linear	50–5,000	
Amylopectin	Homo-	(α 1 $\rightarrow$ 4)Glc, with (α 1 $\rightarrow$ 6)Glc branches every 24–30 residues	Up to 10 ⁶	
Glycogen	Homo-	(α 1 $\rightarrow$ 4)Glc, with (α 1 $\rightarrow$ 6)Glc branches every 8–12 residues	Up to 50,000	Energy storage: in bacteria and animal cells
Cellulose	Homo-	(β 1 $\rightarrow$ 4)Glc	Up to 15,000	Structural: in plants, gives rigidity and strength to cell walls
Chitin	Homo-	(β 1 $\rightarrow$ 4)GlcNAc	Very large	Structural: in insects, spiders, crustaceans, gives rigidity and strength to exoskeletons
Dextran	Homo-	(α 1 $\rightarrow$ 6)Glc, with (α 1 $\rightarrow$ 3) branches	Wide range	Structural: in bacteria, extracellular adhesive
Peptidoglycan	Hetero-; peptides attached	4)Mur2Ac(β 1 $\rightarrow$ 4)GlcNAc(β 1	Very large	Structural: in bacteria, gives rigidity and strength to cell envelope
Hyaluronan (a glycosaminoglycan)	Hetero-; acidic	4)GlcA(β 1 $\rightarrow$ 3)GlcNAc(β 1	Up to 100,000	Structural: in vertebrates, extracellular matrix of skin and connective tissue; viscosity and lubrication in joints

^aEach polymer is classified as a homopolysaccharide (homo-) or heteropolysaccharide (hetero-).

^bThe abbreviated names for the peptidoglycan, agarose, and hyaluronan repeating units indicate that the polymer contains repeats of this disaccharide unit. For example, in peptidoglycan, the GlcNAc of one disaccharide unit is (β 1 $\rightarrow$ 4)-linked to the first residue of the next disaccharide unit.

keratan sulfate) provide viscosity, adhesiveness, and tensile strength to the extracellular matrix.

7.3 Glycoconjugates: Proteoglycans, Glycoproteins, and Glycolipids

In addition to their important roles as fuel stores (starch, glycogen, dextran) and as structural materials (cellulose, chitin, peptidoglycans), polysaccharides and oligosaccharides are information carriers. Some provide communication between cells and their extracellular surroundings; others label proteins for transport to and localization in specific organelles, or for destruction when the protein is malformed or superfluous; and others serve as recognition sites for extracellular signal molecules (growth factors, for example) or extracellular parasites (bacteria or viruses). On almost every eukaryotic cell, specific oligosaccharide chains attached to components of the plasma membrane form a carbohydrate layer (the glycocalyx), several nanometers thick, that serves as an information-rich surface that the cell shows to its surroundings. These oligosaccharides are central players in cell-cell recognition and adhesion, cell migration during development, blood clotting, the immune response, wound healing, and other cellular processes. In most of these cases, the informational carbohydrate is covalently joined to a protein or a lipid to form a **glycoconjugate**, which is the biologically active molecule (Fig. 7-20).

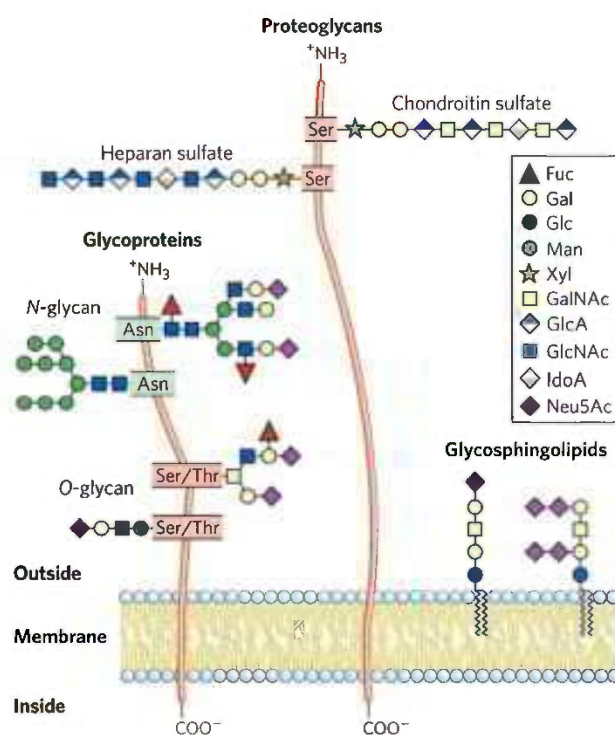


FIGURE 7-20 Glycoconjugates. The structures of some typical proteoglycans, glycoproteins, and glycosphingolipids described in the text.

Proteoglycans are macromolecules of the cell surface or ECM in which one or more sulfated glycosaminoglycan chains are joined covalently to a membrane protein or

a secreted protein. The glycosaminoglycan chain can bind to extracellular proteins through electrostatic interactions between the protein and the negatively charged sugar moieties on the proteoglycan. Proteoglycans are major components of all extracellular matrices.

Glycoproteins have one or several oligosaccharides of varying complexity joined covalently to a protein. They are usually found on the outer face of the plasma membrane (as part of the glycocalyx), in the ECM, and in the blood. Inside cells, they are found in specific organelles such as Golgi complexes (where the oligosaccharide moieties are added to the proteins), secretory granules, and lysosomes.

P6 The oligosaccharide portions of glycoproteins are very heterogeneous and, like glycosaminoglycans, are rich in information, forming highly specific sites for recognition and high-affinity binding by carbohydrate-binding proteins called lectins. Some cytosolic and nuclear proteins can be glycosylated as well.

Glycolipids are plasma membrane components in which the hydrophilic head groups are oligosaccharides. **Glycosphingolipids** are a class of glycolipids with a specific backbone structure that will be considered in more detail in Chapter 10. They have complex oligosaccharide components that act as specific sites for recognition by lectins, as they do in glycoproteins. Neurons are rich in glycosphingolipids, which act in nerve conduction and myelin formation. Glycosphingolipids also play a role in signal transduction in cells.

Proteoglycans Are Glycosaminoglycan-Containing Macromolecules of the Cell Surface and Extracellular Matrix

Mammalian cells can produce dozens of types of proteoglycans. Many are secreted into the ECM, where they act as tissue organizers and influence cellular activities, such as growth factor activation and adhesion. The basic proteoglycan unit consists of a “core protein” with covalently attached glycosaminoglycan(s). The point of attachment is a Ser residue, to which the glycosaminoglycan is joined through a tetrasaccharide linker (Fig. 7-21).

Some proteoglycans are integral membrane proteins (see Fig. 11-9). For example, the basal lamina, a specialized layer of the ECM that separates two organized groups of cells, contains a family of core proteins (M_r 20,000 to 40,000), each with several covalently attached heparan sulfate chains. There are two major families of membrane heparan sulfate proteoglycans. **Syndecans** have a single transmembrane domain and an extracellular domain bearing three to five chains of heparan sulfate and, in some cases, chondroitin sulfate (Fig. 7-22a). **Glypicans** are attached to the membrane by a GPI anchor, a glycosylated derivative of the membrane lipid phosphatidylinositol (see Fig. 11-16). Both syndecans and glypicans can be shed into the extracellular space. A protease in the ECM that cuts proteins close to the membrane surface releases syndecan ectodomains (domains outside the plasma membrane),

and a phospholipase that breaks the connection to the GPI anchor releases glypicans. These mechanisms provide a way for a cell to change its surface features quickly. Shedding is highly regulated and is activated in proliferating cells, such as cancer cells. Proteoglycan shedding

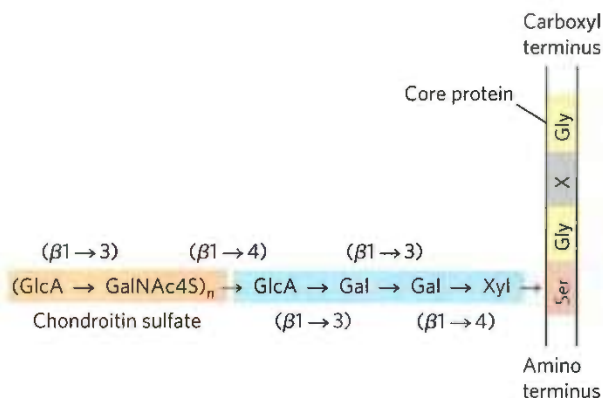


FIGURE 7-21 Proteoglycan structure, showing the tetrasaccharide bridge. A typical tetrasaccharide linker (blue) connects a glycosaminoglycan—in this case, chondroitin sulfate (orange)—to a Ser residue in the core protein. The xylose residue at the reducing end of the linker is joined by its anomeric carbon to the hydroxyl of the Ser residue.

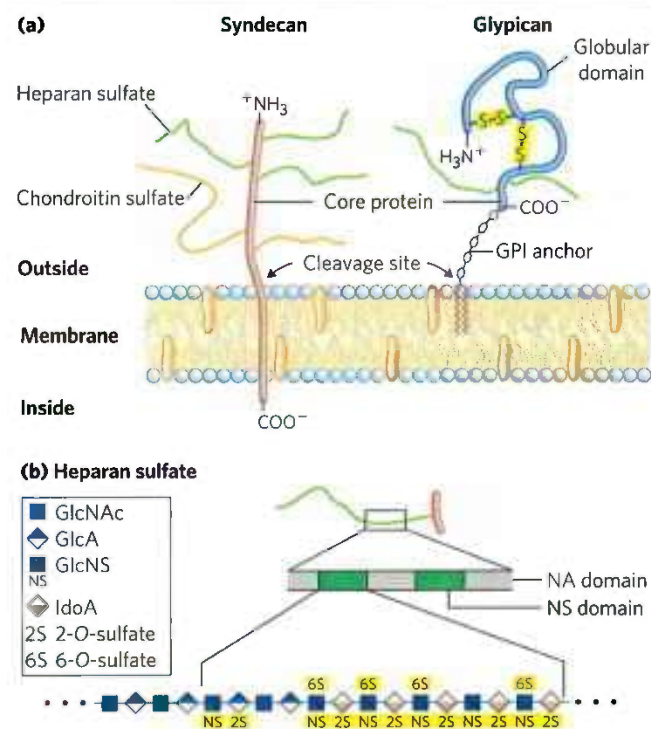


FIGURE 7-22 Two families of membrane proteoglycans. (a) Syndecans and glypicans can be shed from the membrane by enzymatic cleavage near the outer membrane surface. (b) Along a heparan sulfate chain, regions rich in sulfated sugars, the NS domains (green), alternate with regions with chiefly unmodified residues of GlcNAc and GlcA, the NA domains (gray). One of the NS domains is shown in more detail, revealing a high density of modified residues: GlcNAc (N-sulfoglucosamine) and both GlcA and IdoA, with a sulfate ester at C-2. (a) Information from U. Häcker et al., *Nature Rev. Mol. Cell Biol.* 6:530, 2005. (b) Information from J. Turnbull et al., *Trends Cell Biol.* 11:75, 2001.]

is involved in cell-cell recognition and adhesion, and in the proliferation and differentiation of cells. Numerous chondroitin sulfate and dermatan sulfate proteoglycans also exist, some as membrane-bound entities, others as secreted products in the ECM.

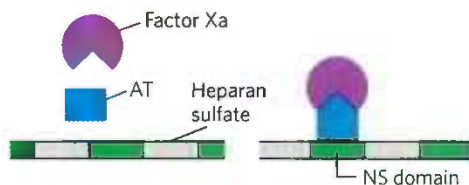
The glycosaminoglycan chains in proteoglycans can bind to a variety of extracellular ligands and thereby modulate the ligands' interaction with specific receptors of the cell surface. Detailed studies of heparan sulfate demonstrate a domain structure that is not random; some domains (typically three to eight disaccharide units long) differ from neighboring domains in sequence and in ability to bind to specific proteins. Highly sulfated domains (called NS domains) alternate with domains having unmodified GlcNAc and GlcA residues (*N*-acetylated, or NA, domains) (Fig. 7-22b). The exact pattern of sulfation in the NS domain depends on the particular proteoglycan; given the number of possible modifications of the GlcNAc–IdoA (iduronic acid) dimer, at least 32 different disaccharide units are possible. Furthermore, the same core protein can display different heparan sulfate structures when synthesized in different cell types.

Heparan sulfate molecules with precisely organized NS domains bind specifically to extracellular proteins and signaling molecules to alter their activities. The change in activity may result from a conformational change in the protein that is induced by the binding (Fig. 7-23a), or it may be due to the ability of adjacent domains of heparan

sulfate to bind to two different proteins, bringing them into close proximity and enhancing protein-protein interactions (Fig. 7-23b). A third general mechanism of action is the binding of extracellular signal molecules (growth factors, for example) to heparan sulfate, which increases their local concentrations and enhances their interaction with growth factor receptors on the cell surface; in this case, the heparan sulfate acts as a coreceptor (Fig. 7-23c). For example, fibroblast growth factor (FGF), an extracellular protein signal that stimulates cell division, first binds to heparan sulfate moieties of syndecan molecules in the target cell's plasma membrane. Syndecan presents FGF to the FGF plasma membrane receptor, and only then can FGF interact productively with its receptor to trigger cell division. Finally, in another type of mechanism, the NS domains interact—electrostatically and otherwise—with a variety of soluble molecules outside the cell, maintaining high local concentrations at the cell surface (Fig. 7-23d).

The protease thrombin, essential to blood coagulation (see Fig. 6-44), is inhibited by another blood protein, antithrombin, which prevents premature blood clotting. Antithrombin does not bind to or inhibit thrombin in the absence of heparan sulfate. In the presence of heparan sulfate or heparin, the binding affinity of thrombin for antithrombin increases 2,000-fold, and thrombin is strongly inhibited. When thrombin and antithrombin are crystallized in the presence of a short (16 residue) segment of

(a) Conformational activation



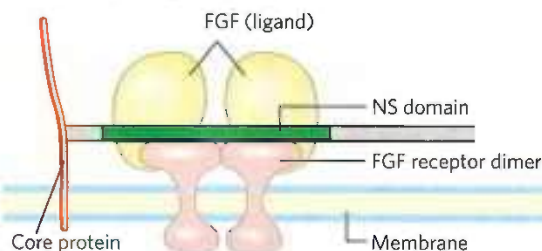
A conformational change induced in the protein antithrombin (AT) on binding a specific pentasaccharide NS domain allows its interaction with blood clotting factor Xa, preventing clotting.

(b) Enhanced protein-protein interaction



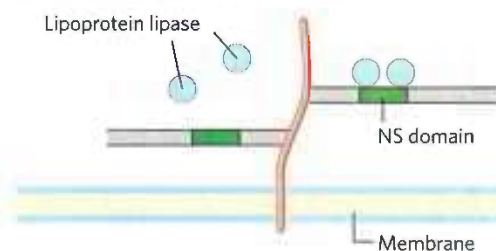
Binding of AT and thrombin to two adjacent NS domains brings the two proteins into close proximity, favoring their interaction, which inhibits blood clotting.

(c) Coreceptor for extracellular ligands



NS domains interact with both the fibroblast growth factor (FGF) and its receptor, bringing the oligomeric complex together and increasing the effectiveness of a low concentration of FGF.

(d) Cell surface localization/concentration



The high density of negative charges in heparan sulfate attracts positively charged lipoprotein lipase molecules and holds them by electrostatic and sequence-specific interactions with NS domains.

FIGURE 7-23 Four types of protein interactions with NS domains of heparan sulfate. [Information from J. Turnbull et al., *Trends Cell Biol.* 11:75, 2001.]

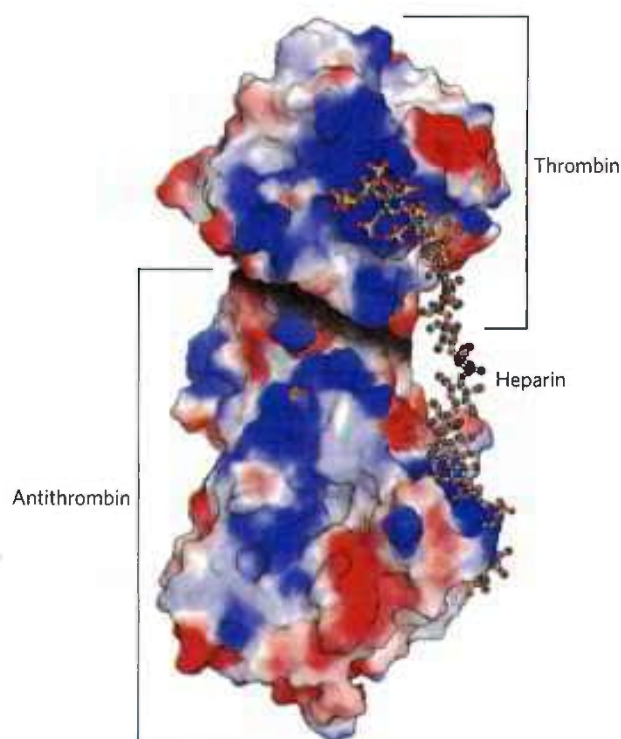


FIGURE 7-24 Molecular basis for heparan sulfate enhancement of the binding of thrombin to antithrombin. In this crystal structure of thrombin, antithrombin, and a 16 residue heparan sulfate-like polymer, all crystallized together, the binding sites for heparan sulfate in both proteins are rich in Arg and Lys residues. These positively charged regions, shown in blue in this electrostatic representation of the proteins, allow strong electrostatic interaction with multiple negatively charged sulfates and carboxylates of the heparan sulfate. Consequently, the affinity of antithrombin for thrombin is three orders of magnitude greater in the presence of heparan sulfate than in its absence. [Data from PDB ID 1TB6, W. Li et al., *Nat. Struct. Mol. Biol.* 11:857, 2004.]

heparan sulfate, the negatively charged heparan sulfate mimic is seen to bridge positively charged regions of the two proteins, causing an allosteric change that inhibits thrombin's protease activity (**Fig. 7-24**). The binding sites for heparan sulfate and heparin in both proteins are rich in Arg and Lys residues; the amino acids' positive charges interact electrostatically with the sulfates of the glycosaminoglycans. Antithrombin also inhibits two other blood coagulation proteins (factors IXa and Xa) in a heparan sulfate-dependent process.

The importance of correctly synthesizing sulfated domains in heparan sulfate is demonstrated in mutant ("knockout") mice lacking the enzyme that sulfates the C-2 hydroxyl of iduronate (IdoA). These animals are born without kidneys and with severe developmental abnormalities of the skeleton and eyes. Other studies demonstrate that membrane proteoglycans are important in the liver for clearing lipoproteins from the blood. Finally, there is growing evidence that proteoglycans containing heparan sulfate and chondroitin sulfate provide directional cues for axon outgrowth, influencing the path taken by developing axons in the nervous system.

The functional importance of proteoglycans and the glycosaminoglycans associated with them can also be seen in the effects of mutations that block the synthesis or degradation of these polymers in humans (**Box 7-3**).

Some proteoglycans can form **proteoglycan aggregates**, enormous supramolecular assemblies of many core proteins all bound to a single molecule of hyaluronan (also called hyaluronate). Aggrecan core protein ($M_r \sim 250,000$) has multiple chains of chondroitin sulfate and keratan sulfate, joined to Ser residues in the core protein through trisaccharide linkers, to give an aggrecan monomer of $M_r \sim 2 \times 10^6$. When a hundred or more of these "decorated" core proteins bind a single, extended molecule of hyaluronan (**Fig. 7-25**), the resulting proteoglycan aggregate ($M_r > 2 \times 10^8$) and its associated water of hydration occupy a volume about equal to that of a bacterial cell! Aggrecan interacts strongly with collagen in the ECM of cartilage, contributing to the development, tensile strength, and resilience of this connective tissue.

Interwoven with these enormous extracellular proteoglycans are fibrous matrix proteins such as collagen, elastin, and fibronectin, forming a cross-linked meshwork that gives the whole ECM strength and resilience. Some of these proteins are multiadhesive, a single protein having binding sites for several different matrix molecules. **Fibronectin**, for example, has separate domains that bind fibrin, heparan sulfate, and collagen. Fibronectin and a number of other proteins in the extracellular matrix contain the conserved RGD sequence (Arg-Gly-Asp), through which they bind to a family of proteins called **integrins**. Integrins mediate signaling between the cell interior and the network of molecules in the ECM. The overall picture of cell-matrix

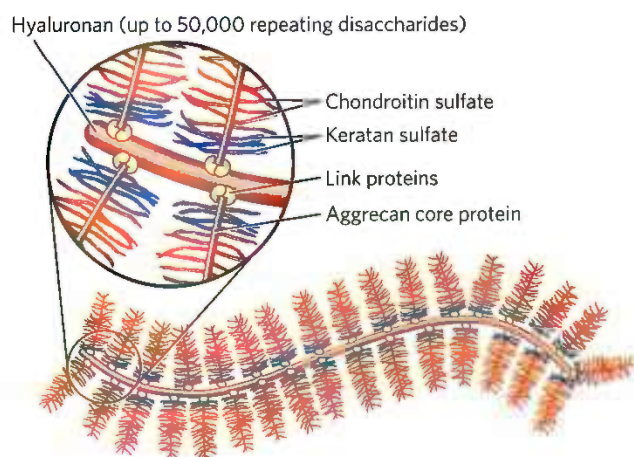


FIGURE 7-25 Proteoglycan aggregate of the extracellular matrix. Schematic drawing of a proteoglycan with many aggrecan molecules. One very long molecule of hyaluronan is associated noncovalently with about 100 molecules of the core protein aggrecan. Each aggrecan molecule contains many covalently bound chondroitin sulfate and keratan sulfate chains. Link proteins at the junction between each core protein and the hyaluronan backbone mediate the core protein-hyaluronan interaction.



Defects in the Synthesis or Degradation of Sulfated Glycosaminoglycans Can Lead to Serious Human Disease

Glycosaminoglycan synthesis requires enzymes that activate monomeric sugars, transport them across membranes, condense the activated sugars into polysaccharides, and add sulfates. Mutations in any of these enzymes in humans can lead to structural defects in the glycosaminoglycan (or in the proteoglycans formed from them). The result can be any of a wide variety of defects in cell signaling, cell proliferation, tissue morphogenesis, or interactions with growth factors (Fig. 1). For example, failure to extend the disaccharide unit GlcNAc-GlcA leads to a bone abnormality in which multiple, large bone spurs develop (Fig. 2).

When the defect occurs in degradative enzymes, the accumulation of incompletely degraded glycosaminoglycans can produce diseases ranging from moderate, as in Scheie

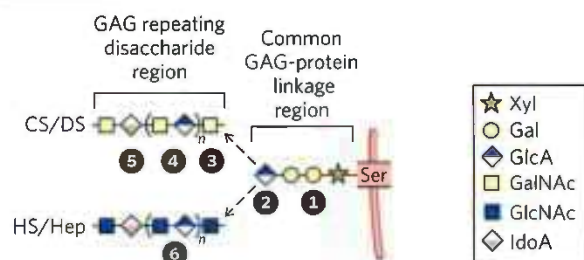


FIGURE 1 A segment of proteoglycan showing the normal structure of the glycosaminoglycans (GAGs) chondroitin sulfate or dermatan sulfate (CS/DS) (top) and heparan sulfate or heparin (HS/Hep) (bottom), attached through the linkage region to a Ser residue in the core protein. When a specific biosynthetic enzyme is absent because of a mutation, the numbered elements cannot be added to the growing oligosaccharide, and the product is truncated. The dysfunctional GAGs result in several types of human disease, depending on the site of truncation: **1** progeroid-type Ehlers-Danlos syndrome — with hyperextensible joints, fragile skin, and premature aging; **2** short stature or frequent joint dislocations; **3** neuropathy (nerve damage); **4** skeletal defects; **5** bipolar disorder or diaphragmatic hernia; **6** and bone deformations in the form of large bone spurs.



FIGURE 2 Bone deformation characteristic of multiple hereditary exostoses, a disease resulting from a genetic inability to add the GlcNAc-GlcA disaccharide to the growing heparan sulfate or heparin chain (see **6** in Fig. 1). The extra bone growth is artificially colored red in this x-ray of the humerus (upper arm bone). [CNRI/Science Photo Library/Science Source.]

syndrome, with joint stiffening but normal intelligence and life span, to severe, as in Hurler syndrome, with enlarged internal organs, heart disease, dwarfism, intellectual disability, and early death. Glycosaminoglycans were formerly called mucopolysaccharides, and diseases caused by genetic defects in their breakdown are often still called mucopolysaccharidoses.

interactions that emerges (Fig. 7-26) shows an array of interactions between cellular and extracellular molecules. These interactions serve not merely to anchor cells to the ECM, providing the strength and elasticity of skin and joints. **P7** They also provide paths that direct the migration of cells in developing tissue and serve to convey information in both directions across the plasma membrane.

Glycoproteins Have Covalently Attached Oligosaccharides

Glycoproteins are carbohydrate-protein conjugates in which the glycans are branched and are much smaller and more structurally diverse than the huge glycosaminoglycans of proteoglycans. The carbohydrate is attached at its anomeric carbon through a glycosidic link to the —OH of a Ser or Thr residue (*O*-linked), or through an

N-glycosyl link to the amide nitrogen of an Asn residue (*N*-linked) (Fig. 7-27). ***N*-glycosyl bonds** join the anomeric carbon of a sugar to a nitrogen atom in glycoproteins and nucleotides (see Fig. 8-1). Some glycoproteins have a single oligosaccharide chain, but many have more than one; the carbohydrate may constitute from 1% to 70% of the glycoprotein by mass. About half of all proteins of mammals are glycosylated.

As we shall see in Chapter 11, the external surface of the plasma membrane has many membrane glycoproteins with arrays of covalently attached oligosaccharides of varying complexity. **Mucins** are secreted or membrane glycoproteins that can contain large numbers of *O*-linked oligosaccharide chains. Mucins are present in most secretions; they are what gives mucus its characteristic slipperiness. Many of the proteins secreted by eukaryotic

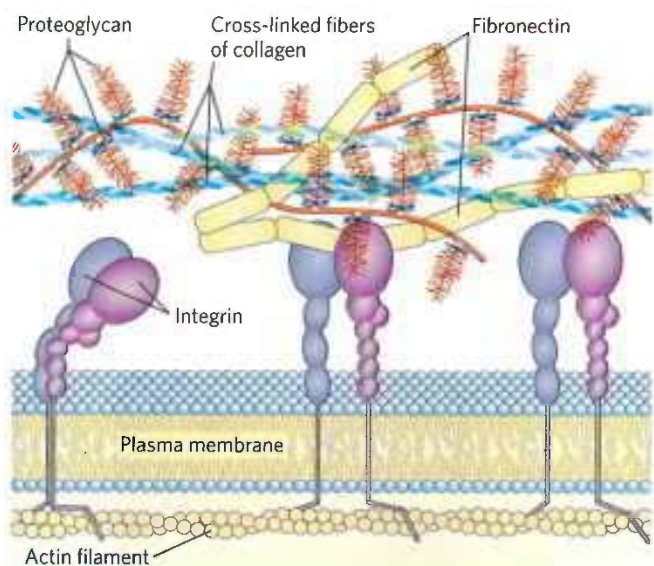


FIGURE 7-26 Interactions between cells and the extracellular matrix. The association between cells and the proteoglycan of the extracellular matrix is mediated by a membrane protein (integrin) and by an extracellular protein (fibronectin, with its RGD motif) with binding sites for both integrin and the proteoglycan. Note the close association of collagen fibers with the fibronectin and proteoglycan.

cells are glycoproteins, including most of the proteins of blood. For example, immunoglobulins (antibodies) and certain hormones, such as follicle-stimulating hormone, luteinizing hormone, and thyroid-stimulating hormone, are glycoproteins. Some of the proteins secreted by the pancreas are glycosylated, as are most of the proteins contained in lysosomes. Many milk proteins, including the major whey protein α -lactalbumin, are glycosylated.

Glycomics is the systematic characterization of all carbohydrate components of a given cell or tissue, including those attached to proteins and to lipids. For glycoproteins, this also means determining which proteins are glycosylated and where in the amino acid sequence each oligosaccharide is attached. This is a challenging undertaking, but worthwhile because of the potential insights it offers into normal patterns of glycosylation and the ways in which they are altered during development or in genetic diseases or cancer. Current methods of characterizing the entire carbohydrate complement of cells depend heavily on sophisticated application of nuclear magnetic resonance and mass spectrometry.

The structures of a large number of O- and N-linked oligosaccharides from a variety of glycoproteins are known; Figures 7-20 and 7-27 show a few typical examples. In Chapter 27, we consider the mechanisms by which specific proteins acquire specific oligosaccharide moieties.

What are the biological advantages of adding oligosaccharides to proteins? The very hydrophilic carbohydrate clusters alter the polarity and solubility of

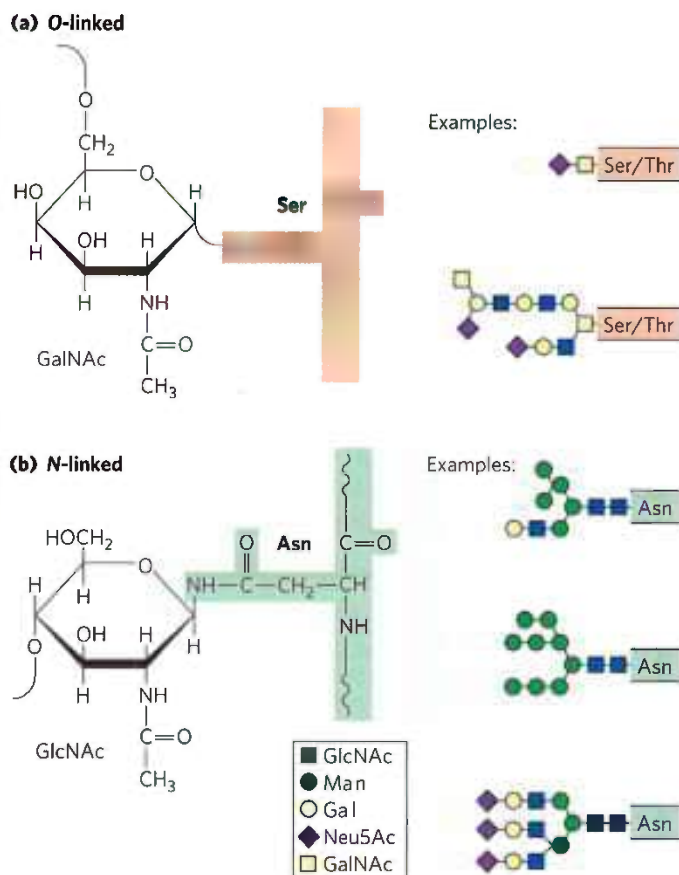


FIGURE 7-27 Oligosaccharide linkages in glycoproteins.

(a) O-linked oligosaccharides have a glycosidic bond to the hydroxyl group of Ser or Thr residues (light red), illustrated here with GalNAc as the sugar at the reducing end of the oligosaccharide. One simple chain and one complex chain are shown. (b) N-linked oligosaccharides have an N-glycosyl bond to the amide nitrogen of an Asn residue (green), illustrated here with GlcNAc as the terminal sugar. Three common types of oligosaccharide chains that are N-linked in glycoproteins are shown. A complete description of oligosaccharide structure requires specification of the position and stereochemistry (α or β) of each glycosidic linkage.

the proteins with which they are conjugated. **P7** Oligosaccharide chains that are attached to newly synthesized proteins in the endoplasmic reticulum (ER) and elaborated in the Golgi complex serve as destination labels and also act in protein quality control, targeting misfolded proteins for degradation (see Figs. 27-38, 27-39). When numerous negatively charged oligosaccharide chains are clustered in a single region of a protein, the charge repulsion among them favors the formation of an extended, rodlike structure in that region. The bulkiness and negative charge of oligosaccharide chains also protect some proteins from attack by proteolytic enzymes.

Beyond these global physical effects on protein structure, there are also more specific biological effects of oligosaccharide chains in glycoproteins (Section 7.4). The importance of normal protein glycosylation is clear from at least 40 different genetic disorders of glycosylation

having been found in humans, all causing severely defective physical or mental development; some of these disorders are fatal.

Glycolipids and Lipopolysaccharides Are Membrane Components

Glycoproteins are not the only cellular components that bear complex oligosaccharide chains; some lipids, too, have covalently bound oligosaccharides. **Gangliosides** are membrane lipids of eukaryotic cells in which the polar head group, the part of the lipid that forms the outer surface of the membrane, is a complex oligosaccharide containing a sialic acid (Fig. 7-9) and other monosaccharide residues. Some of the oligosaccharide moieties of gangliosides, such as those that determine human blood groups (see Fig. 10-13), are identical with those found in certain glycoproteins, which therefore also contribute to blood group type. Like the oligosaccharide moieties of glycoproteins, those of membrane lipids are generally, perhaps always, found on the outer face of the plasma membrane.

Lipopolysaccharides are the dominant surface feature of the outer membrane of gram-negative bacteria such as *Escherichia coli* and *Salmonella typhimurium*. These molecules are prime targets of the antibodies produced by the vertebrate immune system in response to bacterial infection and are therefore important determinants of the serotype of bacterial strains. (Serotypes are strains that are distinguished on the basis of antigenic properties.) The lipopolysaccharides of *S. typhimurium* contain six fatty acids bound to two glucosamine residues, one of which is the point of attachment for a complex oligosaccharide (Fig. 7-28). *E. coli* has similar but unique lipopolysaccharides. The lipid A portion of the lipopolysaccharides of some bacteria is called endotoxin; its toxicity to humans and other animals is responsible for the dangerously lowered blood pressure that occurs in toxic shock syndrome resulting from gram-negative bacterial infections. ■

SUMMARY 7.3 Glycoconjugates: Proteoglycans, Glycoproteins, and Glycolipids

■ Proteoglycans are huge molecules in which one or more large glycans, called sulfated glycosaminoglycans (heparan sulfate, chondroitin sulfate, dermatan sulfate, or keratan sulfate), are covalently attached to a core protein. Bound to the outside of the plasma membrane through a peptide or lipid, proteoglycans provide points of adhesion, recognition, and information transfer between cells, or between a cell and the extracellular matrix. In the extracellular matrix, enormous proteoglycan aggregates form, bound to a long hyaluronan molecule.

■ Glycoproteins contain oligosaccharides covalently linked to Asn, Ser, or Thr residues. The oligosaccharides

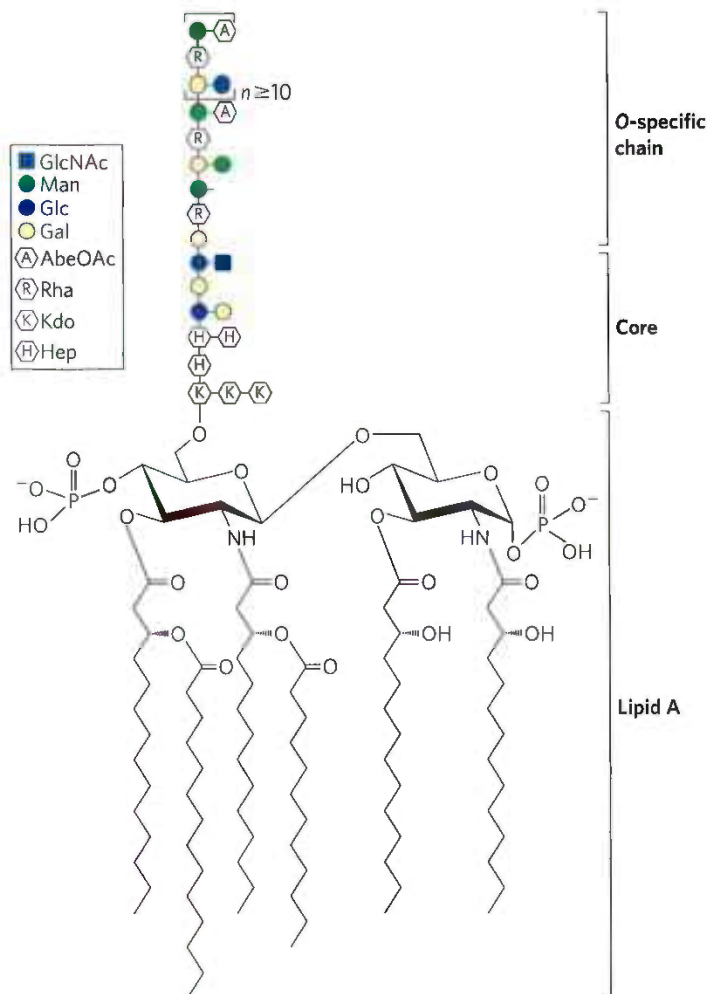


FIGURE 7-28 Bacterial lipopolysaccharides. Schematic diagram of the lipopolysaccharide of the outer membrane of *S. typhimurium*. Kdo is 3-deoxy- α -manno-octulosonic acid (previously called ketodeoxyoctonic acid); Hep is α -glycero- α -manno-heptose; AbeOAc is abequeose (a 3,6-dideoxyhexose) acetylated on one of its hydroxyls. Different bacterial species have subtly different lipopolysaccharide structures, but they have in common a lipid A region, composed of six fatty acid residues and two phosphorylated glucosamines, a core oligosaccharide, and an "O-specific" chain, which is the principal determinant of the serotype (immunological reactivity) of the bacterium. The outer membranes of the gram-negative bacteria *S. typhimurium* and *E. coli* contain so many lipopolysaccharide molecules that the cell surface is almost completely covered with O-specific chains.

are typically branched and smaller than glycosaminoglycans. Many cell surface or extracellular proteins are glycoproteins, as are most secreted proteins. The covalently attached oligosaccharides influence the folding and stability of the proteins, provide critical information about the targeting of newly synthesized proteins, and allow specific recognition by other proteins.

■ Glycolipids and glycosphingolipids in plants and animals and lipopolysaccharides in bacteria are components of the cell envelope, with covalently attached oligosaccharide chains exposed on the cell's outer surface.

7.4 Carbohydrates as Informational Molecules: The Sugar Code

Glycobiology, the study of the structure and function of glycoconjugates, is one of the most active and exciting areas of biochemistry and cell biology. The challenge is to understand how **P6** cells use specific oligosaccharides to encode important information about intracellular targeting of proteins, cell-cell interactions, cell differentiation and tissue development, and extracellular signals. Our discussion uses just a few examples to illustrate the diversity of structure and the range of biological activity of the glycoconjugates. In Chapter 20 we discuss the biosynthesis of polysaccharides, and in Chapter 27, the assembly of oligosaccharide chains on glycoproteins.

Oligosaccharide Structures Are Information-Dense

Improved methods for the analysis of oligosaccharide and polysaccharide structure have revealed remarkable complexity and diversity in the oligosaccharides of glycoproteins and glycolipids. Consider the oligosaccharide chains in Figure 7-27, typical of those found in many glycoproteins. The most complex of those shown contains 14 monosaccharide residues of four different kinds, variously linked as (1→2), (1→3), (1→4), (1→6), (2→3), and (2→6), some with the α and some with the β configuration. Branched structures, not found in nucleic acids or proteins, are common in oligosaccharides. With the reasonable assumption that 20 different monosaccharide subunits are available for construction of oligosaccharides, we can calculate that many billions of different hexameric oligosaccharides are possible; this compares with 6.4×10^7 (20^6) different hexapeptides possible for the 20 common amino acids, and 4,096 (4^6) different hexanucleotides for the four nucleotide subunits. If we also allow for variations in oligosaccharides resulting from sulfation of one or more residues, the number of possible oligosaccharides increases by two orders of magnitude. In reality, only a subset of possible combinations is found, given the restrictions imposed by the biosynthetic enzymes and the availability of precursors. Nevertheless, the enormously rich structural information in glycans does not merely rival but far surpasses that of nucleic acids in the density of information contained in a molecule of modest size. Each of the oligosaccharides represented in Figures 7-20 and 7-27 presents a unique, three-dimensional face—a word in the sugar code—readable by the proteins that interact with it.

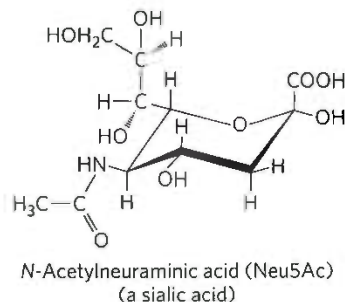
Lectins Are Proteins That Read the Sugar Code and Mediate Many Biological Processes

P6 **Lectins** bind carbohydrates with high specificity and with moderate to high affinity. These proteins serve in a wide variety of cell-cell recognition, signaling, and

adhesion processes and in intracellular targeting of newly synthesized proteins. Plant lectins, abundant in seeds, probably serve as deterrents to insects and other predators. In the laboratory, purified plant lectins are useful reagents for detecting and separating glycans and glycoproteins with different oligosaccharide moieties. Here we discuss just a few examples of the roles of lectins in animal cells.

Some peptide hormones that circulate in the blood have oligosaccharide moieties that strongly influence their circulatory half-life. Luteinizing hormone and thyrotropin (polypeptide hormones produced in the pituitary) have *N*-linked oligosaccharides that end with the disaccharide GalNAc4S(β 1→4)GlcNAc, which is recognized by a lectin (receptor) of hepatocytes. (GalNAc4S is *N*-acetylgalactosamine sulfated on the —OH group at C-4.) Receptor-hormone interaction mediates the uptake and destruction of luteinizing hormone and thyrotropin, reducing their concentration in the blood. Thus the blood levels of these hormones undergo a periodic rise (due to pulsatile secretion by the pituitary) and fall (due to constant destruction by hepatocytes).

Residues of Neu5Ac (a sialic acid) situated at the ends of the oligosaccharide chains of many plasma glycoproteins (Fig. 7-20) protect these proteins from uptake and degradation in the liver. For example, ceruloplasmin, a copper-containing serum glycoprotein, has several oligosaccharide chains ending in Neu5Ac. The mechanism that removes sialic acid residues from serum glycoproteins is unclear. It may be due to the activity of the enzyme neuraminidase (also called sialidase) produced by invading organisms or to a steady, slow release of the residues by extracellular enzymes. The plasma membrane of hepatocytes has lectin molecules (asialoglycoprotein receptors; “asialo-” indicating “without sialic acid”) that specifically bind oligosaccharide chains with galactose residues no longer “protected” by a terminal Neu5Ac residue. Receptor-ceruloplasmin interaction triggers endocytosis and destruction of the ceruloplasmin.



A similar mechanism is apparently responsible for removing “old” erythrocytes from the mammalian bloodstream. Newly synthesized erythrocytes have several membrane glycoproteins with oligosaccharide chains that end in Neu5Ac. In the laboratory, when the sialic acid residues are removed by withdrawing a sample of blood from experimental animals, treating it with

neuraminidase *in vitro*, and reintroducing it into the circulation, the treated erythrocytes disappear from the bloodstream within a few hours; erythrocytes with intact oligosaccharides (withdrawn and reintroduced without neuraminidase treatment) continue to circulate for days.

Selectins are a family of plasma membrane lectins that mediate cell-cell recognition and adhesion in a wide range of cellular processes. One such process is the movement of immune cells (leukocytes) through the capillary wall, from blood to tissues, at sites of infection or inflammation (**Fig. 7-29**). At an infection site, P-selectin on the surface of capillary endothelial cells interacts with a specific oligosaccharide of the surface glycoproteins of circulating leukocytes. This interaction slows the leukocytes as they roll along the endothelial lining of the capillaries. A second interaction, between integrin molecules in the leukocyte plasma membrane and an adhesion protein on the endothelial cell surface, now stops the leukocyte and allows it to move through the capillary wall into the infected tissues to initiate the immune attack. Two other selectins participate in this “lymphocyte homing”: E-selectin on the endothelial cell and L-selectin on the leukocyte bind their cognate oligosaccharides on the leukocyte and endothelial cell, respectively. Several of the selectins essential to lymphocyte homing bind specifically to the tetrasaccharide Neu5Ac-(α 2 $\rightarrow$ 3)-D-Gal-(β 1 $\rightarrow$ 4)(α -L-Fuc-[1 $\rightarrow$ 4])-D-GlcNAc, called sialyl Lewis x or sialyl Le^x.

Human selectins mediate the inflammatory responses in rheumatoid arthritis, asthma, psoriasis, multiple sclerosis, and the rejection of transplanted organs,

and thus there is great interest in developing drugs that inhibit selectin-mediated cell adhesion. Many carcinomas express sialyl Lewis x, which, when shed into the circulation, facilitates tumor cell survival and metastasis. Carbohydrate derivatives that mimic the sialyl Lewis x portion of sialoglycoproteins and compete for specific selectin-binding sites, or that alter the biosynthesis of the oligosaccharide, might prove effective as selectin-specific drugs for treating chronic inflammation or metastatic disease.

Several animal viruses, including the influenza virus, attach to their host cells through interactions with oligosaccharides displayed on the host cell surface. The lectin of the influenza virus, known as the HA (hemagglutinin) protein, is essential for viral entry and infection. After the virus has entered a host cell and has been replicated, the newly synthesized viral particles bud out of the cell, wrapped in a portion of its plasma membrane. A viral sialidase (neuraminidase) trims the terminal sialic acid residue from the host cell's oligosaccharides, releasing the viral particles from their interaction with the cell and preventing their aggregation with one another. Another round of infection can now begin. The antiviral drugs oseltamivir (Tamiflu) and zanamivir (Relenza) are used clinically in the treatment of influenza. These drugs are sugar analogs; they inhibit the viral sialidase by competing with the host cell's oligosaccharides for binding (**Fig. 7-30**). This prevents the release of viruses from the infected cell by sialidase, and also causes viral particles to aggregate, both of which block another cycle of infection.

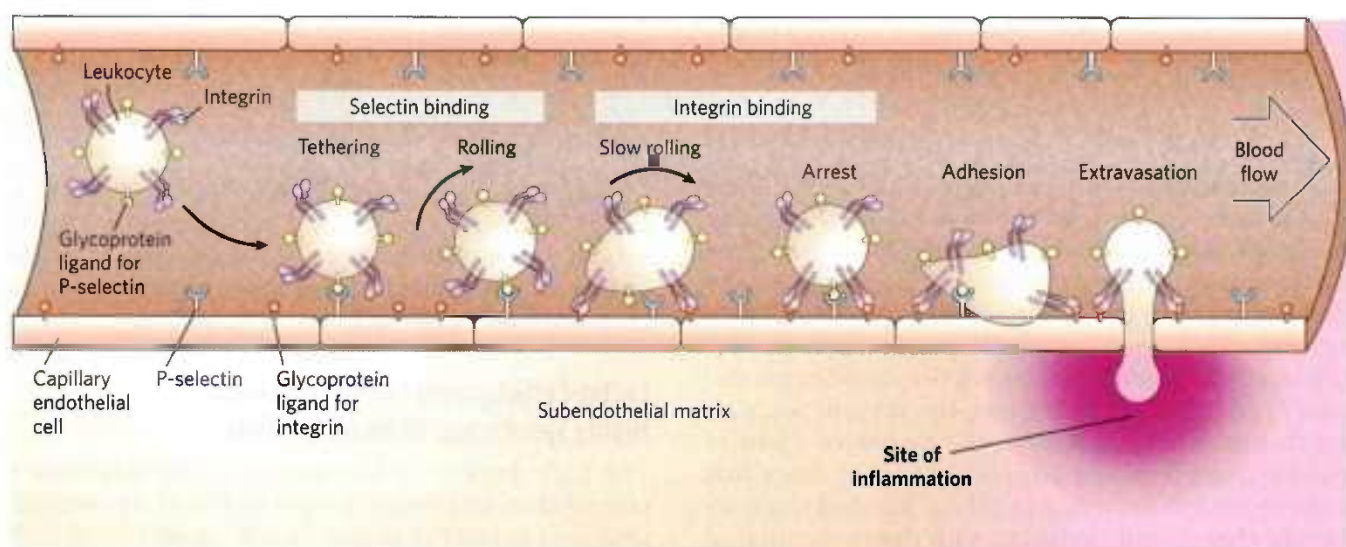
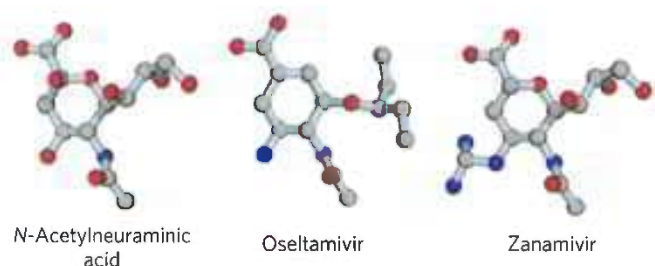
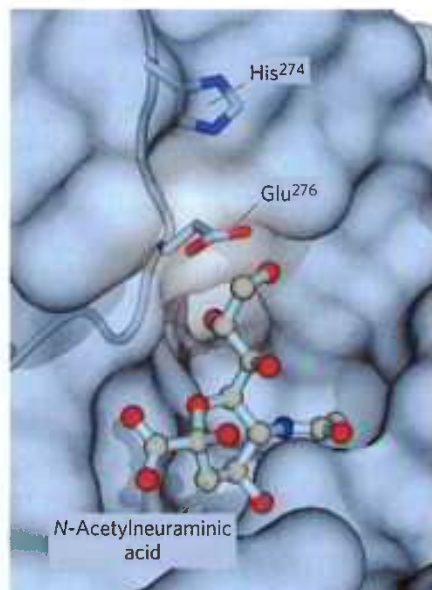


FIGURE 7-29 Role of lectin-ligand interactions in leukocyte movement to the site of an infection or injury. A leukocyte circulating through a capillary is slowed by transient interactions between P-selectin molecules in the plasma membrane of the capillary endothelial cells and glycoprotein ligands for P-selectin on the leukocyte surface. As the leukocyte interacts with successive P-selectin molecules, it rolls along

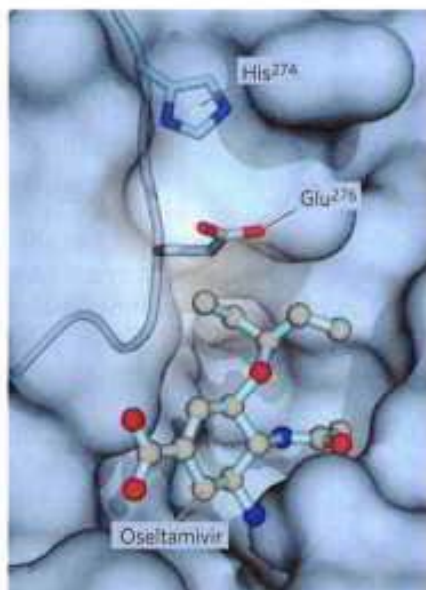
the capillary surface. Near a site of inflammation, stronger interactions between integrin in the leukocyte surface and its ligand in the capillary surface lead to tight adhesion. The leukocyte stops rolling and, under the influence of signals from the site of inflammation (such as sialyl Lewis x), escapes through the capillary wall as it moves toward the region of inflammation.



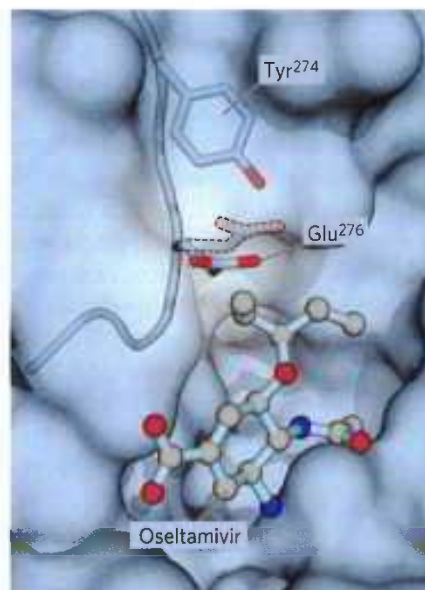
(a)



(b)



(c)



(d)

FIGURE 7-30 Binding site on influenza neuraminidase for *N*-acetylneuraminic acid and an antiviral drug.

(a) The normal binding ligand for influenza neuraminidase is a sialic acid, *N*-acetylneuraminic acid. The drugs oseltamivir and zanamivir occupy the same site on the enzyme, competitively inhibiting it and blocking viral release from the host cell. (b) The normal interaction with *N*-acetylneuraminic acid in the binding site. (c) Oseltamivir can fit into this site by pushing a nearby Glu residue

out of the way. (d) A mutation in the influenza virus's gene for neuraminidase replaces a His near this Glu residue with the larger side chain of a Tyr. Now, oseltamivir is not as effective at pushing the Glu out of its way, and the drug binds much less well to the binding site, making the mutant virus effectively resistant to oseltamivir. [Data from (b) PDB ID 2BAT, J. N. Varghese et al., *Proteins* 14:327, 1992; (c) PDB ID 2HU4, R. J. Russell et al., *Nature* 443:45, 2006; (d) PDB ID 3CL0, P. J. Collins et al., *Nature* 453:1258, 2008.]

Some of the most devastating of the human parasitic diseases, widespread in much of the developing world, are caused by eukaryotic microorganisms that display unusual surface oligosaccharides, which in some cases are known to be protective for the parasites. These organisms include the trypanosomes, responsible for African sleeping sickness (see Box 6-1) and Chagas disease; *Plasmodium falciparum*, the malaria parasite; and *Entamoeba histolytica*, the causative agent of amoebic dysentery. The prospect of finding drugs that interfere with the synthesis of these unusual oligosaccharide chains, and therefore with the replication of the parasites, has inspired much recent work on the biosynthetic pathways of these oligosaccharides. ■

Lectins also act intracellularly, in sorting proteins for transportation to specific cellular compartments (see

Chapter 27). For example, **P7** an oligosaccharide containing mannose 6-phosphate, recognized by a lectin, acts as a molecular “ZIP code” that tags newly synthesized proteins in the Golgi complex for transfer to the lysosome (see Fig. 27-41).

Lectin-Carbohydrate Interactions Are Highly Specific and Often Multivalent

The high density of information in the structure of oligosaccharides provides a sugar code with an essentially unlimited number of unique “words” small enough to be read by a single protein. In their carbohydrate-binding sites, lectins have a subtle molecular complementarity that allows interaction only with their correct carbohydrate binding partners. Often a divalent metal ion such as Ca^{2+} or Mn^{2+}

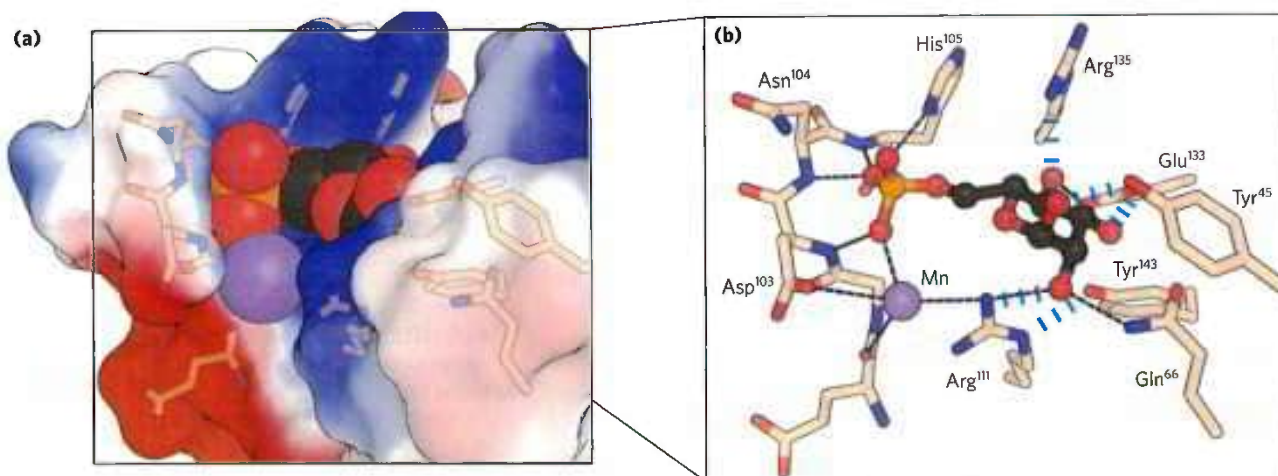


FIGURE 7-31 Details of a lectin-carbohydrate interaction. (a) Structure of the bovine mannanose 6-phosphate receptor complexed with mannanose 6-phosphate. The protein is represented as a surface contour image, showing the surface as predominantly negatively charged (red) or positively charged (blue). Mannose 6-phosphate is shown as a space-filling model; a manganese ion is shown as a violet sphere. (b) An enlarged view of the binding site. Mannose

6-phosphate is hydrogen-bonded to Arg¹¹¹ and coordinated with the manganese ion (shown smaller than its van der Waals radius, for clarity). Each hydroxyl group of mannanose is hydrogen-bonded to the protein. The His¹⁰⁵ hydrogen-bonded to a phosphate oxygen of mannanose 6-phosphate may be the residue that, when protonated at low pH, causes the receptor to release mannanose 6-phosphate into the lysosome. [Data from PDB ID 1M6P; D. L. Roberts et al., *Cell* 93:639, 1998.]

is part of the binding site. Lectin-ligand interactions can have extraordinarily high specificity. The affinity between an oligosaccharide and an individual carbohydrate binding domain (CBD) of a lectin is sometimes modest (micromolar to millimolar K_d values), but the effective affinity is often greatly increased by lectin multivalency, in which a single lectin molecule has multiple CBDs. In a cluster of oligosaccharides—as is commonly found on a membrane surface, for example—each oligosaccharide can engage one of the lectin's CBDs, strengthening the interaction. When cells express multiple lectin receptors, the avidity of the interaction can be very high, enabling highly cooperative events such as cell attachment and rolling (Fig. 7-29).

X-ray crystallographic studies of the structure of the lectin that is the receptor for mannanose 6-phosphate reveal details that explain the specificity of the binding and the role of a divalent cation in the lectin-sugar interaction (Fig. 7-31). When the protein tagged with mannanose 6-phosphate reaches the lysosome (which has a lower internal pH than the Golgi complex), the receptor loses its affinity for mannanose 6-phosphate and the tagged protein is released into the lysosomal matrix.

In addition to such highly specific interactions, there are more general interactions that contribute to the binding of many carbohydrates to their lectins. For example, many sugars have a more polar side and a less polar side (Fig. 7-32); the more polar side hydrogen-bonds with the lectin, while the less polar side undergoes interactions with nonpolar amino acid residues through the hydrophobic effect. The sum of all these interactions produces high-affinity binding and high specificity of lectins for their carbohydrate ligands. This represents

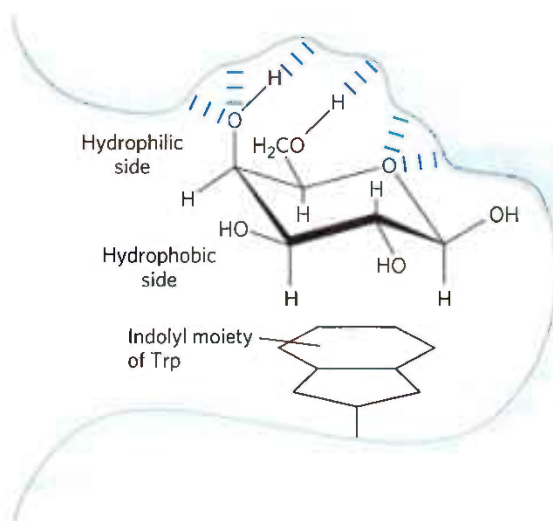


FIGURE 7-32 Interactions of sugar residues due to the hydrophobic effect. Sugar units such as galactose have a more polar side (the top of the chair as shown here, with the ring oxygen and several hydroxyls), available to hydrogen-bond with the lectin, and a less polar side that can interact with nonpolar side chains in the protein, such as the indole ring of Trp residues, through the hydrophobic effect. [Information from a figure provided by Dr. C.-W. von der Lieth, Heidelberg; H.-J. Gabius, *Naturwissenschaften* 87:108, 2000, Fig. 6.]

a kind of information transfer that is clearly central in many processes within and between cells. **Figure 7-33** summarizes some of the biological interactions mediated by the sugar code.

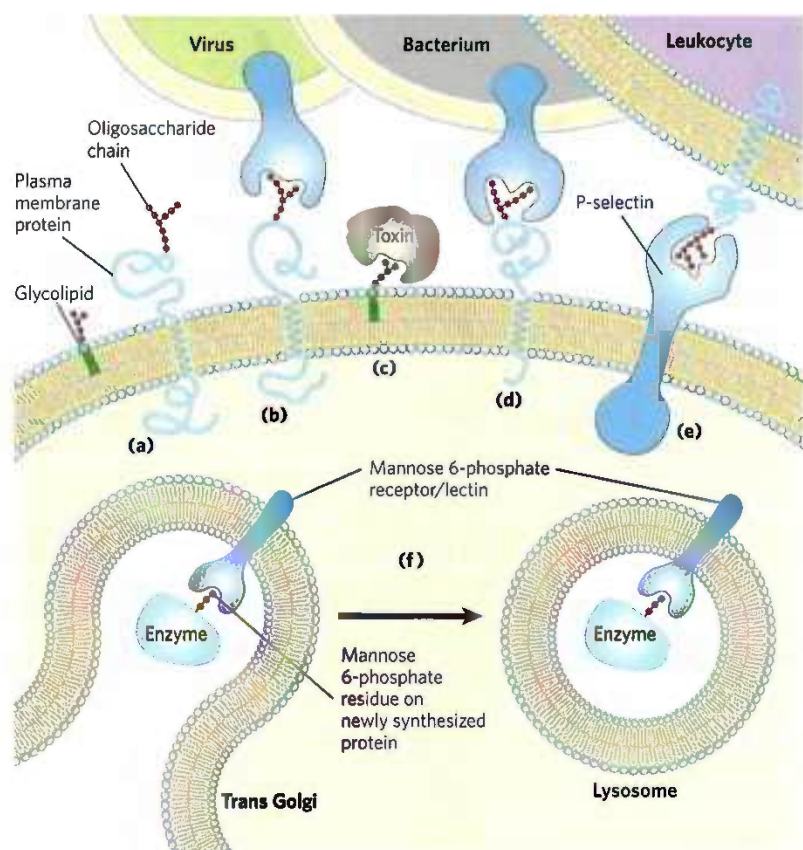


FIGURE 7-33 Role of oligosaccharides in recognition events at the cell surface and in the endomembrane system. (a) Oligosaccharides with unique structures (represented as strings of red hexagons) are components of a variety of glycoproteins or glycolipids on the outer surface of plasma membranes. Their oligosaccharide moieties are bound by extracellular lectins with high specificity and affinity. (b) Viruses that infect animal cells, such as the influenza virus, bind to cell surface glycoproteins as the first step in infection. (c) Bacterial toxins, such as the cholera and pertussis toxins, bind to a surface glycolipid before entering a cell. (d) Some bacteria adhere to and then colonize or infect animal cells. (e) Selectins (lectins) in the plasma membrane of certain cells mediate cell-cell interactions, such as those of leukocytes with the endothelial cells of the capillary wall at an infection site. (f) The mannose 6-phosphate receptor/lectin of the trans Golgi complex binds to the oligosaccharide of lysosomal enzymes, targeting them for transfer into the lysosome. [Information from N. Sharon and H. Lis, *Sci. Am.* 268 (January):82, 1993.]

SUMMARY 7.4 Carbohydrates as Informational Molecules: The Sugar Code

- Monosaccharides can be assembled into an almost limitless variety of oligosaccharides, which differ in the stereochemistry and position of glycosidic bonds, the type and orientation of substituent groups, and the number and type of branches. Glycans are far more information-dense than nucleic acids or proteins.
- The sugar specificity of many plant lectins makes them powerful laboratory reagents in glycobiology. Receptor/lectins in the surface of hepatocytes recognize glycoproteins that have lost their terminal sialic acid residue and mediate the normal uptake (by hepatocytes) and destruction of blood cells and of certain circulating glycoprotein hormones.
- Bacterial and viral pathogens and some eukaryotic parasites adhere to their animal cell targets through binding of lectins on the pathogens to oligosaccharides on the target cell surface, or vice versa. Interactions are often polyvalent.
- Structural studies of lectin-sugar complexes show the detailed complementarity between the two molecules, which accounts for the strength and specificity of lectin interactions with carbohydrates.

7.5 Working with Carbohydrates

Oligosaccharide analysis is complicated by the fact that, unlike nucleic acids and proteins, oligosaccharides can be branched and are joined by a variety of linkages. The high charge density of many oligosaccharides and polysaccharides, and the relative lability of the sulfate esters in glycosaminoglycans, present further difficulties.

For simple, linear polymers such as amylose, the positions of the glycosidic bonds are determined by the classical method of exhaustive methylation: treating the intact polysaccharide with methyl iodide in a strongly basic medium to convert all free hydroxyls to acid-stable methyl ethers, then hydrolyzing the methylated polysaccharide in acid. The only free hydroxyls in the monosaccharide derivatives so produced are those that were involved in glycosidic bonds. To determine the sequence of monosaccharide residues, including any branches that are present, exoglycosidases of known specificity are used to remove residues one at a time from the nonreducing end(s). The known specificity of these exoglycosidases often allows deduction of the position and stereochemistry of the linkages.

For analysis of the oligosaccharide moieties of glycoproteins and glycolipids, the oligosaccharides are released by purified enzymes—glycosidases that specifically cleave *O*- or *N*-linked oligosaccharides, or lipases

that remove lipid head groups. Alternatively, *O*-linked glycans can be released from glycoproteins by treatment with hydrazine.

The resulting mixtures of carbohydrates are resolved into their individual components by a variety of methods (Fig. 7-34), including the same techniques used in protein and amino acid separation: fractional precipitation by solvents, and ion-exchange and size-exclusion chromatography (see Fig. 3-17). Highly purified lectins, attached covalently to an insoluble support, can be used in affinity chromatography of carbohydrates.

Hydrolysis of oligosaccharides and polysaccharides in strong acid yields a mixture of monosaccharides, which can be identified and quantified by chromatographic techniques to yield the overall composition of the polymer.

Oligosaccharide analysis relies heavily on mass spectrometry (see Figs 3-28 and 3-29) and high-resolution NMR spectroscopy (see Fig. 4-31). NMR analysis alone,

especially for oligosaccharides of moderate size, can yield much information about sequence, linkage position, and anomeric carbon configuration. Automated procedures and commercial instruments are used for the routine determination of oligosaccharide structure, but the sequencing of branched oligosaccharides joined by more than one type of bond remains a far more formidable task than determining the linear sequences of proteins and nucleic acids.

Another important tool in working with carbohydrates is chemical synthesis, which has proved to be a powerful approach to understanding the biological functions of glycosaminoglycans and oligosaccharides. The chemistry involved in such syntheses is difficult, but carbohydrate chemists can now synthesize short segments of almost any glycosaminoglycan, with correct stereochemistry, chain length, and sulfation pattern, and oligosaccharides significantly more complex than those

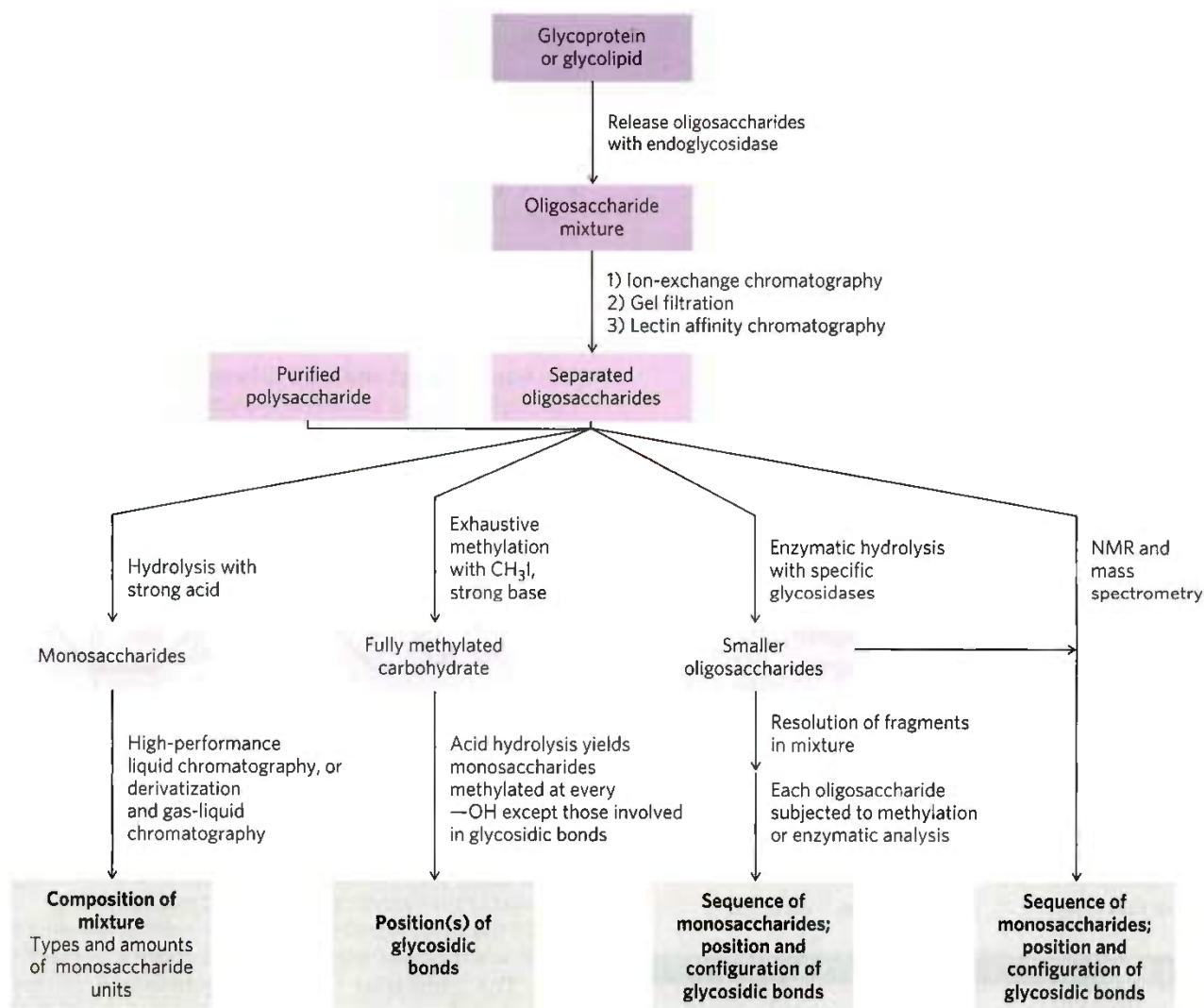


FIGURE 7-34 Methods of carbohydrate analysis. A carbohydrate purified in the first stage of the analysis often requires all four analytical routes for its complete characterization.

shown in Figure 7-27. Solid-phase oligosaccharide synthesis is based on the same principles (and has the same advantages) as peptide synthesis (see Fig. 3-30), but requires a set of tools unique to carbohydrate chemistry: blocking groups and activating groups that allow the synthesis of glycosidic linkages with the correct hydroxyl group. Synthetic approaches of this type currently represent an area of great interest, because it is difficult to purify defined oligosaccharides in adequate quantities from natural sources.

SUMMARY 7.5 Working with Carbohydrates

■ Establishing the complete structure of oligosaccharides and polysaccharides is a more complex problem than protein and nucleic acid analysis. Traditional chemical and enzymatic approaches, as well as mass spectrometry and high-resolution NMR spectroscopy, applicable to small samples of carbohydrate, yield essential information about sequence, configuration at anomeric and other carbons, and positions of glycosidic bonds. Solid-phase synthetic methods yield defined oligosaccharides that are of great value in exploring lectin-oligosaccharide interactions and may prove clinically useful.

KEY TERMS

Terms in bold are defined in the glossary.

carbohydrate 229	hemoglobin glycation 239
monosaccharide 229	glycan 241
oligosaccharide 229	homopolysaccharide 241
disaccharide 229	heteropolysaccharide 241
polysaccharide 229	starch 242
aldose 230	glycogen 242
ketose 230	cellulose 243
enantiomer 232	extracellular matrix (ECM) 244
Fischer projection formulas 232	glycosaminoglycan 245
epimers 232	hyaluronan 245
hemiacetal 234	chondroitin sulfate 245
hemiketal 234	heparan sulfate 246
anomers 234	glyconjugate 247
anomeric carbon 234	proteoglycan 247
pyranose 234	glycoprotein 248
furanose 234	glycolipid 248
Haworth perspective formulas 235	syndecan 248
mutarotation 235	glypican 248
reducing sugar 237	glycomics 252
O-glycosidic bonds 237	ganglioside 253
reducing end 237	lectin 254
	selectins 255

PROBLEMS

1. Sugar Alcohols In the monosaccharide derivatives known as sugar alcohols, the carbonyl oxygen is reduced to a hydroxyl group. For example, D-glyceraldehyde can be reduced to glycerol. Why can the sugar alcohol glycerol no longer be designated D or L?

2. Recognizing Epimers Using Figure 7-3, identify the epimers of (a) D-allose, (b) D-gulose, and (c) D-ribose at C-2, C-3, and C-4.

3. Configuration and Conformation Which bond(s) in α -D-glucose must be broken to change its configuration to β -D-glucose? Which bond(s) must be broken to convert D-glucose to D-mannose? Which bond(s) must be broken to convert one "chair" form of D-glucose to the other?

4. Sugar Structures Compare and contrast the structural features of each pair: (a) Cellulose and glycogen (b) D-Glucose and D-fructose (c) Maltose and sucrose

5. Haworth Structures Draw the Haworth perspective formulas for α -D-mannose and β -L-galactose.

6. Reducing Sugars Draw the structural formula for α -D-glucosyl-(1 $\rightarrow$ 6)-D-mannosamine, and circle the part of this structure that makes the compound a reducing sugar.

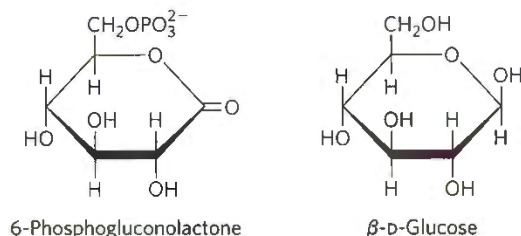


7. Hemoglobin Glycation The measurement of glycated hemoglobin (the HbA1c level) to monitor the regulation of blood glucose (Box 7-2) is not reliable in individuals with certain conditions or diseases. Can you suggest an explanation for a lower-than-normal HbA1c level in a patient who definitely is not diabetic?

8. Hemiacetal and Glycosidic Linkages Explain the difference between a hemiacetal and a glycoside.

9. A Taste of Honey The sweetness of honey gradually decreases at a high temperature. Also, high-fructose corn syrup (a commercial product in which much of the glucose in corn syrup is converted to fructose) is used for sweetening cold drinks but not hot drinks. What chemical property of fructose could account for both of these observations?

10. Gluconolactone and Glucose Oxidation States The cyclic glucose derivative 6-phosphogluconolactone is an intermediate in the pentose phosphate pathway (discussed in Chapter 14). Compare the oxidation state of C-1 for the cyclic form of both gluconolactone and β -D-glucose.



11. Invertase "Inverts" Sucrose As sweet as sucrose is, an equimolar mixture of its constituent monosaccharides, D-glucose and D-fructose, is sweeter. Besides enhancing sweetness, fructose has hygroscopic properties that improve the texture of foods, reducing crystallization and increasing moisture.

In the food industry, hydrolyzed sucrose is called invert sugar, and the yeast enzyme that hydrolyzes it is called invertase. The hydrolysis reaction is generally monitored by measuring the specific rotation of the solution, which is positive (+66.4°) for sucrose but becomes negative (inverts) as more D-glucose (specific rotation = +52.7°) and D-fructose (specific rotation = -92°) form.

From what you know about the chemistry of the glycosidic bond, how would you hydrolyze sucrose to invert sugar nonenzymatically in a home kitchen?

12. Manufacture of Liquid-Filled Chocolates The manufacture of chocolates containing a liquid center is an interesting application of enzyme engineering. The flavored liquid center consists largely of an aqueous solution of sugars rich in fructose to provide sweetness. The technical dilemma is the following: the chocolate coating must be prepared by pouring hot, melted chocolate over a solid (or almost solid) core, yet the final product must have a liquid, fructose-rich center. Suggest a way to solve this problem. (Hint: Sucrose is much less soluble than a mixture of glucose and fructose.)

13. Anomers of Sucrose? Lactose exists in two anomeric forms, but no anomeric forms of sucrose have been reported. Why?

14. Gentioibiose Gentioibiose ($\text{D-Glc}(\beta 1 \rightarrow 6)\text{D-Glc}$) is a disaccharide found in some plant glycosides. Draw the Haworth structure of gentioibiose based on its abbreviated name. Is it a reducing sugar? Does it undergo mutarotation?

15. Identifying Reducing Sugars Is *N*-acetyl- β -D-glucosamine (Fig. 7-9) a reducing sugar? What about D-gluconate? Is the disaccharide $\text{GlcN}(\alpha 1 \rightarrow 1\alpha)\text{Glc}$ a reducing sugar?

16. Physical Properties of Cellulose and Glycogen The almost pure cellulose obtained from the seed threads of *Gossypium* (cotton) is tough, fibrous, and completely insoluble in water. In contrast, glycogen obtained from muscle or liver disperses readily in hot water to make a turbid solution. Despite their markedly different physical properties, both substances are (1 $\rightarrow$ 4)-linked D-glucose polymers of comparable molecular weight. What structural features of these two polysaccharides underlie their different physical properties? Suggest possible biological advantages of their respective properties.

17. Dimensions of a Polysaccharide Compare the dimensions of a molecule of cellulose and a molecule of amylose, each of M_r 200,000.

18. Growth Rate of Bamboo The stems of bamboo, a tropical grass, can grow at the phenomenal rate of 0.3 m/day under optimal conditions. Given that the stems are composed almost entirely of cellulose fibers oriented in the direction of growth, calculate the number of sugar residues per second that must be added enzymatically to growing cellulose chains to account for the growth rate. Each D-glucose unit contributes ~ 0.5 nm to the length of a cellulose molecule.

19. Glycoproteins versus Proteoglycans Which characteristics describe glycoproteins and which describe proteoglycans?

- (a) Exclusively located at the cell surface and in the extracellular matrix
- (b) May contain *N*-linked glycosidic bonds
- (c) Found in Golgi complexes, secretory granules, and lysosomes
- (d) Include the heparan sulfate family
- (e) Sulfated glycosaminoglycan chains can only be covalently linked to a Ser residue
- (f) Form highly specific sites for recognition and high-affinity binding by lectins

20. Relative Stability of Two Conformers Explain why the two structures shown in Figure 7-16 are so different in energy (stability). Hint: See Figure 1-21.

21. Volume of Chondroitin Sulfate in Solution One critical function of chondroitin sulfate is to act as a lubricant in skeletal joints by creating a gel-like medium that is resilient to friction and shock. This function seems to be related to a distinctive property of chondroitin sulfate: the volume occupied by the molecule is much greater in solution than in the dehydrated solid. Why is the volume so much larger in solution?



22. Heparin Interactions Heparin, a highly negatively charged glycosaminoglycan, is used clinically as an anti-coagulant. It acts by binding several plasma proteins, including antithrombin III, an inhibitor of blood clotting. The 1:1 binding of heparin to antithrombin III seems to cause a conformational change in the protein that greatly increases its ability to inhibit clotting. What amino acid residues of antithrombin III are likely to interact with heparin?

23. Permutations of a Trisaccharide Three different hexoses (A, B, and C) can be combined to form a large number of trisaccharides. What structural features of trisaccharides allow so many permutations and combinations?

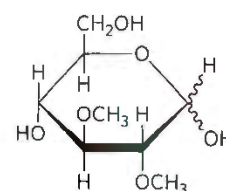
24. Effect of Sialic Acid on SDS Polyacrylamide Gel Electrophoresis Suppose you have four forms of a protein, all with identical amino acid sequence but containing zero, one, two, or three oligosaccharide chains, each ending in a single sialic acid residue. Draw the gel pattern you would expect when a mixture of these four glycoproteins is subjected to SDS polyacrylamide gel electrophoresis (see Fig. 3-18) and stained for protein. Identify any bands in your drawing.

25. Information Content of Oligosaccharides The carbohydrate portion of some glycoproteins may serve as a cellular recognition site. To perform this function, the oligosaccharide(s) must have the potential to exist in a large variety of forms. Which can produce a greater variety of structures: oligopeptides composed of five different amino acid residues, or oligosaccharides composed of five different monosaccharide residues? Explain.



26. Castor Bean Toxin The seed of the castor bean (*Ricinus communis*) contains large amounts of ricin, a poison that is deadly to animals, including humans. One of the two subunits of this toxin is a lectin that binds terminal *N*-acetylgalactosamine residues on glycoproteins on the surface of eukaryotic cells, allowing the other subunit to enter the cell and kill it by preventing proteins from being made. Suggest a possible antidote to prevent or reverse ricin-mediated entry of the toxin.

27. Determination of the Extent of Branching in Amylopectin A biochemist wants to determine the amount of branching in amylopectin, defined by the number of ($\alpha 1 \rightarrow 6$) glycosidic bonds present. First, she treats the sample with methyl iodide, a methylating agent that replaces the hydrogen of every sugar hydroxyl with a methyl group, converting $-\text{OH}$ to $-\text{OCH}_3$. She then hydrolyzes all the glycosidic bonds in the treated sample in aqueous acid and measures the amount of 2,3-di-*O*-methylglucose formed.



2,3-Di-*O*-methylglucose

In this representation of 2,3-di-*O*-methylglucose, the wavy bonds at C-1 indicate that the structure represents both anomers (α and β).

(a) Explain the basis of this procedure for determining the number of (α 1 $\rightarrow$ 6) branch points in amylopectin. What happens to the unbranched glucose residues in amylopectin during the methylation and hydrolysis procedure?

(b) A 258 mg sample of amylopectin that was treated by the methylation and hydrolysis procedure described yielded 12.4 mg of 2,3-di-*O*-methylglucose. Determine what percentage of the glucose residues in the amylopectin contained an (α 1 $\rightarrow$ 6) branch. (Assume that the average molecular weight of a glucose residue in amylopectin is 162 g/mol and the molecular weight of 2,3-di-*O*-methylglucose is 208 g/mol.)

DATA ANALYSIS PROBLEM

28. Determining the Structure of ABO Blood Group Antigens The human ABO blood group system was discovered in 1901, and in 1924 this trait was shown to be inherited at a single gene locus with three alleles. In 1960, W. T. J. Morgan published a paper summarizing what was known at that time about the structure of the ABO antigen molecules. When the paper was published, the complete structures of the A, B, and O antigens were not yet known; this paper is an example of what scientific knowledge looks like "in the making."

In any attempt to determine the structure of an unknown biological compound, researchers must deal with two fundamental problems: (1) If you don't know what *it* is, how do you know if *it* is pure? (2) If you don't know what *it* is, how do you know that your extraction and purification conditions have not changed *its* structure? Morgan addressed problem 1 through several methods. One method is described in his paper as observing "constant analytical values after fractional solubility tests" (p. 312). In this case, "analytical values" are measurements of chemical composition, melting point, and so forth.

(a) Based on your understanding of chemical techniques, what could Morgan mean by "fractional solubility tests"?

(b) Why would the analytical values obtained from fractional solubility tests of a *pure* substance be constant, and those of an *impure* substance not be constant?

Morgan addressed problem 2 by using an assay to measure the immunological activity of the substance present in different samples.

(c) Why was it important for Morgan's studies, and especially for addressing problem 2, that this activity assay be quantitative (measuring a level of activity) rather than simply qualitative (measuring only the presence or absence of a substance)?

The structure of the blood group antigens is shown in Figure 10-13. In his paper, Morgan listed several properties of the three antigens, A, B, and O, that were known at that time (p. 314):

1. Type B antigen has a higher content of galactose than A or O.
2. Type A antigen contains more total amino sugars than B or O.

3. The glucosamine:galactosamine ratio for the A antigen is roughly 1:2; for B, it is roughly 2:5.

(d) Which of these findings is (are) consistent with the known structures of the blood group antigens?

(e) How do you explain the discrepancies between Morgan's data and the known structures?

In later work, Morgan and his colleagues used a clever technique to obtain structural information about the blood group antigens. Enzymes had been found that would specifically degrade the antigens. However, these were available only as crude enzyme preparations, perhaps containing more than one enzyme of unknown specificity. Degradation of the blood type antigens by these crude enzymes could be inhibited by the addition of particular sugar molecules to the reaction. Only sugars found in the blood type antigens would cause this inhibition. One enzyme preparation, isolated from the protozoan *Trichomonas foetus*, would degrade all three antigens and was inhibited by the addition of particular sugars. The results of these studies are summarized in the table below, showing the percentage of substrate remaining unchanged when the *T. foetus* enzyme acted on the blood group antigens in the presence of sugars.

Sugar added	Unchanged substrate (%)		
	A antigen	B antigen	O antigen
Control—no sugar	3	1	1
L-Fucose	3	1	100
D-Fucose	3	1	1
L-Galactose	3	1	3
D-Galactose	6	100	1
N-Acetylglucosamine	3	1	1
N-Acetylgalactosamine	100	6	1

For the O antigen, a comparison of the control and L-fucose results shows that L-fucose inhibits the degradation of the antigen. This is an example of product inhibition, in which an excess of reaction product shifts the equilibrium of the reaction, preventing further breakdown of substrate.

(f) Although the O antigen contains galactose, N-acetylglucosamine, and N-acetylgalactosamine, none of these sugars inhibited the degradation of this antigen. Based on these data, is the enzyme preparation from *T. foetus* an endoglycosidase or an exoglycosidase? (Endoglycosidases cut bonds between interior residues; exoglycosidases remove one residue at a time from the end of a polymer.) Explain your reasoning.

(g) Fucose is also present in the A and B antigens. Based on the structure of these antigens, why does fucose fail to prevent their degradation by the *T. foetus* enzyme? What structure would be produced?

(h) Which of the results in (f) and (g) are consistent with the structures shown in Figure 10-13? Explain your reasoning.

Reference

Morgan, W.T.J. 1960. The Croonian Lecture: a contribution to human biochemical genetics; the chemical basis of blood-group specificity. *Proc. R. Soc. Lond. B Biol. Sci.* 151:308-347.

NUCLEOTIDES AND NUCLEIC ACIDS

- 8.1 Some Basic Definitions and Conventions 263
- 8.2 Nucleic Acid Structure 269
- 8.3 Nucleic Acid Chemistry 278
- 8.4 Other Functions of Nucleotides 294

Nucleotides have a variety of roles in cellular metabolism. They are the energy currency in metabolic transactions, the essential chemical links in the response of cells to hormones and other extracellular stimuli, and the structural components of an array of enzyme cofactors and metabolic intermediates. And, last but certainly not least, they are the constituents of nucleic acids: **deoxyribonucleic acid (DNA)** and **ribonucleic acid (RNA)**, the molecular repositories of genetic information. The structure of every protein, and ultimately of every biomolecule and cellular component, is a product of information programmed into the nucleotide sequence of cellular (or viral) nucleic acids. The ability to store and transmit genetic information from one generation to the next is a fundamental condition for life.

The discussion in this chapter reinforces or introduces five principles:

- P1** **Nucleic acids are both repositories and functional expressions of biological information.** Biological information is one of the required conditions for life, a blueprint for each species transmitted from one generation to the next. RNA can be a functional expression of that information, directing the synthesis of proteins or in some cases acting directly as a signal or a reaction catalyst.
- P2** **The transmission of biological information relies on molecular complementarity.** Chromosomes are the largest molecules in any cell. They are polymers composed of a small set of common nucleotides, with information embedded in the nucleotide sequence. The common nucleotides in RNA and DNA are organized so that two strands

of nucleic acid can maintain a complementary and uniform structure over vast molecular distances. This extended potential for both variable sequence and complementarity, and thus information storage and transmission, is a property shared by no other class of biological molecule.

- P3** **Biological information is subject to natural damage and change.** DNA damage is a constant, and it results in occasional mutation—the raw material for evolution.
- P4** **Biological information can be accessed, interpreted, and altered in the laboratory.** The information embedded in nucleic acids is of singular importance to biochemistry and molecular biology. The techniques for sequencing, synthesizing, and altering nucleic acids are continually advancing.
- P5** **Nucleoside triphosphates occupy a central role in cellular metabolism, serving as an energy currency and as important regulatory signals.** ATP is the ultimate product of catabolic pathways, providing fuel for anabolic pathways.

This chapter provides an overview of the chemical nature of the nucleotides and nucleic acids found in most cells, as well as the tools used to study them. A more detailed examination of the function of nucleic acids is the focus of Part III of this text.

8.1 Some Basic Definitions and Conventions

The amino acid sequence of every protein in a cell, and the nucleotide sequence of every RNA, is specified by a nucleotide sequence in the cell's DNA. A segment of a DNA molecule that contains the information required for the synthesis of a functional biological product, whether protein or RNA, is referred to as a **gene**. A cell typically has many thousands of genes, and DNA molecules, not surprisingly, tend to be very large. **P1** The storage of biological information and the transmission of that information from one generation to the next are the only known functions of DNA.

RNAs have a broader range of functions, and several classes are found in cells. **Ribosomal RNAs (rRNAs)** are components of ribosomes, the complexes that carry out the synthesis of proteins. **Messenger RNAs (mRNAs)** are intermediaries, carrying information for the synthesis of a protein from one or a few genes to a ribosome. **Transfer RNAs (tRNAs)** are adapter molecules that faithfully translate the information in mRNA into a specific sequence of amino acids. In addition to these major classes, there are many RNAs (**noncoding** or **ncRNAs**) with a wide variety of special functions, described in depth in Part III.

Nucleotides and Nucleic Acids Have Characteristic Bases and Pentoses

A **nucleotide** has three characteristic components: (1) a nitrogenous (nitrogen-containing) base, (2) a pentose, and (3) one or more phosphates (Fig. 8-1). The molecule without a phosphate group is called a **nucleoside**. The nitrogenous bases are derivatives of two parent compounds, **pyrimidine** and **purine**. The bases and pentoses of the common nucleotides are heterocyclic compounds.

KEY CONVENTION The carbon atoms and nitrogen atoms in the parent structures are conventionally numbered to facilitate the naming and identification of the many derivative compounds. The convention for the pentose ring follows rules outlined in Chapter 7, but in the pentoses of nucleotides and nucleosides the carbon numbers are given a prime (') designation to distinguish them from the numbered atoms of the nitrogenous bases. ■

The base of a nucleotide is joined covalently (at N-1 of pyrimidines and N-9 of purines) in an *N*- β -glycosyl

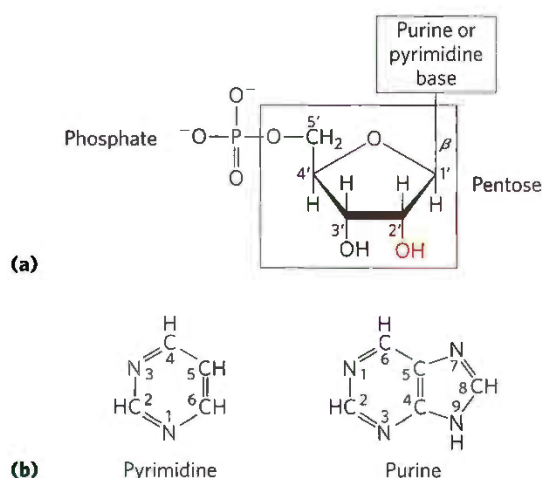


FIGURE 8-1 Structure of nucleotides. (a) General structure showing the numbering convention for the pentose ring. This is a ribonucleotide. In deoxyribonucleotides the —OH group on the 2' carbon (in red) is replaced with —H. (b) The parent compounds of the pyrimidine and purine bases of nucleotides and nucleic acids, showing the numbering conventions.

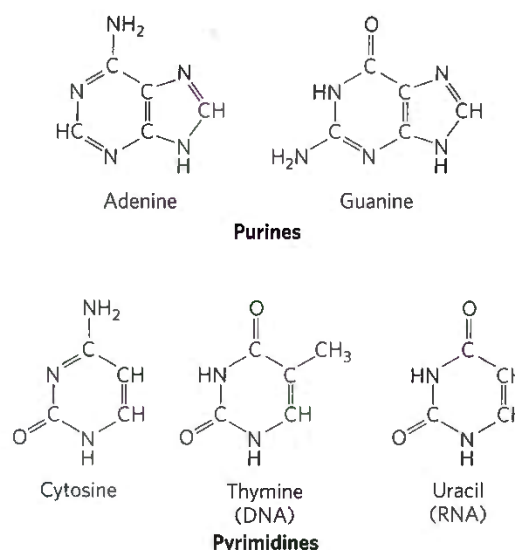


FIGURE 8-2 Major purine and pyrimidine bases of nucleic acids. Some of the common names of these bases reflect the circumstances of their discovery. Guanine, for example, was first isolated from guano (bird manure), and thymine was first isolated from thymus tissue.

bond to the 1' carbon of the pentose, and the phosphate is esterified to the 5' carbon. The *N*- β -glycosyl bond is formed by removal of the elements of water (a hydroxyl group from the pentose and hydrogen from the base), as in *O*-glycosidic bond formation.

Both DNA and RNA contain two major purine bases, **adenine** (A) and **guanine** (G), and two major pyrimidines. In both DNA and RNA, one of the pyrimidines is **cytosine** (C), but the second common pyrimidine is not the same in both: it is **thymine** (T) in DNA and **uracil** (U) in RNA. Only occasionally does thymine occur in RNA or uracil in DNA. The structures of the five major bases are shown in Figure 8-2, and the nomenclature of their corresponding nucleotides and nucleosides is summarized in Table 8-1.

Nucleic acids have two kinds of pentoses. The recurring deoxyribonucleotide units of DNA contain 2'-deoxy-D-ribose, and the ribonucleotide units of RNA contain D-ribose. In nucleotides, both types of pentoses are in their β -furanose (closed five-membered ring) form (Fig. 8-3a). As Figure 8-3b shows, the pentose ring is not planar but occurs in one of a variety of conformations generally described as "puckered."

KEY CONVENTION Although DNA and RNA seem to have two distinguishing features — different pentoses and the presence of uracil in RNA and thymine in DNA — it is the pentoses that uniquely define the identity of a nucleic acid. If the nucleic acid contains 2'-deoxy-D-ribose, it is DNA by definition, even if it contains uracil. Similarly, if the nucleic acid contains D-ribose, it is RNA, regardless of its base composition. The presence of uracil or thymine is not a defining characteristic. ■